A Gauge-Theoretic Framework Toward a Participatory "It From Bit" Program

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Abstract

Wheeler proposed, as a working hypothesis, that every physical quantity derives its significance from registered yes-or-no answers in a universe whose observers participate in what they observe, and he asked how one world arises out of many observer-participants. This paper gives a classical, multi-agent, information-geometric model of four parts of that program. Agents are local sections of a principal bundle at one context point; each holds a probability law in its own frame, and agents compare laws only through explicit transports whose residual relative entropy is the frame-independent part of their disagreement. First, registered change as duration: each agent's clock is the Fisher length of its belief trajectory. Second, observer-dependent geometry: metrics on an index manifold without a physical metric are pulled back from belief sections, a step that uses base derivatives and is developed in a companion supplement. Third, one world out of many: agreement is a stationary point of a consensus free-energy functional whose attention weights are Gibbs distributions over transported divergences. Fourth, participation across scales: exact coarse-graining assembles coherent clusters into meta-agents whose laws act as lagged priors for their members, and the hierarchy closes on a declared root, a fixed top-level prior. A claim-status table separates theorems, derivations, structural correspondences, and conjectures. The model has no irreversible registration and no complementarity, fixes its law in advance, presupposes a continuum and an ordering parameter, and relaxes unless driven by observations; how come the quantum lies outside its scope.

Keywords: Participatory realism · It from bit · Gauge theory · Information geometry · Variational free energy · Multi-agent inference

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Part I: The Construction

1 Introduction

In an essay delivered in 1989 and published in 1990, John Archibald Wheeler returned to "the age-old question, How come existence?" and offered as a working hypothesis that every physical quantity, every "it", derives its significance from "binary yes-or-no indications" elicited by observing devices, in a universe in which observers take part in bringing about what they observe [1, 2]. Among the questions he posed was "How come 'one world' out of many observer-participants?", and he adopted Føllesdal's dictum that meaning is the joint product of all the evidence available to those who communicate [1]. In physics the program has been pursued chiefly as a reconstruction of quantum theory from informational principles. Wheeler himself credited Wootters, whose statistical distance between preparations, defined by the distinguishability of measurement outcomes, coincides up to a constant factor with the Fisher–Rao distance [3]. Zeilinger took as foundational that an elementary system carries one bit [4]; Hardy, Chiribella, D'Ariano and Perinotti, and Masanes and Müller derived the formalism of quantum theory from operational and informational axioms [5–7]; Höhn reconstructed qubit quantum theory from an observer who interrogates a system with binary questions [8]; and Müller derived the appearance of an external world with simple probabilistic laws from observer states alone, under the title "Law without law" [9]. QBism claims Wheeler's participatory program by name as participatory realism [10], relational quantum mechanics relativizes states to arbitrary physical systems [11], and Caticha's entropic program treats space as a statistical manifold and quantum dynamics as entropic inference [12–14]. The present framework is not a reconstruction of quantum theory. It is classical and multi-agent, and it supplies two objects those programs do not address: an explicit law of transport between the reference frames of different agents, and a consensus functional whose stationary points model agreement among them.

Two further lines bear on the construction. The Kant–Helmholtz line reads perception as structured by the perceiver and as unconscious statistical inference [15, 16], and its modern descendant, the free-energy principle, derives perception, learning, and action from one variational free-energy functional [17–19]. For a physicist, a variational free energy is the negative evidence lower bound of a Bayesian model, the Gibbs–Bogoliubov functional minimized in a mean-field approximation, and the framework uses the term only in that technical sense; the consensus functional of Section 5 has the energy-minus-entropy form of a free energy but is a declared composite potential, not such a bound. Independently, spacetime has been argued to be emergent from thermodynamic, holographic, and entanglement structure [20–26]. The framework sits between these readings. Agents with different beliefs pull back different geometries from one shared index space, a presupposed smooth index manifold $\mathcal{C}$ of contexts carrying no physical metric, called noumenal by analogy only (Section 4), and no agent-independent geometry is singled out. The commitment concerns the structure of the map from this index space to what each agent represents; it does not stipulate that anything beyond $\mathcal{C}$ is inaccessible.

The organizing claim of the paper is a correspondence with Wheeler's program, stated together with its limits. Agents are local sections of a principal G-bundle at one base point c_* of $\mathcal{C}$, each carrying a belief q_i, a prior p_i, a frame $\phi_i \in \mathfrak{gl}(K_q)$, and a slow generative model s_i with hyperprior r_i (Section 4); they are coupled through the transport $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$, which acts on covariances by $\Sigma \mapsto \Omega_{ij}\Sigma\Omega_{ij}^{\top}$, and through the consensus free-energy functional (Section 5). On this construction the paper gives a classical, multi-agent, information-geometric model of four parts of Wheeler's program, labeled W1–W4 in Table 1. Observer-relative description, the comparison

of laws only through an explicit transport whose residual divergence is the frame-independent part of a disagreement, underlies all four and is what makes the geometry of W2 observer-dependent. Derived duration and geometry are W1 and W2: the Fisher length of each agent's belief trajectory, and the metrics pulled back to $\mathcal{C}$ from belief sections, a step that uses base derivatives and so leaves the single base point (Section 18). One world out of many observer-participants is W3: intersubjective agreement as a stationary point of the consensus functional, a formal counterpart of Wheeler's use of Føllesdal's dictum. Participation across scales, W4, is modeled by one mechanism: an exact coarse-graining assembles a coherent cluster of level-ℓ agents into a level-$(\ell + 1)$ meta-agent, whose law returns to its members as a lagged predictive prior, and the hierarchy is closed by a declared root, a fixed top-level prior acting as its boundary condition, together with an energy-level apex for the belief channel that pools current beliefs and is not a generative factor (Sections 9–12). The coarse-graining is an exact block-variable pushforward in Kadanoff's sense [27–29] followed by a truncation. It is not a renormalization-group transformation: it involves no rescaling, no flow on a fixed space of couplings, and no fixed point, and on a finite network no typed comparison of interaction spaces across a reducing step exists [30, Ch. 11]. Table 1 lists the five parts W1–W5 of the correspondence with their status, and Table 2 records the status of every load-bearing claim.

The correspondence keeps four of Wheeler's commitments and drops four. It keeps the observer-relativity of description, the formation of one shared world out of many participants, the derivation of metric and duration from informational structure, and the participation of collectives in the priors of their members. It drops, first, the elementary act of registration. Beliefs here are continuous Gaussian laws, and the relative entropy of an update, measured in bits, is a unit of measure that is not additive under refinement of the update path. Section 17 adds a binary registration channel whose expected information gain is at most one bit per registration and adds over conditionally independent registrations, which gives the word bit its physical meaning; even so, the construction contains no irreversible amplification and no complementarity, and its agents receive observations without choosing which question to ask, so "no question, no answer" is not represented. It drops, second, "no continuum": the index manifold $\mathcal{C}$ and the fibers are smooth and presupposed, and the regress that Wheeler answered with a loop is stopped here by an index floor and a declared root, a deliberate departure whose formal residue is discussed in Section 12. It drops, third, "law without law": the functional, the belief family, and the divergence are fixed in advance, and Müller's observer-first program [9] is closer to Wheeler on this point. It drops, fourth, the self-excited circuit. The autonomous dynamics relax to stationary points, observations supply the only persistent drive, and the geometry induced on $\mathcal{C}$ enters no term of the free energy, so the loop from information back to the participants closes only in the space of beliefs; closing it through the induced geometry requires base transport and is deferred to the companion supplement [31]. How come the quantum is outside the scope of the construction.

Interpretive stance. The author's interpretive stance carries status S throughout, and no equation of the paper depends on it. It is a cognitive-first structural monism: the primitives of the construction are the agents' beliefs and generative models, and physics-like structure appears as a derived pullback over a non-cognitive index space $\mathcal{C}$, which plays the structural role of Whitehead's extensive continuum [32]. The stance inverts the Primacy of Physics Constraint of Ladyman and Ross's naturalized metaphysics [33] while sharing the relational emphasis of ontic structural realism [34] (Section 14). It makes four commitments. Every system at every level carries frame structure and takes part in variational dynamics on a statistical manifold suited to its level: a multivariate Gaussian family on $\mathbb{R}^{K_q}$ in the classical construction developed here, and, at level zero, the manifold of density operators with the Umegaki quantum relative entropy as divergence, whose

Table 1: The correspondence between strands of Wheeler's participatory program and the construction. Status codes are those of Section 2; "not claimed" marks what the construction does not deliver.

	Wheeler's strand	Formal counterpart here	Where	Status
W1	Registered change as the measure of duration	Fisher-length clock of each agent; exact clock–dissipation identity along an autonomous single-rate natural-gradient flow; binary registration channel whose expected information gain is at most one bit per registration; identification of the clock with time is postulate P1	Section 17	P (identity), D (registration), S (reading as time)
W2	Observer-dependent "it"	Pullback of the Fisher metric along belief sections; the covariant Fisher block is gauge invariant, so differences between agents come from differences of belief	Section 18	D (pullback), P (invariance), S (reading as space)
W3	One world out of many observer-participants	Stationary points of the consensus free energy; Gibbs attention rows with reduced value $-\tau \log Z$; consensus geometry, whose orbit average needs a regulator only on the frame-twist block	Sections 5 and 20	P (rows, regulator obstruction), S (reading as one world), C (regulated metric)
W4	Participation across scales; no tower of turtles	Exact coarse-graining into meta-agents; lagged predictive priors from parents to members; a declared root closing the hierarchy	Sections 9–12	P (coarse-graining, root), D (lagged prior), C (formation by the agents' own inference)
W5	What is not delivered	Discrete irreversible registration with complementarity; law without law; no continuum; a self-excited circuit; how come the quantum	Sections 2.1 and 24	not claimed

construction is open (Section 24). All levels share the transport structure of a principal bundle with a structure group suited to the level [35], $\mathrm{GL}(K_q, \mathbb{R})$ in the classical construction and $U(\mathcal{H})$ at level zero. The base $\mathcal{C}$ is an index manifold of contexts, not a physical thing. Agents at level zero are ascribed frame structure and variational dynamics, not phenomenal properties, and the combination problem this incurs is owned as open (Section 22). Two remarks fix the status of these commitments. First, the commitments to universal agency and universal transport exclude no system: any dynamics can be given a trivial bundle, trivial frames, and a residual functional of which it is the minimizer, so, as with Kretschmann's objection to general covariance [36, 37], they carry no empirical content by themselves. Second, at the level of kernels the commitment to universal agency is free: observations can always be presented as messages from boundary agents without changing any law, although that presentation removes no conditioning on the record [30, Ch. 5]; what the stance adds, a criterion of agency for every node, is not supplied by the mathematics. On this reading an electron would be an agent at level zero, a molecule a meta-agent assembled from atomic constituents, and a brain a meta-agent at a higher level; against the panpsychism taxonomy

the position is panprotopsychist and constitutive (Section 22). The vocabularies of Wheeler, Kant, and Lahav and Neemeh are borrowed with the departures listed above and in Section 2.1. Symbol conventions, including the few deliberate overloadings, are collected in Appendix J.

2 Claim Status and Scope

Every load-bearing claim carries one of five status codes, defined once here. P (proven) marks a claim proved in this paper or entailed by a cited result. D (derived) marks a result derived here under stated assumptions, such as a Gaussian family, a fixed schedule, or a fixed attention row. S (structural) marks a structural correspondence or an interpretation: a mathematical object is read as the formal counterpart of a physical or philosophical notion, and the reading adds no theorem. E (empirical) marks a measured result; no claim of the present paper has this status, and its multi-scale claims are tested by the companion's pairing-selection conjecture and by the prospective protocol of Section 13, its finite-sample counterpart. C (conjectured) marks a conjecture or an open problem. The numbered inputs P1–P10 of Part II are postulates, registered in Section 16; a postulate is adopted, not derived, and a result built on one inherits its status. Part I declares one further postulate outside this register, the kinetic metric of Section 8.7; no other result uses it, and the inertial reading of precision it expresses has status S. The letter P as a status code and the labels P1–P10 are distinct. Part I is the construction: its results carry status P or D, and its interpretive readings are marked S. Part II is the speculative program: its physical readings carry status S or C and rest on the postulates, while the mathematical results it uses (the clock–dissipation identity, the registration proposition, the pullback and its invariance, the signature constructions) carry status P or D and do not depend on those readings. Table 2 gives each load-bearing claim with its location, its status, and what would change that status. The body states a claim's status at its first statement and otherwise refers to the table.

Table 2: Status of the load-bearing claims. "Theory" denotes the companion manuscript [30] and "Supp. B" its base-transport supplement [31].

Claim	Where	Status	What would change it
Part I: the construction			
$D_{\mathrm{KL}}(q_i \Vert \Omega_{ij} q_j)$ is invariant under a common $\mathrm{GL}(K_q)$ pushforward; in Regime I every frame can be removed from the alignment, attention, and cross-scale terms at one base point	Theorem 1; Proposition 1	P	none (theorem)
At one base point the active symmetry is one global copy of G; per-agent frames are passive re-expressions	Section 4.6.2; Theory Ch. 2	P	none
Regime I cocycle closure and the lattice Bianchi identity with orientation reversal, read as parallels to "the boundary of a boundary is zero"	Section 4.6.5	P (identities), S (parallel)	a conservation law derived from them
The attention row, a softmax over transported divergences, is the Gibbs optimum of the entropy-regularized consensus problem, with reduced value $-\tau \log Z$ and the envelope gradient	Section 5	P	none

Claim	Where	Status	What would change it
On an open mean-field family, live pairwise divergences cannot be the negative evidence bound of one fixed state-level joint; the consensus functional is a declared composite potential	Section 5; Appendix G; Theory Chs. 6 and 12	P	none; the functional stays a declared ansatz
Among real-analytic f-divergences, the relative entropy with the receiver law in the first slot is the unique one giving the geometric-mean best response to frozen neighbors	Section 5.2; Appendix F	P (best response only)	extension to the full functional fails
Observations as couplings to environmental agents	Section 6; Theory Ch. 5	D (Gaussian, full-rank linear readout); P (kernel level)	none
The functional is invariant under the change of reference $U_i \mapsto U_i g$ and, without observations, under the global belief rotation; a fixed observation kernel explicitly breaks the global belief-rotation symmetry	Section 6.5	P (invariances), D (breaking)	none
Adaptive precision as a declared penalty	Section 6	D	a normalized hyperprior
Adiabatic separation of fast beliefs and slow models	Section 7	C	verified Tikhonov–Fenichel hypotheses
The precision Hessian is a stiffness and the Fisher metric the friction tensor of the first-order natural-gradient flow	Section 8.6	D	none
Kinetic-metric postulate $M_K = M$, with its own second-order equation of motion, under which every mode has unit frequency	Section 8.7	S (declared postulate, not among P1–P10)	a second-order dynamics derived from the functional
Moment-projection barycenter: closed form and M-step reading	Section 9	P	none
The frame Karcher mean on $\mathrm{SO}(K_q)$ is unique for rotation angles below $\pi/2$	Section 9	P	none
The parent frame is not identifiable, and no left-equivariant permutation-symmetric map $\mathrm{GL}^+(K_q)^n \to \mathrm{GL}^+(K_q)$ exists	Section 9	P	none
Licensing criterion, degenerate under exact parents (Theory Ch. 9); threshold detector; meta-agents read as belief-bearing agents	Sections 9.2 and 9.3	C (criterion), S (detector, reading)	a derivation of the detector from the criterion
Contraction of the meta-agent weight map at rate $O(\varepsilon)$	Lemma 2	D (sketch)	a complete proof

Claim	Where	Status	What would change it
The cross-scale shadow is the one-episode-lagged predictive prior of the episode-indexed joint under a transported-copy transition; the same-time form is an energy-level simplification and not a message of the augmented joint	Section 10; Theory Ch. 12	D (lagged), S (same-time)	none
Meta-agent pullback geometry is dominated by the constituents' wherever the parent frame introduces no horizontal defect	Section 10; Supp. B Ch. B2	P (conditional)	none; base-direction work deferred
Mean-field solution of the augmented Gaussian joint	Appendix G	P	a proof that it governs the shadow dynamics
Exact coarse-graining: a block-variable pushforward and a truncation, with no rescaling, flow, or fixed point	Section 11; Appendix I; Theory Chs. 9 and 11	P	a comparison map between coupling spaces
A declared root closes the hierarchy, and the apex is an energy-level closure; the root read as the formal residue of "no tower of turtles"	Section 12; Theory Ch. 12	P (root), S (reading)	none
The apex loop has the feedback structure of Wheeler's self-excited circuit, without its self-excitation (W5)	Section 12	S	sustained non-stationary dynamics without exogenous input
Observation injection sustains non-stationarity; without it the flow relaxes	Section 12	C (sustained), P (relaxation)	a proof that observation injection sustains non-stationarity, or a counterexample
Collectives are constituted by their members' inference: pairing selection by slow learning of the attention prior	Section 13; Theory Ch. 14	C	proof or counterexample
Interpretive stance (cognitive-first structural monism)	Section 1	S	none; not a result
Part II: the speculative program			
Clock–dissipation identity along an autonomous single-rate natural-gradient flow	Section 17; Theory Ch. 7	P	none; a driven flow adds $\partial_t \mathcal{F}$ (Eq. (138))
Binary registration: $\mathbb{E}_O[D_{\mathrm{KL}}(p(x\|O)\|p(x))] = I(X;O) \leq H(O) \leq 1$ bit, additive over conditionally independent registrations	Section 17	D	none
The information clock as the agent's time; the ordering parameter is presupposed	Section 17	S (postulate P1)	a calibration against a physical clock
Meta-clock ratio in the relative-entropy currency; a meta-agent accumulates no more Fisher duration than the collective of its constituents; the arc-length ratio	Section 17; Theory Ch. 7	D (ratio), P (contraction), C (arc length)	a proof of the arc-length ratio

Claim	Where	Status	What would change it
The Fisher pullback tensor and its rank; gauge invariance of the covariant Fisher block	Section 18; Supp. B Ch. B2	D (pullback), P (invariance)	none
The pullback read as a formal counterpart of "it from bit"	Section 18	S	the topology or dimension of $\mathcal{C}$ derived from registrations
All-real frame-twist form of signature $(-,+,+,+)$, exhibited under P4; in Regime I a property of a declared reference section with no gauge-invariant content	Section 19	P (under P4); C (selection)	a dynamical selection mechanism
Orbit-averaged consensus metric; its regulator obstruction concerns the frame-twist block and the continuum and noncompact limits	Section 20; Proposition 11	P (obstruction), C (regulated metric)	a constructed regulator
Physical correspondences of Table 3; gravity as curvature of an induced geometry, a placeholder	Sections 20.4 and 21	S	a field equation with a source
Consciousness as frame-dependent presentation	Section 22	S, C	none; interpretive
Open problems C1–C5 are mathematical problems and C6 a derivation target; the physical readings remain interpretive	Section 24	C	solutions of C1–C6
Appendices			
High-precision agent (the rock): the mean back-action is suppressed, while the stationary precision is reduced by a factor tending to $1/(1+\beta)$; the rock read as an observed object	Appendix B	D; S (reading)	none
Precision-linear covariance fixed point: exists if and only if $\rho(L) < 2$, and is then globally stable under the receiver-only flow; its reading as a population equilibrium reproducing the shared prior reflects the doubled entropy weight	Appendix D	P; S (reading)	none
The frozen mean map is a contraction for isotropic couplings	Appendix G	P (isotropic)	fails for anisotropic couplings
Nested state–configuration bound, the two-level case of the Theory's multiscale bound; the one-particle-irreducible and two-particle-irreducible fluctuation program	Appendix H; Theory Ch. 12	P (bound), C (program)	a constructed effective action
Gibbs lift of a normalizable consensus energy to belief configurations	Appendix I	P	none
Covariance of the coarse-graining map under one global g at one base point, extending to left-invariant metrics; mean-sector Schur lemma for mean-only linear couplings	Appendix I.1	P	none

Claim	Where	Status	What would change it
One-loop shift $-\frac{1}{2}T_{\mathrm{cfg}}(\lvert I\rvert-1)\log\det\Sigma_I$ of the parent's entropy weight; force residual measured against the zero-temperature profile, a finite-temperature version failing	Appendix I	D	none
Schematic residual bound for the truncation	Appendix I.1	C	pinned constants and tightness
Detector retention, conditional on a positive saving term A_I whose source is not identified; the body detector is not covered	Appendix I.1	P (conditional)	a source for A_I

2.1 Scope and Limitations

The claims of the paper are delimited as follows. The construction supplies formal counterparts, in the sense of Table 1, for strands of Wheeler's participatory universe, of Kant's distinction between appearance and what appears, of observer-dependent description, and of the Lahav–Neemeh proposal that consciousness is frame-relative. We do not derive quantum mechanics or general relativity. The signature construction of Section 19 exhibits a Lorentzian frame-twist form under postulate P4 but selects nothing, and the orbit-averaged consensus metric of Section 20 is not the only gauge-invariant geometric candidate: the Fisher block of the pullback, computed with each agent's own covariant derivative, is gauge invariant without any orbit average [31, Ch. B2], so the orbit average and its regulator are needed only for the frame-twist block.

Earlier simulations of the meta-agent hierarchy are not used as evidence. Section 13 states a prospective test, which requires an active slow model channel and a control arm in which the attention prior is frozen. The control is needed because, on the four-agent cycle of the companion theory, a collective forms from every Gaussian initial law whether or not the attention prior learns, the random initial source assignment seeding the pair [30, Ch. 14]; the occurrence of formation therefore does not by itself distinguish participation from seeding. Whether the discrete detector of Section 9.3 and the continuous variational criterion, from which the detector is not derived, produce the same organization remains open.

Several further limitations each receive one statement. The framework yields no quantitative physical predictions: it produces no values of physical constants, masses, couplings, or dimensionless ratios, and its induced tensors are dimensionless, so a map from information-geometric quantities to physical units is missing. There is no quantum extension: the framework contains no Hilbert space, no Born rule, and no measurement postulate, so its parallel with QBism and relational quantum mechanics is one of stance (Section 3), and the density-operator construction at level zero is open (Section 24). The objectivity the framework delivers is agreement between the pullback geometries of agents that share a belief family, not access to an agent-independent geometry (Section 20). The discussion of consciousness relocates the hard problem without solving or dissolving it (Section 22).

Testability is tiered. Mathematical claims internal to the framework are settled by proof or counterexample. The participatory reading of the formalism, that collectives are constituted by their members' inference and are not supplied, has one decidable test in the companion theory: a conjecture that slow learning of the attention prior selects which pair of agents on a homogeneous four-agent cycle forms a collective, at the level of the law, against a frozen-prior control whose pairing law stays symmetric [30, Ch. 14]. The conjecture is settled by proof or counterexample, not by observation, and its failure would reduce that reading to hierarchical modeling in that family.

The participatory-physics reading of Part II has no identified falsifying observation and may be unfalsifiable as currently stated (Section 23). The open problems are collected in Section 24.

2.2 Relation to the Companion Theory

A companion manuscript [30] develops the single-base-point core of this program with normalized laws throughout: one collective evidence bound for the agent population [30, Ch. 5], attention as inference over a declared source label [30, Ch. 6], Fisher-length information clocks [30, Ch. 7], exact Markov coarse-graining of the agent network into meta-agents [30, Chs. 9 and 11], the multiscale free energy with its episode-indexed joint and declared root [30, Ch. 12], and formation theorems for declared families [30, Ch. 13]. Its supplements treat the multivariate-Gaussian case [38] and base transport, including the connection-relative pullback geometry behind Section 18 [31]. Where the companion supersedes a construction used here, namely the consensus functional, the barycenter and detector, the same-time cross-scale shadows, and the self-closing apex, we say so at the construction. The companion declines every physical identification made in Part II [30, Chs. 1 and 14]: it identifies no Fisher duration with physical time, gives the context base no spacetime or Lorentzian structure, treats calling a boundary node an agent as a naming decision, and does not read the free energy as a physical action. Declining a claim is not refuting it. Part II adopts these identifications as its postulates, so the two documents agree on the mathematics and differ only in what is postulated here. The companion's vocabulary translates as follows: a recognition law is a variational posterior, the source prior is the attention prior π_{ij}, a declared root is a top-level prior fixed as a boundary condition, and a Fisher duration is the information clock.

3 Related Work

Several programs connect information with spacetime geometry. Jacobson derived Einstein's field equations from the thermodynamics of local causal horizons [20], Padmanabhan developed complementary thermodynamic formulations [21], and Verlinde derived gravity from entropic forces on holographic screens [39]. The present framework shares their information-theoretic starting point, but it works with the classical probability laws of many agents, not with horizon or screen entropies, and it derives no gravitational dynamics. Tensor-network proposals read the geometry of the AdS/CFT correspondence off entanglement [26], and quantum-information approaches recover spatial geometry from the entanglement structure of Hilbert space [40]; the present construction shares their strategy of reading geometry off informational structure and implements it through classical information geometry and gauge transport. Constructor theory reformulates information in physical terms, characterizing information media by which transformations are possible, and rejects information as an a priori mathematical primitive [41]. The closest physics precedents for individual ingredients are the following. The information clock of Section 17 is the natural-gradient analog of thermodynamic length, which bounds dissipation and is tied to Fisher information [42, 43], and its distinguishability unit is Wootters' statistical distance [3]. Caticha's proposals that space, and spacetime, carry the information geometry of a statistical manifold [12, 13] anticipate the pullback of Section 18, and the finding that an anti-de Sitter metric is a generic feature of Fisher geometries with symmetry [44] cautions against over-reading the hyperbolic Gaussian metric. Quantum Darwinism explains the objectivity of classical records by their redundant proliferation among many observers [45]; it is the quantum precedent for the consensus mechanism of Section 20. A free-energy principle for generic quantum systems [46] is the closest published attempt at the level-zero extension listed as open in Section 24.

A second line is gauge-theoretic free energy and Bayesian mechanics. Sengupta, Tozzi, Cooray, Douglas, and Friston recast the variational free-energy principle in gauge-theoretic language, with prediction errors acting as local forces and precision weighting related to gauge fields [47]. It is the closest precursor on this axis; the present construction differs in supplying explicit section fields $\phi_i \in \mathfrak{gl}(K_q)$, the inter-agent transport Ω_{ij}, its action on covariances, and a multi-agent free energy coupled through relative entropies. Geometric Bayesian mechanics extends the free-energy principle to systems bounded by Markov blankets, sets of variables that screen internal from external states [48–52], and it is the precedent for the multi-level blanket reading of agents in Section 4. We adopt such blankets at every level and do not derive them; whether they exist across arbitrary scales is open, since the partition can depend on the coarse-graining. Bruineberg et al. distinguish an epistemic Pearl blanket from an ontological Friston blanket [53]; the blanket here is indexed by the observer by design, which is the Pearl reading. Aguilera et al. show in linear stochastic systems that the Bayesian-mechanics reading holds only at weak solenoidal coupling [54], while the couplings β_{ij} and γ_{ij} here are dense and irreversible, so the belief coupling is motivated as a declared consensus objective (Section 5.2). Biehl et al. show that foundational arguments equivocate between blanket definitions [55]; we anchor the construction to one, the per-agent support χ_i on the domain $\mathcal{U}_i$, under which the statement that internal states parametrize beliefs is a modeling stipulation. Friston and Frith treat the synchronization of beliefs between communicating agents without a law of transport between their frames [56]. Consensus coupled through edge maps on graphs is studied as the connection Laplacian [57] and as sheaf-Laplacian opinion dynamics [58], and gauge-equivariant networks impose frame covariance on learned features [59]; the present construction differs in coupling probability laws through a relative-entropy functional with state-dependent attention weights.

A third line takes the observer as primary. Relational quantum mechanics holds that quantum states and the values of variables are relative to other physical systems, any system serving as an observer, and describes only the information that systems have about each other [11]. With independent links on the interaction graph, network holonomy in the present construction obstructs a global frame, a belief-level counterpart of the question of how the perspectives of relational quantum mechanics are linked [60]. QBism keeps outcomes personal and needs no consensus [61, 62]; Fuchs names Wheeler's "law without law" as its inspiration and calls the resulting view participatory realism [10]. The present framework shares QBism's personalist starting point but not its quantum content, and it models Wheeler's third question, which QBism leaves to communication between agents, by consensus as a stationary point. Lahav and Neemeh propose a relativistic theory of consciousness in which phenomenal experience is frame-relative, and they write the transformation between cognitive frames for functionally identical systems [63, 64]; the frame fields ϕ_i and transports Ω_{ij} supply an explicit map between frames of different agents at the level of belief transport (Section 22.1).

4 One Rung: Statistical Manifolds, Bundles, Gauge Transport, and Working Simplifications

This section assembles the geometry of a single level, the unit on which the hierarchical coarse-graining of Sections 9–12 operates. We build it in one linear pass: the base manifold of contexts, the statistical fibers that carry beliefs and models, the principal and associated bundles that organize the frame freedom, agents as smooth sections, the transport operator that compares them, the three group actions that the construction distinguishes, the connection forms and curvature, the independent links on the interaction graph, and the working simplifications. At one base point

these constructions are the Gaussian, $\mathrm{GL}^+(K_q)$ specialization of those of the companion Theory [30, Ch. 2].

A convention fixed once: Regime I versus Regime II. Two regimes for the links between agents govern every later statement about holonomy. In *Regime I* the links are frame aligned: $\Omega_{ij} = U_i U_j^{-1}$ is built from the agents' frames $U_i = \exp \phi_i$, the links form a cocycle, and every loop product is the identity (Lemma 1). Along the base the frame gradient $A_\mu^{(i)}(c) = U_i^{-1}(c)\partial_\mu U_i(c)$ is pure gauge, and its curvature $F_{\mu\nu}^{(i)}$ vanishes identically by the Maurer-Cartan identity (Section 4.6.3). In *Regime II* the links are independent variables on the oriented edges of the interaction graph, subject to orientation reversal $\Omega_{ji} = \Omega_{ij}^{-1}$ (Section 4.6.5); their loop products are graph holonomies, which can be nontrivial when every agent sits at a single base point. Graph links are not base transport, and graph-link holonomy is neither the holonomy of a base connection nor a property of the topology of the principal bundle [30, Ch. 1]. An independent connection on $\mathcal{C}$, with path-ordered transport along base curves and a nonzero field strength, is a separate extension that belongs to base-transport theory and is deferred to the companion Supplement B [31]. All constructions of Part I use Regime I. The worked signature example of Section 19 also lives in Regime I; its frame-twist form $\mathrm{tr}(A_\mu A_\nu)$ is not a curvature invariant (Section 4.3). Each Regime II passage is marked where it appears.

4.1 The Base Manifold: Parameter / Index Space of Contexts

Agents form beliefs about something, and we call the index domain of those beliefs the base manifold $\mathcal{C}$: the parameter or index space of contexts to which agent fields are referred. The base manifold is presupposed. It is a smooth index manifold carrying no physical metric, called noumenal only by analogy with Kant's things-in-themselves, and it plays a role loosely analogous to Whitehead's extensive continuum [32]. It is neither derived from agentic structure nor an observer-independent physical substrate; in the language of Section 1, $\mathcal{C}$ is a domain of inquiry, not a thing.

Definition 1 (Base Manifold)**.** *The base space $\mathcal{C}$ is a smooth n-dimensional manifold serving as the parameter / index domain over which agent sections (beliefs, priors, gauge frames, generative models, hyper-priors) are defined.*

The base manifold $\mathcal{C}$ is not physical spacetime. Metric structure available to agents enters only through pullbacks of belief sections (Section 18.1), which give agent-relative metric data on a given base; the pullback construction is derived here, and any spacetime reading of it is a structural correspondence (Table 2). The topology, dimension, and smooth structure of $\mathcal{C}$ are inputs. This is a deliberate departure from Wheeler, who answers the regress of structure beneath structure with a loop and counts "no continuum" among his four no's [1]: the construction stops the regress with a presupposed index manifold and keeps a continuum index space (W5). We use local coordinates $c = (c^1, \ldots, c^n)$ on $\mathcal{C}$. The background structures that the construction does place on $\mathcal{C}$, a chart density for integration and a fiducial chart metric, are declared with the working simplifications (Section 4.9).

4.2 Statistical Manifolds: Beliefs and Models

At each point $c \in \mathcal{C}$, agents maintain probability distributions on two information-geometric arenas. The first holds distributions over the latent state k that an agent is trying to infer; the second holds distributions over the parameters m of the agent's generative model. Beliefs and the priors that anchor them live on the first arena, while generative models and their hyper-priors live on the second.

Beliefs q_i are fast variables and generative models s_i are slow variables in the two-timescale typing of the companion Theory [30, Ch. 8]. Priors p_i and hyper-priors r_i are not independent dynamical variables: they are supplied by the level immediately above as one-episode-lagged predictive priors (Section 10).

Definition 2 (Statistical Manifolds)**.** *Let*

$$\mathcal{B}_{\text{state}} = \{\mathcal{N}(\mu, \Sigma) : \mu \in \mathbb{R}^{K_q}, \Sigma \succ 0\}, \qquad \mathcal{B}_{\text{model}} = \{\mathcal{N}(\mu, \Sigma) : \mu \in \mathbb{R}^{K_m}, \Sigma \succ 0\} \tag{1}$$

denote the finite-dimensional manifolds of Gaussian distributions over the latent state $k \in \mathbb{R}^{K_q}$ and over the generative-model parameters $m \in \mathbb{R}^{K_m}$, respectively. Each is equipped with the Fisher-Rao metric, given in the coordinates $\theta = (\mu, \Sigma)$ by

$$g_{\mathcal{B}}(\partial_a, \partial_b) = \mathbb{E}\left[(\partial_a \log \rho_\theta)(\partial_b \log \rho_\theta)\right], \tag{2}$$

with ρ_θ the corresponding Gaussian density and indices a, b ranging over the coordinates (μ, Σ).

The same coordinate formula extends formally to any smooth finite-dimensional parametric family with square-integrable score ($\mathbb{E}_{\rho_\theta}[(\partial_a \log \rho_\theta)^2] < \infty$ for each coordinate, the resulting Gram matrix positive-definite); a rigorous extension to the full nonparametric space of distributions requires additional structure we do not need, since the fibers $\mathcal{B}_{\text{state}}$ and $\mathcal{B}_{\text{model}}$ are Gaussian and the Fisher-Rao metric is evaluated only on them. The Gaussian form of the two fibers is a working choice (Section 4.9), and the construction extends to an exponential-family fiber once a family-preserving pushforward action and a computable divergence are supplied. Non-Gaussian objects appear elsewhere as auxiliary constructions, including the finite Gaussian mixtures of the mixture-of-sources energy (Eq. (35)) and of the moment-matched coarse-graining covariance (Section 9.4), and the categorical attention distributions $\beta(z)$; none of these carries the fibers' Fisher-Rao structure.

Beliefs q_i and priors p_i both take values in $\mathcal{B}_{\text{state}}$, while generative models s_i and hyper-priors r_i both take values in $\mathcal{B}_{\text{model}}$. The two manifolds are generically distinct because the latent state k and the model parameters m are different objects with different sample spaces, and they have dimensions $K_q(K_q + 3)/2$ and $K_m(K_m + 3)/2$.

The Fisher-Rao metric measures how statistically distinguishable nearby distributions are: distributions with sharper peaks lie further apart. It is unique up to scaling as the Riemannian metric on probability spaces invariant under sufficient statistics; Chentsov's original theorem [65] covers finite sample spaces, and the continuous-parametric extension relevant to the Gaussian families used here is supplied by Ay, Jost, Lê, and Schwachhöfer [66]. The relative entropy between two distributions q and p on the same manifold,

$$D_{\text{KL}}(q\|p) = \int q(x) \log \frac{q(x)}{p(x)} dx \geq 0, \tag{3}$$

is not symmetric, $D_{\text{KL}}(q\|p) \neq D_{\text{KL}}(p\|q)$, but its second-order expansion recovers the Fisher metric, $D_{\text{KL}}(\rho_\theta \| \rho_{\theta+d\theta}) = \frac{1}{2} g_{ab} d\theta^a d\theta^b + O(\|d\theta\|^3)$, and the natural gradient $\tilde{\nabla}_q f = g_{\mathcal{B}}^{-1} \nabla_q f$ gives the steepest-descent direction in this metric. For Gaussian distributions the divergence has the closed form

$$D_{\text{KL}}(\mathcal{N}(\mu_1, \Sigma_1) \| \mathcal{N}(\mu_2, \Sigma_2)) = \frac{1}{2}\left[\log \frac{|\Sigma_2|}{|\Sigma_1|} + \text{tr}(\Sigma_2^{-1}\Sigma_1) + (\mu_2 - \mu_1)^\top \Sigma_2^{-1}(\mu_2 - \mu_1) - K_q\right], \tag{4}$$

used throughout for belief and prior alignment. The Fisher information matrix of the Gaussian fiber is recorded in Appendix C, and the fixed-point analysis of the covariance dynamics is in Appendix D.

4.3 Principal Bundles and Gauge Freedom

Different agents use different internal reference frames when representing beliefs and models. A physicist measuring particle spin uses basis vectors in a laboratory frame; another physicist with a rotated laboratory measures the same physics but obtains different numerical values, and the two descriptions are related by a transformation. Agents likewise represent beliefs in internal coordinate systems, which we call their gauge frames. The framework uses these frames in two roles that must be kept separate.

Two roles for the gauge frame. The symbol ϕ_i does double duty, and conflating the two roles produces apparent contradictions, in which a frame called pure gauge in one sentence carries content in another. In its *transport role* (Role A) the frame is redundant. The functional depends on frames only through the links Ω_{ij}, which are unchanged by a common right translation $U_i \mapsto U_i g$ (Section 4.6.2), and in Regime I every frame can be removed from the interaction terms by the change of variables of Proposition 1; an isolated frame carries no physical content. In its *relational role* (Role B) the frame matters only relative to fields held in a fixed frame: an observation channel, a fixed-frame environmental agent (Section 6.5), or a frozen prior. In Regime I at a single base point this exhausts the content of a frame, since every interaction term depends on (q_i, U_i) only through the frame-stripped law $\tilde{q}_i = (U_i^{-1})_{\#} q_i$ (Proposition 1). Quantities built from base derivatives of the frames, such as the frame-twist form $\mathrm{tr}(A_\mu^{(i)} A_\nu^{(i)})$ and the ordinary fixed-frame pullbacks of Section 18.1, depend on the frames relative to the fixed reference section (next paragraph). The interpretive readings of Section 22 use Role B and carry status S. We reserve gauge invariant for quantities unchanged under passive re-expression and change of reference (Section 4.6.2), gauge covariant for quantities transforming by a fixed representation, and frame dependent for quantities that depend on the choice of reference section.

Precedents and the frame-twist form. The Role-A / Role-B treatment of ϕ_i has precedents in gauge theory and in quantum reference frames, cited as motivation and not as constructions reproduced here. Edge modes in gauge theory [67] promote would-be pure-gauge boundary data to physical degrees of freedom when a region's interior is treated as a subsystem. Quantum reference frames [68–70] formalize this for finite quantum systems: a reference frame is gauge from a perspective-neutral viewpoint but a physical system carrying state from its own perspective, and Rovelli's relational interpretation [11] sits in the same family. In each case the frame becomes physical when it is referred to something held fixed, a boundary or a reference system; in the present construction the analogous fixed structures are the observation channel and the fixed-frame environmental agents, which is Role B. We do not construct the corresponding structures, since there is no boundary symplectic form, no conjugate charge, and no theorem forcing Role-B data from a subsystem decomposition. The closer precedent in the physics literature is the dressing-field method [71], discussed after Proposition 1. What the framework does state is which transformation acts on which quantity. The frame-twist form $\mathrm{tr}(A_\mu^{(i)} A_\nu^{(i)})$, with $A^{(i)} = U_i^{-1} dU_i$, is invariant under constant per-agent actions: a constant right translation sends $A \mapsto g^{-1} A g$, which leaves the trace unchanged by cyclicity, and a constant left translation leaves A unchanged. It is not invariant under the local change of reference $U_i(c) \mapsto U_i(c) g(c)$, applied to all agents with one base-dependent g, under which

$$A^{(i)} \longmapsto g^{-1} A^{(i)} g + g^{-1} dg. \tag{5}$$

That transformation leaves every link $\Omega_{ij}(c) = U_i(c) U_j(c)^{-1}$ unchanged, and the functional contains no frame-gradient term (Section 7.4), so the functional is unchanged as well; the choice $g = U_1^{-1}$ sets

the frame-twist form of agent 1 to zero at every point. The frame-twist form therefore measures agent i's frame relative to the fixed reference section. It is a property of a declared reference section, not an observable, and the signature construction of Section 19 and the ordinary fixed-frame pullbacks carry this qualification. The covariant Fisher pullback $G_i^{(q)}$ of Section 18.1, formed with covariant derivatives, is passively gauge invariant without any orbit average, relative to the chosen connection [31, Ch. B2].

Definition 3 (Principal Bundle)**.** *Let $\pi : \mathcal{N} \to \mathcal{C}$ be a principal G-bundle, where $\mathcal{N}$ is the total space of all gauge choices at all base points, $\mathcal{C}$ is the base manifold, and G is the structure group acting freely on the right on $\mathcal{N}$. The projection satisfies $\pi(n \cdot g) = \pi(n)$ for all $g \in G$, so the group action preserves fibers; G acts freely and transitively on each fiber $\pi^{-1}(c)$, so $\mathcal{N}/G \cong \mathcal{C}$; and the bundle is locally trivial, $\mathcal{N}|_{\mathcal{U}} \cong \mathcal{U} \times G$ as G-spaces for sufficiently small open $\mathcal{U} \subseteq \mathcal{C}$.*

The structure group used throughout is $\mathrm{GL}^+(K_q)$, with the compact subgroups $\mathrm{SO}(N)$ as special cases (Section 4.9). Each point $c \in \mathcal{C}$ sits below a fiber $\pi^{-1}(c) \cong G$ of frame choices, and moving within a fiber changes reference frame without changing the base point. For a physicist this is the fiber-bundle description of a gauge potential [35, 72]; here a frame encodes an agent's representational axes.

Local trivializations and the scope of the bundle formalism. A general principal G-bundle does not admit a global section: a globally defined frame field $U : \mathcal{C} \to G$ exists if and only if the bundle is trivializable, and writing such a U as a single exponential $U = \exp \phi$ of a Lie-algebra field is a further restriction, treated in the next paragraph. The framework operates in local trivializing patches $\{\mathcal{U}_i\}$ that cover $\mathcal{C}$, with each $U_i = \exp(\phi_i) : \mathcal{U}_i \to G$ a local section in the fixed reference trivialization, parameterized by the Lie-algebra field $\phi_i : \mathcal{U}_i \to \mathfrak{g}$ and restricted to agent i's domain. On overlaps the operator $\Omega_{ij} = U_i U_j^{-1}$ satisfies the cocycle condition $\Omega_{ij}\Omega_{jk} = \Omega_{ik}$ on triple overlaps (Lemma 1); at a single base point these are the frame comparisons of the companion Theory [30, Ch. 2]. Because each U_i is defined on the whole of its patch, this cocycle is the coboundary of the family $\{U_i\}$, so the reconstructed bundle is canonically trivializable. We accordingly assume a fixed reference trivialization throughout, and the bundle language is organizational, not a source of topological generality. Transition data that is not a coboundary (nontrivial bundle topology) is out of scope. The links Ω_{ij} play a second role that must not be confused with gluing: the agent beliefs q_i are *distinct* sections compared within a common frame, with $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) > 0$ generically, so Ω_{ij} enters as a transport operator between agents and not as a map identifying local representatives of one section. Independent links on the interaction graph (Section 4.6.5) replace the frame-aligned links by edge variables that can carry graph holonomy; they leave the base bundle unchanged.

Two technical caveats accompany this restriction. First, the exponential parameterization $U_i(c) = \exp[\phi_i(c)]$ is only locally bijective and does not cover all of $\mathrm{GL}^+(K_q)$ for $K_q \geq 2$ (Section 4.9). Second, the Maurer-Cartan one-form $\theta = U^{-1}dU$ of Section 4.6.3 is well defined on each local patch whether or not the global bundle is trivializable; the patchwise cocycle structure is what the framework requires, and references to a global frame field ϕ are to be read in the local-trivialization sense unless explicitly noted otherwise.

4.4 Associated Bundles: Attaching Probability Distributions to Gauge Structure

Having established the principal bundle, we attach the two statistical manifolds $\mathcal{B}_{\mathrm{state}}$ and $\mathcal{B}_{\mathrm{model}}$ through the standard associated-bundle construction [35, 73, 74]. A change of frame must transform

the probability distributions accordingly, which requires specifying how G acts on each statistical manifold.

Definition 4 (Representations and Associated Bundles)**.** *Let* $\rho_{\text{state}} : G \to \text{Diff}(\mathcal{B}_{\text{state}})$ *and* $\rho_{\text{model}} : G \to \text{Diff}(\mathcal{B}_{\text{model}})$ *be smooth representations of* G *by diffeomorphisms of the two statistical manifolds. The associated bundles are*

$$\mathcal{E}_{\text{state}} := \mathcal{N} \times_{\rho_{\text{state}}} \mathcal{B}_{\text{state}}, \tag{6}$$

$$\mathcal{E}_{\text{model}} := \mathcal{N} \times_{\rho_{\text{model}}} \mathcal{B}_{\text{model}}, \tag{7}$$

where the equivalence relation $(n \cdot g, b) \sim (n, \rho(g)b)$ *identifies gauge-transformed pairs and the appropriate* ρ *is determined by which manifold* b *inhabits.*

The equivalence relation $(n \cdot g, b) \sim (n, \rho(g)b)$ treats $\rho : G \to \text{Aut}(\mathcal{B})$ as a left action paired with the right principal action $n \mapsto n \cdot g$ on $\mathcal{N}$, the convention of [35] and of the companion Theory. It is equivalent, under $g \mapsto g^{-1}$ and the homomorphism property, to the alternative $(n \cdot g, b) \sim (n, \rho(g^{-1})b)$ found in some texts; we use the former consistently. For Gaussian distributions with $G = \text{GL}(K_q)$ (or any subgroup such as $\text{SO}(N)$), the representation acts by linear transformation of means and congruence of covariances:

$$\rho_{\text{state}}(g) \cdot \mathcal{N}(\mu, \Sigma) = \mathcal{N}(g\mu, g\Sigma g^\top), \tag{8}$$

and analogously for ρ_{model}. For this congruence action each $\rho(g)$ is an isometry of the Fisher-Rao metric: common pushforward leaves every f-divergence invariant (Theorem 1 and the remark following it), and $g_{\mathcal{B}}$ is the Hessian of the relative entropy along the diagonal, so the associated bundles carry a fiberwise Fisher-Rao metric invariant under the structure group and every transport Ω_{ij} acts isometrically on fibers. When the fiber dimensions differ, the congruence form is type-correct on the K_m-dimensional model fiber only if ρ_{model} is a K_m-dimensional representation of G, and no canonical choice exists for generic $K_m \neq K_q$, so we fix it explicitly. For $G = \text{SO}(3)$, ρ_{model} is a direct sum of irreducible representations of total dimension K_m; for $G = \text{GL}(K_q)$ we take $K_m = K_q$ with the fundamental representation. Alternatively, the model fiber may carry its own principal bundle with structure group $\text{GL}(K_m)$ and an independent frame field $\tilde{\phi}_i$, so that $\tilde{\Omega}_{ij} = \exp(\tilde{\phi}_i)\exp(-\tilde{\phi}_j)$ is a transport in its own right and not the image of Ω_{ij} under a representation; this is the independent-frame option of the envelope-gradient discussion following Eq. (69). That the two channels may share one frame is a gauge choice in the companion Theory's treatment [30, Ch. 2]. The associated bundles $\mathcal{E}_{\text{state}}$ and $\mathcal{E}_{\text{model}}$ are fiber bundles over $\mathcal{C}$ with fibers isomorphic to $\mathcal{B}_{\text{state}}$ and $\mathcal{B}_{\text{model}}$, and the four agent fields (q_i, p_i) and (s_i, r_i) are sections of these two bundles, with the prior and hyper-prior supplied by the parent level via Eq. (131).

4.5 Agents as Smooth Sections

We now define agents as sections of these associated bundles, smooth by hypothesis at the primitive level; the companion Theory's definition of an agent is the single-point version [30, Ch. 2]. For the meta-agents of higher levels, whose fields are constructed by coarse-graining and not posited, the available regularity is weaker: smooth on the interior of the formation support under the smooth-bump convention of Section 5.6, and only measurable and piecewise smooth across support boundaries (Section 9).

Definition 5 (Agent)**.** *An agent $\mathcal{A}^i$ at level ℓ with support domain $\mathcal{U}_i \subseteq \mathcal{C}$ consists of two primitive smooth sections, two derived sections, and a gauge-frame field:*

$$q_i : \mathcal{U}_i \to \mathcal{E}_{\text{state}}|_{\mathcal{U}_i}, \qquad \textit{(belief, primitive)} \tag{9}$$

$$s_i : \mathcal{U}_i \to \mathcal{E}_{\text{model}}|_{\mathcal{U}_i}, \qquad \textit{(generative model, primitive)} \tag{10}$$

$$p_i = \Omega_{iI}\big[q_I^{(\ell+1)}\big] \in \Gamma\left(\mathcal{E}_{\text{state}}|_{\mathcal{U}_i}\right), \qquad \textit{(prior, derived)} \tag{11}$$

$$r_i = \tilde{\Omega}_{iI}\big[s_I^{(\ell+1)}\big] \in \Gamma\left(\mathcal{E}_{\text{model}}|_{\mathcal{U}_i}\right), \qquad \textit{(hyper-prior, derived)} \tag{12}$$

$$\phi_i : \mathcal{U}_i \to \mathfrak{g}, \qquad \textit{(gauge frame field)} \tag{13}$$

where $\mathfrak{g} = \mathrm{Lie}(G)$ and $I = \mathrm{pa}(i)$ indexes the level-$(\ell+1)$ meta-agent, if any, containing agent i (the appendices write the parent map as $\pi(i)$). The parent laws $q_I^{(\ell+1)}$ and $s_I^{(\ell+1)}$ are those of the preceding episode, so p_i and r_i are the one-episode-lagged predictive priors of the episode-indexed joint under a transported-copy transition; no generative factor reads a posterior of the same episode [30, Ch. 12]. The derived sections are conditional on a parent existing. The appendices assume a finite tree induced by measurable partitions; whether the formation criterion of Section 9 produces persistent structures satisfying those assumptions is open (Table 2). Without a parent, the prior and hyper-prior are primitive boundary conditions, as in the single-level construction. The four statistical sections form the hierarchy

$$r_i \to s_i \to p_i \to q_i \to \textit{observations},$$

connected across levels by Eq. (131)*: $r_i^{(\ell)}$ regularizes $s_i^{(\ell)}$ because it is the transported parent model, and $p_i^{(\ell)}$ regularizes $q_i^{(\ell)}$ because it is the transported parent belief. At the top level the prior $r_i^{(\ell_{\max})}$ is a declared root, a fixed boundary condition of the hierarchy [30, Ch. 12]; at the bottom level, cross-level relations act only downward.*

Each agent maintains smooth probability distributions $q_i(c)$ and $p_i(c)$ over its domain $\mathcal{U}_i$ (Fig. 1). For Gaussian agents, $q_i(c) = \mathcal{N}(\mu_i(c), \Sigma_i(c))$ and $p_i(c) = \mathcal{N}(\mu_i^p(c), \Sigma_i^p(c))$ live on the latent-state fiber, while $s_i(c) = \mathcal{N}(\mu_i^s(c), \Sigma_i^s(c))$ and $r_i(c) = \mathcal{N}(\mu_i^r(c), \Sigma_i^r(c))$ live on the model fiber. The pairs (μ_i^p, Σ_i^p) and (μ_i^r, Σ_i^r) are obtained from the parent fields by Eq. (131), and $\phi_i(c) = \phi_i^a(c) G_a$ defines the local frame. Different agents may therefore maintain distinct beliefs and frames over overlapping regions of $\mathcal{C}$.

4.6 Cognitive Reference Frames as Gauge Frames

Lahav and Neemeh [63, 64] argue that phenomenal consciousness is observer-relative in the way simultaneity and velocity are observer-relative in special relativity: a cognitive system's experience is a property of that system viewed from its own cognitive frame of reference. Their account writes an explicit transformation only for the degenerate case of functionally identical cognitive systems, where it reduces to a delta-function identification (Eqs. 26 and 35 of the 2022 paper), and leaves the transformation law between distinct cognitive frames unspecified. The construction gives a formal counterpart for the nondegenerate case, as a structural correspondence (status S): each agent i carries a smooth frame field $\phi_i : \mathcal{U}_i \to \mathfrak{g}$ in the role of its cognitive frame of reference, the transport $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ defined below is the transformation between frames, and $D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ is a scalar measuring belief disagreement after transport, invariant under passive re-expression. In Regime I this scalar equals the divergence between the frame-stripped laws (Proposition 1), so the residual disagreement is disagreement between the laws the agents present, and the frames themselves carry no content at a point. The structure group $\mathrm{GL}^+(K_q)$ is a modeling commitment justified on information-geometric grounds (Section 4.9), not by Lahav and Neemeh, who invoke

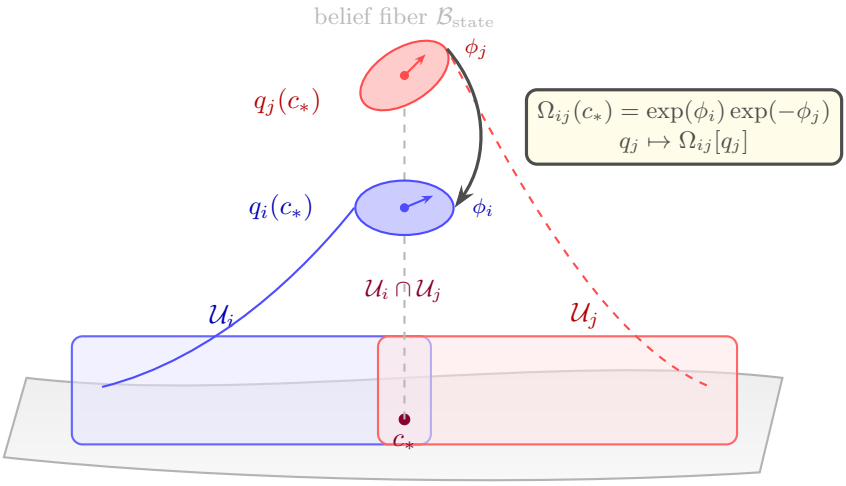


Figure 1: **Overlapping agent sections and gauge transport.** Agents i and j define Gaussian belief sections over neighborhoods $\mathcal{U}_i$ and $\mathcal{U}_j$ of the base manifold $\mathcal{C}$. At $c_* \in \mathcal{U}_i \cap \mathcal{U}_j$, the beliefs occupy the same statistical fiber but use local frames ϕ_i and ϕ_j. The transport $\Omega_{ij}(c_*) = \exp(\phi_i)\exp(-\phi_j)$ maps $q_j(c_*)$ into agent i's frame before $D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ is evaluated. In the zero-dimensional limit the neighborhoods collapse to one base point and the comparison remains entirely within one fiber.

the Lorentz transformation only as a structural analogy; Section 22 develops the correspondence with its caveats, including that invariance places no constraint on which frame produces which phenomenal quale.

The analogy with coordinate changes in general relativity is a working correspondence, not an identity: the transformation group is $\mathrm{GL}^+(K_q)$ acting on probability distributions on a statistical manifold, not a group of coordinate changes of a Lorentzian spacetime. The frame $\phi_i(c)$ specifies the agent's internal axes for representing belief content at base point c, a basis in its representational space, not an orientation in physical space. For example, two agents perceiving the same colored surface may represent color along different internal bases, agreeing about the spectral content of the stimulus but differing in the components in which they represent it; their frames differ by the basis change relating the two schemes, and after transport the residual $D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ is the part of the disagreement that does not reduce to a basis choice.

4.6.1 Transport Operators: Translating Between Frames

When agents i and j overlap at a point $c \in \mathcal{U}_i \cap \mathcal{U}_j$, the inter-agent transport operator is

$$\Omega_{ij}(c) = \exp[\phi_i(c)] \exp[-\phi_j(c)] \in G. \tag{14}$$

This group element carries agent j's representations into agent i's frame, acting on probability distributions through the representation $\rho : G \to \mathrm{Aut}(\mathcal{B})$ as $\Omega_{ij}[q_j](c) := \rho(\Omega_{ij}(c))q_j(c)$. The product of two exponentials is what gives Ω_{ij} its range. A single exponential misses part of $\mathrm{GL}^+(K_q)$ for $K_q \geq 2$ (Section 4.9), whereas the pair $\exp(\phi_i)\exp(-\phi_j)$ attains every element of $\mathrm{GL}^+(K_q)$ for a single pair considered in isolation: by polar decomposition $A = RP$ with $R \in \mathrm{SO}(K_q) = \exp(\mathfrak{so}(K_q))$ and P symmetric positive-definite with $-\phi_j = \log P$. Across the coupled system

the same ϕ_i enters every pair, so the joint transports are constrained to the cocycle family. The same construction makes Ω_{ij} a genuinely relative operator: demanding a single constant, pair-independent transport forces $\Omega^2 = \Omega$ through the cocycle $\Omega_{ij}\Omega_{jk} = \Omega_{ik}$, hence $\Omega = I$ whenever at least three agents are present, and $\Omega = I$ collapses all frames to a common U; for exactly two agents only the weaker involution condition $\Omega^2 = I$ survives, admitting nonidentity involutions such as $\Omega = -I$ for even K_q. For Gaussian beliefs $q_j = \mathcal{N}(\mu_j, \Sigma_j)$, transport acts as $\mu_j \mapsto \Omega_{ij}\mu_j$ and $\Sigma_j \mapsto \Omega_{ij}\Sigma_j\Omega_{ij}^\top$, and the transported belief $\Omega_{ij}[q_j](c)$ is agent j's belief at c expressed in agent i's coordinates, which makes the comparison $D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ meaningful.

The vertex-local form of Eq. (14) carries an immediate cocycle theorem, invoked throughout the manuscript.

Lemma 1 (Vanishing Holonomy)**.** *For gauge transport of the form $\Omega_{ij} = g_i g_j^{-1}$ with vertex-local group elements $g_i \in G$, the holonomy around any closed loop vanishes:*

$$H_{ijk} = \Omega_{ij}\Omega_{jk}\Omega_{ki} = g_i g_j^{-1} g_j g_k^{-1} g_k g_i^{-1} = I. \tag{15}$$

Equivalently, the cocycle condition $\Omega_{ij}\Omega_{jk} = \Omega_{ik}$ holds for all triples (i, j, k).

Proof. Direct computation. The consecutive inverse pairs cancel identically: $g_j^{-1} g_j = I$ and $g_k^{-1} g_k = I$. □

This is an identity of the vertex-local parameterization $\Omega_{ij} = U_i U_j^{-1}$, not an approximation: transport from agent k to agent i is path independent whatever the frames ϕ_i, the direct route Ω_{ik} and the indirect route $\Omega_{ij}\Omega_{jk}$ giving identical results, and it is the graph form of the Regime I flatness stated in Section 4.6.3. Nontrivial loop products require the independent links of Section 4.6.5, of which the cocycle here is the special case $\delta_{ij} = 0$. The converse also holds: independent links whose products around every closed walk are the identity are frame aligned for some choice of frames, unique up to one common right translation on each connected component (the trivializing criterion of [30, Ch. 2]).

In Regime I the frames are more than path independent: they can be removed from every term that compares two agents.

Proposition 1 (Removability of Regime I frames)**.** *Let all links be frame aligned, $\Omega_{ij} = U_i U_j^{-1}$ within a level and $\Omega_{iI} = U_i U_I^{-1}$ between an agent and its parent, and define the frame-stripped laws $\tilde{q}_i := (U_i^{-1})_{\#} q_i$ and $\tilde{q}_I := (U_I^{-1})_{\#} q_I$, with $\tilde{s}_i$ and $\tilde{s}_I$ defined in the same way from the model-channel frames. Then for every pair*

$$D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) = D_{\mathrm{KL}}(\tilde{q}_i \| \tilde{q}_j), \qquad D_{\mathrm{KL}}(q_i \| \Omega_{iI} q_I) = D_{\mathrm{KL}}(\tilde{q}_i \| \tilde{q}_I), \tag{16}$$

and the same identities hold in the model channel. Consequently the alignment energies, the attention and model-coupling rows built from them, and the cross-level prior terms of the functional (49) *are functions of the frame-stripped laws alone. The frames enter the functional only through terms that pair a belief with a field held in a fixed frame, such as an observation likelihood, a fixed-frame environmental agent, or a frozen prior, and after the change of variables they appear there as compensating fields: for a likelihood $\ell_i(k) = p(o_i \mid k)$ written in agent i's coordinates, $\mathbb{E}_{q_i}[-\log \ell_i(k)] = \mathbb{E}_{\tilde{q}_i}[-\log \ell_i(U_i \tilde{k})]$.*

Proof. Relative entropy is invariant under a common measurable bijection of both arguments (Theorem 1 for the Gaussian case and the f-divergence extension following it). Applying U_i^{-1} to both arguments gives $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) = D_{\mathrm{KL}}((U_i^{-1})_{\#} q_i \| (U_i^{-1}\Omega_{ij})_{\#} q_j)$, and $U_i^{-1}\Omega_{ij} = U_j^{-1}$; the parent

case is identical with j replaced by I, and the model channel is identical with the model frames. The rows are functions of these energies and of frame-free source priors, so they inherit the property. The last identity is the change of variables $k = U_i\tilde{k}$. □

Frame content at a point is therefore relative to the fields held fixed (Section 6.5); where every field is co-transformed the frames are pure redundancy. The frame U_i acts here as a dressing field in the sense of the dressing-field method of gauge theory [71]: it transforms as $U_i \mapsto g_i U_i$ under passive re-expression, so the composite $\tilde{q}_i$ is invariant, and the per-agent frame symmetry can be eliminated by a dressing built from the construction's own fields, which is François's criterion for an artificial gauge symmetry [75]. The same substitution removes the parent frame U_I from every constituent prior, which is the non-identifiability of the parent frame discussed in Section 9.

4.6.2 Passive Re-expression, Change of Reference, and the Active Group

Three different group actions are each called a gauge transformation in the literature on constructions of this kind, and we keep them apart, following the companion Theory [30, Ch. 2]. *Passive re-expression* lets each agent change its own frame, $U_i \mapsto g_i U_i$ with an independent $g_i \in G$, its laws changing coordinates by $q_i \mapsto \rho(g_i)q_i$; links transform as $\Omega_{ij} \mapsto g_i \Omega_{ij} g_j^{-1}$ (Eq. (22)), so $\Omega_{ij}q_j$ acquires $\rho(g_i)$ on the left and every divergence $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ is unchanged by Theorem 1. The tuple (g_i) ranges over $\prod_i G$, and these transformations are bookkeeping, a change of coordinates on each fiber. A *change of reference* is a common right translation $U_i \mapsto U_i g$ by one $g \in G$ with the laws held fixed; it leaves every frame-aligned link unchanged, $\Omega_{ij} = U_i U_j^{-1} \mapsto U_i g g^{-1} U_j^{-1} = \Omega_{ij}$, corresponds to moving the fixed reference point of the principal fiber against which frames are recorded, and conjugates every per-edge twist by g (Section 4.6.5). The *active gauge group* at a base point, the group of right-equivariant bijections of the principal fiber over that point, is a single copy of G inside $\prod_i G$, a proper subgroup of dimension $\dim G$ against $|V| \dim G$ for $|V|$ agents [30, Ch. 2]. At one base point the active group is therefore one global G.

Each action is stated at the group level. The Lie-algebra shift $\phi_i \mapsto \phi_i + \xi$ agrees with the group-level right translation $U_i \mapsto U_i \exp \xi$ only when ξ commutes with ϕ_i, since the Zassenhaus formula $\exp(\phi_i + \xi) = \exp(\phi_i)\exp(\xi)\exp(-\frac{1}{2}[\phi_i, \xi]) \cdots$ collapses to $\exp(\phi_i)\exp(\xi)$ only when the commutator vanishes; the additive shift is a gauge transformation only for abelian groups or central ξ, and for the non-abelian groups used here the group-level forms are the correct statements. The attention and meta-level entropies of the functional (49) are invariant under passive re-expression because β and γ are functions of invariant divergences. The functional as a whole is invariant under passive re-expression exactly when all statistical fields, including the observation model and any frozen hyper-prior, are re-expressed together, and it fails to be invariant only when a fixed field is exempted from the action. It is invariant under change of reference with the observation term included, since that term contains no frame. Because the functional depends on frames only through the pointwise links $\Omega_{ij}(c)$, the change of reference may even depend on the base point, $U_i(c) \mapsto U_i(c)g(c)$, which is the fact used in Section 4.3. What survives all three actions at a point is the relative entropy between frame-stripped laws, the per-edge twist classes of independent links, and the conjugacy classes of loop holonomies.

4.6.3 Connection Forms and Parallel Transport

Under the Regime I / Regime II convention fixed at the head of this section, the connection-form discussion below develops the vocabulary of base-direction structure; in Regime I it is dynamically

trivial. The gauge frame ϕ_i induces a Lie-algebra-valued one-form on agent i's domain,

$$\theta_\mu^{(i)}(c) := U_i^{-1}(c)\partial_\mu U_i(c) \in \mathfrak{g}, \tag{17}$$

where $U_i(c) = \exp[\phi_i(c)] \in G$ and $\partial_\mu = \partial/\partial c^\mu$. This is the left Maurer-Cartan form pulled back along the agent's local section $U_i : \mathcal{U}_i \to G$ on the trivializing patch $\mathcal{U}_i$, not an independent gauge potential: its curvature $d\theta + \theta \wedge \theta$ vanishes identically. This is the canonical statement of Regime I flatness, established once here and assumed thereafter. We use the symbol $A_\mu^{(i)}$ for $\theta_\mu^{(i)}$ when the context is the frame-gradient reading; an independent gauge potential in the sense of Yang and Mills [76, 77] would be an independent connection on the base, which is deferred base-transport structure. On overlapping regions the relation between the two agents' frame gradients follows from $U_i = \Omega_{ij} U_j$ and the chain rule,

$$\theta_\mu^{(i)} = \mathrm{Ad}(U_j^{-1})\big[\Omega_{ij}^{-1}\partial_\mu \Omega_{ij}\big] + \theta_\mu^{(j)}, \tag{18}$$

where $\mathrm{Ad}(U_j^{-1})$ is the adjoint action of U_j^{-1} on $\mathfrak{g}$. Equivalently, $\theta^{(i)} = h^{-1}\theta^{(j)}h + h^{-1}dh$ with $h = U_j^{-1}U_i$, the standard transformation law of a connection for a right principal action [35, 72]; the right Maurer-Cartan form $-(dU)U^{-1}$ obeys the same law in the form $A^{(i)} = gA^{(j)}g^{-1} - (dg)g^{-1}$ with $g = \Omega_{ij}$. The two forms differ by the left or right convention, not by any property of pure-gauge connections.

4.6.4 Gauge Field Strength: Path-Dependent Information Transport

The field strength (curvature two-form) measures the path dependence of parallel transport along the base,

$$F_{\mu\nu}^{(i)}(c) = \partial_\mu A_\nu^{(i)} - \partial_\nu A_\mu^{(i)} + [A_\mu^{(i)}, A_\nu^{(i)}] \in \mathfrak{g}. \tag{19}$$

Under Regime I the connection has the pure-gauge form $A = U^{-1}dU$, the Maurer-Cartan identity forces $F_{\mu\nu}^{(i)} = 0$ identically (Section 4.6.3), and transport around any closed base loop returns the original belief unchanged. A nonzero $F_{\mu\nu}^{(i)}$ requires an independent connection on $\mathcal{C}$, and then beliefs transported around closed base loops acquire holonomy, as a vector parallel transported around a closed triangle on a sphere returns rotated. That is base holonomy, distinct from the graph holonomy of Section 4.6.5, and no result of this paper requires it.

4.6.5 Independent Links on the Interaction Graph

Independently of any base connection, the interaction graph at one base point carries independent link variables, whose loop products are graph holonomies [30, Ch. 2]; in the zero-dimensional limit they are the only holonomy available. Beginning from the frame-aligned links $\Omega_{ij} = U_i U_j^{-1}$ of Regime I, one introduces an edge variable $\delta_{ij} \in \mathbb{R}^{n_{\mathrm{gen}}}$ valued in the Lie-algebra coordinates and writes

$$\Omega_{ij} = U_i \exp\big(\delta_{ij} \cdot G\big) U_j^{-1}, \qquad \delta_{ij} \cdot G \equiv \sum_{a=1}^{n_{\mathrm{gen}}} \delta_{ij}^a G_a \in \mathfrak{g}, \tag{20}$$

with $\{G_a\}$ the structure-group generators, subject to orientation reversal

$$\delta_{ji} = -\delta_{ij}, \qquad \text{equivalently} \qquad \Omega_{ji} = \Omega_{ij}^{-1}, \tag{21}$$

which is part of the companion Theory's definition of graph links [30, Ch. 2]. Without it a backtrack $i \to j \to i$ would carry holonomy and a loop and its reverse would carry unrelated traces.

Setting $\delta_{ij} = 0$ recovers the Regime I cocycle and the vanishing-holonomy lemma; allowing $\delta_{ij} \neq 0$ admits nontrivial holonomy on closed cycles. With Eq. (21) imposed, Eq. (20) is the lattice-gauge link variable [78–80] on the edges of an abstract interaction graph, written in a vertex-trivialized parameterization that exposes the Regime I skeleton as a special case. Under passive re-expression $U_i \to g_i U_i$ the link is multiplied on the left by g_i and on the right by g_j^{-1},

$$\Omega_{ij} \longmapsto g_i \Omega_{ij} g_j^{-1}, \tag{22}$$

the lattice-gauge transformation law for link variables [78, Eqs. (3.27)–(3.28)]; see also [80]. The link Ω_{ij} is gauge covariant, not gauge invariant. Its per-edge twist $U_i^{-1}\Omega_{ij}U_j = \exp(\delta_{ij} \cdot G)$ is unchanged by passive re-expression and is conjugated by g under a change of reference $U_i \mapsto U_i g$, so each edge carries the conjugacy class of its twist, invariant under all passive and active changes at the point; for example $\operatorname{tr}\exp(\delta_{ij} \cdot G)$ is a per-edge invariant. Loop holonomies carry the remaining invariant content (the per-edge twist-invariance proposition of [30, Ch. 2]).

Around a 3-cycle $i \to j \to k \to i$ on the interaction graph, the holonomy is

$$H_{ijk} := \Omega_{ij}\Omega_{jk}\Omega_{ki} = U_i \cdot \exp(\delta_{ij} \cdot G) \exp(\delta_{jk} \cdot G) \exp(\delta_{ki} \cdot G) \cdot U_i^{-1}, \tag{23}$$

with the interior vertex factors $U_j^{-1}U_j$ and $U_k^{-1}U_k$ collapsing as in the vanishing-holonomy lemma. Under the transformation (22), $H_{ijk} \to g_i H_{ijk} g_i^{-1}$ is conjugated at the base vertex, and orientation reversal gives $H_{ikj} = H_{ijk}^{-1}$, so the trace

$$W_{ijk} := \operatorname{Re}\operatorname{Tr}(H_{ijk}) = \operatorname{Re}\operatorname{Tr}\big[\exp(\delta_{ij} \cdot G) \exp(\delta_{jk} \cdot G) \exp(\delta_{ki} \cdot G)\big], \tag{24}$$

is the gauge-invariant Wilson observable of the triangle. The corresponding Wilson-style holonomy penalty, driving the links toward Regime I, is

$$S_{\text{Wilson}}[\delta] = \beta_{\text{W}} \sum_{(i,j,k)} \Big(1 - \tfrac{1}{K_q} W_{ijk}\Big), \tag{25}$$

where the sum runs over the triangles of the interaction graph, each with a chosen orientation, β_{W} is the lattice coupling, and K_q is the dimension of the defining representation. For compact G (for example $\mathrm{SO}(N)$) the orientation does not matter, since $\operatorname{Tr}(H^{-1}) = \operatorname{Tr}(H^{\top}) = \operatorname{Tr}(H)$ for orthogonal H; moreover $\operatorname{Re}\operatorname{Tr}(H_{ijk}) \leq K_q$ with equality only at the identity, so the compact action is nonnegative, minimized at $H_{ijk} = I$, and forces $H_{ijk} \to I$ as $\beta_{\text{W}} \to \infty$. This conclusion does not extend to noncompact $\mathrm{GL}^+(K_q)$, where the trace action is unbounded below. The conjugacy-invariant eigenvalue deviation

$$d_{\text{ss}}(H) := \sum_a |\lambda_a(H) - 1| \tag{26}$$

is only a semisimple diagnostic. For

$$H_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad t \neq 0, \tag{27}$$

both eigenvalues equal one although $H_t \neq I$, so $d_{\text{ss}}(H_t) = 0$. The Frobenius penalty detects the same shear, $\|H_t - I\|_F^2 = t^2$, but it is not invariant under general conjugation: for $g = \operatorname{diag}(a, a^{-1})$, $\|gH_tg^{-1} - I\|_F^2 = a^4t^2$. For $K_q \geq 2$, no continuous conjugacy-invariant function of holonomy alone can be both faithful and zero only at I on all of $\mathrm{GL}^+(K_q)$, because a nonidentity unipotent shear, embedded as a 2×2 block when $K_q > 2$, can be conjugated along an orbit that accumulates at the

identity. The one-dimensional group $\mathrm{GL}^+(1) = \mathbb{R}_{>0}$ is an exception: $|H - 1|$ is continuous, conjugacy invariant, and vanishes only at the identity. A nonnegative plaquette penalty on $\mathrm{GL}^+(K_q)$ must therefore give up either faithfulness or conjugation invariance. A scalar relaxation parameter $\epsilon_{\mathrm{rel}} \in [0, 1]$ in $\Omega_{ij}(\epsilon_{\mathrm{rel}}) = U_i \exp(\epsilon_{\mathrm{rel}} \delta_{ij} \cdot G) U_j^{-1}$ interpolates between frame-aligned and independent links. We call Eq. (25) a Wilson-style holonomy penalty on an abstract graph and do not assert a continuum Yang-Mills field theory [76]. A link may also be declared a function of the agents' states, provided it satisfies Eq. (21) and the transformation law (22); a link computed from means expressed in a common ambient frame satisfies the latter only if that frame is itself re-expressed. For the frame-aligned links of Part I every twist is the identity and these diagnostics vanish.

A structural parallel: the boundary of a boundary. Wheeler lists "the boundary of a boundary is zero" among the clues to it from bit [1]. Two identities of this section have that form. Lemma 1 says that a coboundary has trivial coboundary: $\Omega_{ij} = U_i U_j^{-1}$ implies that the product around every triangle is the identity, the lowest nonabelian instance of $\delta \circ \delta = 0$. For independent links with orientation reversal (21), write $P_{abc} := \Omega_{ab}\Omega_{bc}\Omega_{ca}$ for the holonomy of the face (a, b, c) based at a. The four faces of any 3-simplex (i, j, k, l) of the interaction complex satisfy the lattice Bianchi identity

$$P_{ijk} P_{ikl} P_{ilj} \big(\Omega_{ij} P_{jlk} \Omega_{ji}\big) = I, \tag{28}$$

whatever the individual face holonomies are: writing out the product, every edge of the boundary of the boundary is traversed once in each direction, and each pair $\Omega_{ba}\Omega_{ab} = I$ cancels. These are structural parallels only (status S). Wheeler used the principle, through the Bianchi identities of electrodynamics and geometrodynamics, to make conservation laws automatic; that route needs a source coupled to the connection, which the present construction does not have, and no conservation law is derived here.

4.7 The Hierarchy of Transport Operators

The multi-agent, multilevel structure uses several types of transport. Intra-level belief transport $\Omega_{ij} : \mathcal{B}_{\mathrm{state}} \to \mathcal{B}_{\mathrm{state}}$ is the operator of Eq. (14), carrying agent j's belief into agent i's frame within one level; because p and q both inhabit $\mathcal{B}_{\mathrm{state}}$, the same operator transports priors, and we use the single symbol Ω_{ij} for both. The model-coupling term $\gamma_{ij} D_{\mathrm{KL}}(s_i \| \tilde{\Omega}_{ij} s_j)$ uses the model-fiber transport $\tilde{\Omega}_{ij} : \mathcal{B}_{\mathrm{model}} \to \mathcal{B}_{\mathrm{model}}$, equal to Ω_{ij} as a group element in the shared-frame configuration but acting on a different fiber; since hyper-priors r and generative models s both inhabit $\mathcal{B}_{\mathrm{model}}$, $\tilde{\Omega}_{ij}$ also transports hyper-priors. Cross-level transport connects hierarchical levels: bottom-up transport $\Lambda^{\ell \to \ell+1}$ aggregates transported constituent beliefs when meta-agent I forms at level $\ell + 1$ (Section 9), with a model-channel analog aggregating constituent generative models, and top-down transport in the reverse direction supplies the lagged predictive priors of Eq. (131).

The parent-to-constituent transport $\Omega_{iI} = \exp(\phi_i^{(\ell)}) \exp(-\phi_I^{(\ell+1)})$, which carries the parent's laws into agent i's frame for the top-down priors, and its inverse Ω_{Ii}, which carries constituent beliefs into the parent frame for the bottom-up aggregation that builds $\bar{\mu}_I$ and $\bar{\Sigma}_I$, are special cases of the general law $\Omega_{ab} = \exp(\phi_a) \exp(-\phi_b)$ with a and b at different levels. Any parent frame $U_I \in G$ makes Ω_{iI} a well-defined group element, so intra-level and cross-level re-expressions act through the same group. By Proposition 1 the parent frame cancels from every constituent prior in Regime I, so its choice is a gauge convention; the construction of a parent frame from constituent frames, and its noncompact $\mathrm{GL}^+(K_q)$ caveats, are in Sections 9 and 9.2.

4.8 Curvature Structure: Four Notions

Four distinct notions of curvature appear, and only the first is active in Part I. *Fiber curvature*: the fiber $\mathcal{B}$ is a Riemannian manifold, and the manifold of Gaussian distributions under the Fisher-Rao metric has sectional curvature of mixed sign for $K_q \geq 2$ [81]. The SPD covariance sector $\mathbb{S}^+_{K_q}$ is a Riemannian symmetric space of nonpositive curvature, and the univariate case and the location-scale family with fixed correlation reduce to hyperbolic geometry of constant negative curvature, but the full joint mean-covariance manifold admits 2-planes of strictly positive curvature: for orthonormal pure-mean directions the sectional curvature equals $+1/4$, a value that is not the supremum, since mixed mean-covariance planes reach slightly larger values. This fiber curvature governs natural-gradient flow within each fiber. *Group curvature*: the Riemannian curvature of the structure group depends on the metric chosen on it. $\mathrm{SO}(3) \cong \mathbb{RP}^3$ with its bi-invariant metric has constant positive curvature, whereas $\mathrm{GL}^+(K_q)$ for $K_q \geq 2$ carries no bi-invariant Riemannian metric, so any curvature assigned to it depends on a non-canonical choice of left- or right-invariant metric [82]. Noncommutativity is a separate, metric-free property: the order of unrelated transports matters ($\Omega_{ab}\Omega_{cd} \neq \Omega_{cd}\Omega_{ab}$ for disjoint pairs), while for chained transports through a common intermediate agent the cocycle $\Omega_{ij}\Omega_{jk} = \Omega_{ik}$ holds identically whatever the commutators. *Gauge-field curvature*: the field strength $F^{(i)}_{\mu\nu}$ of Eq. (19) vanishes identically in Regime I (Section 4.6.3); a nonzero field strength requires an independent base connection, which is deferred, and the graph holonomy of independent links is a different object. *Base curvature*: $\mathcal{C}$ carries no physical metric, the fiducial chart metric of the working framework is flat, and metrics that represent agent-relative geometry are induced by beliefs through pullback (Section 18.1). Reading induced curvature as gravity is a structural placeholder of Part II (Section 21).

4.9 Working Framework: Simplifications and Scope

One structural fact and four simplifications keep every quantity computable in closed form. The structural fact: beliefs and priors share the latent-state manifold $\mathcal{B}_{\text{state}}$, and generative models and hyper-priors share the model-parameter manifold $\mathcal{B}_{\text{model}}$, both *by construction* from the cross-level prior relation (131) and not as separate assumptions (Section 10). The four simplifications follow. Both fibers are Gaussian. The structure group is $G = \mathrm{GL}^+(K_q, \mathbb{R})$ acting on the belief fiber, with the compact subgroups $\mathrm{SO}(N) \subset \mathrm{GL}(N)$, including $\mathrm{SO}(3)$, as special cases for analytic clarity; a complexified group appears only in the worked signature example of Section 19. The base manifold is flat, $\mathcal{C} = \mathbb{R}^2$ with its Euclidean chart. Gauge frames are smooth, and the field strength vanishes identically in Regime I however fast the frames vary (Section 4.6.3).

These choices give closed-form relative entropies, known curvature of the Gaussian fibers, and transparent group actions. They keep gauge covariance with nontrivial transport operators, information-geometric dynamics through Fisher metrics, multilevel hierarchical structure, curved statistical manifolds, and non-commutative group structure; they give up graph and base holonomy, non-Gaussian belief structures, heterogeneous agent types, curved base geometry, and gauge-field curvature $F_{\mu\nu}$ (Regime I flatness).

Matched bundles on the two-manifold structure. Beliefs and priors share $\mathcal{B}_{\text{state}} =: \mathcal{B}$, the representation $\rho_{\text{state}} =: \rho$, and one associated bundle $\mathcal{E}_{\text{state}} =: \mathcal{E}$, and models and hyper-priors share $\mathcal{B}_{\text{model}}$ and $\mathcal{E}_{\text{model}}$. Setting the slow subsystem aside, each agent i is represented by a single section $\sigma^i : \mathcal{U}_i \to \mathcal{E}$ carrying $\sigma^i(c) = (q_i(c), p_i(c))$. This is reasonable because beliefs and priors are both probability distributions over the same underlying space, distinguished by separate components and not by separate bundle structure; what is given up is the ability to model qualitatively different belief

and prior types, for example Gaussian beliefs with categorical priors. Equivalently, the canonical functional (49) contains no cross-bundle morphism coupling the model and belief channels; the two channels couple only through the frame sector and the within-channel cross-level priors of Eq. (131). In the shared-frame configuration Ω_{ij} and $\tilde{\Omega}_{ij}$ act as the same element of G in two representations; in the independent-frame configuration the model fiber carries its own $\mathrm{GL}(K_m)$ bundle and frame field $\tilde{\phi}_i$ (the option following Eq. (69)), $\tilde{\Omega}_{ij}$ becomes an independent transport, and even the structure-group identification is absent, leaving the cross-level priors as the only inter-channel coupling.

Gaussian probability manifold. The fiber is $\mathcal{B} = \{\mathcal{N}(\mu, \Sigma) : \mu \in \mathbb{R}^{K_q}, \Sigma \in \mathbb{S}^+_{K_q}\}$, with relative entropy in the closed form of Eq. (4). Gaussians are the maximum-entropy distributions for given mean and covariance. They are a simplifying choice (Section 4.2), and a non-Gaussian exponential-family fiber requires a common bijective sample-space action that preserves the family and leaves the chosen divergence computable. The Gaussian theorem below supplies this closure for linear pushforwards of Gaussian states; mixture models, heavy-tailed distributions, and density-operator fibers require separate closure and computation proofs. A complex or quantum fiber with complex-valued amplitudes is a separate program (Section 24).

Gauge group: $\mathrm{GL}(K_q)$ and its compact subgroups. The full Gaussian family on $\mathbb{R}^{K_q}$ is preserved under the affine group $x \mapsto Ax + b$ with $A \in \mathrm{GL}(K_q)$, $b \in \mathbb{R}^{K_q}$, and the relative entropy (indeed every f-divergence) is invariant under any common bijective change of variables, since the Jacobian factors cancel in the density ratio. Restricting to the origin-preserving linear subgroup $G = \mathrm{GL}(K_q) \subset \mathrm{Aff}(K_q)$, equivalently treating mean vectors as frame coordinates and not as elements of an affine space with translation freedom, gives the working gauge group. This is a modeling choice: $\mathrm{GL}(K_q)$ is the linear sector of the Gaussian symmetry, and the translation sector b, which would correspond to frame-dependent shifts of agent means, is available without being used. The invariance theorem below is the standard linear-pushforward statement [83].

Theorem 1 ($\mathrm{GL}(K_q)$ Gauge Invariance)**.** *For any two Gaussian distributions $P = \mathcal{N}(\mu_P, \Sigma_P)$ and $Q = \mathcal{N}(\mu_Q, \Sigma_Q)$ on $\mathbb{R}^{K_q}$, and any invertible linear transformation $\Omega \in \mathrm{GL}(K_q)$, the relative entropy is invariant under simultaneous pushforward:*

$$D_{\mathrm{KL}}(\Omega_* P \| \Omega_* Q) = D_{\mathrm{KL}}(P \| Q), \tag{29}$$

where $\Omega_* P = \mathcal{N}(\Omega\mu_P, \Omega\Sigma_P\Omega^\top)$ denotes the pushforward of P under Ω. We assume $\Sigma_P, \Sigma_Q \succ 0$ throughout; the pushforward $\Omega\Sigma\Omega^\top$ remains positive-definite by Sylvester's law of inertia for Ω invertible. Singular Ω ($\det\Omega = 0$) is excluded by the assumption $\Omega \in \mathrm{GL}(K_q)$, which would otherwise make the pushforward degenerate.

Proof. We use the Gaussian closed form of Eq. (4) (recorded again in Appendix C.2). Under the pushforwards $\Omega_* P = \mathcal{N}(\Omega\mu_P, \Omega\Sigma_P\Omega^\top)$ and $\Omega_* Q = \mathcal{N}(\Omega\mu_Q, \Omega\Sigma_Q\Omega^\top)$, the trace term satisfies

$$\mathrm{tr}\left[(\Omega\Sigma_Q\Omega^\top)^{-1}(\Omega\Sigma_P\Omega^\top)\right] = \mathrm{tr}\left[\Omega^{-\top}\Sigma_Q^{-1}\Omega^{-1}\Omega\Sigma_P\Omega^\top\right] = \mathrm{tr}(\Sigma_Q^{-1}\Sigma_P), \tag{30}$$

the quadratic term satisfies

$$\begin{aligned}&(\Omega\mu_P - \Omega\mu_Q)^\top(\Omega\Sigma_Q\Omega^\top)^{-1}(\Omega\mu_P - \Omega\mu_Q)\\ &= (\mu_P - \mu_Q)^\top\Omega^\top\Omega^{-\top}\Sigma_Q^{-1}\Omega^{-1}\Omega(\mu_P - \mu_Q) = (\mu_P - \mu_Q)^\top\Sigma_Q^{-1}(\mu_P - \mu_Q),\end{aligned} \tag{31}$$

and the log-determinant term satisfies

$$\log \frac{\det(\Omega\Sigma_Q\Omega^\top)}{\det(\Omega\Sigma_P\Omega^\top)} = \log \frac{(\det\Omega)^2 \det\Sigma_Q}{(\det\Omega)^2 \det\Sigma_P} = \log \frac{\det\Sigma_Q}{\det\Sigma_P}, \tag{32}$$

the $(\det\Omega)^2$ factors canceling identically. Since all three components are invariant,

$$D_{\mathrm{KL}}(\Omega_* P \| \Omega_* Q) = D_{\mathrm{KL}}(P\|Q). \tag{33}$$

□

The proof extends to all f-divergences: for any convex f with $f(1) = 0$,

$$D_f(\Omega_* P \| \Omega_* Q) = D_f(P\|Q), \quad \forall \Omega \in \mathrm{GL}(K_q), \tag{34}$$

since the factors $|\det\Omega|$ from the density transformation cancel in the ratio $p(x)/q(x)$. Transport operators therefore need only satisfy $\det(\Omega_{ij}) \neq 0$, and the full $\mathrm{GL}(K_q)$ acts by passive re-expression without changing any divergence. The consequence for Regime I is Proposition 1: applied with $\Omega = U_i^{-1}$, the theorem removes the frames from every pairwise term. The exponential parameterization e^ϕ used throughout restricts to $\det > 0$, since $\det(\exp\phi) = \exp(\mathrm{tr}\,\phi) > 0$, and the canonical statement of its coverage limitation is this: for $K_q \geq 2$ the real matrix exponential $\exp : \mathfrak{gl}(K_q, \mathbb{R}) \to \mathrm{GL}(K_q, \mathbb{R})$ is *not* surjective onto $\mathrm{GL}^+(K_q, \mathbb{R})$. Its image is the set of matrices admitting a real logarithm, a proper subset of the identity component (matrices with negative real eigenvalues of odd algebraic multiplicity, for instance, admit no real logarithm). The parameterization includes a neighborhood of the identity but is not a global parameterization of $\mathrm{GL}^+(K_q, \mathbb{R})$; products of two exponentials are required, and by the polar factorization following Eq. (14) sufficient, for full coverage of GL^+, while the component $\det < 0$ is not reached and is reserved for future study.

The gauge action on Gaussian states is $\rho(\Omega) \cdot (\mu, \Sigma) = (\Omega\mu, \Omega\Sigma\Omega^\top)$, the fundamental representation of $\mathrm{GL}(K_q)$. For belief vectors of dimension $d > 3$ with SO(3) structure, ρ decomposes into irreducibles with $d = \sum_k n_k(2j_k + 1)$, where j_k is the spin of the k-th irreducible block.

Flat base manifold. For base-space constructions we use a flat Euclidean background chart, $\mathcal{C} = \mathbb{R}^2$ with $g_{\mathcal{C}} = \delta_{\mu\nu}$. Every integral over $\mathcal{C}$ in the functional is taken against the Lebesgue density dc of this chart, and the sector thresholds of Section 20 use its metric; these are fixed background structures, so the base is not free of structure that the agents did not produce. The chart metric is a device for integration, differentiation, and support placement, not a property of the index manifold, which carries no physical metric; the metrics that represent agent-relative geometry are the pullbacks induced by beliefs (Section 18.1). Those pullbacks enter no term of the functional, so the arc of Wheeler's loop that runs from the induced geometry back to the agents is not constructed here (W5); closing it would require the induced geometry to enter the dynamics, for example as the integration measure, which is base-transport work deferred to Supplement B [31]. An induced pullback has rank at most $\dim\mathcal{C}$, so on the working base $\mathbb{R}^2$ it has at most two nonzero eigenvalues. Agents occupy compact support regions $\mathcal{U}_i \subset \mathbb{R}^2$. The degenerate case $\dim\mathcal{C} = 0$, with all agents at a single base point c_*, is the setting in which the companion Theory develops the construction [30, Ch. 2]; there coarse-graining is purely informational. These choices give up intrinsic curvature of the base, global topological effects, and compact base manifolds without boundary; curved base manifolds are deferred with the rest of the base-transport theory.

Gauge-covariant relative entropy and the zero-dimensional limit. With these simplifications the transported relative entropy $D_{\mathrm{KL}}[q_i(c)\|\Omega_{ij}(c)q_j(c)]$ is invariant under simultaneous re-expression of both arguments (Theorem 1). When the base manifold collapses to a single point ($\dim\mathcal{C} = 0$), all frames are constants $\phi_i(c) = \phi_i$, the frame gradients and curvature vanish, $A_\mu = F_{\mu\nu} = 0$, and the integral over $\mathcal{C}$ reduces to evaluation at c_*. The active gauge group is then one global copy of G (Section 4.6.2), and in Regime I the frames are removable (Proposition 1), so the construction at a point is a system of interacting probability laws related by passive re-expressions, not a gauge theory in the physicist's sense. Its gauge content at a point consists of the independent links of Regime II, with their per-edge twist classes and graph holonomies, and of the frames' relation to fields held in a fixed frame.

With these simplifications the rung carries gauge-covariant transport between agents, information-geometric dynamics on curved statistical manifolds, and the cross-level structure on which multilevel meta-agent formation and top-down priors are built in Sections 9–12. The results of this section have status P or D (Table 2); the physical readings of the induced metrics (Section 18.1) and of the signature construction carry status S. With the rung's geometry fixed, the next section equips it with the consensus free-energy functional, whose descent drives the dynamics of the rest of the paper.

5 The Single-Scale Consensus Free-Energy Functional

A single agent that minimizes the variational free energy $F[q] = \mathbb{E}_q[\log q(x) - \log p(o, x)] \geq -\log p(o)$ performs approximate Bayesian inference: the bound is tight exactly when q equals the posterior $p(x \mid o)$, and the functional decomposes into a complexity term $D_{\mathrm{KL}}(q\|p)$ that anchors beliefs to the prior and an accuracy term $-\mathbb{E}_q[\log p(o \mid x)]$ that rewards explaining observations [17, 19] (preliminaries in Appendix C.1). This section extends the single-agent functional to a population of agents, obtains the inter-agent coupling and its Gibbs (softmax) attention weights from a source-selection construction, and assembles the complete functional whose descent defines the dynamics of the rest of the paper. We call that object the consensus free-energy functional. It is a composite potential: it is assembled from relative entropies and has the energy-minus-entropy form of a free energy, but unlike $F[q]$ it is not an upper bound on the surprisal of any single normalized model (Section 5.6), and the name "variational free energy" is reserved for the single-agent bound. Its stationary points are the consensus states that serve as the paper's formal model of Wheeler's question of how one world arises out of many observer-participants (W3), a structural correspondence (status S), not a derivation.

5.1 Multi-Agent Extension: From Single Inference to Collective Dynamics

The inter-agent belief-coupling terms $\sum_{ij} \beta_{ij} D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$, the attention-entropy term $\tau\beta_{ij}\log(\beta_{ij}/\pi_{ij})$, and the model-channel analog with γ do not appear in closed variational form with an inter-frame transport law in the standard free-energy-principle literature [17,19]. The closest precedent is federated inference and belief sharing [84], in which an agent treats a neighbor's broadcast posterior as an observation and integrates it by variational updating, with precision deciding whom to attend to. In the flat-transport case $\Omega_{ij} = I$ the alignment block satisfies $\sum_j \beta_{ij} D_{\mathrm{KL}}(q_i\|q_j) = -H(q_i) - \sum_j \beta_{ij}\mathbb{E}_{q_i}[\log q_j]$, so it equals federated inference's treatment of broadcast posteriors as observation likelihoods up to the source-independent belief entropy $-H(q_i)$, with the attention weights in the normalized-precision role; we do not match the two precision dynamics exactly. Relative to federated inference the new ingredient is the $\mathrm{GL}(K_q)$ transport, not the coupling; relative to consensus with edge maps on graphs and to gauge-equivariant networks

(Section 3), it is the coupling of probability laws through a relative-entropy functional with state-dependent attention weights. The multi-agent functional is therefore an engineered consensus energy, not a derivation from the free-energy principle alone; we retain the principle's citations for its conceptual backbone (variational inference, the evidence lower bound, and the complexity and accuracy decomposition).

The cross-scale prior has a different status. Section 10 shows that the prior $p_i^{(\ell)} = \Omega_{iI}[q_I^{(\ell+1)}]$ is exact as the one-episode-lagged predictive prior of an episode-indexed joint [30, Ch. 12]; its same-time use in the energy is an energy-level simplification (Table 2).

The extension introduces several ingredients. Each agent i carries a belief $q_i(c)$ and a prior $p_i(c)$, local sections over the base manifold $\mathcal{C}$ that this paper analyzes at a fixed point; agents carry gauge frames $\phi_i(c) \in \mathfrak{gl}(K_q)$ that encode their internal coordinate systems (Section 4.9); transport operators $\Omega_{ij} \in \mathrm{GL}^+(K_q)$, defined as in (14), translate between frames; and presence functions $\chi_i(c)$ indicate where agents maintain beliefs. The functional combines self terms, which align each belief with its prior through $D_{\mathrm{KL}}(q_i \| p_i)$ and each generative model with its hyper-prior through $D_{\mathrm{KL}}(s_i \| r_i)$; belief coupling $\beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$, which penalizes disagreement after transport; model coupling $\gamma_{ij} D_{\mathrm{KL}}(s_i \| \tilde{\Omega}_{ij} s_j)$, which aligns generative models; and observation terms, through which beliefs explain sensory data.

Pairwise potentials $\psi_{ij}(q_i, \Omega_{ij}[q_j])$ could be chosen freely, but an arbitrary choice fixes neither the pairwise energies nor the weights with which each agent attends to them, and it has no connection to variational inference or information geometry. The construction below fixes both. The weights follow exactly from entropy-regularized minimization of a source-selection relative entropy. The pairwise energy is the relative entropy $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$, with the receiver's law in the first slot (the variational, or I-projection, direction); the scope of its selection by the conditional representation theorem is stated in the status paragraph following the Gibbs solution.

5.2 Attention from a Mixture-of-Sources Consensus Energy

We obtain the inter-agent coupling and its attention weights from a source-selection construction over transported neighbors. The construction is stated at a single base point $c \in \mathcal{C}$, and all distributions and group elements below are evaluated there; integrating the pointwise functional against agent supports gives the functional over $\mathcal{C}$ of Section 5.6.

Mixture-of-sources construction. Agent i (the receiver) updates its belief $q_i(k)$ by attending to a set of source agents j. For the purpose of the alignment energy, the construction treats agent i's latent state k as if drawn from one of the neighboring agents, selected by a categorical latent variable $z \in \{1, \ldots, N\}$:

$$P(k, z) = P(k \mid z) P(z), \tag{35}$$

where $P(z = j) = \pi_j$ is the prior probability of attending to agent j (the source prior), and $P(k \mid z = j) = \mathcal{N}(k; \Omega_{ij}\mu_j, \Omega_{ij}\Sigma_j\Omega_{ij}^\top)$ is agent j's belief transported into agent i's frame by Ω_{ij}. Agent j's belief $q_j(k_j)$ lives in j's local frame, and $\Omega_{ij} q_j$ is agent i's representation of what agent j believes. The model contains no observed variable: the alignment energy below is a relative entropy to the prior joint $P(k, z)$. The attention row obtained from it is the recognition-optimal source row, constant in k, and not the posterior source row $P(z = j \mid k) \propto \pi_j (\Omega_{ij} q_j)(k)$, which varies with k; the two differ by a nonnegative Jensen gap [30, Ch. 6] (the proposition on the cost of the constant-row restriction).

Variational family. We restrict the variational family to the product

$$Q(k, z) = q_i(k)\beta(z) \tag{36}$$

and minimize its relative entropy to $P(k, z)$, where $\beta(z = j) \equiv \beta_{ij} \geq 0$ are attention weights with $\sum_j \beta_{ij} = 1$. The factorization $Q(k \mid z) = q_i(k)$ for all z is a substantive structural restriction and not a tractability simplification: it imposes source independence on the consensus belief, requiring that the one belief q_i serve under every candidate source j. Without it the minimizer of $D_{\mathrm{KL}}(Q\|P)$ over unconstrained $Q(k, z)$ is $Q = P$, with $\beta = \pi$, $Q(k \mid z{=}j) = \Omega_{ij}q_j$, and zero alignment energy, so no divergence-dependent weights and no consensus pressure on a single belief arise. The Gibbs form $\beta_{ij} \propto \pi_j \exp[-D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)/\tau]$ derived below is therefore the exact consequence of the source-independence restriction together with the entropy-regularized relative-entropy energy, and not of the variational principle alone.

Alignment energy. For a fixed row of source templates $P(k \mid z{=}j)$, the alignment component is the relative entropy of the restricted variational law to the source-selection model,

$$\mathcal{F}_{\mathrm{align}} = D_{\mathrm{KL}}\big(Q(k, z)\|P(k, z)\big) = \sum_j \beta_{ij} \int q_i(k) \log \frac{q_i(k)\beta_{ij}}{P(k \mid z{=}j)\pi_j} \mathrm{d}k. \tag{37}$$

Decomposing the logarithm yields

$$\mathcal{F}_{\mathrm{align}} = \sum_j \beta_{ij} \left[\underbrace{\int q_i(k) \log \frac{q_i(k)}{P(k \mid z{=}j)} \mathrm{d}k}_{=D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)} + \log \beta_{ij} - \log \pi_j \right]. \tag{38}$$

The identification of the integral with $D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ follows from the definition of the relative entropy after setting $P(k \mid z{=}j) = (\Omega_{ij}q_j)(k)$. It is an exact row identity for fixed templates; the dependence of the templates on the other agents' beliefs is addressed in the status paragraph below. Defining the alignment energy $E_{ij} \equiv D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ gives

$$\mathcal{F}_{\mathrm{align}} = \sum_j \beta_{ij}\big(E_{ij} + \log \beta_{ij} - \log \pi_j\big). \tag{39}$$

For a uniform source prior $\pi_j = 1/N$ this has the energy-minus-entropy form $\mathcal{F}_{\mathrm{align}} = \langle E\rangle_\beta - H(\beta) +$ const, with $H(\beta) = -\sum_j \beta_{ij} \log \beta_{ij}$ the entropy of the attention distribution. For a non-uniform prior the prior term depends on β, and $\mathcal{F}_{\mathrm{align}} = \langle E\rangle_\beta - H(\beta) - \sum_j \beta_{ij} \log \pi_j = \langle E\rangle_\beta + D_{\mathrm{KL}}(\beta_i\|\pi)$.

Minimization and the Gibbs solution. Minimizing $\mathcal{F}_{\mathrm{align}}$ with respect to β_{ij} subject to $\sum_j \beta_{ij} = 1$ with a Lagrange multiplier,

$$\mathcal{L} = \sum_j \beta_{ij}(E_{ij} + \log \beta_{ij} - \log \pi_j) - \lambda \left(\sum_j \beta_{ij} - 1 \right), \tag{40}$$

the stationarity condition $\partial\mathcal{L}/\partial\beta_{ik} = 0$ gives

$$E_{ik} + \log \beta_{ik} + 1 - \log \pi_k - \lambda = 0. \tag{41}$$

Solving and normalizing gives Gibbs weights over the sources, at unit temperature with energies E_{ik} and prior weights π_k,

$$\beta_{ik} = \frac{\pi_k \exp(-E_{ik})}{\sum_m \pi_m \exp(-E_{im})}. \tag{42}$$

With $\mathrm{softmax}_k(x_k) = e^{x_k} / \sum_m e^{x_m}$, a uniform source prior $\pi_k = 1/N$ gives

$$\boxed{\beta_{ik} = \mathrm{softmax}_k(-E_{ik}) = \mathrm{softmax}_k\big(-D_{\mathrm{KL}}(q_i \| \Omega_{ik} q_k)\big).} \tag{43}$$

A temperature $\tau > 0$ is introduced by writing the entropy-regularized objective as $\sum_j \beta_{ij} E_{ij} + \tau \sum_j \beta_{ij} \log(\beta_{ij}/\pi_j)$, whose minimizer is the Gibbs row at temperature τ,

$$\beta_{ik} = \frac{\pi_k \exp(-E_{ik}/\tau)}{\sum_m \pi_m \exp(-E_{im}/\tau)}. \tag{44}$$

The energy carries the factor $1/\tau$, while the prior π_k enters at unit log-weight. The energies, the temperature, and the weights are dimensionless, with relative entropies measured in nats. For $\tau \neq 1$ the tempered objective is not the relative entropy of the source-selection model above; that identification holds only at $\tau = 1$.

In the special case of a single base point with isotropic Gaussian beliefs of a common covariance and trivial frames, the energies reduce to scaled squared distances between means, and with a separately introduced bilinear compatibility between the compared vectors the Gibbs row reduces to the scaled dot-product attention of machine-learning sequence models [85]. That limit, its variants, and its empirical study are developed in a companion paper [86]; none of it is used in the present paper.

Status of the construction. The mixture-of-sources construction is a consensus-energy ansatz, not a derivation from a fixed state-level generative model: the component laws $P(k \mid z{=}j) = \Omega_{ij} q_j$ depend on the variational beliefs q_j of other agents, so the row model moves when the population state moves. Retaining the entropy and prior term makes the resulting population functional a scalar potential whose row optimum has the log-partition value derived in Section 5.6. The choice of divergence is characterized by the conditional representation theorem (Appendix F, Theorem 2) within the class fixed by that appendix's assumptions (an f-divergence coupling, linear coupling in the weights, and exponential-family closure of the minimizer) and the real-analyticity hypothesis on the generator f. Within that class, the relative entropy $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ is the unique f-divergence for which agent i's best response, with neighbor beliefs and attention held fixed, is the geometric-mean Boltzmann form. In the joint scalar functional, where each belief is also a transported template in its neighbors' rows, no nonzero f-divergence yields free-form exponential-family closure, and a Gaussian family imposes closure by restriction. The theorem therefore does not select the divergence of the joint functional; that divergence is declared here, as it is in the companion theory [30, Chs. 6 and 12]. Theorem 3 proves that the within-level terms cannot, on an open mean-field family, equal the negative evidence lower bound of one fixed joint on the original agent-state variables. The obstruction does not exclude the exact zero-coupling hierarchical joint, the equilibrium-frozen auxiliary construction of Theorem 4, or the belief-configuration Gibbs lift of Theorem 6.

Non-uniform source priors. The source prior π_j in (42) carries structural prior knowledge about which sources are available and how likely each is to be relevant, independently of the current beliefs. Zero prior mass removes a source from the row altogether, so an availability or ordering restriction is a prior with restricted support; a prior that decays with a declared distance

between agents encodes locality; and any such prior enters the logits additively as $\log \pi_j$, outside the factor $1/\tau$. If the self pair is admitted, $\Omega_{ii} = I$ gives $E_{ii} = 0$, the smallest energy in the row, so under a prior uniform on the support the self pair carries the largest weight; a prior that removes or down-weights $j = i$ is a separate modeling choice. The source prior is an input of the model; only the Gibbs form of the row is derived.

5.3 Transport and Frame Redundancy

The transport operators are pointwise frame transformations defined as in (14), with $\Omega_{ij}(c) \in \mathrm{GL}^+(K_q)$ and $\phi_i : \mathcal{U}_i \to \mathfrak{gl}(K_q)$ agent i's frame field. They act on latent vectors through a representation ρ, $\Omega_{ij}[k_j] := \rho(\Omega_{ij})k_j \in \mathbb{R}^{K_q}$ and $\tilde{\Omega}_{ij}[m_j] := \rho(\tilde{\Omega}_{ij})m_j \in \mathbb{R}^{K_m}$; in the fundamental representation this is multiplication by an invertible matrix.

Under a common right translation $U_i(c) \mapsto U_i(c)g(c)$ of all frames every transport operator is unchanged, $\Omega_{ij}(c) \mapsto U_i(c)g(c)g(c)^{-1}U_j(c)^{-1} = \Omega_{ij}(c)$, so this change of reference (Section 4.6.2) changes no term of the functional; the additive form $\phi_i \mapsto \phi_i + \xi$ represents it only when ξ commutes with every ϕ_i. At a single base point the active symmetry is one global copy of G acting on frames and beliefs together, and the Regime I frames can be removed from every alignment term by transporting each belief into a common frame (Proposition 1), since $D_{\mathrm{KL}}(q_i \| U_i U_j^{-1} q_j) = D_{\mathrm{KL}}(U_i^{-1} q_i \| U_j^{-1} q_j)$ by Theorem 1. On a base of dimension at least one the frame fields also induce the connection form of Eq. (17) and the field strength of Eq. (19), flat in Regime I (Section 4.6.3), with path-ordered transport $\Omega_\gamma = \mathcal{P} \exp(-\int_\gamma A_\mu \mathrm{d}x^\mu)$ along a path γ and trivial holonomy for the vertex-local transport (Lemma 1). Transport along the base lies outside the single-point analysis of this paper and is deferred to the companion treatment of covers and connection descent [31, Ch. B4].

5.4 Base Priors and the Population Construction

At intermediate levels each agent's prior p_i and hyper-prior r_i are set by the parent meta-agent through Eq. (131), exact in the one-episode-lagged sense and used at the same time in the energy: $p_i^{(\ell)}(k_i; c) = \mathcal{N}(k_i; \mu_{0,i}^{(q)}(c), \Sigma_{0,i}^{(q)}(c))$ regularizes the latent state k_i with parameters determined by $\Omega_{iI}[q_I^{(\ell+1)}]$, and $r_i^{(\ell)}(m_i; c) = \mathcal{N}(m_i; \mu_{0,i}^{(s)}(c), \Sigma_{0,i}^{(s)}(c))$ regularizes the generative-model parameters m_i with parameters determined by $\tilde{\Omega}_{iI}[s_I^{(\ell+1)}]$. Different agents receive different prior parameters through different transport operators Ω_{iI} and the spatial profile of the parent's belief field, not through independent inductive biases. Because p_i moves with the parent belief, treating it as fixed on the fast timescale requires the parent laws to enter only between episodes (Section 7), a schedule that is assumed and not derived. The population construction at each point c combines the single-agent terms [17, 19] with the two row-wise source-selection alignments derived above: the belief channel uses $\beta_{ij}(c)$ and $\Omega_{ij}(c)$, and the model channel uses $\gamma_{ij}(c)$ and $\tilde{\Omega}_{ij}(c)$. At the top level, where no parent exists, $r_i^{(\ell_{\max})}$ is a declared root (Section 12).

5.5 The Pointwise Consensus Functional

At a single point $c \in \mathcal{C}$ the consensus free-energy functional is

$$\begin{aligned}\mathcal{F}(c) = &\sum_i D_{\mathrm{KL}}(q_i(c)\|p_i(c)) + \lambda_h \sum_i D_{\mathrm{KL}}(s_i(c)\|r_i(c)) \\ &+ \sum_{i,j} \beta_{ij}(c) D_{\mathrm{KL}}\big(q_i(c)\big\|\Omega_{ij}(c)q_j(c)\big) + \tau \sum_{i,j} \beta_{ij}(c) \log \frac{\beta_{ij}(c)}{\pi_{ij}(c)} \\ &+ \sum_{i,j} \gamma_{ij}(c) D_{\mathrm{KL}}\big(s_i(c)\big\|\tilde{\Omega}_{ij}(c)s_j(c)\big) + \tau_\gamma \sum_{i,j} \gamma_{ij}(c) \log \frac{\gamma_{ij}(c)}{\pi^{(s)}_{ij}(c)} \\ &- \sum_i \mathbb{E}_{q_i(c)\otimes s_i(c)}\big[\log p(o_i(c) \mid k_i, m_i)\big],\end{aligned} \tag{45}$$

where $\beta_{ij}(c)$ and $\gamma_{ij}(c)$ are free variational parameters (one normalized row per agent and channel), $\pi_{ij}(c)$ and $\pi^{(s)}_{ij}(c)$ are the source priors of the two channels (Section 5.2), and $\tau > 0$ and $\tau_\gamma > 0$ are the belief-channel and model-channel temperatures. In the observation term $k_i \sim q_i(c)$ is agent i's latent state, $m_i \sim s_i(c)$ its generative-model parameter, and $o_i(c)$ its observation at c; a record shared by all agents is the special case $o_i = o$.

Typing of the observation term. The observation term is typed jointly in (k_i, m_i) because the evidence bound forces this typing. For agent i's private block, with prior $p_i(\mathrm{d}k)r_i(\mathrm{d}m)$, likelihood $p(o_i \mid k, m)$, and any recognition law ζ_i on (k_i, m_i) with marginals q_i and s_i, the negative evidence lower bound is $D_{\mathrm{KL}}(\zeta_i\|p_i \otimes r_i) - \mathbb{E}_{\zeta_i}[\log p(o_i \mid k_i, m_i)]$, the energy form of the collective free energy [30, Ch. 5], and

$$D_{\mathrm{KL}}(\zeta_i\|p_i \otimes r_i) = D_{\mathrm{KL}}(q_i\|p_i) + D_{\mathrm{KL}}(s_i\|r_i) + I_{\zeta_i}(K; M), \tag{46}$$

which follows by writing $\log[\zeta_i/(p_i r_i)] = \log[\zeta_i/(q_i s_i)] + \log(q_i/p_i) + \log(s_i/r_i)$ and taking the ζ_i-expectation. Under the product restriction $\zeta_i = q_i \otimes s_i$ the mutual information vanishes, and the block contributes the two self terms, with $\lambda_h = 1$, and the observation term of (45). The predictive form $-\mathbb{E}_{q_i}[\log \int p(o_i \mid k_i, m) s_i(\mathrm{d}m)]$ lies below the joint term by the Jensen gap $\mathbb{E}_{q_i}[D_{\mathrm{KL}}(s_i\|s_i^{(o,k)})] \geq 0$, where $s_i^{(o,k)}(\mathrm{d}m) \propto p(o_i \mid k, m) s_i(\mathrm{d}m)$; paired with $D_{\mathrm{KL}}(s_i\|r_i)$ it is not the negative evidence lower bound of the block and can fall below the block's surprisal.

Temperature. The temperature may be parameterized as $\tau = \kappa\sqrt{K_q}$ with a dimensionless constant κ. For independent coordinates the spread of the energies E_{ij} across sources grows as $\sqrt{K_q}$, so this choice keeps logit differences of order one as K_q varies; it does not normalize the absolute logits. On the model fiber the same rule gives $\tau_\gamma = \kappa\sqrt{K_m}$, which equals τ only when $K_m = K_q$ or when one temperature is declared for both channels. The temperature is not a variational parameter of $\mathcal{F}$. With $Z_i = \sum_j \pi_{ij} e^{-E_{ij}/\tau}$, the row value at the Gibbs weights is $-\tau \log Z_i$ (Section 5.6), and, using $\log(\beta^*_{ij}/\pi_{ij}) = -E_{ij}/\tau - \log Z_i$,

$$\frac{\partial}{\partial \tau}\big(-\tau \log Z_i\big) = -\log Z_i - \frac{\langle E_i \rangle_{\beta^*_i}}{\tau} = D_{\mathrm{KL}}(\beta^*_i\|\pi_i) \geq 0, \tag{47}$$

the analog of $\partial F/\partial T = -S$ with $S = -D_{\mathrm{KL}}(\beta^*_i\|\pi_i) \leq 0$. Minimizing $\mathcal{F}$ over τ therefore drives $\tau \to 0^+$, where $-\tau \log Z_i \to \min_j E_{ij}$ on the support of π_i and each row collapses onto its lowest-energy source. The constant κ is accordingly declared as part of the model, and an objective that adapts it must be external to $\mathcal{F}$.

5.6 The Complete Consensus Functional

Status of the functional. The functional displayed below is an ansatz, not a theorem. Its form was fixed by requiring every term to transform covariantly under the frame changes of Section 4, together with the source-selection reading of Section 5.2; it is not obtained from a single normalized generative model over the joint population state (Theorem 3). In the terminology of the companion theory it is a composite potential, a descent potential assembled from normalized pieces that carries no evidence interpretation of its own [30, Ch. 12]. Where normalized laws are required the companion supersedes it: its population objective is one collective evidence bound [30, Ch. 5], and pairwise divergences enter an evidence bound there only as terms of a normalized Gibbs prior inside the tower free energy [30, Ch. 12]. Given the functional, the Gibbs attention row (50) and the reduced row value $-\tau \log Z_i$ follow exactly, as shown below and in Appendix A. The geometric-mean Boltzmann form of the belief update holds only for the frozen-neighbor best response of Section 5.2; at a stationary point of the complete functional each belief also responds to the rows in which it is a sender, and the geometric pool is not stationary there. Because the belief entropy enters through the self term and through each of agent i's alignment rows, the functional weights $H(q_i)$ by $1 + B_i$ with $B_i := \sum_j \beta_{ij} = 1$, so the frozen-row best response is a Gibbs state at temperature two in units in which Bayesian updating has temperature one (Appendix D).

The exact probabilistic status of the attention row is established in the companion theory, and it bounds what may be claimed here. On a sample space enlarged by a source label with prior π_i and a record likelihood proportional to $\exp(-D_{ij}/\tau)$, the optimal recognition row held constant in the latent state is the Gibbs row of the expected energies, and the row contributes $D_{\mathrm{KL}}(\beta_i \| \pi_i) + \tau^{-1} \sum_j \beta_{ij} \mathbb{E} D_{ij}$ to the negative evidence lower bound [30, Ch. 6] (the proposition on exact posterior and recognition attention). The row block of Eq. (49) is τ times this quantity, so for $\tau \neq 1$ it is not a sector of one evidence bound alongside the unit-weight self and observation terms [30, Ch. 12] (the proposition on Gibbs rows). A second condition arises when the source records are Gaussian links. Let $y_i \mid y_j, J_i = j \sim \mathcal{N}(\Omega_{ij} y_j, L_{ij})$, with the sources topologically ordered so that the links define a normalized directed joint, and let the recognition be mean-field Gaussian with $q_j = \mathcal{N}(\mu_j, \Sigma_j)$. With $A_{ij} = \Omega_{ij} \Sigma_j \Omega_{ij}^\top$, the coordinate update of the label is

$$\log \beta_{ij} = \log \pi_{ij} - \tfrac{1}{2} \log\det L_{ij} - \tfrac{1}{2} \operatorname{tr}\left[L_{ij}^{-1}(\Sigma_i + A_{ij})\right] - \tfrac{1}{2}(\mu_i - \Omega_{ij}\mu_j)^\top L_{ij}^{-1}(\mu_i - \Omega_{ij}\mu_j) + \text{const}, \quad (48)$$

while $-E_{ij}/\tau = -\frac{1}{2\tau}[\operatorname{tr}(A_{ij}^{-1}\Sigma_i) + (\mu_i - \Omega_{ij}\mu_j)^\top A_{ij}^{-1}(\mu_i - \Omega_{ij}\mu_j) - K_q + \log\det A_{ij} - \log\det \Sigma_i]$. Matching the two as an identity in the unconstrained sender mean forces $L_{ij} = \tau A_{ij}$; the trace terms then agree up to a source-independent constant, and the log-determinants leave the extra source-dependent logit $-\frac{1}{2}(1 - 1/\tau) \log\det(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)$, which vanishes for every configuration only at $\tau = 1$. The row at the operating temperature is therefore the exact coordinate update of no model in this family, and the tie $L_{ij} = \tau A_{ij}$ sets a generative parameter equal to a recognition covariance, so the joint is fixed only if Σ_j is frozen at its current value during the update; the tie is a declared postulate and not a stationary point of the same objective. None of this alters the status of the functional as an ansatz, and none of it is needed for the consequences derived below, which follow from the functional as posited.

Extending to fields over the base manifold $\mathcal{C}$ with presence functions $\chi_i(c)$, we obtain the

canonical functional of this paper:

$$\begin{aligned}\mathcal{F}[\{q_i\},\{p_i\},\{s_i\},\{r_i\},\{\phi_i\},\{\beta_{ij}\},\{\gamma_{ij}\}] = &\sum_i \int_{\mathcal{C}} \chi_i(c) D_{\mathrm{KL}}(q_i(c)\|p_i(c))\mathrm{d}c \\ &+ \lambda_h \sum_i \int_{\mathcal{C}} \chi_i(c) D_{\mathrm{KL}}(s_i(c)\|r_i(c))\mathrm{d}c \\ &+ \sum_i \int_{\mathcal{C}} \chi_i(c) \sum_j \Big[\beta_{ij}(c) D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j) + \tau\beta_{ij}(c)\log\tfrac{\beta_{ij}(c)}{\tilde{\pi}_{ij}(c)}\Big]\mathrm{d}c \\ &+ \sum_i \int_{\mathcal{C}} \chi_i(c) \sum_j \Big[\gamma_{ij}(c) D_{\mathrm{KL}}(s_i\|\tilde{\Omega}_{ij}s_j) + \tau_\gamma\gamma_{ij}(c)\log\tfrac{\gamma_{ij}(c)}{\tilde{\pi}^{(s)}_{ij}(c)}\Big]\mathrm{d}c \\ &- \sum_i \int_{\mathcal{C}} \chi_i(c) \mathbb{E}_{q_i(c)\otimes s_i(c)}\big[\log p(o_i(c)\mid k_i, m_i)\big]\mathrm{d}c,\end{aligned} \tag{49}$$

where the sender-presence-weighted prior is $\tilde{\pi}_{ij}(c) := \chi_j(c)\pi_{ij}(c)/\sum_k \chi_k(c)\pi_{ik}(c)$, defined where $\chi_i(c) > 0$ and the denominator is positive (and analogously $\tilde{\pi}^{(s)}_{ij}$), and the observation term carries the joint typing of the pointwise functional. The weight $\lambda_h > 0$ on the model-channel self term equals 1 when the self and observation terms form the evidence bound of each agent's private block (the typing paragraph above); setting $\lambda_h = 0$ freezes the slow subsystem, a special case that a prospective test must declare explicitly. The same λ_h weights the depth-one model coupling of Eq. (134) and is held fixed across levels by assumption in the hierarchical coarse-graining of Section 11; the exact coarse-graining is not shown to preserve it, and at one loop it shifts a coefficient of the same kind, the parent's entropy weight, in proportion to cluster size (Eq. (278)). The belief self term $D_{\mathrm{KL}}(q_i\|p_i)$ is shown with unit weight, the $\alpha = 1$ specialization of the per-agent self-coupling weight $\alpha_i(c)$; promoting this weight to a state-dependent variational parameter is the generalization developed in Section 6. Differentiating the row Lagrangian $L_i = \int \chi_i \sum_j [\beta_{ij} D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j) + \tau\beta_{ij}\log(\beta_{ij}/\tilde{\pi}_{ij})] - \lambda_i(c)(\sum_j \beta_{ij} - 1)$ and imposing the row-normalization constraint $\sum_j \beta_{ij}(c) = 1$ yields the closed-form Gibbs row

$$\beta^*_{ij}(c) = \frac{\chi_j(c)\pi_{ij}(c)\exp\big[-\frac{1}{\tau}D_{\mathrm{KL}}\big(q_i(c)\big\|\Omega_{ij}q_j(c)\big)\big]}{\sum_k \chi_k(c)\pi_{ik}(c)\exp\big[-\frac{1}{\tau}D_{\mathrm{KL}}\big(q_i(c)\big\|\Omega_{ik}q_k(c)\big)\big]}. \tag{50}$$

The receiver presence $\chi_i(c)$ multiplies the whole row and does not affect its minimizer. The sender presence $\chi_j(c)$ enters multiplicatively in numerator and denominator, so $\beta^*_{ij}(c)$ vanishes where $\chi_j(c) = 0$ and reduces to the prior-weighted Gibbs row where all senders are fully present; for hard masks $\chi_j \in \{0,1\}$ it is the Gibbs row restricted to the active set $\mathcal{J}_i(c) = \{j : \chi_j(c) = 1\}$. The result is the general solution (42) in pair-indexed form, and a uniform prior on the active set recovers (43). The model-channel weights $\gamma^*_{ij}(c)$ have the analogous form, with τ_γ, $\tilde{\Omega}_{ij}$, s_j, and $\pi^{(s)}_{ij}$ in place of τ, Ω_{ij}, q_j, and π_{ij}.

Here $\chi_i(c) \in [0,1]$ is agent i's presence function. Each alignment row is weighted by its receiver's presence, as the self and observation terms are, and the sender presences are carried inside $\tilde{\pi}_{ij}$, so that the row-simplex constraint $\sum_j \beta_{ij}(c) = 1$ is uniform across pairs. Absorbing the full pairwise presence $\chi_i\chi_j$ into the prior instead would cancel χ_i between numerator and denominator and leave the row with unit weight where agent i is absent, which is why the receiver presence multiplies the row explicitly. This bookkeeping is the convention for every later restatement of the alignment terms. We adopt the smooth-section convention $\chi_i : \mathcal{C} \to [0,1]$ in place of a hard $\{0,1\}$ indicator,

because the bundle language of Section 4.3 requires agents to be smooth sections over open domains, and a discontinuous χ_i would make the integrands distributional at support boundaries; the informal $\chi_i \in \{0, 1\}$ is the sharply localized limit.

Setting $\gamma_{ij} = 0$ freezes model alignment, a case that a prospective test must declare separately from an active model channel. The model-alignment term is retained because consensus on generative models, the content of $\gamma_{ij} D_{\mathrm{KL}}(s_i \| \tilde{\Omega}_{ij} s_j)$, is required by the meta-agent and declared-root constructions of Sections 9–12. Stiffness terms (Section 8) are optional additions at a point. A Yang-Mills penalty [76] on the curvature of Eq. (19) vanishes identically for Regime-I frames by the Maurer-Cartan identity (Section 4.6.3); a nonzero lattice action belongs to the independent-link sector of Section 4.6.5; and base-direction regularizers such as frame-field smoothness belong to base transport, deferred to [31, Ch. B4].

Complete functional versus the entropy-suppressed surrogate. The boxed functional (49) includes the attention entropy and prior terms explicitly. For one row, write

$$\mathcal{F}_{i,\mathrm{full}}(\beta, x) = \sum_j \beta_{ij} E_{ij}(x) + \tau \sum_j \beta_{ij} \log \frac{\beta_{ij}}{\tilde{\pi}_{ij}}, \qquad Z_i(x) = \sum_j \tilde{\pi}_{ij} \exp\left[-\frac{E_{ij}(x)}{\tau}\right]. \tag{51}$$

Minimization on the row simplex gives $\beta^*_{ij} = \tilde{\pi}_{ij} \exp(-E_{ij}/\tau)/Z_i$ and the exact identity

$$\mathcal{F}_{i,\mathrm{full}}(\beta^*(x), x) = \mathcal{F}_{i,\mathrm{red}}(x) = -\tau \log Z_i(x). \tag{52}$$

The row stationarity condition makes the explicit $\mathrm{d}\beta^*/\mathrm{d}x$ terms cancel, so

$$\mathrm{d}\mathcal{F}_{i,\mathrm{red}} = \sum_j \beta^*_{ij} \mathrm{d}E_{ij}. \tag{53}$$

The entropy-suppressed quantity $S_i(x) := \sum_j \beta^*_{ij}(x) E_{ij}(x)$ is a different scalar potential. Differentiating every occurrence of x gives $\mathrm{d}S_i = \sum_j \beta^*_{ij} \mathrm{d}E_{ij} - \tau^{-1} \mathrm{Cov}_{\beta^*_i}(E_{ij}, \mathrm{d}E_{ij})$. Its Hessian remains symmetric wherever the energies are twice continuously differentiable; the response term shows only that $S_i \neq \mathcal{F}_{i,\mathrm{red}}$, not that S_i is nonintegrable. A nonintegrable field arises only after an additional asymmetric approximation that keeps the receiver-row derivatives and drops the sender-row derivatives of the same scalar. The dynamics of this paper descend the complete functional: the reduced rows (51) of both alignment channels are added to one scalar, and that scalar is differentiated. Neither the entropy-suppressed surrogate nor the receiver-only field is used.

6 State-Dependent Precision, the Envelope Theorem, Environmental Agents, and Symmetry Breaking

The canonical functional (49) fixes the belief self-coupling weight at unity. This section first promotes that weight to a state-dependent variational parameter, then performs the envelope reduction that substitutes the optimal attention weights back into the functional, then recasts the external observation terms as couplings to environmental agents, and finally identifies the global symmetry that observations break explicitly.

6.1 State-Dependent Prior Precision

The self-coupling term $D_{\mathrm{KL}}(q_i(c) \| p_i(c))$ in Eq. (49) carries an implicit unit weight, equivalent to assuming that each agent's prior precision is fixed and uniform in context. In a generative

hierarchy this is restrictive: the trust an agent should place in its prior depends on how far its current belief has moved from it, and this varies across the manifold and with the agent's beliefs. Promoting the self-coupling weight to a per-agent variational parameter $\alpha_i(c) > 0$ reproduces the precision-weighted prediction-error form of the self-coupling gradient. Here $\alpha_i(c)$ acts as a state-dependent scalar gain on the prior-precision metric, not as the optimized precision matrix itself, so the correspondence with predictive coding is structural and does not recover its precision update exactly.

Precision as a variational parameter. With α_i a per-agent variational parameter and $R(\alpha_i)$ a precision regularizer preventing degenerate $\alpha_i \to 0$ or $\alpha_i \to \infty$ solutions, the self-coupling sector of Eq. (49) generalizes to

$$\mathcal{F}_i^{\text{adaptive}} = \int_{\mathcal{C}} \chi_i(c) \Big[\alpha_i(c) D_{\mathrm{KL}}(q_i(c) \| p_i(c)) + R(\alpha_i(c))\Big] dc + \mathcal{F}_i^{\text{align}} - \int_{\mathcal{C}} \chi_i(c) \mathbb{E}_{q_i(c) \otimes s_i(c)}[\log p(o_i(c) \mid k_i, m_i)] dc, \tag{54}$$

where $\mathcal{F}_i^{\text{align}}$ collects the alignment terms unchanged from Eq. (49). We take the log-barrier regularizer

$$R(\alpha_i) = b_0 \alpha_i - c_0 \log \alpha_i, \qquad b_0 > 0, \quad c_0 > 0, \tag{55}$$

in which the linear term penalizes arbitrarily large precision and the logarithmic term enforces $\alpha_i > 0$ and penalizes vanishing precision. For the rate-parameterized density $p(\alpha_i \mid b_0, c_0) = b_0^{c_0+1} \alpha_i^{c_0} \exp(-b_0 \alpha_i)/\Gamma(c_0 + 1)$, the full negative log-density is

$$-\log p(\alpha_i \mid b_0, c_0) = b_0 \alpha_i - c_0 \log \alpha_i + \log \Gamma(c_0 + 1) - (c_0 + 1) \log b_0. \tag{56}$$

The last two terms may be dropped only when b_0 and c_0 are fixed. Holding α_i fixed, their hyperparameter derivatives are

$$\partial_{b_0}[-\log p] = \alpha_i - \frac{c_0 + 1}{b_0}, \qquad \partial_{c_0}[-\log p] = -\log \alpha_i + \psi(c_0 + 1) - \log b_0, \tag{57}$$

where ψ is the digamma function. Learning b_0 and c_0 would require these terms together with a hyperprior on them, and it is not pursued here: the hyperparameters are fixed throughout. The stationary point (58) derived next is the minimizer of the declared penalty $\alpha_i D_{\mathrm{KL}}(q_i \| p_i) + R(\alpha_i)$ at fixed hyperparameters. It is not the MAP precision of a normalized hierarchical model, because the term $\alpha_i D_{\mathrm{KL}}(q_i \| p_i)$ carries no α_i-dependent normalizer. For comparison, the normalized hierarchical Gaussian model $k_i \sim \mathcal{N}(\mu_{p,i}, \Sigma_{p,i}/\alpha_i)$ with the same Gamma prior has, at stationarity of $\mathbb{E}_{q_i}[-\log p(k_i \mid \alpha_i)] - \log p(\alpha_i \mid b_0, c_0)$, the precision $\alpha_i = (c_0 + K_q/2)/(b_0 + Q_i)$ with $Q_i = \frac{1}{2}\mathbb{E}_{q_i}[(k_i - \mu_{p,i})^\top \Sigma_{p,i}^{-1}(k_i - \mu_{p,i})]$, which differs from Eq. (58) in both numerator and denominator.

Optimal precision. Stationarity of $\mathcal{F}_i^{\text{adaptive}}$ with respect to α_i at fixed beliefs gives $D_{\mathrm{KL}}(q_i \| p_i) + b_0 - c_0/\alpha_i = 0$, with closed-form solution

$$\boxed{\alpha_i^*(c) = \frac{c_0}{b_0 + D_{\mathrm{KL}}(q_i(c) \| p_i(c))}.} \tag{58}$$

The per-agent objective $g(\alpha_i) = (b_0 + D_{\mathrm{KL}}(q_i \| p_i))\alpha_i - c_0 \log \alpha_i$ has $g''(\alpha_i) = c_0/\alpha_i^2 > 0$ for all $\alpha_i > 0$, so it is strictly convex on the positive ray and α_i^* is its unique global minimizer. When an agent's belief is close to its prior, $D_{\mathrm{KL}}(q_i \| p_i) \approx 0$ and $\alpha_i^* \approx c_0/b_0$ takes its maximum value, so the agent trusts its prior strongly and resists displacement by the alignment and observation

terms; when the belief has drifted far, $D_{\mathrm{KL}}(q_i\|p_i)$ is large and $\alpha_i^* \to 0$, so the prior's influence is reduced and the alignment and observation terms dominate. The hyperparameter b_0 controls sensitivity, and the ratio c_0/b_0 sets the maximum precision attained when beliefs match the prior exactly. Substituting the optimum gives the reduced self term in closed form,

$$\min_{\alpha_i>0}\big[\alpha_i D + R(\alpha_i)\big] = c_0 + c_0 \log\frac{b_0+D}{c_0}, \qquad D = D_{\mathrm{KL}}(q_i\|p_i), \tag{59}$$

a logarithmic penalty that is concave in D. Its slope $c_0/(b_0+D) = \alpha_i^*$ decreases as the belief drifts, which is the weakening of the anchor described above stated in one formula.

Adaptive anchoring and the uniform limit. With isotropic prior covariance $\bar{\Sigma}_{p,i} = \sigma_p^2 I$ and narrow Gaussian beliefs $q_i = \mathcal{N}(\mu_i, \sigma_q^2 I)$ with $\sigma_q^2 \ll \sigma_p^2$, the adaptive prior term contributes the μ_i-dependent quadratic anchoring $\frac{\alpha_i}{2\sigma_p^2}\|\mu_i - \mu_{\mathrm{prior}}\|^2$ with per-agent stiffness $\lambda_{p,i} \equiv \alpha_i/\sigma_p^2$. Substituting the optimum (58) makes this stiffness state dependent in the way just described. The constant-precision specialization $\alpha_i \to \alpha = 1$ collapses Eq. (54) to the canonical functional (49) and gives a uniform quadratic penalty with stiffness $\lambda_p = \alpha/\sigma_p^2$, the product of two quantities the framework treats independently, a precision gain and a prior width. This specialization connects to the precision-as-stiffness interpretation of Section 8.

Per-coordinate precision for diagonal Gaussians. For agents whose belief and prior are diagonal Gaussians in a fixed frame, $q_i = \mathcal{N}(\mu_i, \mathrm{diag}(\sigma_{q,i}))$ and $p_i = \mathcal{N}(\mu_{p,i}, \mathrm{diag}(\sigma_{p,i}))$, the joint distributions factorize and the relative entropy inherits the additive decomposition

$$D_{\mathrm{KL}}(q_i\|p_i) = \sum_{k=1}^{K_q} D_{\mathrm{KL}}^{(k)}(q_i^{(k)}\|p_i^{(k)}), \quad D_{\mathrm{KL}}^{(k)} = \tfrac{1}{2}\left[\frac{\sigma_{q,i}^{(k)}}{\sigma_{p,i}^{(k)}} + \frac{(\mu_{q,i}^{(k)} - \mu_{p,i}^{(k)})^2}{\sigma_{p,i}^{(k)}} - 1 + \log\frac{\sigma_{p,i}^{(k)}}{\sigma_{q,i}^{(k)}}\right]. \tag{60}$$

The standard mean-field treatment of a product of univariate Gaussians under Gamma-Normal conjugacy [87, §10.1.3, §2.3.6] places an independent Gamma prior on each coordinate precision with hyperparameters $(c_0^{(k)}, b_0^{(k)})$. Here $\alpha_i^{(k)}$ weights an entire divergence and does not parameterize a Gaussian precision, so the correspondence is structural and is not an exact conjugacy. Promoting the precision parameter from a scalar α_i to a per-coordinate vector $\alpha_i^{(k)} > 0$ replaces the self-coupling term by

$$\alpha_i D_{\mathrm{KL}}(q_i\|p_i) + R(\alpha_i) \longmapsto \sum_{k=1}^{K_q}\Big[\alpha_i^{(k)} D_{\mathrm{KL}}^{(k)} + R_k(\alpha_i^{(k)})\Big], \qquad R_k(\alpha_i^{(k)}) = b_0^{(k)}\alpha_i^{(k)} - c_0^{(k)}\log\alpha_i^{(k)}, \tag{61}$$

with stationary solution

$$\boxed{\alpha_i^{(k)*}(c) = \frac{c_0^{(k)}}{b_0^{(k)} + D_{\mathrm{KL}}^{(k)}(q_i^{(k)}(c)\|p_i^{(k)}(c))}.} \tag{62}$$

This per-coordinate analog of Eq. (58) behaves in the same way coordinate by coordinate, with maximum precision $c_0^{(k)}/b_0^{(k)}$. The scalar form is recovered by tying $(c_0^{(k)}, b_0^{(k)}) = (c_0, b_0)$ across k and using one shared scalar gain on the summed divergence, yielding $\alpha_i^* = c_0/(b_0 + \sum_k D_{\mathrm{KL}}^{(k)})$. The per-coordinate form is an optional refinement; the rest of the paper uses the scalar form with fixed b_0 and c_0.

Product rule and envelope cancellation. The closed form (58) makes α_i^* a function of the belief through $D_{\mathrm{KL}}(q_i\|p_i)$. Differentiating with respect to any belief parameter $\theta \in \{\mu_i, \Sigma_i\}$,

$$\frac{\partial \alpha_i^*}{\partial \theta} = -\frac{(\alpha_i^*)^2}{c_0}\frac{\partial D_{\mathrm{KL}}(q_i\|p_i)}{\partial \theta}, \tag{63}$$

and applying the product rule to the self-coupling term $\alpha_i^* D_{\mathrm{KL}}(q_i\|p_i)$ in isolation gives

$$\frac{\partial}{\partial \theta}\big(\alpha_i^* D_{\mathrm{KL}}(q_i\|p_i)\big) = \alpha_i^*\frac{\partial D_{\mathrm{KL}}}{\partial \theta} - \frac{(\alpha_i^*)^2}{c_0}D_{\mathrm{KL}}\frac{\partial D_{\mathrm{KL}}}{\partial \theta} = (\alpha_i^*)^2\frac{b_0}{c_0}\frac{\partial D_{\mathrm{KL}}}{\partial \theta}. \tag{64}$$

Equation (64) is the derivative of $\alpha_i^* D_{\mathrm{KL}}$ alone, not of the complete adaptive sector, which also carries the regularizer $R(\alpha_i^*)$. Differentiating $R(\alpha_i^*)$ with the same chain rule and using the stationarity identity $b_0 - c_0/\alpha_i^* = -D_{\mathrm{KL}}$ gives $\partial R(\alpha_i^*)/\partial\theta = +(\alpha_i^*)^2 c_0^{-1} D_{\mathrm{KL}}\partial D_{\mathrm{KL}}/\partial\theta$, which combines with Eq. (64) to

$$\frac{\partial}{\partial \theta}\big(\alpha_i^* D_{\mathrm{KL}} + R(\alpha_i^*)\big) = \frac{(\alpha_i^*)^2}{c_0}\big(b_0 + D_{\mathrm{KL}}\big)\frac{\partial D_{\mathrm{KL}}}{\partial \theta} = \alpha_i^*\frac{\partial D_{\mathrm{KL}}}{\partial \theta}, \tag{65}$$

the envelope form, since $b_0 + D_{\mathrm{KL}} = c_0/\alpha_i^*$. The product-rule correction is therefore canceled exactly by the regularizer's θ-dependence: differentiating the complete adaptive functional (54), with R included, gives $\alpha_i^*\partial D_{\mathrm{KL}}/\partial\theta$, and no correction term survives in the precision sector. The canonical attention block has the same cancellation structure: differentiating its complete energy-plus-entropy scalar yields Eq. (53), whereas differentiating the entropy-suppressed scalar S_i yields the separate covariance response term.

6.2 The Envelope Theorem and the Reduced Free Energy

The boxed functional (49) is the complete (un-minimized) consensus functional, in which β_{ij} and γ_{ij} are optimized jointly with the belief and model fields. Substituting the optimal weights $\beta_{ij}^* = \mathrm{softmax}_j(-E_{ij}/\tau + \log\pi_{ij})$ from (50) into the tempered row (51) yields the reduced alignment energy $-\tau\log Z_i$ of Eq. (52), where $Z_i = \sum_j \pi_{ij}\exp(-E_{ij}/\tau)$. For a physicist the row is the Helmholtz free energy of one discrete source variable at temperature τ with reference measure π_i,

$$-\tau\log Z_i = \langle E_i\rangle_{\beta_i^*} - \tau S_i, \qquad S_i = -D_{\mathrm{KL}}(\beta_i^*\|\pi_i), \tag{66}$$

and the softmax is its Gibbs state by the Gibbs variational principle. The unit-temperature row (39) is a different function: evaluated at the same tempered weights it gives $(1-1/\tau)\langle E_i\rangle_{\beta_i^*} - \log Z_i$, so the two rows agree only at $\tau = 1$. Holding the model sections fixed and setting aside the model-channel alignment terms (which carry an analogous structure in s_i and γ_{ij}), the reduced free energy at point c is

$$\boxed{\mathcal{F}_{\mathrm{red}}[\{q_i\}] = \sum_i D_{\mathrm{KL}}(q_i\|p_i) - \tau\sum_i \log Z_i - \sum_i \mathbb{E}_{q_i\otimes s_i}[\log p(o_i \mid k_i, m_i)],} \tag{67}$$

with $E_{ij} = D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ and $Z_i = \sum_j \pi_{ij}\exp(-E_{ij}/\tau)$; the presence weights χ_i are suppressed for readability in the display and restored in the gradient below.

The envelope theorem [88] guarantees that the gradient of $\mathcal{F}_{\mathrm{red}}$ contains only explicit derivatives of the complete functional evaluated at the optimal attention weights, with the sector receiving the derivative matching the variable being varied. With each row weighted by its receiver presence as

in Eq. (49), for belief variables $x_i \in \{\mu_i, \Sigma_i\}$ the model-channel terms do not contribute unless the fibers are explicitly tied, and

$$\begin{aligned}\nabla_{x_i}\mathcal{F}_{\mathrm{red}} = \chi_i(c)\nabla_{x_i}D_{\mathrm{KL}}(q_i\|p_i) + \chi_i(c)\sum_j \beta^*_{ij}(c)\nabla_{x_i}E_{ij} + \sum_\ell \chi_\ell(c)\beta^*_{\ell i}(c)\nabla_{x_i}E_{\ell i} \\ - \chi_i(c)\nabla_{x_i}\mathbb{E}_{q_i\otimes s_i}[\log p(o_i \mid k_i, m_i)],\end{aligned} \tag{68}$$

and for model-fiber variables $y_i \in \{\mu^s_i, \Sigma^s_i\}$,

$$\begin{aligned}\nabla_{y_i}\mathcal{F}_{\mathrm{red}} = \lambda_h\chi_i(c)\nabla_{y_i}D_{\mathrm{KL}}(s_i\|r_i) + \chi_i(c)\sum_j \gamma^*_{ij}(c)\nabla_{y_i}\tilde{E}_{ij} + \sum_\ell \chi_\ell(c)\gamma^*_{\ell i}(c)\nabla_{y_i}\tilde{E}_{\ell i} \\ - \chi_i(c)\nabla_{y_i}\mathbb{E}_{q_i\otimes s_i}[\log p(o_i \mid k_i, m_i)].\end{aligned} \tag{69}$$

For shared frame variables ϕ_i, both channels contribute only when the same frame field induces both Ω_{ij} and $\tilde{\Omega}_{ij}$; with independent frame fields the belief-channel derivative is taken with respect to ϕ_i and the model-channel derivative with respect to $\tilde{\phi}_i$. The envelope reduction applies to the adaptive functional (54) in the same way it applies to (49), with the inner optimum over α_i absorbed by the closed form (58), so $\mathcal{F}_{\mathrm{red}}$ is available for gradient computation without an explicit Lagrange multiplier on α_i.

The stationary formula predicts that $\beta_{ij}(c)$ concentrates where the transported divergence $D_{\mathrm{KL}}(q_i(c)\|\Omega_{ij}(c)[q_j(c)])$ is small relative to competing senders under the same prior and temperature. This is a prediction of the formula, not an observed spatial attention field.

6.3 Interpretation of the Free Energy Terms

Two readings of Eq. (49) are load-bearing. First, the prior p_i that the self-consistency term anchors against is the transported parent law of Eq. (131), $p_i^{(\ell)} = \Omega_{iI}[q_I^{(\ell+1)}]$, not an independent learned object, and it is exact only in the one-episode-lagged reading, as the predictive prior of an episode-indexed joint (Section 10; [30, Ch. 12]); where the energy evaluates it at the current parent law, that is the energy-level simplification of Table 2 (status S) and not a prior of one joint. The agent's own model section s_i is regulated separately by its hyper-prior r_i on the model fiber through $D_{\mathrm{KL}}(s_i\|r_i)$, so s_i does not act through p_i at the same level. Second, the per-agent priors p_i are not coupled across agents in this formulation; inter-agent agreement at the slow level is carried by the model coupling on s_i. The observation term anchors the system to its records and breaks the global belief-rotation symmetry explicitly (Section 6.5): without it, the population could rotate rigidly into a configuration that is self-consistent but agrees with no record.

6.4 Recasting External Observations as Environmental-Agent Couplings

The functional as presented includes observation terms $-\mathbb{E}_{q_i\otimes s_i}[\log p(o_i(c) \mid k_i, m_i)]$, where o_i is a record of external data. This form connects to standard active inference [17, 19] but introduces an asymmetry: agents are internal to the framework while observations come from outside. At the level of kernels the asymmetry can be removed at no cost. Every normalized observation kernel is the message of an operational environment node, exactly for the joint law of latent state and record, and conversely every environment state with a message policy yields a normalized kernel; this presentation keeps the conditioning on the record and is not exact for a variational problem over the enlarged state [30, Ch. 5] (the observation-interaction kernel equivalence theorem). This section gives the Gaussian counterpart inside the consensus functional: observation terms with Gaussian linear sensors are represented by couplings to environmental agents, exactly at the level of the mean gradient and exactly for the full belief flow after one substitution.

6.4.1 Environmental Agents

We replace each observation by an environmental agent $e_k \in \mathcal{E}$ whose belief $q_{e_k}(c)$ is concentrated around the state encoded by observation o_k, coupled to agent i through the standard information-theoretic term $\sum_{k\in\mathcal{E}} \beta_{i,e_k} D_{\mathrm{KL}}(q_i(c)\|\Omega_{i,e_k}[q_{e_k}(c)])$. When environmental agents have sharp beliefs, the coupling drives agent i's belief mean toward the encoded values exactly as observation terms would; the covariance channel differs by the entropy term derived below. Environmental agents enter as a boundary class of constituent agents whose dynamics are fixed by construction ($q_{e_k} = p_{e_k}$, with $\beta_{i,e_k} = 1$ a fixed coupling outside the row-normalized simplex $\sum_j \beta_{ij} = 1$); they carry sensory information and do not minimize $\mathcal{F}$ themselves. The asymmetry between agent and environment is therefore relocated to the choice of which agents are dynamic and which are boundary; it is not removed.

6.4.2 Formal Equivalence

Proposition 2 (Gaussian-sensor equivalence)**.** *Consider observation terms whose likelihood does not depend on the model parameter m_i, so that the s_i-expectation is trivial, with Gaussian linear likelihood $o_k = Hk + w_k$ with $w_k \sim \mathcal{N}(0, R)$, $R \succ 0$, and $\Lambda_o := H^\top R^{-1} H \succ 0$, entering*

$$\mathcal{F}_{obs} = \mathcal{F}_{internal}[\{q_i\}] - \sum_{i,k} \int \chi_{ik}(c)\mathbb{E}_{q_i(c)}[\log p(o_k \mid k_i)]dc. \tag{70}$$

Set $m_k := \Lambda_o^{-1} H^\top R^{-1} o_k$ and define environmental agents $q_{e_k} = p_{e_k} = \mathcal{N}(m_k, \Sigma_o)$ with $\Sigma_o := \Lambda_o^{-1}$, couplings $\beta_{i,e_k} = 1$, and links $\Omega_{i,e_k} = I$, entering

$$\mathcal{F}_{agent} = \mathcal{F}_{internal}[\{q_i\}] + \sum_{i,k} \int \chi_{ik}(c)\beta_{i,e_k}(c) D_{\mathrm{KL}}\big(q_i(c)\|\Omega_{i,e_k}[q_{e_k}(c)]\big)dc. \tag{71}$$

Then for Gaussian beliefs $q_i = \mathcal{N}(\mu_i, \Sigma_i)$: (i) $\partial_{\mu_i}\mathcal{F}_{obs} = \partial_{\mu_i}\mathcal{F}_{agent}$; (ii) at each point and for each coupled environmental agent, the covariance gradients differ by $-\frac{1}{2}\Sigma_i^{-1}$; (iii) replacing each environmental relative entropy by the cross-entropy $-\mathbb{E}_{q_i}[\log q_{e_k}]$ makes $\mathcal{F}_{agent} - \mathcal{F}_{obs}$ independent of $\{q_i\}$, so the two belief flows coincide. The sensor covariance Σ_o is fixed by (H, R) and is not a free parameter.

Proof. Completing the square, $-\log p(o_k \mid k) = \frac{1}{2}(k - m_k)^\top \Lambda_o (k - m_k) + c_k$ with c_k independent of k, so

$$-\mathbb{E}_{q_i}[\log p(o_k \mid k_i)] = \tfrac{1}{2}\big[(\mu_i - m_k)^\top \Lambda_o (\mu_i - m_k) + \mathrm{tr}(\Lambda_o \Sigma_i)\big] + c_k, \tag{72}$$

while for Gaussian q_i in the sensor's frame

$$D_{\mathrm{KL}}\big(q_i\|q_{e_k}\big) = \tfrac{1}{2}\big[(\mu_i - m_k)^\top \Lambda_o (\mu_i - m_k) + \mathrm{tr}(\Lambda_o \Sigma_i) + \log \tfrac{|\Sigma_o|}{|\Sigma_i|} - K_q\big]. \tag{73}$$

Both have mean gradient $\Lambda_o(\mu_i - m_k)$, which is (i). Differentiating with respect to Σ_i gives

$$\partial_{\Sigma_i} D_{\mathrm{KL}}(q_i\|q_{e_k}) = \tfrac{1}{2}\big(\Lambda_o - \Sigma_i^{-1}\big), \tag{74}$$

against $\partial_{\Sigma_i}[-\mathbb{E}_{q_i} \log p(o_k \mid k_i)] = \frac{1}{2}\Lambda_o$, which is (ii); the discrepancy comes from the entropy term $-\frac{1}{2}\log|\Sigma_i|$ inside the relative entropy. The cross-entropy $-\mathbb{E}_{q_i}[\log q_{e_k}] = \frac{1}{2}[(\mu_i - m_k)^\top \Lambda_o (\mu_i - m_k) + \mathrm{tr}(\Lambda_o \Sigma_i)] + \frac{1}{2}\log|2\pi\Sigma_o|$ differs from the expected negative log-likelihood by a constant independent of q_i, which is (iii). □

The scope of the proposition is sharp. If $\operatorname{rank} H < K_q$, the induced precision $H^\top R^{-1} H$ is singular, q_{e_k} is not a normalized density, and the relative entropy is undefined. For a nonlinear or non-Gaussian likelihood, such as a readout $o = \tanh(2k) + w$, the expected mean gradient of the observation term is in general not affine in μ_i, whereas the mean gradient $\Lambda_o(\mu_i - m)$ of every Gaussian environmental coupling is affine, so no Gaussian environmental agent reproduces it. Outside the Gaussian linear case the exact statement is the kernel-level equivalence cited above.

Construction. Since $q_i(c)$ is a density over the latent fiber $\mathbb{R}^{K_q}$, the environmental agent is specified in fiber variables, and $m_k \in \mathbb{R}^{K_q}$ is the fiber encoding of o_k under the sensor model (for identity readout, $m_k = o_k$). The active-inference special case $\mathcal{B}_{\text{state}} \cong \mathcal{C}$ with $K_q = \dim \mathcal{C}$ and $m_k = c_k$ recovers the reading in which beliefs range over noumenal locations directly, whereas any configuration with $K_q \neq \dim \mathcal{C}$ lies outside that case. A Dirac choice $q_{e_k}(c) = \delta(k - m_k)$ gives an infinite relative entropy, since a belief with a density is not absolutely continuous with respect to a point mass, so environmental agents carry finite sensory precision and the standard active-inference identification is recovered in the high-precision limit $\Sigma_o \to 0$. For each observation o_k at noumenal location c_k the support χ_{e_k} is concentrated near c_k. The equality $p_{e_k} = q_{e_k}$ is stipulated as a boundary condition and is not derived through the cross-scale relation of Eq. (131); environmental agents do not inherit a prior from a meta-agent, and we do not specify primitive model or hyperprior sections for them, so the environmental sub-class is a degenerate special case of the agent definition (Section 4.5). The choice $\Omega_{i,e_k} = I$ declares that the sensor's frame coincides with the receiving agent's frame, that is, that the observation kernel is written in agent i's own coordinates. It is a choice of coordinates for a datum that is then held fixed, and it changes no physical quantity; Section 6.5 shows that data held fixed in this way act as external fields.

The cross-entropy form of part (iii),

$$\mathcal{F}^{\text{xent}}_{\text{agent}}(q_i, q_{e_k}) := -\mathbb{E}_{q_i}[\log q_{e_k}] = D_{\text{KL}}(q_i \| q_{e_k}) + H(q_i), \tag{75}$$

recovers the negative log-likelihood up to a constant when q_{e_k} is the Gaussian sensor density, and equals the relative entropy plus the agent's entropy $H(q_i) = \frac{1}{2} \log |2\pi e \Sigma_i|$. It is structurally identical to the observation operator $-\mathbb{E}_{q_i \otimes s_i}[\log p(o_i(c) \mid k_i, m_i)]$ of the canonical functional (49) [17,19], so the substitution identifies the environmental coupling with the existing observation slot. Under this substitution every term in the free energy is written as an agent-to-agent or agent-to-environmental-agent coupling, with no explicit observation symbol o. The dependence on external information is not eliminated, since q_{e_k} carries the location c_k in exactly the role o_k played, and Section 6.5 identifies environmental agents as external fields that break the global belief-rotation symmetry. The Markov-blanket-of-sensors picture, in which sensory blankets are themselves composed of sensory blankets down to single bits, is a heuristic composition with the cross-scale machinery of Eq. (131); an equation-level cross-scale construction with e_k carrying a level label $\ell' < \ell$ is not given here.

Like physical action principles, the consensus functional is a scalar integrated over a manifold, invariant under the redundancies of Section 4.6.2, and rendered stationary by the dynamics. The analogy holds at the level of variational stationarity only. The dynamics here is a first-order natural-gradient flow, not the second-order Euler-Lagrange system of classical mechanics, general relativity, or Yang-Mills theory [76], and $\mathcal{F}$ is dimensionless in nats while a physical action carries units of action.

6.5 Explicit Breaking of the Global Belief-Rotation Symmetry

The functional has two global invariances at fixed structural data, and observations break only one of them. In the trichotomy of Section 4.6.2 [30, Ch. 2], per-agent frame changes $U_i \mapsto g_i U_i$ with every datum of agent i re-expressed are passive re-expressions, a common right translation

$U_i \mapsto U_i g$ is a change of reference, and the active group at one base point is a single global copy of G.

The right translation $U_i \mapsto U_i g$ leaves every transport $\Omega_{ij} = U_i U_j^{-1}$, every belief, every model section, and every transported prior unchanged. The observation term contains no frame, so it is invariant as well, and no term of the functional breaks this action: configurations related by it are the same physical state, and the action is a labeling redundancy. It can be broken only by exempting some frame from the transformation, for instance the frame U_{e_k} of an environmental agent, which then becomes a fixed background field; the links $\Omega_{i,e_k} = U_i U_{e_k}^{-1}$ change under $U_i \mapsto U_i g$ when U_{e_k} is held fixed.

The symmetry that observations do break is the diagonal left action

$$U_i \mapsto hU_i, \qquad q_i \mapsto h_{\#} q_i, \qquad p_i \mapsto h_{\#} p_i, \qquad s_i \mapsto h_{\#} s_i, \qquad r_i \mapsto h_{\#} r_i, \qquad h \in G, \tag{76}$$

with one h for all agents and the model-channel pushforwards taken through ρ_{model}. It maps $\Omega_{ij} \mapsto h\Omega_{ij}h^{-1}$, so by the invariance of relative entropy under a common pushforward (Theorem 1) every self, alignment, and attention term is unchanged, and the observation-free functional is invariant. The action rotates the collective belief configuration rigidly and moves the consensus belief, so configurations related by it are physically distinct states that the observation-free functional cannot tell apart. A fixed observation kernel $p(o \mid k)$, written in agent i's coordinates and not transformed with the agents, is not invariant, since $\mathbb{E}_{h_{\#} q_i}[\log p(o \mid k)] \neq \mathbb{E}_{q_i}[\log p(o \mid k)]$ in general. It therefore breaks this global symmetry explicitly, in the sense of an external field in statistical mechanics. The same holds for every other datum held fixed while the agents are rotated: the frame and belief of an environmental agent, and a declared root prior at the top of the hierarchy that is not transformed. If the observation kernel were instead re-expressed along with the agents, the transformation would reduce to a passive change of coordinates and nothing would be broken; the physical content of the breaking is the relation of the population to data held fixed. This also separates explicit breaking from gauge fixing: setting $\Omega_{i,e_k} = I$ in the environmental-agent construction chooses coordinates in which the sensor's frame coincides with agent i's frame and changes no physical quantity.

These statements concern the functional. They do not establish that numerical trajectories converge to a common orbit of the belief-rotation symmetry. Orbit membership must be tested by an orbit-sensitive residual, such as the minimum pairwise Procrustes residual over a declared compact subgroup, since equality of belief norms does not suffice. No spontaneous-symmetry-breaking or Goldstone claim follows without an internal selection mechanism and an analysis of the fluctuation spectrum, and whether source-induced specialization yields stable multi-child clusters is left to the prospective validation of Section 13.

7 Two-Timescale Dynamics and the Adiabatic Hypothesis

7.1 An Imposed Time-Scale Separation

Physical systems often exhibit widely separated timescales, and such separation licenses hierarchical descriptions in which fast variables equilibrate on the frozen background of slow ones. We adopt this structure for the consensus functional as a declaration: the functional does not choose the fast-slow split, the rate ratio, or the block mobilities [30, Ch. 8]. Following the typing of that chapter, the fast block consists of the beliefs $\{q_i(c)\}$, which respond to observations, together with the attention rows held at their Gibbs optimum; the slow block consists of the model sections $\{s_i(c)\}$ and the prior parameters. The priors $\{p_i(c)\}$ and hyper-priors $\{r_i(c)\}$ are prior parameters in this sense: in the lagged reading of Section 10 they are fixed within an episode and updated between episodes

from the parent laws of the previous episode, and they have no dynamics of their own beyond that schedule. At the top level the hyper-prior is a fixed boundary condition, the declared root of the hierarchy (Section 12). At one base point the frames are not a separate block (Section 7.4).

7.2 Fast Subsystem: Belief Dynamics

On the fast timescale, agents update beliefs q_i to explain new observations while the priors p_i, the model sections, and the frames are held fixed. The relevant free energy is

$$\begin{aligned}\mathcal{F}_{\text{fast}}[\{q_i\}] = &\sum_i \int \chi_i(c) D_{\text{KL}}(q_i(c) \| p_i(c)) dc \\ &+ \sum_{ij} \int \chi_i(c) \beta_{ij}(c) D_{\text{KL}}(q_i(c) \| \Omega_{ij}[q_j(c)]) dc \\ &- \sum_i \int \chi_i(c) \mathbb{E}_{q_i \otimes s_i}[\log p(o_i(c) \mid k_i, m_i)] dc, \end{aligned} \tag{77}$$

where the attention weights β_{ij} are held at the softmax optimum β^*_{ij} of Eq. (50) during the belief updates, each row is weighted by its receiver presence χ_i, and sender presence is carried inside the attention prior $\tilde{\pi}_{ij}$ as in Section 5.6. Under this block-coordinate convention the attention-entropy block $\tau \beta_{ij} \log(\beta_{ij}/\tilde{\pi}_{ij})$ of the canonical functional (49) is independent of q_i and is omitted from the displayed scalar. The omission leaves the belief gradient unchanged, so the fast flow is block-coordinate descent of Eq. (49) in q at fixed $\beta = \beta^*$, which by the envelope theorem (Section 6.2) coincides with the gradient of the reduced free energy $\mathcal{F}_{\text{red}}$. The displayed $\mathcal{F}_{\text{fast}}$ is therefore gradient-equivalent bookkeeping at fixed β^*, not the entropy-suppressed surrogate of Section 5.6, whose $\beta^*(q)$ dependence is differentiated through. We omit explicit base-space integrals in the equations below for clarity, but they are understood to be present.

Natural-gradient descent on beliefs follows $\partial q_i/\partial t = -\eta_q \tilde{\nabla}_{q_i} \mathcal{F}_{\text{fast}}$ (we write t for the flow time throughout this section, reserving τ for the attention temperature), where $\tilde{\nabla}_{q_i} = g_{\mathcal{B}}^{-1} \nabla_{q_i}$ is the natural gradient with respect to the Fisher-Rao metric $g_{\mathcal{B}}$ on $\mathcal{B}_{\text{state}}$. For Gaussian beliefs $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ the Fisher metric in the moment parameterization (μ_i, Σ_i) is block-diagonal with $g_{\mu\mu} = \Sigma_i^{-1}$ on the mean block and $g_{\Sigma\Sigma}[V, W] = \frac{1}{2}\text{tr}(\Sigma_i^{-1} V \Sigma_i^{-1} W)$ on the covariance block [83]. The block rate η_q is a mobility: it fixes the unit of flow time, and only ratios of block rates enter the timescale argument below. This subsystem performs variational inference, with agents adjusting beliefs to explain observations while respecting their current priors and coordinating with nearby agents.

7.3 Slow Subsystem: Model Learning

On longer timescales, agents update their model sections s_i to reflect long-term statistical structure, regularized by the model hyper-priors r_i, coupled to neighbors through the model-alignment term, and tied to the records through the jointly typed observation term:

$$\mathcal{F}_{\text{slow}}[\{s_i\}] = \lambda_h \sum_i \int \chi_i(c) D_{\text{KL}}(s_i(c) \| r_i(c)) dc + \sum_{ij} \int \chi_i(c) \gamma_{ij}(c) D_{\text{KL}}(s_i(c) \| \tilde{\Omega}_{ij}[s_j(c)]) dc - \sum_i \int \chi_i(c) \mathbb{E}_{q_i \otimes s_i}[\log p(\tag{78}$$

where r_i are the model hyper-priors, supplied by the parent's model section at the previous episode (Section 10) and by the fixed root at the top level. The same block-coordinate and absorbed-prior conventions apply: sender presence is carried inside $\tilde{\pi}^{(s)}_{ij}$ (Section 5.6), and γ_{ij} is held at its softmax optimum γ^*_{ij} during the s-updates, so the omitted entropy block $\tau_\gamma \gamma_{ij} \log(\gamma_{ij}/\tilde{\pi}^{(s)}_{ij})$ is independent of

s_i and the displayed scalar is gradient-equivalent to Eq. (49) in s at fixed $\gamma = \gamma^*$. Natural-gradient descent on models follows $\partial s_i/\partial t = -\eta_s \tilde{\nabla}_{s_i} \mathcal{F}_{\text{slow}}$ with block rate $\eta_s = \epsilon \eta_q$, $\epsilon \ll 1$, on which agents adjust their expectations about how the world works and move toward shared models.

Within an episode the fast subsystem sees p_i and r_i as constants. Whether an episode is long enough for the fast block to relax is a hypothesis on the schedule and is not derived. A same-time assignment, in which p_i follows the current parent belief during the fast updates, is not adopted for the dynamics, because for coherent clusters the parent belief moves at a rate comparable to that of its constituents (Proposition 7), so that assignment would close the loop $q_i \to q_I \to p_i \to q_i$ on the fast timescale and remove the separation exactly when meta-agents exist.

7.4 Gauge Frames at One Base Point

Frames $\phi_i(c)$ may also be evolved to lower the functional, but at one base point they are not an independent slow variable. In Regime I every alignment, attention, and transported-prior term depends on the pair (q_i, U_i) only through the frame-free law point $(U_i^{-1})_{\#} q_i$, by the invariance of relative entropy under a common pushforward (Theorem 1). Evolving a frame at fixed coordinate belief therefore moves the law point along its G-orbit, and the frame flow acts as an additional anisotropic mobility on beliefs. Frame content at a point is relative to data held fixed. The observation kernel reads q_i in agent i's own coordinates, so U_i is the link between the agent's observation leg and its interaction legs, and evolving frames amounts to learning that link; the record kernels of [30, Ch. 5] give this its typing, since a record may carry its own frame and a link on each of its legs. For independent links on the interaction graph (Regime II), the links are genuine interaction variables, and whether they are learned, and on which schedule, is part of the declaration. In either case the frame rate η_ϕ is a block mobility, and a rate ratio between blocks is an anisotropic mobility that can change the selected orbit, not only the speed along it [30, Chs. 7 and 8]. Its value is a modeling choice with consequences.

When frames are evolved, no frame regularizer is included: they move only through the terms of $\mathcal{F}$ in which they appear, under the metric-preconditioned group flow $\dot{U}_i = -\eta_\phi U_i \xi_i$, where ξ_i is the left-trivialized Lie-algebra velocity of Eq. (83). A gradient penalty $\lambda_\phi \sum_i \int \|\nabla \phi_i(c)\|^2 dc$ would couple frames at different base points; it belongs to base-transport theory and is deferred [31]. The chart form $\partial \phi_i/\partial t = -\eta_\phi v_{\phi_i}$ in the parameterization $U_i = \exp(\phi_i)$ agrees with the group flow to first order in the step, as shown below.

7.5 Timescale Separation and the Adiabatic Hypothesis

We declare the rate ratio $\eta_s/\eta_q = \epsilon$ with $\epsilon \ll 1$ between the slow and fast blocks. Write f for the fast coordinates, s for the slow ones, and G_f, G_s for their block metrics. For every $\epsilon > 0$ the rate-separated natural gradient, $\epsilon \dot{f} = -\mathsf{G}_\mathrm{f}^{-1} \partial_f \mathcal{F}$ and $\dot{s} = -\mathsf{G}_\mathrm{s}^{-1} \partial_s \mathcal{F}$ in flow time measured in slow units, decreases the functional. On the graph $\mathcal{S}^\star = \{(f^\star(s), s)\}$ of a branch of conditional minimizers its fast component vanishes and its slow component is the natural gradient of the reduced functional $\mathcal{F}_{\text{red}}(s) = \min_f \mathcal{F}(f, s)$, by the envelope identity $d\mathcal{F}_{\text{red}} = \partial_s \mathcal{F}(f^\star(s), s)$ [30, Ch. 8] (the envelope lemma and the rate-separated natural-gradient proposition). This holds for every ϵ, but the orbit can depend on ϵ, and it does not by itself say that trajectories starting off $\mathcal{S}^\star$ approach it.

That tracking statement is the adiabatic reduction proper, and it requires Tikhonov's hypotheses: a compact slow region $\mathcal{K}$ on which a C^1 branch of conditional minimizers satisfies the uniform fast-curvature bound $\partial_{ff}^2 \mathcal{F}(f^\star(s), s) \succeq c_1 \mathsf{G}_\mathrm{f}$ for some $c_1 > 0$, a reduced slow orbit that stays in the interior of $\mathcal{K}$, and a fast initial state in the basin of the frozen fast flow. Under these hypotheses the fast variables track the conditional minimizer as $\epsilon \to 0$ [89], and for smooth data a locally

invariant slow manifold exists $O(\epsilon)$-close to $\mathcal{S}^\star$ [90]. The effective small parameter is the fastest slow relaxation rate divided by the slowest fast relaxation rate, that is, the block rates multiplied by the spectra of $\mathsf{G}^{-1}\partial^2\mathcal{F}$ in each block, and not the bare ratio η_s/η_q.

For the recognition families of this paper these hypotheses are not established, and the Theory declares the reduction open at one base point [30, Ch. 8] (the open problem on slow-manifold reduction). The uniform fast-curvature bound is certified only in restricted regimes: zero within-scale coupling, where the cross-scale joint is a proper Gaussian with a unique optimum, and frozen isotropic couplings with $\mathrm{SO}(N)$ transport, where the mean update is a contraction; no interior fixed point is established for the fully coupled, state-dependent $\mathrm{GL}^+(K_q)$ system (Appendix G.5). The bound also fails generically where attention switches. After the rows are optimized, the curvature of the reduced row $-\tau \log Z_i$ is $\sum_j \beta^*_{ij}\nabla^2 E_{ij} - \tau^{-1}\,\mathrm{Cov}_{\beta^*_i}(\nabla E_i)$, and the covariance correction can make it indefinite even when every frozen-weight curvature is positive [30, Ch. 6] (the attention-reduced potential proposition). A one-agent example exhibits the resulting fold. With a slow prior mean m, anchor strength $\alpha = 0.2$, two unit-precision neighbors at $x = \pm 1$, and $\tau = 0.05$, the reduced energy

$$\mathcal{F}_{\mathrm{red}}(\mu; m) = \frac{\alpha}{2}(\mu - m)^2 - \tau \log \sum_{j=\pm 1} \exp\Big[-\frac{(\mu - x_j)^2}{2\tau}\Big] \tag{79}$$

has two fast minima and a saddle for moderate m, and at $m \approx 4.22$ the left minimum and the saddle merge in a saddle-node, so the fast state must jump. At such folds the reduction must be replaced by jump or relaxation-oscillation analysis. Outside the restricted regimes above the adiabatic hierarchy is therefore a conjecture (status C in Table 2); where the hypotheses hold, it supports a perturbative expansion in powers of ϵ. The Born-Oppenheimer separation of electronic and nuclear motion [91] is an analogy for the slaving of fast to slow variables, with a different mechanism (slaving to instantaneous eigenstates).

7.6 Fisher Natural Gradient on Beliefs and Metric-Preconditioned Frame Retraction

The system evolves by Fisher-Rao natural-gradient flow on the Gaussian belief sector and, when frames are evolved, by a metric-preconditioned group retraction on the frame sector. On the product manifold $\mathcal{M} = (\mathbb{R}^{K_q} \times \mathbb{S}^+_{K_q} \times G)^N$, the block-Fisher form for Gaussian beliefs (Appendix C.3) gives $\tilde{\nabla}_\mu \mathcal{F} = \Sigma \nabla_\mu \mathcal{F}$ and $\tilde{\nabla}_\Sigma \mathcal{F} = 2\Sigma(\nabla_\Sigma \mathcal{F})\Sigma$, whose sandwich keeps covariance updates within $\mathbb{S}^+_{K_q}$ under an affine-invariant retraction. For the group sector write $U_i = \exp(\phi_i)$ and let $\xi_i \in \mathfrak{g}$ denote the left-trivialized Lie-algebra velocity consumed by the group equation. The local flow is

$$\frac{d\mu_i}{dt} = -\eta_\mu \tilde{\nabla}_\mu \mathcal{F}, \tag{80}$$

$$\frac{d\Sigma_i}{dt} = -2\eta_\Sigma \Sigma_i (\nabla_\Sigma \mathcal{F})\Sigma_i, \tag{81}$$

$$\frac{dU_i}{dt} = -\eta_\phi U_i \xi_i, \tag{82}$$

with the pullback-preconditioned chart vector and its corresponding group velocity defined by

$$v_{\phi_i} := \mathcal{G}^{-1}_{\mathrm{pull}}(\phi_i)\nabla_{\phi_i}\mathcal{F}, \qquad \xi_i := \mathrm{dexp}^L_{\phi_i}(v_{\phi_i}), \tag{83}$$

where $\mathrm{dexp}^L_\phi(v) = \exp(-\phi) D\exp_\phi[v]$ is the left-trivialized differential of the exponential map, pairing with the right-multiplication retraction of Eq. (85). With separate rates $\eta_\mu \neq \eta_\Sigma$ the belief

flow is the natural gradient of the rate-weighted metric $G_\eta := (\Sigma_i^{-1}/\eta_\mu) \oplus (g_{\Sigma\Sigma}/\eta_\Sigma)$ and coincides with a single-rate Fisher natural gradient only when $\eta_\mu = \eta_\Sigma$. Identities that assume one scalar rate in the Fisher metric, such as the clock-dissipation identity of Section 17, hold exactly in the rate-weighted metric and block by block, and in the Fisher metric itself only at a single rate.

A chart gradient $\nabla_{\phi_i}\mathcal{F}$, or even the chart vector v_{ϕ_i}, is not inserted directly into a group retraction unless it has been converted to the trivialized velocity ξ_i. On a compact semisimple Lie algebra one may use the negative Killing form, which is bi-invariant and positive-definite, to precondition the left-trivialized gradient directly. For a compact algebra with central or abelian factors, the Killing form is degenerate on those directions and must be supplemented by a separately chosen positive Ad-invariant inner product, such as the negative trace form in the defining matrix representation of $\mathfrak{so}(2)$. For non-compact G the Killing form is indefinite, and $\mathrm{GL}^+(K_q)$ admits no bi-invariant Riemannian metric [82, Lemma 7.5], so the positive-definite substitute used in exponential coordinates is the position-dependent pullback of the ambient flat Frobenius metric on $\mathrm{GL}^+(K_q)$ through the exponential map,

$$\begin{aligned} \mathcal{G}_{\mathrm{pull},ab}(\phi) &= \langle D\exp_\phi[T_a], D\exp_\phi[T_b]\rangle_F \\ &= \mathrm{tr}\left(\Psi_R(\mathrm{ad}_\phi)(T_a)^\top \Psi_R(\mathrm{ad}_\phi)(T_b)\exp(\phi)\exp(\phi)^\top\right), \\ \Psi_R(z) &= \frac{e^z-1}{z}, \end{aligned} \tag{84}$$

where $\langle\cdot,\cdot\rangle_F$ is the ambient Frobenius inner product and $D\exp_\phi[H] = \Psi_R(\mathrm{ad}_\phi)(H)\exp(\phi)$ is the right-trivialized differential of the exponential map, so the weight $\exp(\phi)\exp(\phi)^\top = UU^\top$ is the right $\exp(\phi)$ factor of that differential. This form is positive definite exactly where dexp_ϕ^L is regular. As the pullback of a flat metric by a local diffeomorphism it is flat, and it is invariant under neither left nor right translations by non-orthogonal elements of G. By contrast, for variations $\delta U = U\xi$, the quantity $U^\top\nabla_U\mathcal{F}$ is the left-trivialized, or body, covector. Identifying it with a velocity through a fixed Frobenius inner product on the algebra induces a left-invariant metric on $\mathrm{GL}^+(K_q)$. That metric is also right invariant only if the inner product is Ad-invariant, which no positive inner product on $\mathfrak{gl}(K_q)$ is; a right-invariant metric by itself exists on any Lie group and gives the mirror convention below. The position-dependent ambient pullback above is a third, distinct construction.

Proposition 3 (dexp regularity)**.** *The operator* dexp_ϕ^L *is singular exactly when* $\mathrm{spec}(\mathrm{ad}_\phi)$ *contains a nonzero element of* $2\pi i\mathbb{Z}$*; equivalently, the pullback metric of Eq.* (84) *is positive definite exactly away from that spectral locus.*

Clipping $\|\mathrm{ad}_\phi\|$ below 2π is a sufficient guard. On the full $\mathfrak{gl}(K_q)$ chart a skew frame of rotation angle θ has ad_ϕ-eigenvalues 0, $\pm i\theta$, and $\pm 2i\theta$, so dexp_ϕ^L is singular already at $\theta = \pi$; the familiar 2π locus applies only to the chart restricted to $\mathfrak{so}(3)$. The mirror right-trivialized velocity convention would instead pair with a left-multiplication retraction $\exp(-\eta_\phi\xi_i)U_i$; we use the left-trivialized velocity because Eq. (82) and Eq. (85) multiply the update on the right. The discrete analog of Eq. (82) is the group-level retraction

$$U_i^{t+1} = U_i^t \exp\big(-\eta_\phi \xi_i^t\big), \tag{85}$$

which keeps $U_i \in G$ exactly at every step and is the canonical group update once ξ_i has been computed.

For $G = \mathrm{GL}^+(K_q)$, which is open in $\mathfrak{gl}(K_q)$, the metric of Eq. (84) is the exact pullback of the ambient Frobenius metric and $D\exp_\phi$ is surjective wherever dexp_ϕ^L is regular. The pullback-preconditioned chart vector $v_{\phi_i} = \mathcal{G}_{\mathrm{pull}}^{-1}\nabla_{\phi_i}\mathcal{F}$ therefore pushes forward to the ambient gradient,

$D\exp_{\phi_i}[v_{\phi_i}] = \nabla_{U_i}\mathcal{F}$, and its left-trivialized velocity telescopes exactly to $\xi_i = \mathrm{dexp}^L_{\phi_i}(v_{\phi_i}) = U_i^{-1}\nabla_{U_i}\mathcal{F}$. On the full $\mathfrak{gl}(K_q)$ this preconditioning reproduces ambient Euclidean descent on U_i, $U_i\exp(-\eta_\phi U_i^{-1}\nabla_{U_i}\mathcal{F}) = U_i - \eta_\phi\nabla_{U_i}\mathcal{F} + O(\eta_\phi^2)$, and it carries no Fisher content. In computation ξ_i is formed directly from $\nabla_{U_i}\mathcal{F}$; the $K_q^2 \times K_q^2$ metric $\mathcal{G}_{\mathrm{pull}}$ need never be assembled, and inverting it explicitly would square the condition number of $D\exp_\phi$. On a proper subgroup such as $\mathrm{SO}(K_q)$, $D\exp_\phi$ maps onto $T_{U_i}G$ only, and the same construction gives the tangent projection $\xi_i = U_i^{-1}P_{T_{U_i}G}\nabla_{U_i}\mathcal{F}$. The additive chart step $\phi_i^{t+1} = \phi_i^t - \eta_\phi v_{\phi_i}$ and the group step (85) agree to $O(\eta_\phi^2)$ at every ϕ_i, by definition of dexp^L; the conversion is needed only when a chart vector is inserted into the group exponential. The retraction stays in G at every discrete step, but global existence of the continuous frame flow in $\mathrm{GL}^+(K_q)$ is not established. The functional is constant along the noncompact orbits of the change-of-reference redundancy, so its sublevel sets are not compact, and nothing in the ambient flow keeps U_i away from singular matrices. A gauge slice, a coercive frame prior, or an invariant bounded set is required (Appendix G.5).

Frame-flow covariance. The Gaussian sector thus has an intrinsic Fisher-Rao natural gradient, while the frame sector has a coordinate- and trivialization-dependent metric preconditioning. Because $\mathrm{GL}^+(K_q)$ has no bi-invariant Riemannian metric, no single frame metric respects both the per-agent left translations and the global right translation. The ambient-pullback flow, which equals the Euclidean flow $\dot{U}_i = -\eta_\phi\nabla_{U_i}\mathcal{F}$, is equivariant only under orthogonal left and right translations. The left-invariant (body) flow $\dot{U}_i = -\eta_\phi U_i U_i^\top\nabla_{U_i}\mathcal{F}$ is equivariant under per-agent left translations with beliefs co-transformed, and under right translations only by orthogonal g. The right-trivialized flow $\dot{U}_i = -\eta_\phi(\nabla_{U_i}\mathcal{F}U_i^\top)U_i$, paired with the left-multiplication retraction, is equivariant under the change-of-reference redundancy $U_i \mapsto U_i g$, so its trajectories descend to the quotient by that redundancy, at the cost of left equivariance. Under the first two choices, gauge-equivalent initial data evolve to physically different transports, and the linearized frame relaxation rates are not invariant under $U_i \mapsto U_i g$. In a two-agent example with $K_q = 2$ and $\mathcal{F} = D_{\mathrm{KL}}(q_1\|\Omega_{12}q_2)$ at a minimizer, the slowest nonzero frame rate of the ambient flow is $1.07\eta_\phi$ in one gauge and $0.023\eta_\phi$ after $g = \mathrm{diag}(10, 1)$, with $\mathcal{F}$ unchanged, whereas the right-invariant metric gives the rates $(0.93, 4.01, 6.29, 7.04)\eta_\phi$ in every gauge. Any timescale assigned to the frame block is therefore gauge invariant only with a right-invariant frame metric. The choice of frame metric is a declared option, and a covariant flow exists for either quotient. Gauge-covariant trajectory claims for the other choices require compact (Ad-invariant), abelian, or near-identity hypotheses; under them the Gaussian update transforms covariantly as $\Sigma_i(\nabla_\Sigma\mathcal{F})\Sigma_i \to \rho(g)[\Sigma_i(\nabla_\Sigma\mathcal{F})\Sigma_i]\rho(g)^\top$.

8 Statistical Precision as Stiffness and Friction

The natural-gradient flow of Eq. (82) descends the consensus free-energy functional $\mathcal{F}$ of Section 5, which is therefore the potential of the agent dynamics. This section computes the second variation of $\mathcal{F}$ in the Gaussian belief sector and derives the two roles of statistical precision in that first-order dynamics (status D, Table 2). The Hessian is a stiffness tensor whose blocks are built from precisions: the prior term contributes the prior precision $\bar{\Lambda}_{p_i}$, and the interaction terms contribute transported neighbor precisions. The Fisher metric, which in the mean sector is the belief precision, is the friction tensor of the flow, and no term of the dynamics carries an inertia. The only velocity-quadratic form in the dynamics is the Rayleigh dissipation function $\mathcal{R}$, which is half the free-energy dissipation rate and fixes the rate of the information clock of Section 17 through $(d\vartheta/dt)^2 = 2\eta\mathcal{R}$ (Section 8.6). An inertial reading of precision needs a separate kinetic-metric postulate with its own second-order equation of motion, stated in Section 8.7.

Stiffness, friction, and inertia. For a damped oscillator $m\ddot{x}+\gamma\dot{x}+kx=0$ the three coefficients are independent data: the Hessian of the potential gives the stiffness k, the dissipation gives the friction γ, and the kinetic energy gives the mass m. The Hessian of $\mathcal{F}$ derived below supplies k, the Fisher metric of the natural-gradient flow supplies γ, and neither $\mathcal{F}$ nor the flow supplies m. We hold the prior parameters $(\bar{\mu}_i, \bar{\Sigma}_i)$ fixed, the quasi-static approximation in which they evolve slowly relative to the beliefs (μ_i, Σ_i).

8.1 Setup and Notation

Each agent i maintains a belief distribution $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ anchored to a prior $p_i = \mathcal{N}(\bar{\mu}_i, \bar{\Sigma}_i)$ and receives observations o_i through the identity readout $p(o_i \mid k_i, m_i) = \mathcal{N}(o_i; k_i, \Sigma_{o_i})$, whose likelihood does not depend on the model parameter m_i and is written $p(o_i \mid k_i)$ below. We define precision matrices

$$\Lambda_{q_i} = \Sigma_i^{-1} \qquad \text{(belief precision)} \tag{86}$$

$$\bar{\Lambda}_{p_i} = \bar{\Sigma}_i^{-1} \qquad \text{(prior precision)} \tag{87}$$

$$\Lambda_{o_i} = \Sigma_{o_i}^{-1} \qquad \text{(observation precision)}, \tag{88}$$

with transported quantities $\tilde{\mu}_k = \Omega_{ik}\mu_k$ and $\tilde{\Sigma}_k = \Omega_{ik}\Sigma_k\Omega_{ik}^\top$, where $\Omega_{ik} \in \mathrm{GL}(K_q)$ is the transport of the interaction term: in Regime I the frame quotient $e^{\phi_i}e^{-\phi_k}$, and in Regime II an independent link on the interaction graph with orientation reversal $\Omega_{ki} = \Omega_{ik}^{-1}$. The block formulas below use only the transports and hold in both regimes. The transported precision is the inverse of the transported covariance,

$$\tilde{\Lambda}_{q_k} := (\Omega_{ik}\Sigma_k\Omega_{ik}^\top)^{-1} = \Omega_{ik}^{-\top}\Lambda_{q_k}\Omega_{ik}^{-1}, \tag{89}$$

which differs from the covariance transport law because precision is the inverse of covariance. For orthogonal transport $\Omega \in \mathrm{O}(K_q)$ the two laws coincide ($\Omega^{-\top} = \Omega$), and Eq. (89) reduces to $\Omega\Lambda\Omega^\top$; for general $\mathrm{GL}(K_q)$ the dual transport is essential, and we use Eq. (89) throughout. The boxed identity $\Omega_{ji}^\top\tilde{\Lambda}_{q_i}^{(j)}\Omega_{ji} = \Lambda_{q_i}$ used below holds in $\mathrm{GL}(K_q)$ because $\Omega^\top\Omega^{-\top} = I$ algebraically; the secondary simplifications $\tilde{\Lambda}_k\Omega_{ik} = \Omega_{ik}\Lambda_k$ and $\Omega_{ki}^\top\tilde{\Lambda}_i = \Lambda_i\Omega_{ki}^\top$ hold only under $\mathrm{O}(K_q)$; the general-linear identity is given below.

8.2 The Extended Free Energy Functional

The extended working functional at one base point, the fixed-β^* block-coordinate form of Eq. (49) with explicit sensory evidence (the attention-entropy block is omitted as constant in the beliefs; this is not the entropy-suppressed surrogate of Section 5.6), is

$$\mathcal{F}[\{q_i\}] = \sum_i D_{\mathrm{KL}}(q_i \| p_i) + \sum_{i,k} \beta_{ik} D_{\mathrm{KL}}(q_i \| \Omega_{ik}[q_k]) - \sum_i \mathbb{E}_{q_i}[\log p(o_i \mid k_i)]. \tag{90}$$

The three terms represent prior anchoring (deviation from the prior p_i, which in the hierarchy of Section 10 is the one-episode-lagged predictive prior supplied by the parent level), interaction (alignment with other agents through the transports), and sensory evidence. Under the product recognition law $q_i \otimes s_i$ and the identity readout, the jointly typed observation term $-\mathbb{E}_{q_i\otimes s_i}[\log p(o_i \mid k_i, m_i)]$ of Eq. (49) reduces to the q_i-expectation shown, and at fixed s_i the model-fiber terms are constant in (μ_i, Σ_i) and drop out of every variation below. Under the frozen-β^* convention adopted for this section (the differentiation-convention paragraph below), the omitted attention-entropy term $\tau\beta_{ik}\log(\beta_{ik}/\pi_{ik})$ of the canonical functional is an additive constant in (μ, Σ), so the first and second variations taken under that convention coincide with the frozen-β variations of Eq. (49).

8.3 Component Free Energies for Gaussians

For $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$, the relative entropy is given by the closed-form expression of Eq. (4), applied with the substitution $(\mu_1, \Sigma_1, \mu_2, \Sigma_2) \to (\mu_q, \Sigma_q, \mu_p, \Sigma_p)$.

For the Gaussian likelihood $p(o_i \mid k_i) = \mathcal{N}(o_i; k_i, \Sigma_{o_i})$, the expected log-likelihood evaluates to

$$-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)] = \frac{1}{2}(o_i - \mu_i)^\top \Lambda_{o_i}(o_i - \mu_i) + \frac{1}{2}\operatorname{tr}(\Lambda_{o_i}\Sigma_i) + \text{const.} \tag{91}$$

8.4 First Variations (Gradient)

Prior Term

$$\frac{\partial D_{\mathrm{KL}}(q_i \| p_i)}{\partial \mu_i} = \bar{\Lambda}_{p_i}(\mu_i - \bar{\mu}_i), \tag{92}$$

$$\frac{\partial D_{\mathrm{KL}}(q_i \| p_i)}{\partial \Sigma_i} = \frac{1}{2}(\bar{\Lambda}_{p_i} - \Lambda_{q_i}). \tag{93}$$

Interaction Term With respect to receiver i:

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \mu_i} = \tilde{\Lambda}_{q_k}(\mu_i - \tilde{\mu}_k), \tag{94}$$

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \Sigma_i} = \frac{1}{2}(\tilde{\Lambda}_{q_k} - \Lambda_{q_i}). \tag{95}$$

With respect to sender k (using the general-linear precision transport of Eq. (89)):

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \mu_k} = \Lambda_{q_k}\Omega_{ik}^{-1}(\tilde{\mu}_k - \mu_i), \tag{96}$$

$$\frac{\partial D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial \Sigma_k} = \frac{1}{2}\Omega_{ik}^\top \left[\tilde{\Lambda}_{q_k} - \tilde{\Lambda}_{q_k}\big(\Sigma_i + \tilde{d}_{ik}\tilde{d}_{ik}^\top\big)\tilde{\Lambda}_{q_k}\right]\Omega_{ik}, \tag{97}$$

where $\tilde{d}_{ik} := \tilde{\mu}_k - \mu_i = \Omega_{ik}\mu_k - \mu_i$ is the receiver-frame mean residual. The quadratic-form contribution $-\tilde{\Lambda}_{q_k}\tilde{d}_{ik}\tilde{d}_{ik}^\top\tilde{\Lambda}_{q_k}$ originates in differentiating the Mahalanobis term $\tilde{d}_{ik}^\top\tilde{\Lambda}_{q_k}\tilde{d}_{ik}$ with respect to $\tilde{\Sigma}_k$ via $\partial\tilde{\Lambda}_{q_k}/\partial\tilde{\Sigma}_k = -\tilde{\Lambda}_{q_k} \otimes \tilde{\Lambda}_{q_k}$, then chain-ruled through $\tilde{\Sigma}_k = \Omega_{ik}\Sigma_k\Omega_{ik}^\top$. The outer $\Omega_{ik}^\top(\cdot)\Omega_{ik}$ structure follows from the same chain rule applied to the leading $\tilde{\Lambda}_{q_k}$ and $\tilde{\Lambda}_{q_k}\Sigma_i\tilde{\Lambda}_{q_k}$ contributions; for the mean gradient, the same chain rule yields the explicit inverse Ω_{ik}^{-1}, which for orthogonal Ω would equal $\Omega_{ik}^\top$. The $\tilde{d}_{ik}\tilde{d}_{ik}^\top$ term vanishes at consensus ($\tilde{d}_{ik} = 0$), so the boxed stiffness blocks of Eqs. (102)–(109), evaluated at consensus, are unchanged. Off consensus the term contributes a rank-one stiffness aligned with $\tilde{d}_{ik}$, so any second-variation analysis away from the consensus point must retain it.

Sensory Term

$$\frac{\partial}{\partial \mu_i}\left[-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)]\right] = \Lambda_{o_i}(\mu_i - o_i), \tag{98}$$

$$\frac{\partial}{\partial \Sigma_i}\left[-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)]\right] = \frac{1}{2}\Lambda_{o_i}. \tag{99}$$

Total Gradient

$$\frac{\partial \mathcal{F}}{\partial \mu_i} = \bar{\Lambda}_{p_i}(\mu_i - \bar{\mu}_i) + \sum_k \beta_{ik}\tilde{\Lambda}_{q_k}(\mu_i - \tilde{\mu}_k) + \sum_j \beta_{ji}\Lambda_{q_i}\Omega_{ji}^{-1}(\tilde{\mu}_i^{(j)} - \mu_j) + \Lambda_{o_i}(\mu_i - o_i), \tag{100}$$

where $\tilde{\mu}_i^{(j)} = \Omega_{ji}\mu_i$ is agent i's mean transported into agent j's frame.

8.5 Second Variations: The Stiffness Tensor

The frozen-β Hessian of the free energy, $M = \partial^2\mathcal{F}/\partial\xi\partial\xi$ with the attention weights held at β^* (see the differentiation-convention paragraph below), is a symmetric bilinear form on belief configuration space, the stiffness tensor of the potential $\mathcal{F}$. It is positive definite in the covariance sector at the reciprocal-consensus point established below, and it is not in general a metric: the reduced-functional correction and the state-dependent prior weight discussed below both add negative-semidefinite terms away from consensus. The Hessian is taken in the flat affine chart $(\mu_i, \Sigma_i) \in \mathbb{R}^{K_q} \times \mathrm{Sym}(K_q)$. A coordinate Hessian is a tensor, independent of the chart, only at a critical point of $\mathcal{F}$, and consensus is a critical point only when the prior and sensory terms are also stationary there. Away from critical points the chart matters: for $f(\Sigma) = D_{\mathrm{KL}}(\mathcal{N}(0,\Sigma)\|\mathcal{N}(0,\Sigma_p)) + \frac{1}{2}\operatorname{tr}(\Lambda_o\Sigma)$ the second variations along the same tangent vectors in the Σ chart and in the precision chart $\Lambda = \Sigma^{-1}$ agree at the critical point but differ by 49 percent at a noncritical point. A chart-free alternative is the covariant Hessian of the Fisher Levi-Civita connection, which differs from the coordinate Hessian by a Christoffel-symbol term proportional to the gradient and therefore coincides with it at critical points. The full state vector is $\xi = (\mu_1, \ldots, \mu_N, \Sigma_1, \ldots, \Sigma_N)$, yielding the block structure

$$M = \begin{pmatrix} M_{\mu\mu} & C^{\mu\Sigma} \\ (C^{\mu\Sigma})^\top & M_{\Sigma\Sigma} \end{pmatrix}, \tag{101}$$

where each block is itself an $N \times N$ matrix of sub-blocks.

Notation hierarchy. We use three related but distinct objects throughout: (i) $M_{\mu\mu}$ (a matrix), the full Hessian of the mean sector defined in (101); (ii) $\bar{\Lambda}_{p_i} = \bar{\Sigma}_i^{-1}$ (a matrix), the prior precision of agent i, which is the whole of $[M_{\mu\mu}]_{ii}$ when inter-agent coupling and observation likelihoods are absent ($\beta_{ij} = 0$, $\Lambda_{o_i} = 0$); and (iii) the scalar stiffness $\bar{M}_i := \operatorname{tr}([M_{\mu\mu}]_{ii})/K_q$, used when a one-number summary is needed. When unqualified, "the stiffness tensor" refers to (i) and "stiffness" as a scalar refers to (iii).

Differentiation convention. The second-variation expressions below treat the attention weights β_{ij} as fixed at their softmax-equilibrium values β_{ij}^*. The envelope theorem licenses this frozen-β convention at first order only: for the reduced free energy with β eliminated, the true Hessian adds the correction $-\tau^{-1}\mathrm{Cov}_{\beta^*}(\nabla E_{ij}, \nabla E_{ij})$ to the frozen-β blocks (the reduced-Hessian identity (274) of Appendix I), a negative-semidefinite term that vanishes exactly at consensus, where the per-edge gradients ∇E_{ij} coincide, but persists at generic equilibria in which prior, interaction, and sensory pulls balance with nonzero per-edge gradients, so the frozen-β blocks overstate the curvature of the reduced functional away from consensus. Differentiating through the softmax rows produces these cross terms through $\partial\beta_{ij}/\partial\mu$ and $\partial\beta_{ij}/\partial\Sigma$. The correction is bounded: writing $D_i = \max_{j,k}\|\nabla E_{ij} - \nabla E_{ik}\|$ for the per-edge gradient diameter, the identity $\mathrm{Cov}_\beta(X) = \frac{1}{2}\sum_{j,k}\beta_j\beta_k(X_j - X_k)(X_j - X_k)^\top$ gives the spectral bound $\|\tau^{-1}\mathrm{Cov}_{\beta^*}(\nabla E_{ij})\| \le D_i^2/(2\tau)$. At nondegenerate energy gaps the covariance decays exponentially in the gap over τ, so the correction vanishes in the sharp-attention limit away from ties and is large precisely near attention switching points, where per-edge energies

nearly coincide at small τ; this is the rapid-reorganization regime in which the frozen-β blocks are unreliable. The boxed forms below are written for general $\beta^*_{ij} \geq 0$ under this convention and reduce to the isolated-agent limit by setting $\beta^*_{ij} = 0$ in the sums. The first variations of the preceding subsection need no such convention, since by the envelope theorem the frozen-β gradient equals the gradient of the reduced functional at $\beta = \beta^*$; the convention is stated here because it first matters at second order.

Mean Sector

Diagonal blocks ($i = k$)**:** From the prior we obtain $\partial^2 D_{\mathrm{KL}}(q_i \| p_i)/\partial\mu_i \partial\mu_i^\top = \bar{\Lambda}_{p_i}$. From the interaction term as receiver, $\partial^2 D_{\mathrm{KL}}(q_i \| \tilde{q}_k)/\partial\mu_i \partial\mu_i^\top = \tilde{\Lambda}_{q_k}$. From the interaction term as sender to agent j, $\partial^2 D_{\mathrm{KL}}(q_j \| \tilde{q}_i)/\partial\mu_i \partial\mu_i^\top = \Omega_{ji}^\top \tilde{\Lambda}_{q_i}^{(j)} \Omega_{ji} = \Lambda_{q_i}$. From sensory evidence, $\partial^2[-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)]]/\partial\mu_i \partial\mu_i^\top = \Lambda_{o_i}$. The total diagonal stiffness block is

$$\boxed{[M_{\mu\mu}]_{ii} = \bar{\Lambda}_{p_i} + \sum_k \beta_{ik} \tilde{\Lambda}_{q_k} + \sum_j \beta_{ji} \Lambda_{q_i} + \Lambda_{o_i}.} \tag{102}$$

Off-diagonal blocks ($i \neq k$)**:** From $D_{\mathrm{KL}}(q_i \| \tilde{q}_k)$, using Eq. (89),

$$\frac{\partial^2 D_{\mathrm{KL}}(q_i \| \tilde{q}_k)}{\partial\mu_i \partial\mu_k^\top} = -\tilde{\Lambda}_{q_k} \Omega_{ik} = -\Omega_{ik}^{-\top} \Lambda_{q_k}, \tag{103}$$

where the second equality follows from $\tilde{\Lambda}_{q_k} \Omega_{ik} = \Omega_{ik}^{-\top} \Lambda_{q_k} \Omega_{ik}^{-1} \Omega_{ik} = \Omega_{ik}^{-\top} \Lambda_{q_k}$. From $D_{\mathrm{KL}}(q_k \| \tilde{q}_i)$ (if k also attends to i),

$$\frac{\partial^2 D_{\mathrm{KL}}(q_k \| \tilde{q}_i)}{\partial\mu_i \partial\mu_k^\top} = -\Lambda_{q_i} \Omega_{ki}^{-1}. \tag{104}$$

The sensory term does not couple different agents. Therefore

$$\boxed{[M_{\mu\mu}]_{ik} = -\beta_{ik} \Omega_{ik}^{-\top} \Lambda_{q_k} - \beta_{ki} \Lambda_{q_i} \Omega_{ki}^{-1} \quad (i \neq k).} \tag{105}$$

Under $\mathrm{O}(K_q)$ this reduces to $-\beta_{ik} \Omega_{ik} \Lambda_{q_k} - \beta_{ki} \Lambda_{q_i} \Omega_{ki}^\top$.

Off-diagonal block caveats. As the Hessian of the scalar functional $\mathcal{F}$, the block matrix $M_{\mu\mu}$ is automatically symmetric in the bilinear-form sense $M_{\mu\mu}^\top = M_{\mu\mu}$, equivalently $[M_{\mu\mu}]_{ik} = [M_{\mu\mu}]_{ki}^\top$, by equality of mixed partial derivatives (Schwarz's theorem). The block-transpose identity is exactly the condition for $M_{\mu\mu}$ to define a quadratic form $\frac{1}{2}\delta\mu^\top M_{\mu\mu} \delta\mu$ on the stacked variable $(\mu_i)_i$, the second-order term of $\mathcal{F}$; no stronger blockwise-equality condition is required, and an individual block $[M_{\mu\mu}]_{ik}$ need not be symmetric. Where the block-transpose identity appears not to hold in displayed expressions, the cause is that the displayed $[M_{\mu\mu}]_{ik}$ has been restricted to a subset of the contributions to the Hessian (for example a single sender-only or receiver-only contribution), and the symmetry is restored once both contributions are summed.

Row-asymmetric attention does not by itself destroy the scalar potential: fixed asymmetric weights still define the scalar $\sum_{ij} \beta_{ij} E_{ij}$, and state-dependent Gibbs weights in the canonical entropy-retaining objective reduce to $-\tau \sum_i \log Z_i$. In both cases the full receiver-plus-sender Hessian is symmetric, and the full flow is a gradient flow for every fixed β, reciprocal or not. A nonpotential running rule appears only after the additional receiver-only truncation of Eq. (250), which discards the sender-row derivatives. For that optional field the bilinear form $M_{\mu\mu}$ is not the

linearization of the dynamics; the linearization is the Jacobian of the truncated field, which is not symmetric and can have complex eigenvalues. The boxed forms of this section describe the full scalar $\mathcal{F}$ and do not describe the receiver-only field.

Covariance Sector For matrix-valued variables, we use the directional derivative convention

$$\frac{\partial^2 f}{\partial\Sigma\partial\Sigma}[V,W] = \lim_{\epsilon\to 0}\frac{1}{\epsilon}\left[\left.\frac{\partial f}{\partial\Sigma}\right|_{\Sigma+\epsilon W} - \left.\frac{\partial f}{\partial\Sigma}\right|_{\Sigma}\right][V], \tag{106}$$

with the identity $\partial(\Sigma^{-1})/\partial\Sigma = -\Sigma^{-1}\otimes\Sigma^{-1}$.

Diagonal blocks ($i = k$)**:** From the prior,

$$\frac{\partial^2 D_{\mathrm{KL}}(q_i\|p_i)}{\partial\Sigma_i\partial\Sigma_i}[V,W] = \frac{1}{2}\operatorname{tr}[\Lambda_{q_i}V\Lambda_{q_i}W]. \tag{107}$$

In tensor notation, $\partial^2 D_{\mathrm{KL}}(q_i\|p_i)/\partial\Sigma_i\partial\Sigma_i = \frac{1}{2}(\Lambda_{q_i}\otimes\Lambda_{q_i})$. From the interaction term as receiver and as sender, identical contributions arise *at consensus* (see the remark below). The sensory term $\frac{1}{2}\operatorname{tr}(\Lambda_{o_i}\Sigma_i)$ is linear in Σ_i, so its second derivative vanishes. Therefore

$$\boxed{[M_{\Sigma\Sigma}]_{ii} = \frac{1}{2}(\Lambda_{q_i}\otimes\Lambda_{q_i})\cdot\left(1+\sum_k\beta_{ik}+\sum_j\beta_{ji}\right)\quad\text{at consensus.}} \tag{108}$$

The sensory precision Λ_{o_i} contributes to the mean-sector stiffness but not to the covariance-sector stiffness.

Remark on the consensus simplification. The factor $(1+\sum_k\beta_{ik}+\sum_j\beta_{ji})$ in Eq. (108) arises because the sender contribution from $D_{\mathrm{KL}}(q_j\|\tilde{q}_i)$ involves both a negative term $-\frac{1}{2}\Lambda_{q_i}\otimes\Lambda_{q_i}$ from $\partial^2\ln|\tilde{\Sigma}_i|/\partial\Sigma_i^2$ and a positive trace-term contribution that, away from consensus, equals $\frac{1}{2}\partial^2\operatorname{tr}((\Omega_{ji}\Sigma_i\Omega_{ji}^\top)^{-1}\Sigma_j)/\partial\Sigma_i^2$ and couples Σ_i to Σ_j. At consensus ($\Sigma_j = \Omega_{ji}\Sigma_i\Omega_{ji}^\top$) this trace-term contribution simplifies to $+\Lambda_{q_i}\otimes\Lambda_{q_i}$, and the sender net contribution collapses to $+\frac{1}{2}\Lambda_{q_i}\otimes\Lambda_{q_i}$, recovering Eq. (108). Off consensus, the sender contribution carries an explicit Σ_j-dependent cross-coupling and the diagonal block does not factor into a multiple of $\Lambda_{q_i}\otimes\Lambda_{q_i}$; the boxed form above is the consensus-point restriction.

Off-diagonal blocks ($i \neq k$)**:** From $D_{\mathrm{KL}}(q_i\|\tilde{q}_k)$, varying both Σ_i and Σ_k, with $\tilde{\Lambda}_{q_k}$ defined by Eq. (89),

$$\boxed{\begin{aligned}[M_{\Sigma\Sigma}]_{ik} = &-\frac{1}{2}\beta_{ik}\left(\Omega_{ik}^{-\top}\Lambda_{q_k}\otimes\Omega_{ik}^{-\top}\Lambda_{q_k}\right)\\ &-\frac{1}{2}\beta_{ki}\left(\Omega_{ki}^{-\top}\Lambda_{q_i}\otimes\Omega_{ki}^{-\top}\Lambda_{q_i}\right)^\top\quad (i\neq k).\end{aligned}} \tag{109}$$

The block is a pure Kronecker square of the two mean-sector transport factors. Each contribution is independent of the native covariance of the agent whose divergence produces it, a structural property any candidate form of this block must satisfy: agent i's native covariance Σ_i enters $D_{\mathrm{KL}}(q_i\|\tilde{q}_k)$ only linearly through the trace $\frac{1}{2}\operatorname{tr}(\tilde{\Lambda}_{q_k}\Sigma_i)$, so differentiating in Σ_i removes Σ_i entirely and leaves $\frac{1}{2}\tilde{\Lambda}_{q_k}$, and the subsequent differentiation in Σ_k through the precision transport of Eq. (89)

produces the Kronecker square $\Omega_{ik}^{-\top}\Lambda_{q_k} \otimes \Omega_{ik}^{-\top}\Lambda_{q_k}$, which depends on Σ_k through Λ_{q_k} but carries no residual Σ_i. By the same argument the sender contribution is independent of Σ_k and depends on Σ_i through Λ_{q_i}, so the assembled block depends on both precisions while each term is free of its own agent's covariance. The two-term structure mirrors the mean-sector off-diagonal block of Eq. (105): the first term is the receiver contribution from $D_{\mathrm{KL}}(q_i\|\tilde{q}_k)$ when i attends to k, built from $A_{ik} = \Omega_{ik}^{-\top}\Lambda_{q_k}$; the second is the sender contribution from $D_{\mathrm{KL}}(q_k\|\tilde{q}_i)$ when k also attends to i, built from $B_{ki} = \Omega_{ki}^{-\top}\Lambda_{q_i}$. The transpose on the second Kronecker factor is essential. The sender mixed partial is naturally assembled with the index ordering (Σ_k, Σ_i), whereas the block $[M_{\Sigma\Sigma}]_{ik}$ requires the ordering (Σ_i, Σ_k); since $\Omega_{ki}^{-\top}\Lambda_{q_i} \otimes \Omega_{ki}^{-\top}\Lambda_{q_i}$ is not symmetric, the index swap is exactly the vec-space transpose, and it is what makes the assembled block satisfy the Schwarz block-transpose identity $[M_{\Sigma\Sigma}]_{ik} = [M_{\Sigma\Sigma}]_{ki}^{\top}$. As in the mean sector, this identity holds for every fixed β, reciprocal or not, while an individual block $[M_{\Sigma\Sigma}]_{ik}$ need not be symmetric. Under $\mathrm{O}(K_q)$ the factors reduce to $\Omega_{ik}\Lambda_{q_k} \otimes \Omega_{ik}\Lambda_{q_k}$.

At the reciprocal-consensus point, the covariance-sector Hessian restricted to the symmetric tangent space on which Σ lives is positive definite for all $\beta \geq 0$ and arbitrary precision heterogeneity. Take Regime I transports $\Omega_{ik} = \exp(\phi_i)\exp(-\phi_k)$, so that the cocycle holds and the transported neighbor $\tilde{q}_k$ coincides with q_i, with reciprocal β and each prior equal to the native belief. Every relative entropy then sits at its global zero, contributing a positive-semidefinite Hessian, while the strictly convex prior term $\frac{1}{2}(\Lambda_{q_i} \otimes \Lambda_{q_i})$ on each diagonal block makes the total strictly positive definite. Four scope conditions are required. The result is established only at the consensus point; off consensus the Σ_j cross-coupling re-enters and positive definiteness is not established there. It uses the frozen-β (envelope-theorem) convention of the differentiation paragraph above. It requires a global consensus through the cocycle, not merely pairwise reciprocity; the full $K_q^2 \times K_q^2$ vec operator can be indefinite on the non-symmetric complement, which is not the physical object, since Σ is constrained to the symmetric subspace. And the point is a critical point of $\mathcal{F}$ only when the sensory term is absent: the sensory term adds no covariance curvature but contributes the constant covariance gradient $\frac{1}{2}\Lambda_{o_i}$, so with sensory evidence the statement concerns the flat-chart second variation at a noncritical point, with the chart dependence described at the start of this subsection.

Cross Mean-Covariance Blocks The prior term gives $\partial^2 D_{\mathrm{KL}}(q_i\|p_i)/\partial\mu_i\partial\Sigma_i = 0$. The sensory free energy decomposes into a term quadratic in μ_i and a term linear in Σ_i; these are independent, so $[C^{\mu\Sigma}]_{ii}^{\mathrm{sensory}} = 0$.

For the interaction term, from $\partial D_{\mathrm{KL}}(q_i\|\tilde{q}_k)/\partial\mu_i = \tilde{\Lambda}_{q_k}(\mu_i - \tilde{\mu}_k)$ with $\tilde{\Lambda}_{q_k} = \Omega_{ik}^{-\top}\Lambda_{q_k}\Omega_{ik}^{-1}$ (Eq. (89)), varying Σ_k gives $\delta\tilde{\Lambda}_{q_k} = -\Omega_{ik}^{-\top}\Lambda_{q_k}V\Lambda_{q_k}\Omega_{ik}^{-1}$ for $V = \delta\Sigma_k$, and therefore

$$\frac{\partial^2 D_{\mathrm{KL}}(q_i\|\tilde{q}_k)}{\partial\mu_i\partial\Sigma_k}[V] = -\Omega_{ik}^{-\top}\Lambda_{q_k}V\Lambda_{q_k}\Omega_{ik}^{-1}(\mu_i - \tilde{\mu}_k). \tag{110}$$

In components,

$$\frac{\partial^2 D_{\mathrm{KL}}}{\partial(\mu_i)_a\partial(\Sigma_k)_{bc}} = -[\Omega_{ik}^{-\top}\Lambda_{q_k}]_{ab}[\Lambda_{q_k}\Omega_{ik}^{-1}(\mu_i - \tilde{\mu}_k)]_c. \tag{111}$$

Under $\mathrm{O}(K_q)$ this reduces to $-\Omega_{ik}\Lambda_{q_k}V\Lambda_{q_k}\Omega_{ik}^{\top}(\mu_i - \tilde{\mu}_k)$. The cross block vanishes at consensus $(\mu_i = \tilde{\mu}_k)$:

$$\boxed{[C^{\mu\Sigma}]_{ik} = 0 \quad \text{when } \mu_i = \tilde{\mu}_k.} \tag{112}$$

8.6 Precision as Stiffness and the Fisher Metric as Friction

The mean-sector stiffness of agent i, in the frozen-β convention, is

$$M_i = \underbrace{\bar{\Lambda}_{p_i}}_{\text{prior}} + \underbrace{\sum_k \beta_{ik}\tilde{\Lambda}_{q_k}}_{\text{receiver}} + \underbrace{\sum_j \beta_{ji}\Lambda_{q_i}}_{\text{sender}} + \underbrace{\Lambda_{o_i}}_{\text{sensory}} . \tag{113}$$

The prior block resists deviation from the prior p_i, the receiver block resists deviation from the transported neighbors that agent i attends to, the sender block resists moves that would disturb the agents attending to i, and the sensory block anchors the agent to its observations. The stiffness is dimensionless on the representational space; no identification with a physical mass is made.

Constant versus state-dependent self-coupling. The boxed stiffness blocks of this section take the prior self-coupling weight at its canonical default $\alpha_i = 1$; the prior term of Eq. (90) carries unit weight. Under the state-dependent generalization of Section 6.1, where $\alpha_i^*(c) = c_0/(b_0 + D_{\mathrm{KL}}(q_i\|p_i))$ is recomputed from the current belief at each inner step, the prior block scales as $\alpha_i^*\bar{\Lambda}_{p_i}$. Because α_i^* co-varies with the belief, the second variation acquires a rank-one residual that the first-order envelope cancellation of Eq. (65) does not remove,

$$[M_{\mu\mu}]_{ii}^{\text{prior}} = \alpha_i^*\bar{\Lambda}_{p_i} - \frac{(\alpha_i^*)^2}{c_0}\left(\bar{\Lambda}_{p_i}\Delta\mu_i\right)\left(\bar{\Lambda}_{p_i}\Delta\mu_i\right)^\top, \qquad \Delta\mu_i = \mu_i - \bar{\mu}_i, \tag{114}$$

which vanishes only at $\mu_i = \bar{\mu}_i$. This is the prior-sector analog of the off-consensus rank-one stiffness retained for the interaction term in the first-variation analysis above. For a mean displacement at matched covariance, where $D_{\mathrm{KL}}(q_i\|p_i) = \frac{1}{2}\Delta\mu_i^\top\bar{\Lambda}_{p_i}\Delta\mu_i$, the regularized prior block stays positive definite until $D_{\mathrm{KL}} = b_0$, whereas the unregularized weight $\alpha_i^* D_{\mathrm{KL}}$ alone would lose definiteness already at $D_{\mathrm{KL}} = b_0/3$; the log-barrier delays it. Beyond that bound the prior contribution to the stiffness is indefinite. The boxed forms of this section are therefore the quasi-static $\alpha_i = 1$ specialization.

Friction and relaxation. The flow of Eq. (82) is first order. In the Gaussian sector it reads

$$\eta_b^{-1}G_b\dot{\xi}_b = -\nabla_{\xi_b}\mathcal{F}, \qquad G_\mu = \Lambda_q, \qquad G_\Sigma[V,W] = \tfrac{1}{2}\operatorname{tr}(\Sigma^{-1}V\Sigma^{-1}W), \tag{115}$$

for each block $b \in \{\mu, \Sigma\}$ with its rate η_b, where G_b is the Fisher metric of the block. This is the form $\gamma\dot{q} = -\nabla V$ of an overdamped system with friction tensor $\gamma = \eta^{-1}G$ and mobility ηG^{-1}, and it is the Rayleigh form $\partial\mathcal{R}/\partial\dot{\xi} = -\nabla\mathcal{F}$ with dissipation function

$$\mathcal{R}(\dot{\xi}) = \sum_b \frac{1}{2\eta_b}G_b(\dot{\xi}_b, \dot{\xi}_b), \qquad \frac{d\mathcal{F}}{dt} = \nabla\mathcal{F}\cdot\dot{\xi} = -2\mathcal{R}. \tag{116}$$

With a common rate $\eta_b = \eta$ the squared Fisher speed is $G(\dot{\xi},\dot{\xi}) = (d\vartheta/dt)^2 = 2\eta\mathcal{R}$, which is the clock-dissipation identity of Eq. (137) for an autonomous single-rate flow. The block-descent law and the Fisher clock are established in general in the companion theory [30, Chs. 7 and 8]. Linearized about a critical point ξ^*, the flow is $\delta\dot{\xi} = -\eta G^{-1}M\delta\xi$. Since G is positive definite and M is symmetric, $G^{-1}M$ is similar to $G^{-1/2}MG^{-1/2}$, so its spectrum is real and the linearized flow has no oscillatory modes; complex eigenvalues arise only in the receiver-only truncation of Eq. (250). For an isolated agent at $q_i = p_i$ the stiffness and the friction coincide, $[M_{\mu\mu}]_{ii} = \bar{\Lambda}_{p_i} = \Lambda_{q_i} = G_\mu$ and $[M_{\Sigma\Sigma}]_{ii} = \frac{1}{2}\Lambda_{q_i}\otimes\Lambda_{q_i} = G_\Sigma$, so the linearized mean and covariance relaxation rates are exactly

η_μ and η_Σ, independent of the precision. Under a Euclidean gradient flow $\dot{\xi} = -\eta\nabla\mathcal{F}$ the rates would instead be η times the eigenvalues of M, so a precise agent would relax faster. In neither flow is a high-precision agent slow; large precision produces large stiffness and small static compliance M^{-1}.

High-precision agents. Consider an agent with very sharp prior self-localization, $\bar{\Sigma}_i = \delta I$ with $\delta \ll 1$ a dimensionless precision scale to which no physical units are assigned (the macroscopic-object example of Appendix B.1.1). In the isolated-agent limit of Eq. (113), where the interaction blocks switch off ($\beta_{ik} = \beta_{ji} = 0$) and the sensory block is negligible ($\Lambda_{o_i} \to 0$), M_i reduces to the prior precision $\delta^{-1}I$; the interaction and sensory terms add further stiffness when the agent is embedded. Such an agent has large stiffness and compliance of order δ, so a bounded force displaces it by an amount of order δ; under the natural-gradient flow its linear relaxation rate is still η.

The stiffness above runs entirely on the state-fiber prior precision $\bar{\Lambda}_{p_i}$ and is therefore specific to the prior-pullback tier in the sense of Section 18.2. Under the structural-tier reading of Section 21.1, perceived geometry is carried instead by the model-fiber generative-model precision $\Sigma_{s,i}^{-1}$, so the stiffness of this section and the pulled-back geometry are distinct objects at that tier; relating them would require a dynamical link between $\bar{\Lambda}_{p_i}$ and $\Sigma_{s,i}^{-1}$ that the present construction does not supply.

8.7 A Declared Kinetic-Metric Postulate

Postulate (kinetic metric). In addition to $\mathcal{F}$ and its first-order flow, one may postulate a kinetic energy $\mathcal{T} = \frac{1}{2}\dot{\xi}^\top M_K \dot{\xi}$ with inertia tensor M_K and adopt the second-order equation of motion

$$M_K\ddot{\xi} + \eta^{-1}G\dot{\xi} = -\nabla\mathcal{F}, \tag{117}$$

the Euler-Lagrange equation of $\mathcal{T} - \mathcal{F}$ with the Rayleigh dissipation of Eq. (116). Its overdamped limit, in which the inertial times are short compared with the relaxation times, is the natural-gradient flow of Eq. (115). Neither $\mathcal{F}$ nor the flow of Eq. (82) supplies M_K. The inertial reading of precision that the postulate expresses has status S (Table 2), and no other result of the paper uses Eq. (117). The natural candidate takes $M_K = M$, the frozen-β stiffness at consensus. Its consequences are fixed by linearizing about a critical point. Without dissipation the linearized equation is $M\delta\ddot{\xi} = -M\delta\xi$, so $\delta\ddot{\xi} = -\delta\xi$: every linear mode has the same frequency, $\omega = 1$ in flow-time units, whatever the precision. With the dissipation retained, an isolated agent at $q_i = p_i$, where $M = G$, obeys $\delta\ddot{\xi} + \eta^{-1}\delta\dot{\xi} + \delta\xi = 0$, with damping ratio $1/(2\eta)$; the motion is overdamped for $\eta < 1/2$, and for $\eta \ll 1/2$ the slow root is $-\eta$, the natural-gradient rate. Under the postulate $M_K = M$ the stiffness and the inertia coincide, so no precision-dependent frequency, and in particular no scaling $\omega^2 \propto 1/\bar{M}_i$, follows from it. A precision-dependent frequency requires an inertia tensor that is not the stiffness tensor, which the construction does not supply. For example, the different postulate $M_K = \mathrm{diag}_i(\bar{\Lambda}_{p_i})$ in the mean sector, with $K_c := M_{\mu\mu} - M_K$ the interaction-plus-sensory stiffness, gives $\omega^2 = \mathrm{eig}(M_K^{-1}M_{\mu\mu}) = 1 + \mathrm{eig}(M_K^{-1}K_c)$, a testable relation between frequencies and coupling-plus-sensory stiffness, not an identity; we do not adopt it. A dynamical test of any such postulate requires a trajectory generated by Eq. (117), for example underdamped ringing, which the first-order dynamics used throughout this paper cannot produce, because its linearized spectrum is real; a static-response measurement of the compliance M^{-1} tests the stiffness and says nothing about M_K.

Under the postulate $M_K = M$ the kinetic energy takes the form

$$\mathcal{T} = \frac{1}{2}\dot{\mu}^\top M_{\mu\mu}\dot{\mu} + \frac{1}{2}M_{\Sigma\Sigma}[\dot{\Sigma},\dot{\Sigma}] + \frac{1}{2}G_\phi(\phi)[\dot{\phi},\dot{\phi}], \tag{118}$$

where the frame-sector coefficient G_ϕ is a further choice, fixed by cases. For $G = \mathrm{SO}(K_q)$ with $K_q \geq 2$, the frame $\phi \in \mathfrak{so}(K_q)$ carries the bi-invariant inner product $G_\phi[\dot\phi, \dot\phi] = \langle \dot\phi, \dot\phi \rangle_{\mathfrak{g}} = -\operatorname{tr}(\dot\phi^2)$. When $K_q \geq 3$, $\mathfrak{so}(K_q)$ is compact semisimple and this inner product is $1/(K_q - 2)$ of the negative Killing form, since $B(X, X) = (K_q - 2)\operatorname{tr}(X^2)$. At $K_q = 2$, $\mathfrak{so}(2)$ is abelian and its Killing form vanishes, so the positive Ad-invariant negative trace form is a separate choice, not a Killing-form construction. Central or abelian factors in any other compact algebra require the same separate choice of positive Ad-invariant inner product. For the general $\mathrm{GL}(K_q)$ case the Killing form on $\mathfrak{gl}(K_q)$ is degenerate (its radical is the center) and indefinite on the semisimple part $\mathfrak{sl}(K_q)$, and G_ϕ would be supplied instead by the position-dependent pullback $\mathcal{G}_{\mathrm{pull}}$ of the ambient Frobenius metric through the exponential map, Eq. (84), under the $\operatorname{dexp}_\phi^L$-regularity hypothesis recorded at Eq. (83). That form is flat, being the pullback of a flat metric by a local diffeomorphism, and its kinetic term $\frac{1}{2}\mathcal{G}_{\mathrm{pull}}(\dot\phi, \dot\phi) = \frac{1}{2}\|\dot U\|_F^2$ is neither left nor right invariant. Under the common right translation $U_i \to U_i g$, which leaves every Regime I transport and hence $\mathcal{F}$ unchanged, the ratio $\|\dot U g\|_F^2 / \|\dot U\|_F^2 = \operatorname{tr}(\dot U g g^\top \dot U^\top) / \operatorname{tr}(\dot U \dot U^\top)$ ranges over the interval between the smallest and largest eigenvalues of $gg^\top$, so the term is invariant for every velocity only when g is orthogonal. Since $\mathrm{GL}^+(K_q)$ admits no bi-invariant Riemannian metric [82], any frame-sector kinetic term for noncompact frames breaks part of the redundancy that $\mathcal{F}$ respects, and in Regime I, where the frames are removable passive re-expressions (Section 4), the frame sector of Eq. (118) is an additional, frame-dependent postulate.

The second term of Eq. (118) couples covariance evolution to the covariance-sector stiffness, which at an isolated equilibrium is the Fisher (affine-invariant) metric on $\mathbb{S}^+_{K_q}$. The inter-agent off-diagonal blocks contribute

$$\mathcal{T}_{\mathrm{couple}} = \sum_{i<k} \left[-\beta_{ik} \dot\mu_i^\top \Omega_{ik}^{-\top} \Lambda_{q_k} \dot\mu_k - \beta_{ki} \dot\mu_i^\top \Lambda_{q_i} \Omega_{ki}^{-1} \dot\mu_k \right], \tag{119}$$

which under the postulate couples the accelerations of the agents. By the block-transpose symmetry of the off-diagonal block caveats paragraph above, this quadratic form is well defined for any β, reciprocal or not. Equality of the two displayed coefficient blocks is a separate requirement: it holds if and only if $\beta_{ik} = \beta_{ki}$ and the transport identity $\Omega_{ik}^{-\top} \Lambda_{q_k} = \Lambda_{q_i} \Omega_{ki}^{-1}$ hold, and under orientation reversal $\Omega_{ki} = \Omega_{ik}^{-1}$ the latter reduces to the consensus condition $\Omega_{ik}^{-\top} \Lambda_{q_k} \Omega_{ik}^{-1} = \Lambda_{q_i}$ (the transported precisions agree), which reciprocity alone does not enforce.

9 Coarse-Graining I: The Meta-Agent as Gauge-Covariant Barycenter

A meta-agent is the gauge-covariant coarse-graining of a coherent cluster of agents into a single agent one level up. Its anatomy, used throughout the manuscript, has four parts. The meta-agent is *defined* by its state, a variational barycenter whose frame-free part is the defining datum. *When* a cluster should be replaced by such a parent is the question of a free-energy licensing criterion (status C). *How* candidate clusters are found is a threshold detector, a coherence filter (status S) not derived from the criterion. Finally, gauge covariance makes the barycenter a frame-consistent coarse-graining map, not an exact one. The companion theory supersedes both the barycenter and the detector: its parents are exact pushforwards of the fine laws under declared coarse channels [30, Chs. 9 and 11], and its formation results are witnesses for declared families [30, Ch. 13]. The barycenter is retained here as a working construction; its relation to the exact parent is stated at the end of Section 9.1, and the exact coarse-graining step of this manuscript is developed in Section 11 and Appendix I.1.

9.1 The Meta-Agent as Gauge-Covariant Variational Barycenter

The meta-agent's belief is the M-projection barycenter of its transported constituents. Conditional on a parent existing, treating the parent as a variational distribution and minimizing the alignment energy along the child-parent edges over q_I gives

$$q_I^* = \arg\min_{q_I} \sum_{i \in I} w_i^I D_{\mathrm{KL}}(\Omega_{Ii} q_i \| q_I) + \lambda D_{\mathrm{KL}}(q_I \| p_I), \tag{120}$$

a moment-matching barycenter, with the meta-belief q_I in the second slot, regularized toward a prior p_I on the parent. The belief coupling $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ of Eq. (50) places the receiver's law in the first slot (the I-projection direction), so the two constructions use opposite slot orders. Here $w_i^I \geq 0$ is the cluster-aggregation weight of constituent i in cluster I; it is a distinct object from the per-agent precision α_i of Section 6.1, which weights the agent's own belief-prior coupling $D_{\mathrm{KL}}(q_i \| p_i)$. For Gaussian beliefs $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ with transported moments $\tilde{\mu}_i = \Omega_{Ii}\mu_i$ and $\tilde{\Sigma}_i = \Omega_{Ii}\Sigma_i\Omega_{Ii}^\top$, the closed form (with $W_I := \sum_{i \in I} w_i^I$ and $\lambda \to 0$ for legibility) is

$$\mu_I^* = \tfrac{1}{W_I} \textstyle\sum_i w_i^I \tilde{\mu}_i, \tag{121}$$

$$\Sigma_I^* = \tfrac{1}{W_I} \textstyle\sum_i w_i^I \big[\tilde{\Sigma}_i + (\mu_I^* - \tilde{\mu}_i)(\mu_I^* - \tilde{\mu}_i)^\top\big]. \tag{122}$$

The mean is the w^I-weighted average of the transported constituent means; the covariance is the w^I-weighted average of the transported covariances plus a dispersion term that records how much the constituents disagree about the mean. The dispersion term vanishes at strict consensus, when all $\tilde{\mu}_i$ coincide. In the high-coherence regime it is $\mathcal{O}(\varepsilon)$ in the post-transport pairwise dispersion, and dropping it is an approximation, not an identity (the error bound and the working forms are given in Section 9.4).

The M-projection is not the only equivariant variational barycenter. The I-projection $\arg\min_q \sum_i w_i^I D_{\mathrm{KL}}(q \| \Omega_{Ii} q_i)$, whose Gaussian solution is the precision average $\Lambda_I = \sum_i w_i^I \tilde{\Lambda}_i / W_I$, is also a strict minimizer and is equally covariant under common pushforward. We adopt the M-projection for two reasons: it is the exact M-step coordinate of a random-effects model for the constituents (Section 9.2), and it pools the constituents' uncertainty, whereas the precision average sharpens it.

Proposition 4 (Existence, uniqueness, and equivariance of the belief barycenter)**.** *Let $w_i^I \geq 0$ with $W_I = \sum_{i \in I} w_i^I > 0$, and let the transported covariances $\tilde{\Sigma}_i = \Omega_{Ii}\Sigma_i\Omega_{Ii}^\top$ be positive definite. Then the minimizer of Eq.* (120) *with $\lambda \to 0$ over the Gaussian family exists and is unique, given by Eqs.* (121)–(122)*, and the map $(\{q_i\}, \{U_i\}, U_I) \mapsto q_I^*$ commutes with the action of one global $g \in G$ on beliefs and frames (Theorem 9).*

Proof. In the natural parameters θ of q_I, each term $D_{\mathrm{KL}}(\Omega_{Ii} q_i \| q_\theta)$ is, up to a θ-independent constant, the Bregman divergence $B_\psi(\theta, \cdot)$ of the Gaussian log-partition function ψ, which is strictly convex on the open natural-parameter domain, so the objective equals $\sum_{i \in I} w_i^I [\psi(\theta) - \langle \tilde{\eta}_i, \theta \rangle] + \text{const}$ with $\tilde{\eta}_i$ the expectation parameter of the transported constituent $\Omega_{Ii} q_i$. Strict convexity gives a unique stationary point $\nabla\psi(\theta^*) = W_I^{-1} \sum_{i \in I} w_i^I \tilde{\eta}_i$, which is exactly the moment matching of Eqs. (121)–(122), and the minimizer exists because a convex combination of interior expectation parameters lies in the open expectation domain. Equivariance is the statement of Theorem 9. □

The same closed form, together with its minimized score $\frac{1}{2}[\log\det\Sigma_I^* - \sum_i w_i^I \log\det\tilde{\Sigma}_i]$ for normalized weights, is the full-Gaussian barycenter proposition of [38, Ch. G8]. For $\lambda > 0$ the regularized objective adds the term $\lambda D_{\mathrm{KL}}(q_I \| p_I)$, with the parent in the first slot, for which coercivity yields existence; uniqueness is claimed only for the $\lambda \to 0$ statement above.

The parent's gauge frame requires separate treatment, because it is not identifiable from anything the parent does. The next proposition states this and a companion obstruction.

Proposition 5 (The parent frame is pure gauge; no symmetric equivariant frame mean)**.** *(i) Let a parent with frame U_I and law $\mathcal{N}(\mu_I, \Sigma_I)$ in that frame supply constituent i the transported law $(\Omega_{iI})_{\#}\mathcal{N}(\mu_I, \Sigma_I)$ with $\Omega_{iI} = U_i U_I^{-1}$, as in the cross-scale prior of Eq.* (131)*. This law is $\mathcal{N}(U_i m, U_i S U_i^{\top})$ with*

$$m = U_I^{-1}\mu_I, \qquad S = U_I^{-1}\Sigma_I U_I^{-\top},$$

so every constituent law depends on (U_I, μ_I, Σ_I) only through the frame-free pair (m, S). The barycenter of Eqs. (121)–(122)*, computed with $\Omega_{Ii} = U_I U_i^{-1}$ and weights built from transported divergences, gives a pair (m, S) that does not depend on U_I. (ii) For $K_q \geq 2$ and $n \geq 2$ there is no map $M : \mathrm{GL}^+(K_q)^n \to \mathrm{GL}^+(K_q)$ that is symmetric under permutations of its arguments and left equivariant, $M(gU_1, \ldots, gU_n) = gM(U_1, \ldots, U_n)$ for all g; no continuity is assumed, and the same holds with $\mathrm{SO}(K_q)$ in place of $\mathrm{GL}^+(K_q)$.*

Proof. (i) The pushforward of $\mathcal{N}(\mu_I, \Sigma_I)$ by Ω_{iI} is $\mathcal{N}(U_i U_I^{-1}\mu_I, U_i U_I^{-1}\Sigma_I U_I^{-\top} U_i^{\top})$, which is the stated law. For the barycenter, $\tilde{\mu}_i = U_I(U_i^{-1}\mu_i)$ and $\tilde{\Sigma}_i = U_I(U_i^{-1}\Sigma_i U_i^{-\top})U_I^{\top}$, so Eqs. (121)–(122) give $m = W_I^{-1}\sum_i w_i^I U_i^{-1}\mu_i$ and $S = W_I^{-1}\sum_i w_i^I[U_i^{-1}\Sigma_i U_i^{-\top} + (m - U_i^{-1}\mu_i)(m - U_i^{-1}\mu_i)^{\top}]$, in which U_I does not appear; transported divergences are invariant under the common pushforward by U_I^{-1}, so the weights do not depend on U_I either. (ii) Let $v \in \mathrm{SO}(K_q)$ be a rotation by $2\pi/n$ in one coordinate plane, so that v has order n, and take $U_j = v^{j-1}$ for $j = 1, \ldots, n$. Left multiplication by v permutes this tuple cyclically, so symmetry gives $M(vU_1, \ldots, vU_n) = M(U_1, \ldots, U_n)$, while equivariance gives $M(vU_1, \ldots, vU_n) = vM(U_1, \ldots, U_n)$. Hence $(v - I)M = 0$, and since M is invertible, $v = I$, a contradiction. □

Part (i) says that U_I ranges over a K_q^2-dimensional orbit along which nothing observable varies, and only (m, S) is determined. A search for the correct parent frame is therefore a search for a canonical value of a pure-gauge quantity: any rule fixing U_I is a gauge convention to be declared, and the open problem recorded in Section 24 concerns the choice of such a convention, not a theorem. Part (ii) is a global statement. Frame means that are symmetric and equivariant exist on restricted domains of sufficiently concentrated frames, which exclude the cyclic tuples of the proof.

The default convention is the Riemannian center of mass of the constituent frames,

$$U_I^* = \arg\min_{U \in G} \sum_{i \in I} w_i d_G(U, U_i)^2. \tag{123}$$

For a compact gauge group with bi-invariant metric $d_G(U, V)^2 = \|\log(U^{-1}V)\|_F^2$, the construction is that of Grove and Karcher [92], with estimates by Karcher [93] (see also [94]). The minimizer exists, is unique as the global minimizer over G, and lies in the ball, whenever the weighted frames lie in an open geodesic ball of radius $\rho < r_{\mathrm{cx}} = \frac{1}{2}\min\{\mathrm{inj}\, G, \pi/\sqrt{\Delta}\}$, with Δ an upper bound on the sectional curvature [95, Thm. 2.1]. For $\mathrm{SO}(K_q)$ with this metric, $r_{\mathrm{cx}} = \pi/\sqrt{2}$ in d_G, which for SO(3) is a rotation angle of $\pi/2$ (for SO(3), $\Delta = 1/8$ and $\mathrm{inj} = \sqrt{2}\pi$; for $K_q \geq 4$, $\Delta = 1/4$ with the same injectivity radius). Kendall [96] obtains the larger radius $\min\{\mathrm{inj}\, G, \pi/(2\sqrt{\Delta})\}$ only for the mean defined with the distance restricted to the ball, and that radius does not govern the global argmin (123): three rotations by 0, $2\pi/3$, and $4\pi/3$ about one axis lie inside Kendall's radius, and the uniform-weight objective attains its minimum $16\pi^2/27$ at each of the three. On the convex ball the Karcher convention is left and right equivariant, since d_G is bi-invariant. The frame condition $\rho < r_{\mathrm{cx}}$ is not detectable by the consensus score of Section 9.3, which is built from

gauge-invariant divergences: if the three constituents above carry the beliefs $(U_j)_{\#}q_0$ of one law q_0, every transported divergence vanishes and the belief-coherence factor equals one.

The natural variational-inference gauge group $\mathrm{GL}^+(K_q)$ is noncompact and admits no bi-invariant Riemannian metric [82, Lemma 7.5], a connected Lie group carrying one exactly when it is isomorphic to a product of a compact group and a vector space. The Ad-invariant trace form $\langle X, Y\rangle = \mathrm{tr}(XY)$ on $\mathfrak{gl}(K_q)$ is bi-invariant and nondegenerate, so a bi-invariant *pseudo*-Riemannian metric does exist, of signature $(K_q(K_q+1)/2, K_q(K_q-1)/2)$. Being indefinite, it induces no distance function and supplies no Fréchet objective. The expression $\|\log(U^{-1}V)\|_F$ is not a distance on $\mathrm{GL}^+(K_q)$: the principal logarithm is undefined on an open set of the group, and the triangle inequality fails. The obstruction is to a bi-invariant variance, not to a bi-invariant mean. The exponential barycenter of the canonical Cartan-Schouten connection, the solution U_I of $\sum_i w_i \log(U_I^{-1}U_i) = 0$, is left-, right- and inversion-equivariant and exists uniquely for sufficiently concentrated frames [97, Thms. 11–12]; it is not the minimizer of any Riemannian variance. Under a left-invariant Riemannian metric instead [82, 98], the group is complete and homogeneous, so Afsari's theorem applies and the Riemannian center of mass exists and is unique below r_{cx}, but it is only left-equivariant. The affine-invariant route through the polar decomposition [99] is a further option. By Proposition 5, the choice among these is a choice of gauge convention with no effect on the parent laws.

Gauge covariance of the barycenter map (Theorem 9) makes the belief barycenter and a declared frame convention frame-consistent coarse-graining maps: under the theorem's hypotheses they commute with the global action on beliefs and frames, so the parent is not an artifact of the frames in which the constituents happen to be written. Covariance does not single these maps out, since the I-projection above is equally covariant, and it does not make the barycenter an exact coarse-graining. The exact parent of a cluster is the pushforward of the fine joint law, posterior, and recognition law under one declared coarse kernel [30, Ch. 9] (the pointwise parent-datum corollary). The Gaussian barycenter with its dispersion term is the moment-matched projection of the random-selection mixture of the transported constituents [38, Ch. G8], and the pooled form without the dispersion term, used in Section 9.4, is already for two scalar constituents with equal weights the image of no parameter-independent Markov kernel [38, Ch. G8] (the normal-barycenter kernel proposition). The evidence-preservation and data-processing statements of the exact construction therefore do not transfer to the working form.

Definition 6 (Meta-Agent)**.** *Let $I = \{i, j, \ldots\} \subseteq \mathcal{I}^{(\ell)}$ be a cluster of level-ℓ agents with (1) non-empty multi-child support*

$$\mathcal{U}_I = \mathrm{ConnComp}\Big(\big\{c \in \mathcal{C} : \textstyle\sum_{i\in I} \chi_i(c) \geq 2\big\}\Big) \neq \varnothing, \tag{124}$$

a connected region in which at least two constituents maintain beliefs simultaneously, with $\chi_i : \mathcal{C} \to [0,1]$ agent i's support function (introduced with the continuous-field free energy in Section 5.6; Appendix I.1 takes $\chi_i \in \{0,1\}$), and (2) minimum cluster size $|I| \geq N_{\min}$ (typically $N_{\min} = 2$). The meta-agent $\mathcal{A}^{(\ell+1)}$ of I is the measurable, piecewise-smooth section $(q^{(\ell+1)}, p^{(\ell+1)}, s^{(\ell+1)}, r^{(\ell+1)}, \phi^{(\ell+1)})$ over $\mathcal{U}_I$ whose belief is the barycenter (120) *and whose model is the analogous barycenter on the model fiber. Its defining datum is the frame-free pair (m, S) of Proposition 5; the frame $\phi^{(\ell+1)}$, equivalently U_I, is a declared gauge convention, by default the Karcher mean* (123)*, which is single-valued when the constituent frames at c lie in a geodesic ball of radius below r_{cx}. The averaged fields are smooth wherever the active constituent set and the weights vary smoothly, but can fail to be smooth across support boundaries where $\sum_{i\in I}\chi_i$ crosses the threshold in* (124)*, so we claim only the measurability that the coarse-graining of Appendix I.1 requires. Under the smooth-bump convention of Section 5.6 (smooth χ_i), the averaged fields are*

smooth on the interior of $\mathcal{U}_I$, and it is in this interior sense that the meta-agent re-enters the free energy as an agent at level $\ell + 1$. The stricter m-fold variant $\sum_{i \in I} \chi_i(c) \geq m$ for $m > 2$, up to the global-intersection limit $m = |I|$, is available where applications require deeper simultaneous coverage. A meta-agent is licensed *when, in addition, the criterion* (125) *of Section 9.2 holds; the threshold detector of Section 9.3 is a filter for candidate clusters and is not part of the definition.*

The slow model channel s carries the persistent identity across this definition: belief consensus shifts on the perceptual timescale, while the generative model changes only on the learning timescale. On an interpretive reading (status S), a culture, a school of thought, or a scientific community is a cluster whose members share enough of a common slow generative model, after transport into one another's frames, to render their mutual disagreements intelligible. Members belong to the same meta-agent not when $s_i = s_j$, a strict identity no nontrivial culture exhibits, but when $s_i \approx \tilde{\Omega}_{ij} s_j$, with the transport $\tilde{\Omega}_{ij}$ distinguishing intra-cultural disagreement (translatable across frames) from radical incommensurability (not translatable). The fast-channel beliefs q_i may diverge so long as the models remain mutually translatable. The active-inference counterpart of the construction is the group-level generative model of Waade et al. [100], who propose that a collective can constitute a group-level agent if it maintains a group-level Markov blanket; the detector used here is a similarity threshold and does not test that conditional-independence condition. That coherent collective structure can emerge from each agent minimizing variational free energy with no hand-coded alignment rules is the central finding of Heins et al. [101], the closest external precedent for the emergent reading, while the complementary fragmented limit, in which confirmation-biased free-energy-minimizing sampling produces echo chambers tunable by prior precision, is the active-inference account of Albarracin et al. [102]. The stronger reading, on which meta-agents are dynamical agents in their own right and not statistical summaries of their constituents, is an interpretive commitment of status S (Table 2); Section 9.2 shows that the exact variational reading of the barycenter treats the parent as a fitted hyperparameter.

9.2 Licensing: The Free-Energy Improvement Criterion

The definition above specifies what a meta-agent is, conditional on its existing. The proposed rule for when a cluster should be replaced by one is a variational free-energy improvement criterion. A candidate meta-agent I is licensed when the optimized free energy with the parent inserted, plus an organizational overhead $C(I)$, falls below the optimized free energy of the disaggregated constituents,

$$\mathcal{F}^*[q_I, p_I, s_I, r_I, \{q_i, s_i\}_{i \in I}] + C(I) < \mathcal{F}^*[\{q_i, p_i, s_i, r_i\}_{i \in I}]. \tag{125}$$

Here $C(I)$ is the cost of carrying the additional organizational structure (the parent's variational parameters and its place in the cross-scale hierarchy). It is not specified here; entropy or minimum-description-length terms in the parent's primitive variables are candidates, so the criterion contains a free function. The criterion is the $\Delta_I > 0$ special case of the description-length retention rule of Appendix I (Eq. (283)), where Δ_I is the net retention gain at an unspecified complexity penalty $C(I)$; the general rule, its penalty choices, and the closure ansatz under which it is admissible are deferred to Appendix I, with statements and proofs in Appendix I.1. The criterion has status C for two reasons.

First, the source of any saving is not identified. Which intra-cluster coupling terms are removed when the parent is inserted is part of a post-merge functional that the present formalism leaves unspecified. At frozen attention weights, the row constraint $\sum_j \beta_{ij} = 1$ caps the energetic intra-cluster alignment $\sum_i \sum_{j \in I} \beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ at $|I|\varepsilon$ when the pairwise post-transport divergences are bounded by ε (not the quadratic $|I|(|I| - 1)\varepsilon$ an unweighted pair count would give). This

bound does not measure a saving of $\mathcal{F}^*$. At Gibbs-optimal rows each row sits at the value $A_i = -\tau \log(Z_{\text{in}} + Z_{\text{out}})$, with Z_{in} and Z_{out} the prior-weighted partition sums over sources inside and outside the cluster, and deleting the intra-cluster sources raises it by $\tau \log(1 + Z_{\text{in}}/Z_{\text{out}}) > 0$ when the prior is left unnormalized; with the prior rescaled to unit mass over the survivors the change is still positive whenever the intra-cluster energies are the smaller ones, which is the high-coherence case. Any net gain must therefore come from the rows of agents outside the cluster, which couple to one parent in place of $|I|$ constituents, or from the parent's prior term, and this is not derived here.

Second, the criterion is degenerate when the parent is exact. If the parent laws are the push-forward of the fine laws under a declared coarse channel, the fine free energy equals the coarse one plus the discarded conditional information, $\mathcal{F}_{\text{fine}} = \mathcal{F}_{\text{coarse}} + \Delta_A$ with $\Delta_A \geq 0$ for every channel, at unchanged evidence [30, Ch. 9] (the exact coarse free-energy theorem and its evidence-preserving corollary). Read with exact parents, Eq. (125) reduces to $\Delta_A > C(I)$: it licenses whichever cluster discards the most conditional information, and the drop in free energy is neither model improvement nor evidence ascent. A usable licensing rule needs a post-merge functional that is not monotone under refinement and has a nondegenerate optimizer; neither this paper nor [30, Ch. 9], where the selection of partitions is an open problem, supplies one.

The barycenter itself has an exact variational reading, once the generative model it belongs to is stated. Let the constituents' latent states be independent draws $k_i \sim \mathcal{N}(U_i m, U_i S U_i^\top)$ from a random-effects model whose hyperparameter $\theta = (m, S)$ is a point parameter, the frame-free parent pair of Proposition 5. In $\sum_i w_i^I \mathbb{E}_{q_i}[\log p_\theta(k_i)]$ the transport cancels from both the trace and the quadratic form, leaving a weighted Gaussian maximum-likelihood problem for the transported moments whose maximizer is exactly Eqs. (121)–(122), dispersion term included. The barycenter is therefore the M-step, or type-II maximum-likelihood (empirical-Bayes), coordinate for θ at fixed recognition laws, and alternating it with updates of the q_i descends one two-argument objective [103]; it is the Gaussian restriction of the aggregate-prior proposition of [30, Ch. 12]. In this model the parent is a fitted hyperparameter: it carries no belief of its own, q_I is a point estimate, not a posterior, and Σ_I is the population covariance of the constituents. A different generative model, with one parent latent and the constituents as noisy copies of it (Appendix G), gives Σ_I the meaning of posterior uncertainty; there the mean-field coordinate update of the parent is precision addition, which sharpens the parent as $\mathcal{O}(1/|I|)$, does not depend on the constituent covariances, and runs opposite to the covariance pooling of Eq. (129). The two readings are distinct models, and only the random-effects reading makes the barycenter an exact coordinate. The derivation fixes the form and not the weights: any weight vector yields a valid M-step coordinate, so the coherence weight $\exp[-D_{\text{KL}}]$ of Section 9.4 is a declared modeling choice.

A further proposal treats the parent state as a predictive sufficient statistic of its constituents in the sense of the information bottleneck [104]: reading $C(I)$ as the bottleneck rate term and the free-energy change as a proxy for the predictive term, the information-bottleneck Lagrangian [105] would select the coarse-graining that retains maximal predictive information at fixed rate. This is an analogy, not a derivation. The closed-form Gaussian-IB solution of [106] does not apply directly, because the parent state is a pair of Gaussian-parameterized laws on associated bundles, not a jointly Gaussian vector, and the deterministic [107] and agglomerative [108] variants are not worked out here.

9.3 The Threshold Detector: A Coherence Filter for Candidate Clusters

Evaluating (125) requires the optimized free energy under both configurations, which is expensive, and the criterion is itself conjectural. Candidate clusters are therefore found with a threshold

detector built from a coherence proxy that combines post-transport belief alignment, post-transport model alignment, and spatial overlap. Each factor is a bounded exponential of a dispersion, so the detector lives in $[0,1]$. Belief coherence is the bounded exponential of the average post-transport belief dispersion,

$$C_q(I,x) = \exp\left[-\frac{1}{\tau_q |I|^2} \sum_{i,j\in I} D_{\mathrm{KL}}\big(q_i^{(\ell)}(x) \| \Omega_{ij}[q_j^{(\ell)}](x)\big)\right] \in [0,1]. \tag{126}$$

Model coherence is defined analogously as $C_s(I,x) = \exp[-V_I^{(s)}(x)/\tau_s]$ with $V_I^{(s)}(x) = |I|^{-2}\sum_{i,j\in I} D_{\mathrm{KL}}(s_i^{(\ell)}(x)\|\tilde{\Omega}_{ij}[s_j^{(\ell)}](x))$, the superscript labeling the model channel and the subscript the cluster. Presence measures spatial overlap via $P(I,x) = |I|^{-1}\sum_{i\in I}\chi_i(x) \in [0,1]$. The dispersion temperatures $\tau_q, \tau_s > 0$ set the belief and model resolution at which coherence is read off. The consensus score is the product $\Gamma(I,x) = P(I,x)C_q(I,x)C_s(I,x) \in [0,1]$, and a cluster is a candidate when $\Gamma(I,x) > \Gamma_{\min}$ and $|I| \geq N_{\min}$. Boundedness matters because D_{KL} is unbounded above, so a $1 - D_{\mathrm{KL}}$ surrogate would be signed and could yield misleadingly positive products from two negative factors. The same exponential detector reappears in Appendix I.1, written with the barycenter-relative dispersions $V_I^{(q)}$, $V_I^{(s)}$ and strengthened by a spectral-gap condition. Numerical values quoted below use $\Gamma_{\min} = 1/2$, $N_{\min} = 2$, and $\tau_q = 0.05$ for illustration; they are not theoretical requirements.

The detector's relation to the licensing criterion is partial. Because $P, C_s \leq 1$, the pass band $\Gamma > \Gamma_{\min}$ forces the mean belief dispersion below $\tau_q \log(1/\Gamma_{\min})$, which selects the high-coherence end, including the limit of vanishing dispersion. The criterion has no derived optimum against which this band could be compared (Section 9.2), and we do not establish that the product $P{\cdot}C_q{\cdot}C_s$ tracks any free-energy change, even in the high-coherence limit. The strengthened detector of Appendix I.1 additionally requires a positive constrained spectral gap $m_I > m_{\min}$; the canonical specification above does not check this condition, so by the appendix's own classification it operates as a similarity threshold and not as a closure surrogate. The appendix's retention theorem (Theorem 10) concerns the strengthened detector with the gap condition, not the one specified here.

9.4 Gauge-Covariant Belief Construction

When a meta-agent I forms, its belief is the barycenter of (120). The working form drops the dispersion term in (122), an approximation for high-coherence clusters. The dropped term is the constituent mean-spread $\sum_i w_i(\tilde{\mu}_i - \bar{\mu}_I)(\tilde{\mu}_i - \bar{\mu}_I)^\top$. With normalized weights and $\bar{\varepsilon}_w = \sum_{i,j} w_i w_j D_{\mathrm{KL}}(q_i\|\Omega_{ij}[q_j])$, it obeys the operator-norm bound

$$\Big\| \sum_i w_i(\tilde{\mu}_i - \bar{\mu}_I)(\tilde{\mu}_i - \bar{\mu}_I)^\top \Big\|_{\mathrm{op}} \leq \max_j \lambda_{\max}(\tilde{\Sigma}_j)\bar{\varepsilon}_w, \tag{127}$$

which follows from the pairwise-difference identity for weighted spreads, $\sum_i w_i(\tilde{\mu}_i - \bar{\mu}_I)(\tilde{\mu}_i - \bar{\mu}_I)^\top = \frac{1}{2}\sum_{i,j} w_i w_j(\tilde{\mu}_i - \tilde{\mu}_j)(\tilde{\mu}_i - \tilde{\mu}_j)^\top$, the Gaussian mean-term lower bound $D_{\mathrm{KL}}(\mathcal{N}_1\|\mathcal{N}_2) \geq \frac{1}{2}(\Delta\mu)^\top\Sigma_2^{-1}(\Delta\mu)$, and cocycle invariance of the transported divergences, which holds in Regime I (trivial holonomy). The bound is tight for collinear equal-covariance configurations. The detector controls the unweighted mean dispersion $\bar{\varepsilon} = |I|^{-2}\sum_{i,j} D_{\mathrm{KL}}(q_i\|\Omega_{ij}[q_j])$, not $\bar{\varepsilon}_w$, and the two are related by $\bar{\varepsilon}_w \leq |I|^2(\max_i w_i)^2\bar{\varepsilon}$. Inside the pass band the right side of (127) is therefore at most $|I|^2(\max_i w_i)^2\tau_q\log(1/\Gamma_{\min})\max_j\lambda_{\max}(\tilde{\Sigma}_j)$, which reduces to $\tau_q \log 2 \max_j \lambda_{\max}(\tilde{\Sigma}_j) \approx 0.035\max_j\lambda_{\max}(\tilde{\Sigma}_j)$ for uniform weights at the illustrative values. For uniform weights the truncation is thus a moment-matching approximation with a controlled error; for nonuniform weights the

factor $|I|^2(\max_i w_i)^2$ must be carried, and at order-one dispersion temperatures the mean-dispersion bound degrades to $\log 2 \approx 0.69$ nats, a regime in which the dispersion term should be reinstated. The working forms keep the barycenter mean and the transported-covariance pooling and drop only the constituent mean-spread:

$$\bar{\mu}_I(x) = \frac{\sum_{i\in I} w_i(x)\Omega_{Ii}[\mu_i^{(\ell)}](x)}{\sum_{i\in I} w_i(x)}, \tag{128}$$

$$\bar{\Sigma}_I(x) = \frac{\sum_{i\in I} w_i(x)\Omega_{Ii}[\Sigma_i^{(\ell)}](x)\Omega_{Ii}^\top}{\sum_{i\in I} w_i(x)}. \tag{129}$$

The weights depend on presence and coherence through $w_i^I(x) = \chi_i(x)\exp[-D_{\mathrm{KL}}(\Omega_{Ii}[q_i^{(\ell)}]\|\bar{q}_I^{(\ell)})]$, with the comparison taken after transport into the meta-agent's frame. The scalar weight is gauge invariant, being built from a transported divergence and a support mask that are each invariant under common pushforward, as the covariance theorem of Appendix I.1 requires. It is a declared modeling choice (Section 9.2), not the approximation of an optimal weight. Because $\bar{q}_I^{(\ell)}$ appears on the right-hand side through the weights, the formula is implicit and is solved by fixed-point iteration. Its convergence is guaranteed when the maximum pairwise transported divergence satisfies $\varepsilon < \varepsilon_0$ (Lemma 2). The detector's pass band bounds only the mean dispersion, which implies $\max_{i\neq j} D_{\mathrm{KL}}(q_i\|\Omega_{ij}[q_j]) \le |I|^2\tau_q\log(1/\Gamma_{\min})$ and not $\varepsilon < \varepsilon_0$, so for large clusters a separate check of the maximum divergence is required. The check is not a formality: with identity covariances, 609 coincident members and one member at distance 4.6 have mean dispersion $0.0346 < \tau_q \log 2$ at $\tau_q = 0.05$, maximum pairwise divergence 10.6, and a weight map with two stable fixed points, one of which gives the outlier 0.974 of the total weight. A non-convergence test cannot detect this bistability, because the iteration converges.

Lemma 2 (Contraction of the weight map at high coherence)**.** *Fix $0 < \lambda_{\min} \le \lambda_{\max}$, assume Regime I transport, and suppose the transported constituents satisfy $\lambda_{\min} I \preceq \tilde{\Sigma}_i \preceq \lambda_{\max} I$, the supports satisfy $\sum_{i\in I}\chi_i > 0$, and the pairwise transported divergences satisfy $\max_{i\neq j} D_{\mathrm{KL}}(q_i\|\Omega_{ij}[q_j]) \le \varepsilon$. Let T send a Gaussian $\bar{q} = \mathcal{N}(\bar{\mu},\bar{\Sigma})$ with $\lambda_{\min} I \preceq \bar{\Sigma} \preceq \lambda_{\max} I$ to the pair of Eqs.* (128)–(129) *with weights $w_i(\bar{q}) = \chi_i\exp[-D_{\mathrm{KL}}(\Omega_{Ii}[q_i]\|\bar{q})]$. In the coordinates (μ,Σ) with the Euclidean and Frobenius norms, T maps this bounded-spectrum set into the convex hull H of the moments $\{(\tilde{\mu}_i,\tilde{\Sigma}_i)\}$, and on H it is Lipschitz with constant $C(\lambda_{\min},\lambda_{\max})\varepsilon$. For $\varepsilon < \varepsilon_0 = 1/C$ the fixed point is unique and the iteration converges geometrically from any start in the bounded-spectrum set. Every admissible constant satisfies $\varepsilon_0 \le 2$ when $\lambda_{\min} = \lambda_{\max} = 1$.*

Proof sketch. With normalized weights $\bar{w}_i$, T is the convex combination $\sum_i \bar{w}_i\zeta_i$ of the moments $\zeta_i = (\tilde{\mu}_i,\tilde{\Sigma}_i)$ (not of the expectation parameters, which would reinstate the dispersion term), so it maps into H, and H lies in the bounded-spectrum set. Writing g_i for the gradient of $D_{\mathrm{KL}}(\Omega_{Ii}[q_i]\|\bar{q})$ in $(\bar{\mu},\bar{\Sigma})$ and $\bar{g} = \sum_i \bar{w}_i g_i$, the derivative is $dT[h] = -\sum_i \bar{w}_i\langle g_i - \bar{g}, h\rangle(\zeta_i - \bar{\zeta})$, a weighted covariance between moment differences and gradient differences. By cocycle invariance, $D_{\mathrm{KL}}(\Omega_{Ii}[q_i]\|\Omega_{Ij}[q_j]) = D_{\mathrm{KL}}(q_i\|\Omega_{ij}[q_j]) \le \varepsilon$, so the Gaussian divergence bounds on the bounded-spectrum set give $\|\zeta_i - \zeta_j\| = \mathcal{O}(\sqrt{\varepsilon})$. The mean part of $g_i - g_j$ is $\bar{\Sigma}^{-1}(\tilde{\mu}_j - \tilde{\mu}_i)$, independent of $\bar{q}$, and the covariance part is $\mathcal{O}(\sqrt{\varepsilon})$ on the bounded set H; hence $\|dT\| = \mathcal{O}(\varepsilon)$ on the convex set H, and the Banach fixed-point theorem applies there. For the bound on ε_0, take two members $\mathcal{N}(\pm a, 1)$ in one dimension, with pairwise divergence $\varepsilon = 2a^2$. After one step $\bar{\Sigma} = 1$, and the mean map is $m \mapsto a\tanh(am)$, with slope $\varepsilon/2$ at $m = 0$; for $\varepsilon > 2$ it has three fixed points, so uniqueness fails. □

The defining property of the construction is gauge consistency: all constituent beliefs are first transported to the meta-agent's frame via Ω_{Ii} before averaging, so the scalar weights are gauge

invariant while the averaged belief $\bar{q}_I$ transforms by pushforward with the meta-agent's frame. Without transport into a common frame the average would depend on the agents' passive choices of frame. By Proposition 5 the parent frame cancels from every constituent law, so the licensing argument inherits no frame error when one convention for U_I is used consistently.

The meta-agent's gauge frame and its independent model-frame analog are, in the working form, the Lie-algebra-additive averages

$$\begin{aligned}\phi_I^{(\ell+1)}(x) &= \frac{\sum_{i\in I} w_i(x)\phi_i^{(\ell)}(x)}{\sum_{i\in I} w_i(x)},\\ \tilde{\phi}_I^{(\ell+1)}(x) &= \frac{\sum_{i\in I} w_i^{\mathrm{mod}}(x)\tilde{\phi}_i^{(\ell)}(x)}{\sum_{i\in I} w_i^{\mathrm{mod}}(x)}.\end{aligned} \tag{130}$$

The first is the first fixed-point iterate, from the identity, of the exponential barycenter $\sum_i w_i \log(U_I^{-1}U_i) = 0$ [97]: the iteration $U \mapsto U\exp[\sum_i w_i \log(U^{-1}U_i)]$ started at $U = I$ returns $\exp(\sum_i w_i\phi_i)$ when $U_i = \exp\phi_i$. It coincides with the exponential barycenter for commuting ϕ_i, and in particular for abelian G [109]. It is not gauge-equivariant: its defect under $U_i \mapsto gU_i$ is controlled by the magnitude of the gauge transformation and by the spread of the ϕ_i, not by $\|\phi_i\|$. The model-fiber weight w_i^{mod} is built from model-fiber transported divergences and solved by an independent fixed point on the model fiber, so the cross-scale model transport $\tilde{\Omega}_{iI}$ is well-defined at meta levels; in the tied-frame configuration $\tilde{\phi}_I$ is the ρ_{model}-image of ϕ_I and no separate average is required. Reinstating the dispersion term in $\bar{\Sigma}_I$ recovers the strict moment-matched mixture covariance at modest computational cost.

9.5 The Consensus Foil: Pointwise Consensus

One limiting case brackets the construction and clarifies what it is not. In pointwise consensus, transported beliefs and models coincide, $\Omega_{ij}[q_j](c) = q_i(c)$ and $\tilde{\Omega}_{ij}[s_j](c) = s_i(c)$ for all c in the shared support. Every inter-agent coupling term then attains its minimum value zero on both channels, and the couplings exert no force on beliefs or models, so learning is driven only by observations. Pointwise consensus is not the criterion for meta-agent formation, and applied across all agents it would be a static state in which the agents' distinct perspectives no longer register in the coupling terms. Formation requires bounded, not vanishing, internal disagreement modulo transport, with closure relative to the outside. Pointwise consensus nonetheless has a constructive role at the boundary: constituents that have collapsed to common values up to transport can be replaced by a single representative without information loss. The construction requires only that the system as a whole not reach pointwise consensus, from which it is kept away by continual observation and by hierarchical feedback that perturbs constituents heterogeneously.

10 Coarse-Graining II: Cross-Scale Shadows as Lagged Predictive Priors

Coarse-graining I built a single meta-agent from a coherent cluster, the upward leg of the construction. This section supplies the downward leg: how the meta-agent, once formed, sets the priors of its constituents. In the construction the prior $p_i^{(\ell)}$ and the hyper-prior $r_i^{(\ell)}$ at level ℓ are not independent dynamical fields. They are the transported belief and the transported model of the meta-agent I at level $\ell+1$ that contains agent i as a constituent,

$$p_i^{(\ell)}(c) = \Omega_{iI}\big[q_I^{(\ell+1)}\big](c), \qquad r_i^{(\ell)}(c) = \tilde{\Omega}_{iI}\big[s_I^{(\ell+1)}\big](c), \tag{131}$$

where Ω_{iI} is the gauge-covariant transport into agent i's frame on the latent-state fiber and $\tilde{\Omega}_{iI}$ is its model-fiber counterpart. The state transport is $\Omega_{iI} = U_i U_I^{-1}$ with $U = \exp(\phi)$ (Section 4.6.1); the model transport is the ρ_{model}-image of $U_i U_I^{-1}$ when frames are shared, or an independent transport $\tilde{\Omega}_{iI} = \tilde{U}_i \tilde{U}_I^{-1}$ with $\tilde{U} = \exp(\tilde{\phi})$ when model frames are independent. Under this assignment beliefs and priors share the latent-state manifold $\mathcal{B}_{\text{state}}$, models and hyper-priors share $\mathcal{B}_{\text{model}}$, and the four-field hierarchy $r \to s \to p \to q$ posited at the agent definition reduces to two primitive fields (q, s) at each level connected by cross-scale transport: r regularizes s because r is the meta-agent's model transported down, and p regularizes q because p is the meta-agent's belief transported down.

Read as an assignment at one instant, Eq. (131) sets a prior equal to a transported posterior of the same instant. That is not a factor of one normalized joint: a generative kernel that takes a recognition law as its argument is a map from recognition laws to kernels, and the free energy built from it is not an evidence bound [30, Chs. 4 and 12]. Appendix G makes the failure concrete. There a joint with Gibbs cross-scale factors is a well-defined density on the product space of all agent-level latents (Lemma 5), and Lemma 7 computes the rigid-link variational predictive density of a child under that joint, which retains the transported parent covariance and is not a message of the joint (Remark 2). Evaluated against the full parent posterior, the shadow reuses each constituent's own evidence: in the rigid-link limit the child precision obtained from shadow times likelihood exceeds the exact child posterior precision by exactly the precision of the child's own observation, which has entered the parent posterior and then enters the child a second time. For the same reason the same-time shadow functional is not bounded below by the surprisal of the augmented joint.

Read across time, Eq. (131) is exact. Index episodes by $n = 0, 1, \ldots$, and let the tower of episode $n+1$ be generated from the parent latent z_I of episode n by a declared transition kernel, with a declared prior on the tower of episode 0. This episode-indexed joint is normalized and acyclic, every kernel reads only structural data and latents of earlier episodes, and the prior of the tower at episode $n+1$ given the records $o_{0:n}$ is the filtered parent law pushed through the transition [30, Ch. 12, the delayed-feedback proposition]. Choose the transported-copy transition, which places child i's latent at episode $n+1$ at $\Omega_{iI} z_I$. The predictive prior of child i is then

$$p_i^{(\ell),n+1} = (\Omega_{iI})_{\#} \mathbb{P}\big(z_I^{(n)} \in \cdot \mid o_{0:n}\big),$$

which is $\Omega_{iI}[q_I^{(\ell+1)}]$ whenever $q_I^{(\ell+1)}$ is the filtered parent law of episode n, and is the corresponding variational predictive density when $q_I^{(\ell+1)}$ approximates it (status D; the display is the pushforward of the filtered law under a deterministic kernel). A noisy transition $\mathcal{N}(\Omega_{iI} z_I, \Omega_{iI} \Sigma_\epsilon \Omega_{iI}^\top)$ adds $\Omega_{iI} \Sigma_\epsilon \Omega_{iI}^\top$ to the transported parent covariance. No evidence is counted twice, because child i's record of episode $n+1$ has not entered the filter that produced the parent law of episode n. Eq. (131) is therefore the one-episode-lagged predictive prior of the episode-indexed joint, and this lagged construction of the companion theory supersedes the same-time shadow. The same-time form used in the energy of the following sections is an energy-level simplification of the lagged scheme (status S), not a limit of it.

Two further readings are consistent with an evidence bound within one episode. The standard hierarchical scheme [19, 110] derives the level-ℓ prior from a declared conditional $p(z_\ell \mid z_{\ell+1})$ of the generative model; its variational term is the conditional coupling $\mathbb{E}_{q_I} D_{\text{KL}}(q_i \| \chi(\cdot \mid k_I))$ of a downward kernel χ [30, Ch. 12], and the corresponding downward message is the cavity predictive $\int \chi(\cdot \mid k_I) q_I^{\setminus i}(dk_I)$, which uses the parent posterior with constituent i's contribution removed. The second is a random-effects model: the constituents are independent draws $k_i \sim \mathcal{N}(\Omega_{iI} m_I, \Omega_{iI} S_I \Omega_{iI}^\top)$ with a point hyperparameter $\theta_I = (m_I, S_I)$, so that q_I is a point estimate, S_I is the population covariance, and minimizing the free energy over (q, θ_I) is type-II maximum likelihood, whose exact M-step is the strict barycenter with its dispersion term (Section 9.2). Appendix G's single parent

latent with noisy children is a different model, in which Σ_I is posterior uncertainty and not population spread; its rigid-link lemma illustrates the predictive density and does not supply the shadow as a factor.

The single rung extends to a tower. Meta-agents at level $\ell+1$ can themselves cohere into meta-meta-agents at level $\ell+2$, so that individuals form groups, groups form communities, and communities form societies up to a maximum level $\ell_{\max}$, with information flowing bottom-up through consensus formation and top-down through predictive priors. This bidirectional coupling has the feedback structure of Wheeler's participatory loop (W4 in Table 1; status S). That structure alone does not distinguish the construction from hierarchical modeling with learned priors [30, Ch. 14]; formation of collectives by the agents' own inference would, and Section 13 states the test of that reading. The hierarchy has two boundaries. At the top level there is no enveloping meta-agent, and the hyper-prior $r_i^{(\ell_{\max})}$ is a fixed declared prior, the root of the hierarchy (Section 12); at the bottom level $\ell = 0$ there are no constituents, so cross-scale propagation acts only downward from above. A prospective single-level comparison may treat p_i, r_i as primitive boundary conditions, while a multi-level run must test the cross-scale relation of Eq. (131).

11 Coarse-Graining III: Exact Coarse-Graining and Closure

The coarse-graining of the preceding sections is an exact block-variable pushforward of a posited configuration ensemble followed by a truncation to the five-term family, a block transformation in Kadanoff's sense [27] without a rescaling step. It is motivated by, but is not, the renormalization group of statistical physics [111–113]. No rescaling, coupling-space flow, or fixed point is constructed, and on a finite network none can be: a step that reduces the number of agents admits no comparison identification of the interaction spaces before and after it, and with blocking ratio at least $b > 1$ the tower has at most $\log_b N$ levels [30, Ch. 11, the finite-network obstruction proposition]. The partitions are chosen by a state-dependent detector, so the map is not a homogeneous flow, and the attention weights β_{ij} are state variables (the softmax of the current edge energies), not coupling constants; the couplings proper, the temperature τ, the weight λ_h, and the declared precisions, are held fixed across levels by assumption, and the exact pushforward would in general change them; the one exception computed here is the parent's entropy weight, which shifts at one loop (below).

The status of the construction rests on three claims. First, the coarse-graining map admits an exact pushforward, and it is gauge covariant (Theorem 9) for any Lie group with a left-invariant Riemannian metric when the frames of each cluster lie inside the convexity radius; compactness enters only through the normalizability of the Haar reference of the Gibbs ensemble, which fails for noncompact $\mathrm{GL}^+(K_q)$. For an exact channel the fine free energy equals the coarse free energy plus the discarded conditional information, with the same evidence at both levels [30, Ch. 9, the exact coarse free-energy theorem], and the coarse action is the conditional-partition formula of [30, Ch. 11]. Second, closure of the level-$(\ell+1)$ free energy in the same five-term family is conditional and posited, not derived. Third, the consensus detector is used as a candidate filter in place of the variational formation criterion; it is not derived from that criterion (Section 9.3). One coupling does change under the step. At one loop in the mean sector, at coherence, the fluctuation term $\frac{T_{\mathrm{cfg}}}{2}\log\det' H_I^{\perp}$ of the internal modes equals, up to a Y-independent constant, $-\frac{T_{\mathrm{cfg}}}{2}(|I|-1)\log\det\Sigma_I$, so integrating out the internal modes shifts the coefficient of $\log\det\Sigma_I$, the entropy weight of the parent, in proportion to the cluster size (Eq. (278)); this induced renormalization vanishes at $T_{\mathrm{cfg}} = 0$, where the step reduces to constrained minimization.

Several precedents organize coarse-graining in these terms. Bény and Osborne [114] formulate

coarse-graining as a family of channels and pull the Fisher metric back onto the space of models, so that the metric contraction measures the information destroyed per step; Machta et al. [115] trace effective theories to the compression of Fisher-information eigenvalues under coarse-graining; Berman et al. [116] read Bayesian updating as a flow on the Fisher manifold; and Mehta and Schwab [117] construct an exact mapping between Kadanoff's variational renormalization group [28, 29] and restricted Boltzmann machines. Exact form preservation of a coarse-graining step in the strict sense is the classic solvable-model fact (for example one-dimensional Ising decimation), not a claim we attribute to any of these references. The closure statement here is a parametric form preservation: the level-$(\ell+1)$ functional has the same five-term shape as the level-ℓ functional, with coarse belief couplings $\beta^{(\ell+1)} = f_\beta(\{\beta_{ij}^{(\ell)}\})$ pooled by the raw-conductance rule of Eq. (279), in which the pooled intra-cluster conductance acts as a parent self-loop, and model couplings $\gamma^{(\ell+1)} = f_\gamma(\{\gamma_{ij}^{(\ell)}\})$ left unspecified; the working forms, the conductance pooling, and the Schur elimination behind them are in Appendix I. Lemma 8 (mean-sector Gaussian elimination) concerns only the linear coarse-graining of the means at fixed covariances, frames, and couplings, where the exact pushforward of the mean-sector Gibbs density is Gaussian with the Schur-complement precision. A map that retains the parent covariance has bounded fibers and lies outside that statement; a level-dependent log-determinant generates non-quadratic terms at the next level, so the class is not closed under iteration, and exact closure generically requires hyperedges [30, Ch. 11, the hyperedge-generation proposition]. A merely local quadratic expansion retains the anharmonic remainder and is only a Laplace approximation.

The belief-mean sector shows which operation preserves the family. There a functional of the five-term family is a block-diagonal positive part plus a matrix-weighted Laplacian with positive definite edge weights, so each off-diagonal block satisfies $\operatorname{sign}\det\Lambda_{ij} = (-1)^{K_q}$ and the triple product around any triangle carries the same sign, a necessary condition involving neither frames nor transport. Integrating the within-cluster deviations out at fixed barycenter violates this condition on sparse fine graphs: on a chain of pairs it generates a coupling between non-adjacent clusters with the wrong sign, the standard way a decimation leaves a nearest-neighbor ferromagnetic ansatz. The condition is only necessary, and it is weak at even K_q, so it certifies failure and not success. The general statement is in the companion supplement: unrestricted matrix-weighted Kron reduction leaves the Laplacian-plus-self-term family, and on a fixed flat graph closure for every weight assignment holds exactly when every hidden component has at most two retained terminals [38, Ch. G7]; the scalar-weight theory of Kron reduction and graph coarsening is in [118–120]. Identifying constituents preserves the family exactly. The congruence $S^\top \Lambda S$ is the Galerkin coarse form of aggregation-based algebraic multigrid; the within-cluster Laplacian contributions cancel identically, the coarse coupling between two clusters is the sum of the fine weights on the edges cut between them, which supplies $\beta^{(\ell+1)}$ directly without the conductance ansatz, and successive aggregations compose [38, Ch. G9]. Tying has two distinct probabilistic readings. As a coarse model it is conditioning on zero within-cluster disagreement, with natural parameters $(S^\top \mathsf{h}, S^\top \Lambda S)$ [38, Ch. G7]; this is precomposition, which defines a new model to which data processing and evidence preservation do not transfer [30, Ch. 9]. As a restriction of the recognition family on the means it keeps the fine joint and costs bound. The coarse couplings $\beta^{(\ell+1)}$ above are those of the first reading. Exact marginalization, a Markov pushforward, moves neither the evidence nor the bound. The rigorous literature on position-space coarse-graining maps shows that block-spin and decimation images of Gibbs measures can fail to be Gibbsian [121]; its comparison between schemes does not settle the direction the framework's intuition assumes, so no analogy in either direction is asserted, and the lattice treatment of spatial coarse-graining is deferred to [31, Chs. B7–B9].

The construction quantifies cross-scale information propagation through the transported relative

entropy between constituent and meta-agent beliefs,

$$\mathcal{I}_{\ell\to\ell+1} = \sum_{I\in\mathrm{level}(\ell+1)} \sum_{i\in I} D_{\mathrm{KL}}\big(q_i^{(\ell)} \big\| \Omega_{iI}\big[q_I^{(\ell+1)}\big]\big), \tag{132}$$

a divergence between two distributions on the same fiber after transport of the meta-agent belief into the constituent's frame, which vanishes when they agree after transport. We do not write this quantity as a mutual information, since a mutual information requires a joint distribution on (k_i, k_I) and not a pair of marginals. In the lagged reading of Section 10 each summand is the divergence of a constituent's belief from its predictive prior. In a functioning participatory system we expect $\mathcal{I}_{\ell\to\ell+1}$ to be small but nonzero for all $\ell < \ell_{\max}$: zero indicates constituents and meta-agent in exact agreement after transport, pointwise consensus (Section 9.5), while large values indicate a meta-agent that has decoupled from its constituents. The quantity is distinct from the change in the alignment portion of $\mathcal{F}$ when the meta-agent is inserted: that change is signed, whereas $\mathcal{I}_{\ell\to\ell+1}$ is the nonnegative sum of the added child-parent divergences alone.

The complementary top-down channel can be quantified in the same way. Writing the active parent-child relations as a dynamically evolving graph $\mathcal{G} = \{(i, I) : i \in I\}$ whose edges are the channels along which a meta-agent sets a constituent's prior, the per-agent diagnostic

$$\Delta p_i(n) = \int_{\mathcal{C}} D_{\mathrm{KL}}\big(p_i^{(\ell)}(c; n) \big\| p_i^{(\ell)}(c; n-1)\big)\mu_{\mathcal{C}}(dc), \tag{133}$$

with $\mu_{\mathcal{C}}$ the declared background density on $\mathcal{C}$ and $\Delta p_i > \theta_{\min}$ flagging active top-down feedback, measures how far the evolved parent has moved the constituent's predictive prior between successive episodes.

The construction also makes structural predictions about meta-agents at level $\ell + 1$ relative to level ℓ. In the mean-sector uniform-weight case the meta-agent law is a fixed Markov image of its constituents' laws, and wherever the meta-agent's frame introduces no horizontal defect, the meta-agent's pulled-back information geometry is dominated by its constituents', with equality exactly along directions whose fine score the meta-agent retains [31, Ch. B2, the coarse or meta-agent perceived-geometry corollary]. A horizontal defect can reverse the inequality [31, Ch. B2, the exact signed positivity criterion], so the prediction that meta-agent beliefs vary more smoothly over $\mathcal{C}$ than constituent beliefs, with short-wavelength structure averaged out at higher levels, is conditional on frame compatibility, whose exact form is an isotropy condition; this comparison involves base directions and belongs to the base-transport work deferred to Supplement B. Meta-agent covariances follow the convex-combination pooling of Eq. (129), so the construction predicts a parent covariance comparable to the weighted average of the transported constituent covariances, with the strict barycenter adding a positive-semidefinite dispersion term that can only enlarge it. A more confident parent would follow only from the precision-weighted product-of-experts pooling that the meta-agent state construction declines to adopt (Section 9.4). Each prediction requires a multi-seed measurement before any emergence claim can be made (Section 13).

12 Coarse-Graining IV: The Ouroboros Tower and Its Declared Root

Assembling the rungs of the preceding sections yields the Ouroboros tower, the multi-level structure in which each level is a full free-energy functional and adjacent levels are coupled by the cross-scale shadows of Eq. (131), read as lagged predictive priors. Wheeler's participatory universe envisions

observer-participants whose registrations give tangible reality to what is observed [1,2]. The tower gives a formal counterpart of one strand of that picture (W4): when coherent clusters are licensed (Section 9.2; status C), they form meta-agents that set the priors of their constituents. Agents are organized by level $\ell \in \mathbb{N}$: level-0 base agents cohere into level-1 meta-agents, which cluster into level-2 meta-meta-agents, and so on. Each level-ℓ agent is a full section $\sigma_I^{(\ell)}$ carrying a primitive belief $q_I^{(\ell)}$, a primitive generative model $s_I^{(\ell)}$, and a frame $\phi_I^{(\ell)}$, with its prior and hyper-prior given by the cross-scale shadows of the level-$(\ell+1)$ meta-agent through Eq. (131). The tower is capped at a finite $\ell_{\max}$, and on a finite network a cap is forced in any case (Section 11). The reading on which the resulting meta-agents are agents on a par with their constituents, not summaries of them, is an interpretation (status S); the derivation of Section 9.2 supplies instead the reading of the parent as a fitted shared prior.

Constituents receive their shadow from the immediate parent alone. A normalized hierarchy is depth-Markov: each level's factor reads the complete next-level state and no deeper variable, and for an adaptive tower, generated bottom-up by the coarse-graining maps, this holds automatically by Bayes reversal [30, Ch. 12]. A direct grandparent term is then not a factor of the tower law, and multiplying it in counts information already transmitted through the parent a second time; in a separately declared top-down hierarchy it would require its own normalized kernels. We therefore keep only the depth-one model coupling,

$$\mathcal{F}_{\text{ouro}}\big[s_i; s_I^{(\ell+1)}\big] = \lambda_h D_{\mathrm{KL}}\big(s_i \big\| \tilde{\Omega}_{iI}\big[s_I^{(\ell+1)}\big]\big), \tag{134}$$

which is the model hyper-prior term of Eq. (49) with $r_i = \tilde{\Omega}_{iI}[s_I^{(\ell+1)}]$, the depth-one model shadow; deeper ancestors act on s_i only through the chain of depth-one couplings. The cross-scale belief coupling is carried by the self-term $D_{\mathrm{KL}}(q_i \| p_i)$ of Eq. (49), whose prior p_i is the depth-one belief shadow $\Omega_{iI}[q_I^{(\ell+1)}]$, and is not re-added here.

A genuine tower predicts that meta-agents at level $\ell+1$ evolve more slowly than their constituents at level ℓ. This cross-scale slowdown, $t^{(\ell)} < t^{(\ell+1)}$ at matched field type, is a conjecture (status C), not an imposed split. It rests on two mechanisms: meta-agents integrate over constituent beliefs, so high-frequency components of constituent drift are averaged out at the parent level, and changing a meta-agent state requires coherent change across several constituents through the gauge-covariant barycenter, which introduces consensus energy barriers that resist rapid updates. Parent stiffness supplies no third mechanism: with row-normalized β the incoming relational stiffness is a convex combination of constituent precisions and does not grow with $|I|$ under the adopted covariance pooling. The within-level separation $t_{\text{belief}}^{(\ell)} < t_{\text{model}}^{(\ell)}$, by contrast, is imposed by the adiabatic hypothesis of Section 7, in which beliefs and attention rows form the fast block and models and prior parameters the slow block [30, Ch. 8]. The fully interleaved ordering $t_{\text{model}}^{(\ell)} < t_{\text{belief}}^{(\ell+1)}$ that a single chain would assert is supported by no mechanism here and is in tension with the prior-inheritance structure, under which level-ℓ priors inherit their timescales from the parent. The cross-scale slowdown is analogous to collective variables equilibrating more slowly than microscopic ones and to institutional change being slower than individual opinion change. It is not built into the construction, which uses one step size at every level; testing it requires a declared per-level timescale measurement.

The tower terminates at its top level. There is no parent meta-agent, and the top-level hyper-prior $r_i^{(\ell_{\max})}$ is a declared prior, the boundary condition of the hierarchy. For the belief channel an energy-level closure is available: top-level agent i may take as its prior the coherence-weighted pool of the transported beliefs of all active agents at all levels. Mixtures of transported Gaussians leave the Gaussian family on which the closed-form divergences rely, so the pool is taken as the

Gaussian moment projection of the mixture $\bar{Q}_i = \sum_j \bar{w}_j (\Omega_{ij})_{\#} q_j$, the minimizer of $D_{\mathrm{KL}}(\bar{Q}_i \| p)$ over Gaussians p,

$$\mu_i^{(\mathrm{top})} = \sum_j \bar{w}_j \Omega_{ij} \mu_j, \qquad \Sigma_i^{(\mathrm{top})} = \sum_j \bar{w}_j \Big[\Omega_{ij} \Sigma_j \Omega_{ij}^\top + \big(\Omega_{ij}\mu_j - \mu_i^{(\mathrm{top})}\big)\big(\Omega_{ij}\mu_j - \mu_i^{(\mathrm{top})}\big)^\top \Big], \quad (135)$$

with normalized weights $\bar{w}_j = w_j / \sum_k w_k$ and $w_j \propto \exp(-\bar{D}_j)$, where $\bar{D}_j$ is agent j's mean transported belief divergence from the collective, so that the weights favor agents coherent with the collective. The between-mean dispersion term is retained: dropping it, as the working form Eq. (129) of the cluster barycenter does, would depart from the moment projection, and because the apex pool is not gated by coherence, the dispersion term is what records a globally incoherent collective as a broad prior. The model channel is not pooled.

The apex closes the tower at the level of the consensus energy and its update schedule, not as a generative model. Every finite acyclic generative model in which each record has a latent parent leaves at least one latent whose prior is declared and not constituted by the agents it organizes [30, Ch. 12, the declared-root proposition] (status P), and a reciprocal same-time structure is a probability model only as a globally normalized joint with its partition function [30, Ch. 4]. The pooled prior reads the current beliefs of the agents it organizes, so it is not a factor of one joint. The declared root is the formal residue of Wheeler's "no tower of turtles" [1], a reading of status S: the regress of priors stops at a declared prior and does not close on itself, and the companion theory accordingly replaces a self-closing apex by the declared root. The participatory content survives in three typed forms, each a sequence of fixed normalized joints [30, Ch. 12]: the inference loop, in which parent and child updates read each other under block coordinate ascent, as in any connected hierarchical model; slow learning of a declared top prior, whose optimum over an unrestricted family is the aggregate of the recognition laws it summarizes and, within the Gaussian family, their moment match including the dispersion retained above; and delayed feedback, in which the parent inferred at one episode sets its children's prior at the next (Section 10). The apex pool is the same-time, energy-level counterpart of the second: the companion theory aggregates the recognition laws of one top latent across episodes by a slow parameter update, whereas the apex pools across agents of all levels at one time. It has the feedback structure of Wheeler's self-excited circuit without being derived from it or evidence for it (status S). It is not that circuit, which runs through time, is constituted by registrations, and sustains itself [2]: the dynamics here relax to stationarity unless observations are injected, and the top-level model channel is a fixed boundary condition.

A participatory system must keep evolving. Stationarity of the specified dynamics means stationarity of $\mathcal{F}$, at which beliefs, priors, and frames cease evolving. Generic stationary points balance alignment against prior attachment, observation fit, and the attention entropy, and therefore retain persistent disagreement, so perfect consensus is one special stationary configuration and not the generic one; what makes a stationary state non-participatory is the cessation of evolution itself. The notion is non-stationarity of a deterministic descent. It is not the statistical-physics sense of distance from equilibrium, which refers to broken detailed balance or entropy production in a driven stochastic dynamics; the dynamics here is relaxational. Three indicators track it, each rendered dimensionless against a declared early-run reference window before any combination: an energy flux $\Phi_E(t) \propto |d\mathcal{F}/dt|$, an information flux $\Phi_I(t)$ tracking the magnitude of entropy change across belief and prior, and a gradient variance $V_\nabla(t) \propto \mathrm{Var}_i(\|\nabla_{q_i}\mathcal{F}\|^2)$ taken across agents at fixed t. They combine into the non-stationarity score $S_{\mathrm{ns}}(t) = (\Phi_E + \Phi_I + V_\nabla)/3$, whose value depends on the reference window; values such as 0.5 and 1 are candidate thresholds to preregister and calibrate, not validated ones. Without observation drive the flow relaxes. The reduced functional is bounded

below and real-analytic in the Gaussian parameters, so by the Łojasiewicz gradient inequality [122] every natural-gradient trajectory with precompact image converges to a stationary point and has finite total Fisher length [30, Ch. 7, the finite-duration proposition]; the corresponding statement for discrete descent iterates is [123]. Of the mechanisms that might sustain evolution, only observation injection supplies persistent exogenous drive. Heterogeneous top-down feedback and spatial separation shape how the system relaxes without preventing relaxation, and frame differences cannot maintain disagreement by themselves, since Regime I frames are removable at a point. Whether observation injection sustains non-stationarity is open (status C).

Every per-level free-energy term is integrated over $\mathcal{C}$ against the same declared background density, the volume form of the background chart (Section 4.9). That density is an input, not the geometry the agents induce. The induced geometry, the pulled-back metrics of Section 18.1, enters no free-energy term, so the participation loop of the tower closes only within information, from belief to meta-belief to predictive prior.

13 Prospective Validation of Multi-Scale Formation

The construction of Sections 9–12 defines how coherent clusters may form meta-agents and how their beliefs and models may propagate across levels. No simulation result is claimed (Section 2.1); this section states the experiment that would test the multi-level claims. The formation rule under test is the threshold detector and barycenter of Section 9; the companion theory replaces both by normalized marked parents [30, Ch. 13], which is the rule a later protocol should test.

Formation alone cannot discriminate participation from seeding. On the homogeneous four-agent cycle of the companion theory, a collective forms from every Gaussian initial law with probability bounded below uniformly in the regime parameter, whether or not the source prior learns, because the random initial source assignment seeds the pair [30, Ch. 14, the seeded-contrast-formation proposition]. The source prior there is the attention prior π_{ij} of this paper. Stable multi-child clusters, reproduction across seeds, and an active model channel would therefore all be met by a system whose attention prior never learns. The decidable form of the participatory reading is therefore the companion theory's pairing-selection conjecture (status C): slow learning of the attention prior selects which pair forms a collective, at the level of the law, against a frozen-prior control whose pairing law stays symmetric [30, Ch. 14, the pairing-selection conjecture]. The protocol below is its finite-sample counterpart in the present construction; neither bears on the participatory-physics readings of Part II.

The protocol has three arms at one declared configuration, differing only as named. Arm A has a learning attention prior and an active, untied model channel with a declared nonzero γ_{ij}, so that the term $\gamma_{ij} D_{\mathrm{KL}}(s_i \| \tilde{\Omega}_{ij}[s_j])$ is exercised and not inferred from a frozen or tied channel. Arm B, the frozen-attention-prior control, is identical except that π_{ij} does not learn. Arm C is identical to arm A except that the model channel is frozen ($\lambda_h = 0$ and $\gamma_{ij} = 0$). The arms share seeds: each seed fixes the initial means and the initial assignment draws for all three arms, so that comparisons between arms are paired.

The following quantities are declared here and may be revised only before execution, with any revision reported. The formation event at a recorded step is the existence of a connected block I with $2 \le |I| \le N - 1$ whose internal belief and model transported relative entropies are all at most $\delta_{\mathrm{in}} = 10^{-2}$ nats and whose boundary belief transported relative entropies all exceed $\delta_{\mathrm{out}} = 1/2$ nat, the form of the event in the companion proposition with its internal bound relaxed. The persistence window has flow-time length $W = 10/\eta$, ten relaxation times of an isolated belief mode under the natural-gradient flow of rate η, and ends at the declared horizon. The primary statistic

for each seed and arm is the indicator H that one block satisfies the formation event throughout the window, together with the identity of that block. Each arm runs $n = 20$ seeds. The formation claim passes in arm A if $H = 1$ in at least 16 of the 20 seeds. The participatory claim requires a difference between arms A and B: it passes only if an exact two-sided McNemar test on the paired indicators rejects at level 0.05, or if an exact permutation test at level 0.05 on the paired block identities rejects equality of the laws of the formed blocks. With twenty paired seeds no single aberrant seed decides the outcome, and the McNemar test needs at least six discordant pairs in one direction ($2 \cdot 2^{-6} = 0.031$). The model-channel claim passes if arms A and C differ by the same McNemar criterion.

The formation claim is refuted for the tested configuration if arm A fails its pass rule. The participatory claim is unsupported if arms A and B do not differ by the pass rule, whatever arm A shows. A result that appears only with the model channel frozen, only under thresholds chosen after inspection, or only in one seed validates nothing. Two conditions are preconditions for a valid measurement and not refutation criteria: population accounting must close at every recorded step, $N_{\text{total}}(t) = \sum_\ell N_\ell(t)$ with $N_\ell(t)$ the number of active agents at level ℓ, with every meta-agent's constituents and parent recorded; and the raw per-step states (beliefs, models, priors, hyper-priors, frames, transported pairwise divergences, β_{ij}, γ_{ij}, detector scores, assignments, energy components, and cross-scale links) must be retained so that every reported statistic can be recomputed. A failure of either invalidates the run as a measurement; it does not refute the tower.

Until this protocol is completed, the Ouroboros tower supplies a mathematical construction and a set of falsifiable predictions (status C), not evidence that the dynamics generate a persistent hierarchy or sustained non-stationarity.

14 Discussion

This section places the construction of Part I relative to the dynamics of active inference and to existing participatory and structural positions in the foundations of physics. The dynamical statements of the first subsection are derived in Section 8 (status D); the positioning in the second and third subsections is interpretive (status S). Physical and phenomenal readings that rest on the postulates P1 to P10 are collected in Part II.

14.1 First-Order Dynamics: Stiffness, Friction, and the Kinetic-Metric Postulate

The dynamics the construction defines is first order: the natural-gradient flow of Eq. (82), $\dot{\xi} = -G_\eta^{-1}\nabla\mathcal{F}$, where G_η is the Fisher metric with each block divided by its rate (η_μ for the means, η_Σ for the covariances). It has the form $\gamma\dot{q} = -\nabla V$ of the overdamped reduction of a mechanical particle $m\ddot{q} + \gamma\dot{q} = -\nabla V$, with the friction coefficient replaced by the tensor G_η and no term in the inertial slot (Eq. (115)). The comparison illustrates the regime only: the flow is a gradient flow on the statistical manifold and is not obtained as the overdamped limit of a Newtonian stochastic differential equation. First-order natural-gradient descent is also the regime of standard active inference, whose neuronal dynamics approximate natural-gradient descent on the variational free energy [124], an intrinsic Riemannian gradient flow in the sense of Amari [125]; for cortical systems, where synaptic time constants dominate any putative inertial term, the overdamped regime is the appropriate one. Section 8 derives what precision does in this flow. The Hessian of $\mathcal{F}$ at consensus is a stiffness, so a high-precision agent has small compliance and a given force displaces it little. At an isolated equilibrium $q_i = p_i$ the stiffness equals the Fisher metric, so the linear relaxation rates are η_μ and η_Σ whatever the precision. The linearized spectrum of the gradient flow is real, so

there is no oscillation frequency to measure; complex eigenvalues appear only in the receiver-only truncation, whose field is not the gradient of $\mathcal{F}$. The only velocity-quadratic form of the dynamics is the Rayleigh dissipation of Eq. (116), and for a single rate its square root is the rate of the information clock (Eq. (137); [30, Ch. 7]).

The kinetic-metric postulate of Section 8.7 adds the inertia that the flow lacks, with its own second-order equation of motion, Eq. (117), whose overdamped limit is the natural-gradient flow. It is adopted only under the reading of the variational principle as a candidate fundamental scaffolding, and the free-energy-principle canon does not imply it: Friston, Trujillo-Barreto and Daunizeau [126] obtain second-order behavior through generalized coordinates of motion and a first-order variational filter, and Friston et al. [127] decompose the flow into solenoidal and dissipative parts of a first-order stochastic vector field; neither writes a velocity-quadratic kinetic energy whose inertia tensor is the potential Hessian. Under the postulate as stated, with inertia M_K equal to the stiffness M at consensus, every undamped linear mode has unit frequency in flow-time units, so the scaling $\omega^2 \propto 1/\bar{M}_i$ is not a consequence of the postulate. A precision-dependent frequency requires an inertia tensor different from the stiffness tensor. Taking the bare prior precision $\bar{\Lambda}_p$ as inertia and the full Hessian $M = \bar{\Lambda}_p + K_{\mathrm{c}}$ as stiffness, with K_{c} the coupling contribution, gives $\omega^2 \in \mathrm{spec}(\bar{\Lambda}_p^{-1} M) = 1 + \mathrm{spec}(\bar{\Lambda}_p^{-1} K_{\mathrm{c}})$, which is a second and different postulate, testable and not an identity. On either postulate the Fisher information enters the dynamics the construction actually runs as a stiffness and a friction tensor, and belief configurations of large Fisher information are hard to displace, not hard to accelerate.

14.2 Relation to Existing Participatory Approaches

Of the strands of Wheeler's program in Table 1, the construction models W3 most directly: Wheeler's third question, how come "one world" out of many observer-participants, read with Føllesdal's dictum that meaning is "the joint product of all the evidence that is available to those who communicate" [1]. Agents compare laws only through the explicit transport Ω_{ij}, the residual $D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ is the frame-independent part of their disagreement, and intersubjective agreement is a stationary point of the consensus free-energy functional of Section 5; reading this as Wheeler's strand has status S. The companion theory describes the multiscale participation loop in the same register, as resembling Wheeler's proposal without being derived from it or evidence for it, and notes that none of its three typed forms of participation separates the program from a hierarchical model with learned priors, while formation could [30, Ch. 14].

Rovelli's relational quantum mechanics [11], an informational theory in Wheeler's spirit that relativizes states and values to arbitrary physical systems (Section 3), is the closest relative of the construction, which relativizes laws to agents and makes the comparison map between perspectives explicit. The construction is compatible with the relational reading without establishing that physical laws are observer-dependent, and its observer-dependence is one of representation and belief, not of fact. With Regime I frames the cocycle closes (Lemma 1), a global frame exists, and every transported law lives in one classical probability space, so the construction satisfies the classical assumptions that Brukner's no-go theorem for observer-independent facts [128] exploits and does not model the situation it addresses. The precise point of contact with the cross-perspective-links debate within relational quantum mechanics [60] arises with independent links on the interaction graph (Section 4.6.5). There a nontrivial loop holonomy obstructs a global frame, and a normalized record whose law separates loop-holonomy classes exists at one base point, by the companion theory's theorem on an operational record sensitive to network-loop holonomy [30, Ch. 10]. This is a belief-level counterpart of the question whether the perspectives of different relational observers can be linked.

QBism [10, 61] reads quantum probabilities as an agent's personalist degrees of belief and, as participatory realism, claims Wheeler's program by name (Section 3). The construction shares the personalist starting point, since each agent carries its own q_i, and in this respect, together with the relational comparison through transport, it is of the relational and QBist type. It departs from QBism in two places. It has no quantum content, and it adds a transport law between agents and a consensus functional whose stationary points model intersubjective agreement, whereas QBism keeps measurement outcomes personal to each agent [62] and leaves agreement to communication between agents. Read classically, the consensus mechanism is precision-weighted opinion pooling of the kind DeGroot analyzed [129]; its closest quantum-foundations relative is quantum Darwinism, in which objectivity arises from records redundantly available to many observers [45].

Müller's observer-first program [9] formalizes Wheeler's law without law [2] directly. Observer states are primitive, no laws are presupposed, and an external world with simple probabilistic laws, together with agreement between observers, emerges as an asymptotic statistical phenomenon through algorithmic probability. That program is more faithful to Wheeler than the present construction on law without law, on the refusal of a continuum, and on the primacy of registration; the present construction is more explicit about multi-agent transport, gauge structure, and consensus dynamics. Law without law is not modeled here. The functional, the belief family, and the divergence are fixed in advance, so their stationarity conditions are laws in the ordinary sense, and the $\mathrm{GL}(K_q)$ invariance of the divergence is already present for a single agent, a consistency requirement that consensus must respect (Section 23 and Remark 1). A law-without-law version would need the functional itself to be selected by the population dynamics, which is open (status C).

The mechanism of Page and Wootters [130] derives dynamics from correlations between a clock subsystem and the rest of a stationary global state, and so addresses the timelessness behind Wheeler's "no time". The construction does not do this. Its histories are ordered by an algorithmic flow parameter, and the information clock of Section 17 supplies duration along that order, not the order itself. A classical relational analog is available in the construction's own terms: parametrize the other agents' histories by one agent's Fisher length, which is strictly increasing wherever that agent's free-energy gradient is nonzero and constant at its stationary points [30, Ch. 7], so that the relational clock fails exactly where that agent stops updating. This is a structural parallel (status S), not a derivation of time.

14.3 Kantian and Structural-Realist Positioning

This subsection places the author's interpretive stance, cognitive-first structural monism (Section 1), relative to Kant and to structural realism; none of it is derived. The construction recasts Kant's distinction between phenomena and noumena [15] in a structural-realist register closest to Cassirer's neo-Kantian variant [131]. Worrall's epistemic structural realism [132], which suspends judgment about the nature of the relata, is a related precursor. The ontic structural realism of Ladyman [34] and of Ladyman and Ross [33] is a neighbor whose naturalism the position departs from: the cognitive-first order of derivation reverses the Primacy of Physics Constraint, a methodological principle of Ladyman and Ross's naturalized metaphysics, not the definition of ontic structural realism. On this reading information geometry on the fibers is the structure available to agents. The base manifold $\mathcal{C}$ is a presupposed smooth index manifold carrying no physical metric, called noumenal by analogy only. Along an agent's belief section a Riemannian metric on $\mathcal{C}$ is pulled back from the Fisher metric; computed with the covariant derivative of the agent's own connection it is unchanged under a passive change of frame [31, Ch. B2], so differences between agents' pulled-back metrics reflect differences in their beliefs. What measurement protocols label distance and duration then correspond to induced metric intervals and accumulated Fisher length, and what they label

inertia corresponds at most to the precision that acts as a stiffness in Section 8. The frame-invariant data of the construction, namely the relative-entropy and evidence-bound values, the conjugacy classes of link holonomies, and the per-edge twist classes, are compatible with epistemic, moderate, and ontic structural realism alike, and nothing in the mathematics selects among these positions or the neo-Kantian reading [30, Ch. 14]. The treatment of beliefs and frames as agent-internal constructs, not structural representations of an external reality, is likewise compatible with the enactive reading of active inference advanced by Ramstead, Kirchhoff and Friston [133], although that reading rests on an action-perception coupling that the present construction does not model. On the Kantian side the position disavows the apodictic a priori, forms fixed and valid for all time, and keeps a revisable constitutive role for the requirement that shareable descriptions be invariant under transport (Section 23), in the sense of the relativized a priori that Friedman [134] develops from Reichenbach's 1920 analysis of relativity. Whether the within-agent pullback can carry a Lorentzian geometry is taken up under the postulates of Part II (Section 19), where an indefinite form is exhibited and its selection remains open.

15 Summary of Part I

Part I has constructed the classical, multi-agent, information-geometric objects on which the Wheeler correspondence of Table 1 rests. Agents are local sections of associated bundles over a presupposed smooth index manifold $\mathcal{C}$ carrying no physical metric; each carries a belief q_i on the state fiber and a generative model s_i on the model fiber, expressed in its own frame, and agents compare laws only through the transport operators Ω_{ij} and $\widetilde{\Omega}_{ij}$. At one base point the active symmetry is one global copy of the structure group and per-agent frames are passive re-expressions. Regime I frames are removable by the relative-entropy invariance of Theorem 1, whereas independent links on the interaction graph, with orientation reversal imposed, carry loop holonomy that no choice of frames removes. The consensus functional couples agents through transported relative entropies; its attention rows are Gibbs optima with reduced value $-\tau \log Z$, and intersubjective agreement is a stationary point of the consensus free-energy functional, which is the formal content of W3. The dynamics is a first-order natural-gradient flow in which precision is a stiffness and the Fisher metric a friction tensor (Section 8).

Participation across scales (W4) rests on three constructions. The first is hierarchical coarse-graining. A meta-agent is the gauge-covariant M-projection barycenter of its constituents, the moment-matching minimizer of the weighted sum of $D_{\mathrm{KL}}(\Omega_{Ii} q_i \| q_I)$, and it obeys the covariance theorem (Theorem 9). The coarse-graining map is an exact block-variable pushforward followed by a truncation to the Gaussian family, with no rescaling, coupling-space flow, or fixed point, and with blocking ratio $b > 1$ it admits at most $\log_b N$ steps on a network of N agents (Section 11). The second is the cross-scale prior of Eq. (131), read across episodes: when the transition of an episode-indexed joint copies the transported parent latent into the child, the predictive prior of the companion theory's delayed-feedback proposition is exactly this formula, with the parent law filtered from the previous episode in the role of q_I [30, Ch. 12]. The same-time form used in the energy is an energy-level simplification (status S), not a prior of one joint. The third is a declared root (Section 12), a fixed top-level prior that closes the hierarchy as its boundary condition; every finite acyclic generative model of this type needs one [30, Ch. 12], the belief-channel apex of that section is an energy-level closure and not a generative factor, and the root is the formal residue of Wheeler's "no tower of turtles".

What Part I does not supply is listed under W5 in Table 1; in particular, its multiscale feedback relaxes to stationarity unless observations drive it and so is not a self-excited circuit. Part II uses

these objects to build the information clock (W1) and the pulled-back geometry (W2), adds a binary registration channel that gives the bit its registration meaning, and states the postulates P1 to P10 under which these objects receive physical readings. Whether the construction describes physical reality remains open.

Part II: The Speculative Program

16 Status of the Speculative Program

Part II takes structures built in Part I (agent frames, transport, the consensus free energy, the lagged predictive prior supplied by a parent) and asks what they would mean if read physically: as time, as geometry, as observer-relative reality. Its claims have status S (structural correspondence or interpretation) or C (conjectured or open) in the scheme of Table 2, and each rests on one of the numbered postulates P1 to P10 below. None of it makes a quantitative prediction, and in its central domains (the Lorentzian signature, gravity, consciousness) the material is interpretation, not falsifiable physics. The mathematics that supports the readings has its own status and does not depend on them: the clock-dissipation identity of Eq. (137) (P, for an autonomous single-rate flow), the binary registration bound of Proposition 6 (D), the bit-production ratio of Eq. (140) (D), the covariant pullback of Eq. (142) and the passive gauge invariance of its covariant Fisher block (D for the pullback, P for the invariance, which is proved in the companion Supplement B [31, Ch. B2]), the signature constructions of Section 19 (P under their postulates), and the regulator obstruction of Section 20.1 (P), which concerns only the frame-twist block. An existence demonstration of structural compatibility is never presented as a derivation. Whether the participatory reading of the formalism admits a decidable test is recorded in Table 2; the physical readings of Part II do not.

The companion theory declines, without refuting, each physical identification made below [30, Chs. 1 and 14] (Section 2.2); Part II adopts these identifications as the postulates P1 to P10. The extensions stay in this paper because each bears on a question Wheeler's participatory program raises: how duration, geometric signature, observer-relativity, and dimensional structure relate to information-geometric data. Read constructively, Part II fixes the program's next constructions in the order C1 through C6 of Section 24.

The postulate register follows. Every entry is a postulate, not a consequence of free-energy minimization, and the sections below cite entries by number. The detailed assumptions of the algebraic signature route are numbered P4.1 to P4.6 as sub-items of entry P4, so no second numbering scheme arises.

P1 (clock). An agent's internal time is the Fisher-Rao arc length ϑ_i along its belief history, whose order and orientation are supplied by the shared flow parameter t; the bit count of a continuous update is a fixed-granularity diagnostic, and Wheeler's bit is represented separately by binary registrations (Section 17).

P2 (bundle metric). The associated bundle carries the fiber-respecting tensor with vertical block the covariant Fisher-Rao pullback and horizontal block $\kappa\,\mathrm{tr}(A_\mu A_\nu)$ with a declared coupling κ (Section 18).

P3 (tier identification). Perceived ambient spatial structure is the structural-tier pullback $G^{(s)}$ (Section 18.2).

P4 (signature, algebraic route). On a declared reference section, a commuting, mutually trace-orthogonal generator set with one temporal direction of negative trace-square and three spatial directions of positive trace-square (all-real route); on the complex route, instead, a single generator of nonzero trace-square together with the bilinear pairing $\mathrm{tr}(AB)$, the imaginary temporal coefficient, and the real-part projection (Postulates P4.1 to P4.6 of Section 19).

P5 (signature, causal route). A coarse-grained shared update parameter t used as a base coordinate, a positive-definite spatial information metric h_{ab}, an influence region that is an h-ball, and a finite maximal epistemic propagation speed $c_{\mathcal{I}}$ (Section 19.3).

P6 (sector hierarchy). The eigenvalue hierarchy $\lambda_{\mathrm{obs}} \gg \lambda_{\mathrm{subthresh}} \gg \lambda_{\mathrm{internal}}$ with constant sector ranks, and $|\mathcal{D}_{\mathrm{obs}}| = 4$ for human agents, which requires $\dim \mathcal{C} \geq 4$ (Section 20.2).

P7 (dimensional monism). Nats are dimensionally fundamental; physical dimensions are consensus labels (Section 20.4).

P8 (structural curvature). Gravitational phenomenology, if any, is curvature of the structural pullback $G^{(s)}$; the equivalence principle requires an undelivered link between Σ_p^{-1} and Σ_s^{-1} (Section 21.1).

P9 (phenomenal correlate). Phenomenal character corresponds to the geometric structure of the induced metric, with the frame-to-quale assignment conjectured fixed by the cross-scale composition (Section 22).

P10 (shareability). Scientific content must be statable frame-independently; this at most motivates general covariance (Section 23).

17 Time as Registered Change: The Information Clock

This section develops W1 of Table 1, registered change as a clock. The dynamics of Part I are written in a flow parameter t shared by all agents, the continuous limit of the update index, and t supplies the order and the orientation of every belief history. No physical time is presupposed, but an ordering is: the construction measures each agent's progress along a common ordering and does not recover the ordering from correlations, which is the step that Page–Wootters and thermal-time proposals take. The clock ϑ_i of agent i is the Fisher-Rao arc length of its belief history, an additive length that is invariant under refinement and reparameterization and has a unit-speed form [30, Ch. 7]. Postulate P1 reads it as the agent's internal time, an identification the companion theory declines [30, Ch. 1].

A belief update carries the information gain $\Delta I_i = D_{\mathrm{KL}}(q_i^{\mathrm{new}} \| q_i^{\mathrm{old}})$, in nats (division by $\ln 2$ converts to bits), and a step of Fisher length $\delta\vartheta$ carries $\delta\vartheta^2/(2\ln 2)$ bits to leading order by Eq. (136) below. This count measures belief change in bits; it does not model Wheeler's elementary yes-no registration, which Section 17.2 treats separately. The unit of change is set by statistical distinguishability: two beliefs a parameter step $\delta\theta$ apart are distinguishable at the level of one standard error from N independent observations when $N g_{\mathcal{B}}(\delta\theta, \delta\theta) \gtrsim 1$, with $g_{\mathcal{B}}(\delta\theta, \delta\theta)$ the squared Fisher-Rao length [3, 135, 136]. This is Wootters' statistical distance, which Wheeler cited as a candidate link between information and geometry [1], and it is the information-geometric reading of Bateson's "a difference which makes a difference" [137]. For a Gaussian belief and a pure mean step at fixed covariance,

$$D_{\mathrm{KL}}(q^{\mathrm{new}} \| q^{\mathrm{old}}) = \tfrac{1}{2}\delta\mu^{\top}\Lambda\delta\mu = \tfrac{1}{2}ds^2, \tag{136}$$

exactly, with $\Lambda = \Sigma^{-1}$ the precision and ds^2 the squared length of the straight constant-covariance mean path; the Fisher-Rao distance of the full Gaussian family between the same endpoints can be shorter [30, Ch. 7]. Because the increment is quadratic in the step, the bit count is not additive under refinement: a path of unit Fisher length taken in one step carries 0.7213 bits, and the same path taken in 100 equal substeps carries 0.0072 bits in total, while its arc length stays 1. The bit count is therefore defined only relative to a fixed update granularity, which is why P1 takes the arc length as primary.

Different agents accumulate different arc lengths. No time function common to all agents is built into the construction. Along a history of the population with block-diagonal metric the squared speeds add, so the collective clock ϑ obeys $\max_i \vartheta_i \le (\sum_i \vartheta_i^2)^{1/2} \le \vartheta \le \sum_i \vartheta_i$; a clock that is a function of the population state and orthogonal to the descent direction exists only under an exactness condition that fails generically, although unit-speed clocks built from a global cross-section can exist; and the clock of a gauge-invariant free energy is itself gauge invariant, since every natural-gradient history is then orthogonal to the gauge orbits [30, Ch. 7]. The closest precedent is Caticha's entropic time, in which time is introduced as a device to keep track of change in an information-updating dynamics [138]; what the present construction adds is a clock per agent, read in that agent's frame, and the meta-agent clock of Section 17.3. The clock also differs from proposals that read time off an entangled clock subsystem [130], off the modular flow of the system's own global statistical state [139], off computational steps [140], or off horizon thermodynamics and entanglement structure [20, 24]: here it is the length of each agent's own belief history [1, 141].

17.1 Fisher Arc Length Along Belief Trajectories

An agent following a belief trajectory $q(\lambda)$, with λ any orientation-preserving parameter, accumulates the Fisher-Rao arc length $\Delta\vartheta = \int \sqrt{g_{\mathcal{B}}(q'(\lambda), q'(\lambda))}d\lambda$, a reparameterization-invariant Riemannian length, and the clock ϑ is the arc-length parameter it defines. This clock is not relativistic proper time. Proper time is the Lorentzian length of a timelike worldline and is largest for an unaccelerated clock at rest, whereas the Fisher clock accumulates only while the belief changes and is stationary for an unchanging belief: it is a change-based duration. Any connection to physical time dilation would require the indefinite-metric extension of Section 19, and none is established here.

The clock and the dynamics are linked by an exact identity. Let the belief sector follow an autonomous, single-rate Fisher natural-gradient flow $\dot{x} = -\eta G^{-1}\nabla\mathcal{F}$, with G the Fisher metric and η one scalar rate. Then $G(\dot{x}, \dot{x}) = \eta^2 \nabla\mathcal{F}^\top G^{-1}\nabla\mathcal{F}$ and $d\mathcal{F}/dt = \nabla\mathcal{F}^\top \dot{x} = -\eta\nabla\mathcal{F}^\top G^{-1}\nabla\mathcal{F}$, hence

$$\left(\frac{d\vartheta}{dt}\right)^2 = \eta\left(-\frac{d\mathcal{F}}{dt}\right), \tag{137}$$

so along such a flow the information clock ticks exactly while free energy is dissipated. The identity is the natural-gradient counterpart of the thermodynamic-length relations between Fisher speed and dissipation [42, 43, 142], and the companion theory states it in rate form together with its reparameterization law [30, Ch. 7]. For one discrete step $x_{k+1} = x_k - \eta G^{-1}\nabla\mathcal{F}$ with dimensionless step size η, the information gain of Eq. (136), in nats, is $\Delta I = (\eta/2)|\Delta\mathcal{F}|$ to leading order in η.

Three qualifications fix the scope of Eq. (137). First, when the free energy depends explicitly on time through the data, as for an agent receiving observations $o(t)$, $d\mathcal{F}/dt = \nabla\mathcal{F}^\top\dot{x} + \partial_t\mathcal{F}$ and the identity becomes

$$\left(\frac{d\vartheta}{dt}\right)^2 = \eta\left(\partial_t\mathcal{F} - \frac{d\mathcal{F}}{dt}\right), \tag{138}$$

so the clock can tick while $\mathcal{F}$ rises. Second, the belief flow of Eq. (82) carries separate rates η_μ and η_Σ; for such a rate-separated flow the identity holds block by block and holds with unit rate in the rate-weighted metric $\bigoplus_b G_b/\eta_b$, while the unweighted Fisher clock obeys only the bound $|d\mathcal{F}/d\vartheta| \le \|\operatorname{grad}^G \mathcal{F}\|_G$ [30, Chs. 7 and 8]. The frame retraction of Eq. (85) is preconditioned by a metric that is not Fisher, so frame motion contributes a length but not an information clock. Third, for a real-analytic free energy an autonomous flow that converges has finite total clock reading, whereas a smooth free energy admits a converging flow of infinite duration; the companion theory proves the first statement and exhibits the counterexample [30, Ch. 7]. Sustained clock advance therefore requires sustained driving by observations, and Eq. (138), not Eq. (137), governs an observing agent. On the P1 reading, an agent's internal time along an autonomous single-rate flow is accumulated free-energy dissipation measured in the Fisher metric, and along a driven flow it is that dissipation plus the explicit change of $\mathcal{F}$ produced by the data.

17.2 Binary Registrations and Wheeler's Bit

Wheeler's bit is an event, a registered answer to a yes-no question elicited by a device [1], not a unit of measure on a continuous update. The framework can represent that event directly by an observation channel with binary outcomes, and the expected belief change it produces is then bounded by one bit and additive over registrations.

Proposition 6 (Information budget of binary registrations)**.** *Let X be a latent variable with prior law $p(x)$ and let $O \in \{0,1\}$ be a yes-no observation with likelihood $P(O=1 \mid x)$. The expected belief update equals the mutual information and is at most one bit,*

$$\mathbb{E}_O\left[D_{\mathrm{KL}}\left(p(x \mid O)\|p(x)\right)\right] = I(X;O) \le H(O) \le \ln 2. \tag{139}$$

For registrations $O_1,\dots,O_n$, the expected updates add by the chain rule, $I(X;O_1,\dots,O_n) = \sum_{k=1}^n I(X;O_k \mid O_{<k})$, where the k-th term is the expected update produced by O_k from the belief $p(x \mid O_{<k})$. If the registrations are conditionally independent given X, each term is at most $I(X;O_k) \le \ln 2$, so n registrations change the belief by at most n bits in expectation.

Proof. By definition,

$$\begin{aligned}\mathbb{E}_O D_{\mathrm{KL}}(p(x \mid O)\|p(x)) &= \sum_o P(o) \int p(x \mid o) \ln \frac{p(x \mid o)}{p(x)} dx \\ &= \int \sum_o p(x,o) \ln \frac{p(x,o)}{p(x)P(o)} dx = I(X;O).\end{aligned}$$

Since O is discrete, $H(O \mid X) \ge 0$, so $I(X;O) = H(O) - H(O \mid X) \le H(O) \le \ln 2$. The chain rule for mutual information [143, Thm. 2.5.2] gives the sum, and applying the first part to the conditional laws given $O_{<k}$ identifies each term as an expected update. Under conditional independence, $I(O_k;X,O_{<k}) = I(O_k;X)$, and expanding the left side by the chain rule gives $I(X;O_k \mid O_{<k}) = I(X;O_k) - I(O_k;O_{<k}) \le I(X;O_k)$. □

The bound holds in expectation only: a single surprising outcome can move a belief by more than one bit. For a standard normal prior and a sharp logistic channel whose positive answer has probability 1.4×10^{-3}, the positive outcome moves the belief by 9.36 bits, while the expected update is 0.0145 bits; for symmetric channels of low and high sharpness the expected update is 0.135 and 0.962 bits. The registration count, and the information budget it bounds, are additive, whereas the Fisher clock of Section 17.1 is a path length whose continuous-update bit count depends on

the granularity of the update. On this construction a continuous Gaussian belief is an idealization of accumulated registrations, which is Wheeler's order of derivation; that reading is status S. The channel models the inferential half of Wheeler's observer-participant only. No agent chooses which question to pose, the registration is not irreversible in any physical sense, and a classical fiber has no complementarity, so "no question, no answer" and the elementary quantum phenomenon remain outside the model.

17.3 The Scale-Dependent Meta-Agent Clock

A meta-agent that aggregates many constituents does not advance its clock at the constituent rate, because a difference registers on the meta-clock only to the extent that it is shared across the cluster whose gauge-covariant barycenter the meta-belief is. Decompose the constituent mean updates into common and idiosyncratic parts, $\delta\mu_i = \delta c + \epsilon_i$, with $\mathbb{E}[\epsilon_i] = 0$, the ϵ_i uncorrelated with one another and with δc, and a common second moment $\mathbb{E}\|\epsilon_i\|_\Lambda^2 = \mathbb{E}\|\epsilon\|_\Lambda^2$, and define the coherent fraction $f_{\mathrm{coh}} := \mathbb{E}\|\delta c\|_\Lambda^2 / (\mathbb{E}\|\delta c\|_\Lambda^2 + \mathbb{E}\|\epsilon\|_\Lambda^2)$ in the precision norm $\|v\|_\Lambda^2 = v^\top \Lambda v$.

Proposition 7 (Meta-clock slowdown)**.** *In the uniform-weight mean-sector special case of the coarse-graining construction (all coherence weights equal in Eq.* (128)*, with the averaging covariance convention of Eq.* (129)*), the meta-to-constituent ratio of expected per-step bit production is*

$$Z_I := \frac{\mathbb{E}[\Delta I_I]}{\mathbb{E}[\Delta I_i]} = f_{\mathrm{coh}} + \frac{1 - f_{\mathrm{coh}}}{N} = f_{\mathrm{coh}} + O(1/N), \tag{140}$$

with ΔI the information gain of Eq. (136)*.*

Proof. Under the averaging convention the meta-agent's precision is Λ and its mean update is $\delta c + N^{-1}\sum_i \epsilon_i$, so by Eq. (136) and the stated correlations $2\mathbb{E}[\Delta I_I] = \mathbb{E}\|\delta c\|_\Lambda^2 + N^{-1}\mathbb{E}\|\epsilon\|_\Lambda^2$, while $2\mathbb{E}[\Delta I_i] = \mathbb{E}\|\delta c\|_\Lambda^2 + \mathbb{E}\|\epsilon\|_\Lambda^2$. □

The ratio runs from $Z_I \to 1$ for fully coherent activity to $Z_I \to 1/N$ for fully incoherent activity, so registering one meta-tick on the bit clock requires of order N bits of constituent activity. The bit is a fixed reference unit held constant across scales; the slowdown resides in the rate at which bits are produced, and the $1/N$ suppression is the law of large numbers acting on the barycenter, not a dynamical critical exponent. The proposition is stated in the relative-entropy currency. Its square root $\sqrt{Z_I}$ is the ratio of root-mean-square Fisher speeds, not the ratio of expected arc lengths, because the expected ratio of arc lengths is not the square root of the expected ratio of relative entropies; the arc-length ratio on the primary clock of P1 is an open problem of the companion theory [30, Ch. 7], of which Proposition 7 answers the relative-entropy version.

A comparison that does not depend on the covariance convention is available. In this special case the meta-agent law is the image of the constituents' joint law under a fixed Markov kernel for either convention: the averaging convention is the image under $y = N^{-1}\sum_i x_i + \xi$ with independent Gaussian noise ξ of covariance $\Sigma - \Sigma/N$, and the precision-pooling convention $\Lambda_I = N\Lambda$, which the meta-agent state construction declines to adopt (Section 9.4), is the deterministic averaging image. By the same-path record-clock contraction of the companion theory, a law that is the image of a history under one parameter-independent Markov kernel accumulates no more Fisher duration than that history, with defect equal to the expected conditional variance of the score [30, Ch. 7]. The meta-agent is therefore never faster than the collective of its constituents under either convention. Against the joint history of the N constituents, the precision-pooled parent has expected squared-speed ratio $f_{\mathrm{coh}} + (1 - f_{\mathrm{coh}})/N$, and the averaging parent has that ratio divided by N. Only the comparison with a single constituent depends on the convention: the averaging convention gives

the slowdown Z_I of Eq. (140), and precision pooling gives $Nf_{\mathrm{coh}} + (1 - f_{\mathrm{coh}})$, a speedup relative to one constituent. Each of these ratios follows from the second-moment computation in the proof of Proposition 7. The ratio itself follows from the consensus coupling already present in the free energy: to leading order the belief-coupling term reduces to a graph-Laplacian quadratic form $\mu^\top (L \otimes \Lambda)\mu$ in which the barycenter is the zero mode that takes a $1/N$ share of incoherent injected power, while the algebraic connectivity λ_2 of the coupling graph [144] sets the rate $2\eta_\mu \lambda_2$ at which constituent disagreement relaxes and the detector of Section 9.3 triggers.

18 Observer-Dependent Geometry: The Pullback Construction

18.1 The Pullback Mechanism: From Belief Sections to Metric Data

Wheeler proposed, as a working hypothesis, that every physical quantity, "even the spacetime continuum itself", derives its significance from binary yes-or-no indications elicited by observing devices [1]. This section develops W2 of Table 1: a classical information-geometric model in which metric data on a base that carries no physical metric are derived from agents' belief sections, so that geometry is observer-relative. The pullback tensor has status D and the gauge invariance of its covariant Fisher block status P; their identification with physical geometry is status S and rests on postulates P2 and P3. The geometry is derived from continuous belief sections, not from discrete registrations; Section 17.2 gives the registration channel.

The base manifold $\mathcal{C}$ is a presupposed smooth index manifold carrying no physical metric. Its topology, dimension, and smooth structure are inputs, and only metric data are derived, so the construction models Wheeler's thesis for the metric alone and not his rejection of a presupposed continuum, the "no continuum" among his four no's [1]. The literature indicates a route that would address the gap: for a distinguishing spacetime already known to be Lorentzian, the causal order fixes topology, smooth structure, and conformal metric [145, 146], and causal set theory recovers the volume element by counting [147]. Building the base from agent data along such lines is work on the base itself and is deferred to the companion Supplement B [31].

Agent i carries its belief as a section q_i over $\mathcal{C}$, expressed in its frame $U_i(c)$, with Regime I connection $A_\mu^{(i)} = U_i^{-1}\partial_\mu U_i$ and field strength $F_{\mu\nu} = 0$ identically. The covariant first jet of the section is

$$D_\mu q_i = \partial_\mu q_i - \rho_* \left(\partial_\mu U_i U_i^{-1}\right) q_i, \tag{141}$$

which for a Gaussian section reads $D_\mu(\mu_i, \Sigma_i) = (\partial_\mu \mu_i - X_\mu \mu_i, \partial_\mu \Sigma_i - X_\mu \Sigma_i - \Sigma_i X_\mu^\top)$ with $X_\mu = \partial_\mu U_i U_i^{-1}$. Equivalently, $\rho(U_i^{-1}) D_\mu q_i = \partial_\mu \hat{q}_i$ for the frame-stripped section $\hat{q}_i = \rho(U_i^{-1}) q_i$, and because $\rho(U_i^{-1})$ is a Fisher isometry the pullback below is the ordinary pullback of $\hat{q}_i$. Writing $\nabla_\mu^{(i)} \log q_i$ for the score of $D_\mu q_i$, the covariant Fisher pullback is

$$\boxed{G^{(q)}_{i,\mu\nu}(c) = \mathbb{E}_{q_i(c)}\left[(\nabla_\mu^{(i)} \log q_i)(\nabla_\nu^{(i)} \log q_i)\right],} \tag{142}$$

a positive semi-definite tensor whose rank is the rank of the jet [31, Ch. B2]. For constant frames $D_\mu = \partial_\mu$ and Eq. (142) is the ordinary pullback of the Fisher-Rao metric. Postulate P2 adds a horizontal frame-twist block with a declared coupling κ,

$$\widetilde{G}^{(q)}_{i,\mu\nu}(c) = \kappa \,\mathrm{tr}\left(A_\mu^{(i)} A_\nu^{(i)}\right) + G^{(q)}_{i,\mu\nu}(c). \tag{143}$$

The coupling is needed because the two blocks are unrelated in normalization: the trace block scales with the normalization of the Lie-algebra generators, the Fisher block does not, and without

κ the signature of the sum would depend on an arbitrary generator normalization (Section 19). The trace form is Ad-invariant but indefinite on real $\mathfrak{gl}(K_q)$: for $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\mathrm{tr}(J^2) = -2$, whereas for $D = \mathrm{diag}(1, 0)$, $\mathrm{tr}(D^2) = 1$. The horizontal block is therefore an indefinite trace tensor, not a Riemannian metric, and the compact alternative $-\mathrm{tr}(AB)$ is positive only on a skew-symmetric compact subalgebra. The construction transfers to the prior, generative-model, and hyper-prior sections, yielding $G_i^{(q)}, G_i^{(p)}, G_i^{(s)}, G_i^{(r)}$.

The two blocks behave differently under the redundancies of the functional. Under a passive local re-expression of agent frames, $U_i \mapsto g_i(c)U_i$ with $q_i \mapsto \rho(g_i(c))q_i$ and $\Omega_{ij} \mapsto g_i \Omega_{ij} g_j^{-1}$, the frame-stripped section $\hat{q}_i$ is unchanged, so $G_i^{(q)}$ is invariant with no orbit average; this is the passive gauge covariance of covariant pullbacks proved in the companion Supplement B [31, Ch. B2], where the pullback is also shown to be relative to the chosen connection, here the connection defined by the declared reference section common to all agents. The ordinary fixed-frame pullback of q_i is not invariant under this re-expression, since it acquires frame-gradient terms, and the frame-twist block is invariant under neither local action: the change of reference $U_i(c) \mapsto U_i(c)g(c)$ leaves every Ω_{ij}, and hence the free energy, unchanged, yet with $g = U_i^{-1}$ it sets $\mathrm{tr}(A^{(i)}A^{(i)})$ to zero for any chosen agent. At a fixed reference section, then, the differences $G_i^{(q)} \neq G_j^{(q)}$ between agents are differences between their beliefs expressed in a common reference, not differences of frame choice, and the regulator obstruction of Section 20.1 concerns only the frame-twist block. The covariant construction uses base derivatives of sections and frames and so belongs to the base-transport material of Supplement B; within the single-base-point scope of the companion theory the base tangent space is zero and every pullback tensor vanishes.

Both tensors are dimensionless: the Fisher pullback is invariant under a change of fiber units and scales quadratically under a rescaling of base coordinates, so $G^{(q)}_{i,\mu\nu} dc^\mu dc^\nu$ is a pure number of nats in every chart, and the frame-twist block is dimensionless because the frame fields are Lie-algebra coordinates. A physical line element would require a length scale L_0, with $ds^2_{\mathrm{phys}} = L_0^2 G$, and a Lorentzian reading would further require a constant conversion between timelike and spacelike normalizations; the algebraic route of Section 19 supplies neither, since its cone slope is field-dependent, and the causal route postulates the speed.

The pulled-back tensors enter no term of the free energy; the integration measure on $\mathcal{C}$ is the background chart density. The construction therefore supplies the arc of Wheeler's loop from information to geometry but not the arc from geometry back to the participants. Closing the loop would require the induced geometry to enter the dynamics, for example as the integration measure or as the coupling neighborhood, which is base-transport work deferred to Supplement B. Information-geometric models of physical space and of spacetime were proposed by Caticha [12,13], and Fisher-metric constructions of an Einstein tensor exist [148]; the lineage runs back to Wootters' statistical distance [3]. The contribution here is the agent-indexed family of pullbacks and its gauge behavior, not the idea of Fisher geometry as spatial geometry.

18.2 Three Tiers of Induced Geometry: Epistemic, Expectational, Structural

The four induced tensors carry different interpretations, organized by the timescale on which their underlying section evolves. The belief pullback $G_i^{(q)}$ is epistemic geometry, an instantaneous uncertainty map fluctuating on the perceptual timescale. The prior pullback $G_i^{(p)}$ is expectational geometry: its section is the prior supplied by the parent through the lagged predictive prior of Section 10, which for a transported-copy transition is $p_i = \Omega_{iI}[q_I^{(\ell+1)}]$ from the previous episode, so in its covariant form it inherits the profile of the parent's belief and evolves more slowly than the epistemic tier while remaining a state-fiber object. The generative-model pullback $G_i^{(s)}$, with

its hyper-prior partner $G_i^{(r)}$, is structural geometry on the model fiber, quasi-static on the learning timescale. Postulate P3 identifies what an agent experiences as ambient spatial structure with the structural tier $G_i^{(s)}$. The identification is chosen by timescale matching: perceived ambient space is invariant across changing percepts on perceptual timescales, and among the four tiers only the model-fiber pair is quasi-static on that timescale, with $G_i^{(s)}$ the member carrying learned structure. The reading echoes Kant's space as a form of intuition, Wheeler's observer-participatory universe, and Hoffman's interface theory of perception [149]; it places the interface in the slow generative model, whereas predictive-processing accounts usually identify perceptual content with the posterior [150, 151]. Agents with different generative models therefore carry different structural geometries on the same base, and within this interpretive frame there is no agent-independent metric on $\mathcal{C}$, only the family of agent-relative induced tensors. A control in which the slow channel is frozen would make $G_i^{(s)}$ and $G_i^{(r)}$ constant and leave the dynamical content on the fast channel; no such experiment is reported here.

18.3 Example: Gaussian Agents on $\mathbb{R}^2$

For agents with constant frames maintaining Gaussian beliefs $q_i(c) = \mathcal{N}(\mu_i(c), \Sigma_i(c))$ over a flat base $\mathcal{C} = \mathbb{R}^2$, the induced tensor is

$$G^{(q)}_{i,\mu\nu}(c) = (\partial_\mu \mu_i)^\top \Sigma_i^{-1} (\partial_\nu \mu_i) + \tfrac{1}{2} \operatorname{tr}\left(\Sigma_i^{-1}(\partial_\mu \Sigma_i)\Sigma_i^{-1}(\partial_\nu \Sigma_i)\right), \tag{144}$$

whose first term captures mean-gradient and second term covariance-gradient contributions; with nonconstant frames the derivatives become the covariant jets of Eq. (141). In the isotropic case $\Sigma_i(c) = \sigma(c)^2 I$ it reduces to

$$G^{(q)}_{i,\mu\nu}(c) = \sigma^{-2}\left[(\partial_\mu \mu_i) \cdot (\partial_\nu \mu_i) + 2K_q (\partial_\mu \sigma)(\partial_\nu \sigma)\right], \tag{145}$$

positive definite only where the jet $(d\mu_i, d\sigma)$ has rank two. The factor σ^{-2} acts as a conformal factor on the mean term, so where σ varies, confident regions appear larger and uncertain regions smaller, and the term $2K_q d\sigma^2/\sigma^2$ adds a contribution from gradients of uncertainty; for constant σ the factor is a global rescaling and no region is larger than another. The fiber metric behind this example, the Fisher metric of the isotropic Gaussian family, is hyperbolic with constant curvature $-1/(2K_q)$. Hyperbolic Fisher geometry is generic for families with such symmetries [44], so a geometry obtained from Fisher information is not by itself evidence of physics.

The bundle-pullback formulation is the sense in which the model has the structure Wheeler asks for at the level of metric data. A metric on the base is the pullback of a fiber-respecting metric on an information-bearing bundle along an agent's section; metric structure is derived from belief data, not posited; the pulled-back tensors are observer-relative because beliefs differ; and there is no privileged geometry, only the family of agent-relative geometries induced on the same base. The construction does not derive the base, its dimension, or a Lorentzian signature.

19 The Lorentzian Signature Problem

The central unsolved problem for any spacetime reading is that observed spacetime has Lorentzian signature $(-,+,+,+)$, whereas the Fisher pullbacks $G_i^{(q)}, G_i^{(p)}$ are positive semi-definite, being expectations of outer products of score vectors. When a signature appears in this framework it is a postulate, not a derivation: it lives in the Lie-algebra trace form $\operatorname{tr}(A_\mu A_\nu)$ of the connection sector, not in the Fisher-Rao metric on the belief fiber, and we make no claim that free-energy minimization

selects the configuration that produces it. Earlier work treats the signature as a dynamical phase and finds a Lorentzian stationary point singled out in four dimensions [152]; nothing comparable is attempted here, and the companion theory gives the base no Lorentzian structure [30, Ch. 1]. Two caveats govern everything below. First, in Regime I the frames are removable at a point (Proposition 1): by the KL invariance theorem (Theorem 1) the functional depends on frames only through the transports Ω_{ij}, and the change of reference $U_i(c) \mapsto U_i(c)g(c)$ leaves every Ω_{ij} unchanged while it can set $\mathrm{tr}(A_\mu^{(i)} A_\nu^{(i)})$ to zero for any chosen agent (Proposition 11(b)). The frame-twist form is therefore a property of a declared reference section and has no gauge-invariant content. Second, the signature results concern the frame-twist block alone, while the clock, the sectors, perceived space, and the gravity reading are readings of the positive semi-definite Fisher block.

Sector split. The structure group is enlarged from $\mathrm{GL}(K_q, \mathbb{R})$ to $\mathrm{GL}(K_q, \mathbb{C})$ on the connection sector only. The Gaussian belief fiber remains a real distribution on $\mathbb{R}^{K_q}$, and belief transport $\rho(\Omega) \cdot (\mu, \Sigma) = (\Omega\mu, \Omega\Sigma\Omega^\top)$ stays restricted to $\Omega \in \mathrm{GL}(K_q, \mathbb{R})$: a genuinely complex Ω would render $\Omega\Sigma\Omega^\top$ non-Hermitian and the Gaussian relative entropy complex-valued, so real Gaussian probability, Fisher-Rao geometry, nonnegativity of relative entropy, and $\mathcal{F} \geq 0$ all live on the real sector. The complexification enters only through the frame field $\phi : \mathcal{C} \to \mathfrak{gl}(K_q, \mathbb{C})$. The construction stays inside Regime I ($F_{\mu\nu} = 0$ throughout), so the signature mechanism runs through the pure-gauge frame-twist form and not through any curvature invariant. A stronger route that replaces the probability fiber itself by a complex or quantum statistical object (density matrices, Born-rule amplitudes, signed measures) would require a quantum-information formulation of the variational principle and is listed as future work in Section 24.

Where an indefinite form can live. Compact structure groups such as $\mathrm{SO}(N)$ force Riemannian geometry on the fiber, since $\Omega\Sigma\Omega^\top$ is then a similarity transform preserving eigenvalues. Non-compactness alone is necessary but not sufficient: for real invertible Ω and positive-definite Σ the congruence $\Omega\Sigma\Omega^\top$ stays positive definite by Sylvester's law of inertia, so real $\mathrm{GL}(K_q, \mathbb{R})$ on real fibers still produces positive-definite transport. The signature resides in the trace form $\mathrm{tr}(A_\mu A_\nu)$, which on $\mathfrak{gl}(K_q, \mathbb{R})$ is already indefinite, negative definite on the skew subalgebra $\mathfrak{so}(K_q)$ and positive definite on the symmetric complement, with signature $(K_q(K_q+1)/2, K_q(K_q-1)/2)$. An indefinite restriction is therefore available on every $\mathfrak{gl}(K_q, \mathbb{R})$ with $K_q \geq 2$; what the constructions below add is an explicit frame field that produces a chosen split.

19.1 The Postulates for an Indefinite Frame-Twist Form

A Lorentzian-compatible signature on the horizontal block follows from the postulates below, each a postulate and not a consequence of free-energy minimization; register entry P4 summarizes the set. The indefinite form is a pullback only in the sense of the complex-bilinear trace form pulled back along the frame map; it is distinct from the positive semi-definite Fisher pullback of Eq. (142). Postulates P4.2, P4.3, and P4.4 belong to the complex route only, and the all-real construction of Proposition 10 dispenses with them. Two further conditions are not independent postulates: flatness is automatic once the frame is confined to an abelian subalgebra (Postulate P4.5), since then $A_\mu = \partial_\mu \phi$ and $F_{\mu\nu} = 0$ identically, and a separable ansatz for the frame components only diagonalizes the chart, as the worked example shows.

Postulate P4.1 (Generator with nonzero trace-square)**.** *The temporal and spatial frame directions use generators T with $\mathrm{tr}(T^2) \neq 0$; in the two-dimensional example, the symmetric traceless $T =$*

$\mathrm{diag}(1,-1) \in \mathfrak{sl}(K_q,\mathbb{R})$ *with* $\mathrm{tr}(T^2) = 2$. *A compact generator (*$\mathrm{tr}(T^2) < 0$*) exchanges which direction is timelike, and only a nilpotent generator (*$\mathrm{tr}(T^2) = 0$*) removes the form.*

Postulate P4.2 (Complex-bilinear pairing; complex route only)**.** *The form* $\langle A, B\rangle = \mathrm{tr}(AB)$ *on* $\mathfrak{gl}(K_q,\mathbb{C})$*, in place of the sesquilinear Frobenius pairing* $\mathrm{tr}(AB^\dagger)$*, which is positive definite and would give no sign change; the bilinear, non-Hermitian character is load-bearing.*

Postulate P4.3 (Imaginary temporal frame; complex route only)**.** *The assignment* $\phi_0 = i\psi_0 T$ *along the distinguished base direction. This is an imaginary coefficient in the Lie algebra; it is not a Wick rotation, which continues a base coordinate and never meets a real-part projection.*

Postulate P4.4 (Real-part projection; complex route only)**.** *Extraction of* $G^{\mathrm{Lor}}_{\mu\nu} := \mathrm{Re}(G_{\mu\nu})$*, a rank-changing operation (rank one complex to rank two real) for which no principle in the construction mandates discarding the imaginary off-diagonal part.*

Postulate P4.5 (Single-generator collapse)**.** $T_0 = T_1 = T$*, confining* ϕ *to the abelian subalgebra* $\mathbb{C}T$*.*

Postulate P4.6 (Mutual trace-orthogonality of the spatial generators)**.** *In any multi-generator extension the spatial generators are mutually trace-orthogonal,* $\mathrm{tr}(T_aT_b) = 0$ *for* $a \neq b$*, so that the spatial block of the frame-twist form is diagonal at the abelian-exact order; the commuting disjoint-support generators of Proposition 10 satisfy this by construction.*

Each step is mathematically well defined. In the two-dimensional example the complex route is forced by abelian exactness: a single real generator has a trace form of one sign, so within a one-generator abelian subalgebra a sign change can enter only through $i^2 = -1$. The indefiniteness then comes from Postulates P4.2, P4.3, and P4.4 together, since $\mathrm{Re}((ia+b)^2) = b^2 - a^2$ is the hyperbolic quadratic form on $\mathbb{C} = \mathbb{R}^2$, and choosing which base direction receives the factor i is choosing which direction is timelike. A real frame mixing skew and symmetric generators reaches an indefinite form without complexification, and Proposition 10 carries this out in four dimensions.

Proposition 8 (Conditional (1+1) signature)**.** *Under Postulates P4.1 to P4.5 on a two-dimensional base, the real part of the frame-twist form* $\mathrm{tr}(A_\mu A_\nu)$ *has rank two and signature* $(-,+)$ *wherever* $d\psi_0$ *and* $d\psi_1$ *are linearly independent, and at each such point the orthonormal-frame transformations form* $\mathrm{O}(1,1)$*, whose identity component is* $\mathrm{SO}^+(1,1)$*.*

Proof. The computation of Section 19.2. □

The proposition is a conditional result with status P under its postulates; any physical reading of its conclusion is status S, and whether the dynamics select its hypotheses is open.

19.2 Worked Example: $\mathrm{GL}(2,\mathbb{C})$ Frames and (1+1) Lorentzian Signature

On a two-dimensional base $\mathcal{C}$ with coordinates (x^0, x^1), take $T = \mathrm{diag}(1,-1)$ with $\mathrm{tr}(T^2) = 2$ (Postulate P4.1) and the imaginary temporal component of Postulate P4.3,

$$\phi(x^0,x^1) = i\psi_0(x^0,x^1)T + \psi_1(x^0,x^1)T. \tag{146}$$

Under the single-generator collapse (Postulate P4.5), ϕ lies in the one-dimensional abelian subalgebra $\mathbb{C}T$, so the connection is $A_\mu = \partial_\mu\phi$ exactly (the example is abelian-exact, not linearized) and

$F_{\mu\nu} = 0$. For general components, $G_{\mu\nu} = \mathrm{tr}(A_\mu A_\nu) = 2(i\partial_\mu\psi_0 + \partial_\mu\psi_1)(i\partial_\nu\psi_0 + \partial_\nu\psi_1)$, a complex form of rank one with vanishing determinant, and

$$\mathrm{Re}\, G = 2\,(d\psi_1 \otimes d\psi_1 - d\psi_0 \otimes d\psi_0)\,, \qquad \det \mathrm{Re}\, G = -4\mathcal{J}^2, \tag{147}$$

with $\mathcal{J}$ the Jacobian determinant of (ψ_0, ψ_1), so the signature is $(-,+)$ wherever $\mathcal{J} \neq 0$. In the separable chart $\psi_0 = \psi_0(x^0)$, $\psi_1 = \psi_1(x^1)$ the components are

$$G_{00} = i^2(\partial_0\psi_0)^2\,\mathrm{tr}(T^2) = -2(\partial_0\psi_0)^2 < 0, \tag{148}$$

$$G_{11} = (\partial_1\psi_1)^2\,\mathrm{tr}(T^2) = +2(\partial_1\psi_1)^2 > 0, \tag{149}$$

$$G_{01} = i(\partial_0\psi_0)(\partial_1\psi_1)\,\mathrm{tr}(T^2) \in i\mathbb{R}, \tag{150}$$

and the real-part projection $G^{\mathrm{Lor}}_{\mu\nu} := \mathrm{Re}(G_{\mu\nu})$ sets $G^{\mathrm{Lor}}_{01} = 0$ and yields the rank-two real metric

$$\boxed{ds^2 = -2(\partial_0\psi_0)^2(dx^0)^2 + 2(\partial_1\psi_1)^2(dx^1)^2.} \tag{151}$$

Separability therefore only diagonalizes the chart: a non-separable frame gives, for example, $\mathrm{Re}\, G_{00} = +6$ with eigenvalues -0.081 and 7.901, still Lorentzian. With a compact generator $T = J$ ($\mathrm{tr}(J^2) = -2$) the real part is $2(d\psi_0 \otimes d\psi_0 - d\psi_1 \otimes d\psi_1)$, again of signature $(1,1)$ with the timelike role moved to x^1, and with a nilpotent generator the form vanishes. Calling Eq. (151) a physical metric requires both the imaginary frame of Postulate P4.3 and the real-part projection of Postulate P4.4, and this is a derivation gap: writing $\chi = i\psi_0 + \psi_1$, the projected form is the difference of squares of the differentials of two scalar functions, obtainable on any manifold with no gauge theory, so the gauge apparatus contributes only the normalization $\mathrm{tr}(T^2)$ and the sign conventions. The ± 2 coefficients are artifacts of the unnormalized T; under $\mathrm{tr}(T^2) = 1$ they become ± 1, and the construction fixes the metric only up to a conformal class. At each non-degenerate point the orthonormal-frame transformations form $\mathrm{O}(1,1)$, whose identity component $\mathrm{SO}^+(1,1)$ is the group of (1+1) boosts, as the relation $\Lambda^\top \eta \Lambda = \eta$ with $\eta = \mathrm{diag}(-1,+1)$ shows.

Proposition 9 (Rank obstruction of the single-generator collapse)**.** *The single-generator collapse does not extend to four dimensions: with one shared generator and* $\mathrm{tr}(T^2) = 2$ *the frame-twist form is a rank-one outer product whose real-part projection raises the rank only to two, with eigenvalues* $\{-2(\partial_0\psi_0)^2, 2[(\partial_1\psi_1)^2 + (\partial_2\psi_2)^2 + (\partial_3\psi_3)^2], 0, 0\}$ *in the separable chart and degenerate signature* $(-,+,0,0)$*, whose non-degenerate part again carries only* $\mathrm{O}(1,1)$*.*

A rank-four non-degenerate $(-,+,+,+)$ form, with local frame group $\mathrm{O}(1,3)$ and identity component $\mathrm{SO}^+(1,3)$, requires abandoning the collapse in favor of four independent generators, and complexification is dispensable for this existence question: the economical choice is real.

Proposition 10 (Four-dimensional all-real signature construction)**.** *Let* $K_q \geq 6$*. Choose a real skew generator* S *supported on two coordinates with* $\mathrm{tr}(S^2) = -2$*, and three traceless, mutually trace-orthogonal real diagonal generators* T_1, T_2, T_3 *supported on the remaining coordinates (Postulate P4.6). All four generators commute, so* $\phi = \psi_0 S + \sum_a \psi_a T_a$ *lies in an abelian subalgebra,* $A_\mu = \partial_\mu\phi$ *exactly (Regime I,* $F_{\mu\nu} = 0$*), and the construction stays inside* $\mathrm{GL}(K_q, \mathbb{R})$*. Because* $\mathrm{tr}(ST) = 0$ *identically for skew* S *and symmetric* T*, all temporal-spatial cross terms vanish without projection, and the frame-twist form* $G_{\mu\nu} = \mathrm{tr}(A_\mu A_\nu)$ *is real, of rank four and signature* $(-,+,+,+)$ *wherever the four differentials* $d\psi_0, d\psi_1, d\psi_2, d\psi_3$ *are linearly independent, with orthonormal-frame group* $\mathrm{O}(1,3)$ *at each such point. No imaginary unit, complex pairing, or real-part projection enters: Postulates P4.2, P4.3, and P4.4 are not required.*

Proof. S has support disjoint from the T_a and so commutes with them, and the T_a commute with one another because they are diagonal; hence ϕ is abelian and $A_\mu = \partial_\mu \phi$ exactly. Normalizing $\mathrm{tr}(T_a T_b) = 2\delta_{ab}$ gives $G_{\mu\nu} = -2(\partial_\mu \psi_0)(\partial_\nu \psi_0) + 2\sum_a (\partial_\mu \psi_a)(\partial_\nu \psi_a)$. For skew S and symmetric T, $\mathrm{tr}(ST) = \mathrm{tr}((ST)^\top) = -\mathrm{tr}(ST)$ forces $\mathrm{tr}(ST) = 0$, so no cross terms survive. Thus $G = 2\psi^* \eta$ with $\eta = \mathrm{diag}(-1, 1, 1, 1)$, and where $d\psi$ is invertible Sylvester's law of inertia gives the stated signature. □

Several limits bound what the proposition delivers. The metric is $G = 2\psi^*\eta$, the pullback of the Minkowski metric along $\psi = (\psi_0, \psi_1, \psi_2, \psi_3) : \mathcal{C} \to \mathbb{R}^{1,3}$, hence coordinate-flat: the proposition exhibits signature availability, not curvature, and the derivation-gap critique of the two-dimensional example transfers, since the gauge apparatus again contributes only the trace normalization. The signature is fixed by the choice of one skew and three diagonal generators, which is the content of P4, so availability is immediate from the indefinite trace form and selection of the 1+3 split (C4 of Section 24) is the open problem. The bound $K_q \geq 6$ is the minimum under the disjoint-support hypothesis with traceless spatial generators, and $K_q \geq 5$ suffices under that hypothesis without tracelessness; without disjoint supports, commuting trace-orthogonal sets exist at $K_q = 5$ with traceless spatial generators (for instance $S = J \oplus 0$ with $T_a = a_a I_2 \oplus \mathrm{diag}(c_a, d_a, e_a)$, $2a_a + c_a + d_a + e_a = 0$) and at $K_q = 4$ without tracelessness. The irreducible content is the generator split: one compact and three non-compact directions, mutually trace-orthogonal and commuting.

As stated at the head of this section, the signature is a property of a declared reference section. A gauge-invariant candidate exists in the relational form $\mathrm{tr}((A_i - A_j)_\mu (A_i - A_j)_\nu) = \mathrm{tr}((\Omega_{ij}^{-1}\partial_\mu \Omega_{ij})(\Omega_{ij}^{-1}\partial_\nu \Omega_{ij}))$ on agent pairs, which is indefinite on $\mathfrak{gl}(K_q, \mathbb{R})$ for the same reason as $\mathrm{tr}(A_\mu A_\nu)$; it requires base derivatives of frames and is deferred to Supplement B. Because the indefinite form lives on the horizontal block alone while the clock, the sectors, and the gravity reading of Section 21.1 are readings of the Fisher block, the signature that matters for them is that of the combined tensor of Eq. (143), which depends on the relative size of the two blocks: in the four-generator frame with one Gaussian mean $\mu = a\psi_0$ of variance σ^2, the combined tensor in the chart ψ is $\mathrm{diag}(-2\kappa + a^2/\sigma^2, 2\kappa, 2\kappa, 2\kappa)$, Riemannian whenever $a^2/\sigma^2 > 2\kappa$. We do not identify a single tensor that is both Lorentzian and the carrier of the clock, the sectors, and the gravity reading. The open problem accordingly sharpens: not whether an indefinite frame-twist form can be exhibited in four dimensions, but whether a gauge-invariant form can be, and whether free-energy dynamics select one compact and three non-compact commuting frame directions.

19.3 Alternative Route: Lorentzian Conformal Class from Finite-Speed Epistemic Causality

A second route reaches a Lorentzian form without complexifying the connection, using a disjoint postulate set (register entry P5). It assumes a coarse-grained shared update parameter t used as a base coordinate, a positive-definite spatial information metric h_{ab}, and a finite maximal propagation speed $c_{\mathcal{I}}$ for epistemic perturbations, with the region of possible influence the h-ball $h_{ab}dx^a dx^b \leq c_{\mathcal{I}}^2 dt^2$. The boundary of possible influence is then the null cone of $g_{\mu\nu}dc^\mu dc^\nu = -c_{\mathcal{I}}^2 dt^2 + h_{ab}dx^a dx^b$, which is of Lorentzian signature $(-, +, \ldots, +)$ by inspection and of signature $(-, +, +, +)$ for three spatial dimensions. The quadratic cone requires the isotropy assumption; a direction-dependent speed bound gives a Finsler cone. The shared parameter t is a preferred foliation, so Lorentz boosts are symmetries of the cone but not of the first-order dynamics, as in analog gravity [153]. The cone fixes g only up to a conformal factor and does not select the dimension count. This is as it must be: the causal-order theorems cited in Section 18.1 [145, 146] fix the metric only

up to a conformal factor and presuppose the signature, so they do not produce one from a non-Lorentzian order, and causal set theory supplies the missing scale by counting elements [147]. The speed $c_{\mathcal{I}}$ is postulated and carries no units. The finite-speed postulate is in tension with the first-order natural-gradient dynamics only in the continuum limit, where the naive limit is parabolic with infinite signal speed; on a bounded-degree agent graph, first-order dynamics already has an effective light cone of Lieb-Robinson type: for linear dynamics $\dot{x} = -Lx$ with a unit-weight graph Laplacian of maximal degree Δ, $(L^k)_{xy}$ vanishes for k below the graph distance $d(x, y)$, so $|(e^{-tL})_{xy}| \leq \sum_{k \geq d(x,y)} (2\Delta t)^k / k!$ [154, 155]. Relating that graph cone to a causal structure on $\mathcal{C}$ is base-transport work deferred to Supplement B. The two routes are parallel existence statements with disjoint open problems (dynamical selection of one compact and three non-compact commuting frame directions, and gauge-invariant content, for the algebraic route; the conformal scale, the dimension count, the isotropy assumption, and the continuum limit of the dynamics for the causal route), and neither derives the signature from variational dynamics.

19.4 Temporal Direction from Belief Trajectories

An agent's belief trajectory singles out a line in its tangent space through the Gram-Schmidt decomposition

$$T_q\mathcal{B} = \mathbb{R}\dot{q} \oplus \dot{q}^{\perp}, \tag{152}$$

but assigns that line neither a negative norm nor an orientation; the orientation is inherited from the flow parameter t, equivalently from the decrease of $\mathcal{F}$ along the flow. Writing

$$ds^2 = -g_{\mathcal{B}}(\dot{q}, \dot{q})dt^2 + \sum_i g_{\mathcal{B}}(e_i, e_i)dx_i^2 = -d\vartheta^2 + \sum_i g_{\mathcal{B}}(e_i, e_i)dx_i^2, \tag{153}$$

the leading minus sign is imposed by ansatz on the fiber tangent space $T_q\mathcal{B}$, and no construction in this paper produces it. The signature constructions of Section 19 give indefinite forms on the base tangent space built from the connection, not a sign on $g_{\mathcal{B}}$, and the relative entropy between real Gaussian beliefs is unaffected by any real-versus-imaginary choice in the present formulation. Equation (153) records what a Lorentzian reading of the belief trajectory would need; it is not a derivation.

The boxed item below records what is derived and what is postulated.

Belief Tangent-Space Decomposition and the Spacetime Reading: What Is Derived and What Is Postulated

What is derived: A belief trajectory $q(t)$ through the fiber $\mathcal{B}$ has tangent $\dot{q} \in T_q\mathcal{B}$ with the Gram-Schmidt decomposition

$$T_q\mathcal{B} = \underbrace{\mathbb{R}\dot{q}}_{\substack{\text{trajectory} \\ \text{1 dim}}} \oplus \underbrace{\dot{q}^{\perp}}_{\substack{\text{orthogonal} \\ \text{complement}}}$$

of Eq. (152), which singles out the trajectory line without assigning it a sign or an orientation, together with the spectral decomposition of the pullback tensor $G_i(c)$ on the base tangent space $T_c\mathcal{C}$ relative to a declared background chart metric (Section 20.2). Time and the sectors live on the base: under the identification $q(t) = \sigma_i(c(t))$ of the trajectory with the section evaluated along a base curve, the temporal direction is the tangent $\dot{c}$ of that curve, whose pushforward $d\sigma_i(\dot{c}) = \dot{q}$ recovers the fiber trajectory direction, and the observable, subthreshold, and internal sectors $\mathcal{D}_{\text{obs}}, \mathcal{D}_{\text{subthresh}}, \mathcal{D}_{\text{internal}}$ are eigen-direction classes of $G_i(c)$ on $T_c\mathcal{C}$, not of any operator on $\dot{q}^{\perp}$, on which no eigenvalue problem is defined.

What is postulated: (i) the eigenvalue hierarchy $\lambda_{\text{obs}} \gg \lambda_{\text{subthresh}} \gg \lambda_{\text{internal}} \approx 0$ on the spectrum of $G_i(c)$, including the choice of four large eigenvalues in $\mathcal{D}_{\text{obs}}$, one temporal and three spatial (register entry P6), together with any three-way refinement of $\dot{q}^{\perp}$ into fiber subspaces $V_{\text{obs}} \oplus V_{\text{subthresh}} \oplus V_{\text{internal}}$ mirroring the base sectors, with $\mathbb{R}\dot{q}$ mirroring the temporal and V_{obs} the three spatial directions of $\mathcal{D}_{\text{obs}}$; (ii) Lorentzian signature $(-,+,+,+)$ on $\mathbb{R}\dot{q} \oplus V_{\text{obs}}$, as the ansatz of Eq. (153) with no construction behind it; (iii) on the complex route, Postulates P4.2, P4.3, and P4.4 of Section 19 (the all-real construction of Proposition 10 dispenses with them), acting on the frame-twist block of a declared reference section; (iv) in any multi-generator extension, the mutual trace-orthogonality of the spatial generators, Postulate P4.6, exact at the abelian-exact order and satisfied by commuting disjoint-support generators in Proposition 10, while beyond that order the dexp expansion of $A_a = U^{-1}\partial_a U$ mixes generators and exact diagonality would additionally require commuting frame components; (v) that one tensor carries both the signature and the clock and sectors, which no construction here shows. None of (i) to (v) is derived from free-energy minimization in the present formulation.

An interpretive reading: The above structure permits a three-sector reading consisting of a $(1+3)$ spacetime sector (temporal plus observable spatial), a subthreshold informational sector with moderate squared information rate, and an internal sector where that rate is negligible. We do not claim to have derived physical spacetime, the Lorentz signature, or the dimension count from the framework; the construction shows that the framework is compatible with such a reading under the postulates listed above.

20 One World out of Many: Collective Geometry, Observable Sectors, and Observer-Dependent Reality

Wheeler's third question asks how "one world" arises out of many observer-participants, and in answering it he adopts Føllesdal's dictum that meaning is "the joint product of all the evidence that is available to those who communicate" [1]; a claimed measurement, he adds, counts only once it can be checked with one's fellows. This section develops the paper's formal counterpart of that strand, item W3 of Table 1. The agents of Part I communicate only through the transported

relative entropies $\beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ of the consensus functional, Eq. (49), and each agent also carries its own observation term. At a stationary point the attention rows are the Gibbs rows of Section 5.6, and for fixed neighbors each agent's optimal belief is the normalized geometric mean of its prior and the transported beliefs it attends to (Theorem 2), reweighted by a power of the agent's own likelihood; for Gaussian beliefs and a linear Gaussian likelihood it is a precision-weighted average. The residual transported divergence is the frame-independent part of the disagreement that remains. The stationarity conditions are derived (status D); reading the stationary point as "one world" is an interpretation (status S; Table 2). The model has the structure Wheeler names, a shared description assembled from the evidence available to agents who communicate, and it departs from his proposal in three ways: the evidence is continuous, the channel is a declared divergence, and agreement is graded, never a registered yes-no fact.

20.1 The Consensus Metric and Its Regulator Obstruction

Each agent induces its own metric on the base manifold, and the question is how these individual geometries combine into shared structure. The answer depends on which block of the combined tensor of Eq. (143) is averaged. The Fisher block $G_i^{(q)}$ of Eq. (142), computed with the covariant jet of Eq. (141), is the ordinary pullback of the frame-stripped law $\hat{q}_i = \rho(U_i^{-1}) q_i$, and a passive local re-expression $U_i \mapsto g_i(c) U_i$, $q_i \mapsto \rho(g_i(c)) q_i$ leaves it unchanged (Section 18.1 and [31, Ch. B2]). The ordinary Fisher block of q_i is likewise unchanged by the change-of-reference action $U_i \mapsto U_i g(c)$ applied to all agents, which moves neither beliefs nor transports. The presence-weighted consensus metric

$$\bar{G}^{\mathrm{cons}}_{\mu\nu}(c) = \sum_i w_i(c) G^{(q)}_{i,\mu\nu}(c) \tag{154}$$

is therefore gauge invariant with no orbit average, although it is relative to the agents' connections. It is not the only invariant. Relative-entropy values, the conjugacy classes of loop holonomies, and the per-edge twists modulo one simultaneous conjugation survive every change of frame [30, Ch. 14], and the Fisher clock of a natural-gradient history of an invariant free energy equals its gauge-quotient duration [30, Ch. 7]. An orbit average is needed only for the frame-twist block $\mathrm{tr}(A^{(i)}_\mu A^{(i)}_\nu)$, with $A^{(i)} = U_i^{-1} dU_i$, which is where the signature constructions of Section 19 live, and there it meets the following obstruction.

Proposition 11 (Regulator obstruction for the frame-twist block)**.** *Let $T_{i,\mu\nu} = \mathrm{tr}(A^{(i)}_\mu A^{(i)}_\nu)$. (a) T_i is invariant under constant right translation $U_i \mapsto U_i g$, so a Haar average over one copy of G is redundant. (b) Under the c-dependent right translation $U_i \mapsto U_i g(c)$ of all agents, which leaves every transport Ω_{ij} and every belief unchanged, and hence every alignment, attention, and cross-scale term, $A^{(i)} \mapsto g^{-1} A^{(i)} g + g^{-1} dg$, and the choice $g = U_i^{-1}$ gives $A^{(i)} = 0$ and $T_i = 0$ for any one chosen agent. The frame-twist block has no gauge-invariant content. (c) After discretization to finitely many base points, the orbit average over c-dependent g exists for compact G, since the product Haar measure on G^N is a probability measure. It has no continuum limit: the group of maps $\mathcal{C} \to G$ is infinite-dimensional and not locally compact, so the Haar construction does not apply, and at lattice spacing ε the discrete Maurer-Cartan term between independent Haar-distributed neighbors is of order ε^{-1}, so the average of T_i grows in magnitude as ε^{-2}. (d) For noncompact G, including $\mathrm{GL}(K_q)$ and $\mathrm{SO}(1,3)$, the Haar measure has infinite total mass already for a single copy, and the average is undefined without a further regulator.*

Proof. (a) $\mathrm{tr}(g^{-1} A_\mu g g^{-1} A_\nu g) = \mathrm{tr}(A_\mu A_\nu)$ by cyclicity; this is the residual-subgroup statement of Section 4.3. (b) $(U_i g)^{-1} d(U_i g) = g^{-1}(U_i^{-1} dU_i) g + g^{-1} dg$ and $(U_i g)(U_j g)^{-1} = U_i U_j^{-1}$; with $g = U_i^{-1}$

the transformed frame of agent i is the identity at every c. (c) A product of finitely many compact groups is compact, and its normalized Haar measure is the product measure. For independent Haar-distributed $g_x, g_{x+\varepsilon}$ the element $g_x^{-1} g_{x+\varepsilon}$ is Haar distributed and its logarithm is of order one almost surely, so the lattice connection $\varepsilon^{-1} \log(g_x^{-1} g_{x+\varepsilon})$ is of order ε^{-1} and its trace square, which is negative definite on the compact algebra, has expectation proportional to ε^{-2}. (d) A noncompact locally compact group has infinite Haar measure. □

In the flat Regime I every curvature invariant vanishes, since $F = 0$. With independent links on the interaction graph (Regime II) the invariants are the conjugacy classes of loop holonomies listed above, and their operational detection is item C1 of Section 24. The construction transfers verbatim from the state fiber to the model fiber.

20.2 Observable, Subthreshold, and Internal Sectors

The covariant Fisher block $G_i(c) := G_i^{(q)}(c)$ is a positive semidefinite tensor on $T_c\mathcal{C}$ whose rank equals the rank of $d\sigma_i(c)$ and is therefore at most $\dim \mathcal{C}$. Its eigendirections split $T_c\mathcal{C}$ into three sectors against thresholds: an observable sector $\mathcal{D}_{\text{obs}}$ of directions with large eigenvalue, read as perceived spacetime, a subthreshold sector $\mathcal{D}_{\text{subthresh}}$ carrying non-negligible but imperceptible information rate, and an internal sector $\mathcal{D}_{\text{internal}}$ along which the section is nearly constant, with $\lambda_{\text{obs}} \gg \lambda_{\text{subthresh}} \gg \lambda_{\text{internal}} \approx 0$ assumed to hold smoothly with constant sector ranks. Because a chart change acts by congruence, $\tilde{G} = J^\top G J$, absolute eigenvalue thresholds are chart dependent; a chart-invariant statement uses the generalized eigenvalue problem $G_i(c)v = \lambda g_0(c)v$ against a declared fiducial base tensor g_0 that transforms by the same congruence. We take g_0 to be the flat chart metric of the working framework (Section 4.9). It is a declared computational background, not a physical metric, and the sectors are defined relative to it. A background-free alternative would use the consensus metric of Eq. (154) in place of g_0 where that metric is positive definite; it degenerates wherever the agents' sections share a kernel, and we do not develop it.

An operational definition would replace the stipulated threshold by the observation channel, but it needs a declared map from $\mathcal{C}$ to the latent space. The natural choice through the means gives, for a Gaussian likelihood of precision Λ_o, the registered-information tensor $J^{(o)}(c) = d\mu_i^\top \Lambda_o d\mu_i$. This tensor is blind to covariance variation and degenerate wherever $d\mu_i$ has a kernel, so the pencil $G_i v = \lambda J^{(o)} v$ is generically singular, with infinite eigenvalues on directions in $\ker J^{(o)}$ outside $\ker G_i$. A well-posed version assigns $\ker J^{(o)}$ to the unregistered sectors and applies the pencil on a g_0-orthogonal complement of $\ker J^{(o)}$, so that observable directions are those along which belief variation is registered at the rate it occurs. No spatial likelihood channel is specified in this paper, so we keep the threshold form.

Register entry P6 postulates, for human cognitive agents, an observable sector with $|\mathcal{D}_{\text{obs}}| = 4$, one temporal and three spatial directions, with $|\mathcal{D}_{\text{obs}}| \ll |\mathcal{D}_{\text{subthresh}}| \ll |\mathcal{D}_{\text{internal}}|$. Since $\text{rank} G_i \leq \dim \mathcal{C}$, the postulate requires $\dim \mathcal{C} \geq 4$, and the stated hierarchy requires $\dim \mathcal{C}$ much larger. Neither holds on the working base $\mathcal{C} = \mathbb{R}^2$, where at most two sector eigenvalues are nonzero, so the postulate has no instance in this paper. The tensor G_i is positive semidefinite, so nothing in its spectrum distinguishes a temporal direction; a Lorentzian reading needs register entry P4 or P5 (Section 19). The framework neither establishes that the observable distribution is Frobenius-integrable, which a $(1 + 3)$ foliation would require, nor makes a quantitative prediction about measured spacetime.

Two senses of invisibility operate here and should not be conflated. At the base level, only the observable eigendirections of $G_i(c)$ are read as phenomenal spacetime, the remaining $\dim \mathcal{C} - |\mathcal{D}_{\text{obs}}|$ directions carrying subthreshold or internal eigenvalues. At the fiber level, the differential $d\sigma_i|_c$:

$T_c\mathcal{C} \to T_{q_i(c)}\mathcal{B}$ samples at most $\dim \mathcal{C}$ directions of the $K_q(K_q+3)/2$-dimensional Gaussian belief manifold, so the unsampled fiber directions contribute nothing to $G_i(c)$ whatever the threshold.

20.3 Observer-Dependent Reality

Different agents pull back different metrics, $G_i \neq G_j$, over the same base. With the covariant Fisher block this difference has one source only, the difference between the agents' frame-stripped beliefs; with the ordinary fixed-gauge pullback, part of $G_i - G_j$ is gauge artifact, since a local frame change adds frame-gradient terms. Observer-dependence of the perceived geometry is therefore a statement about belief, not about gauge. It is not solipsism: agents remain coupled through the attention weights $\beta_{ij} \propto \exp[-D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])]$, and at a consensus stationary point their covariant pullbacks agree to the extent that their beliefs do. The resulting position combines within-frame validity with an intersubjective consensus that holds within a population of similar agents, not absolutely (status S).

The framework is classical and admits a global frame in Regime I, so it models observer-dependent representation and belief, not the observer-dependent facts at issue in Brukner's no-go theorem [128] and the Frauchiger-Renner argument [156]. Its relation to Brukner's theorem, to the cross-perspective-links question of relational quantum mechanics [60], to QBism [10, 62], and to Müller's observer-first program [9] is set out in Section 14.

"No view from nowhere" has a limited formal meaning here. The base $\mathcal{C}$ is a presupposed smooth index manifold carrying no physical metric: it carries a background density and a fiducial chart metric, used for integration and for the sector thresholds above, and the metrics that represent perceived geometry are the induced pullbacks. Its dimension, topology, and smooth structure are inputs, so the construction derives metric data only and keeps the continuum that Wheeler's "no continuum" rejects [1]. Calling $\mathcal{C}$ noumenal is an analogy only (Section 14).

20.4 Physical Quantities as a Derivation-Target Program

Both induced tensors are dimensionless (Section 18.1). A physical line element requires an external length scale L_0, and a Lorentzian reading requires in addition a constant conversion speed, which the algebraic signature route does not supply and the causal route postulates without units (register entry P5).

Register entry P7 postulates that information, measured in nats by the free energy and the relative entropies, is the dimensionally fundamental quantity, and that physical dimensions are labels agents construct through consensus. The postulate reframes the dimensional question; it does not answer it, since every dimensionful constant, beginning with L_0, remains an external input that no construction in this paper produces. On the interpretive reading, information geometry on the fibers is the structure available to agents and physical measurements are the labels they assign. The falsifiability cost of the commitment is assessed in Section 23. The structural parallels to physical concepts (time from information updates, stiffness read as mass under the kinetic-metric postulate, action from the free-energy functional) are formal analogies with status S, summarized in Table 3.

Physical Quantity	Framework Analog	Missing ingredient
Time t	Information clock ϑ_i	A presupposed ordering parameter is not removed (register entry P1); indefinite-signature extension (Section 19)
Mass m	Stiffness (Hessian of $\mathcal{F}$) under the kinetic-metric postulate	A second-order dynamics in which inertia and stiffness are independent; the flow used here is first order, with the Fisher metric as friction tensor (Section 8)
Energy E	Free energy density	An energy scale and an identification of $\mathcal{F}$ with a physical energy
Entropy S	Information entropy $H[q]$	Identification of q with the physical ensemble, Boltzmann's constant k_B, and a phase-space measure [157]; the Gaussian $H[q]$ is a coordinate-dependent differential entropy
Action	Free energy functional	Legendre structure (stationarity correspondence only)
$\hbar$	Minimal action per bit	Independently defined minimal action per bit
c	Information propagation speed	Independently defined maximal epistemic speed with units (register entry P5)

Table 3: Correspondences between physical quantities and information-theoretic structures (status S), each labeled by the ingredient whose supply would upgrade it from analogy to derivation.

21 Speculative Physical Readings: Gravity, Measurement, and Macroscopic Objects

21.1 A Pullback-Curvature Reading of Gravity

By the three-tier reading of Section 18.2, the geometry an agent experiences as ambient is the structural pullback $G_i^{(s)}$ on the model fiber, and register entry P8 reads gravitational effects, if any, as curvature of this metric (status S). One exact statement holds: for constant isotropic model covariance $\Sigma_s = \sigma_s^2 I$ the structural metric is $G^{(s)} = \sigma_s^{-2} h$, with h the first fundamental form of the immersion $\mu_s : \mathcal{C} \to \mathbb{R}^{K_m}$, and a constant conformal rescaling leaves the Levi-Civita connection unchanged and multiplies sectional curvature by σ_s^2, so the curvature is the intrinsic curvature of the immersion, given by the Gauss equation, rescaled by the model variance. Nothing further is derived: there is no field equation, source coupling, Newtonian limit, or gravitational constant, the structural metric is Riemannian, and no gravitational mass is defined, so the equivalence principle cannot yet be posed, and the reading is a placeholder, not a model of gravity. By contrast, Jacobson derives the Einstein equation as an equation of state from the Clausius relation on local Rindler horizons, with the entropy density fixing Newton's constant [20], and Carlip sets out the obstructions that any emergent-gravity proposal must meet [158].

21.2 Macroscopic Objects as Precision Attractors

On this reading everyday objectivity has a classical mechanism. A high-precision agent, such as a macroscopic object with prior covariance $\Sigma_{\mathrm{ob}} \approx \delta I$ for small dimensionless δ (the precision scale of Section 8), enters each coupling $\beta_{i,\mathrm{ob}} D_{\mathrm{KL}}(q_i \| \Omega_{i,\mathrm{ob}}[q_{\mathrm{ob}}])$ with such large precision that, where the coupling dominates, the alignment term pulls every observer's belief toward the object's transported belief, so high-precision systems anchor all observers. The anchoring is one-sided in a limited sense. Appendix B shows that the back-action of the observers on the object's mean is suppressed by precision dominance, of order δ in absolute terms and $\delta^{1/2}$ in the object's Fisher metric, whereas the back-action on its precision is not: at stationarity the object's precision is reduced by a factor that tends to $1/(1+\beta)$ as $\delta \to 0$ and stays away from one at every δ. The precision δ carries no physical units, and the dimensional bridge is the open question of Section 20.4. Macroscopic objects are not modeled as agents anywhere in this paper, so the description extends the meta-agent construction by analogy (status S). Its structure, objectivity arising because many observers couple to the same high-precision record, is the classical counterpart of quantum Darwinism [45].

21.3 An Inferential-Consensus Analogy for Measurement

Measurement can be read as the reverse of the macroscopic-object case: observers who hold different priors about a system couple to a high-precision apparatus, which dominates the free energy and pulls them into agreement. Classically this mechanism is precision-weighted opinion pooling [129, 159], and its closest quantum-foundations relative is the quantum Darwinism noted above [45]; environment-induced superselection [160] needs a basis-selection structure that a framework without a Hilbert space lacks. The analogy is therefore suggestive only (status S): with no Hilbert space, Born rule, or superposition, the framework does not address the measurement problem. The closest inference-route constructions are Caticha's entropic dynamics, which derives Schrödinger evolution with the Born rule built into the representation [14, 161, 162], and Reginatto's minimum-Fisher-information derivation, in which adding the Fisher term to the classical Hamilton-Jacobi action yields the Schrödinger equation [163]. We adopt Fisher information only as the metric that prices belief change, not Frieden's program of deriving physical law from extreme physical information [164], whose claims of universality Lavis and Streater criticize [165]. Whether anything resembling quantum measurement can arise awaits the density-operator fiber of item C5 (Section 24).

22 Consciousness: An Interpretive Stance

Every statement in this section has status S or C (Table 2). No equation or proposition elsewhere depends on it, and a reader who rejects it keeps the mathematics intact. It records the author's interpretive stance so that its costs are visible.

22.1 A Candidate for the Cognitive-Frame Transformation Law

Lahav and Neemeh write the cognitive-frame transformation law only for functionally identical cognitive systems and leave the law between distinct frames unspecified [63,64]. Section 4.6 proposes an explicit candidate, with the frame U_i as cognitive frame of reference, the transport Ω_{ij} as the transformation between frames, and $D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j])$ as the frame-invariant residual. The match is one of structural function, and its scope is narrow: at one base point Regime I frames are removable (Proposition 1), so a frame carries content only relative to fields held fixed, such as

the observation channel. The correspondence departs from Lahav and Neemeh in two respects. They are relational physicalists, for whom physical reality is observer-relative but ontologically prior, whereas the stance here denies an observer-independent substrate beneath the agents; and neither account supplies a second postulate analogous to the constancy of the speed of light, since the signature constructions of Section 19.2 are postulated, not selected. The qualia-indeterminacy worry that the relational reading dissolves is relocated, not removed: the only candidates for a frame-independent label are the invariants of Section 20.1.

22.2 Qualia, the Hard Problem, and Meta-Agents

On the interpretive reading of register entry P9, the formal correlate of phenomenal character is the geometry of an agent's covariant pullback metric G_i. That tensor depends on the agent's belief section, not on its frame, so an inverted spectrum modeled as a passive re-expression of one agent's frame [166] changes no invariant quantity; the formalism represents such an inversion as no difference at all. The stance relocates Chalmers' hard problem [167, 168] without solving it: the explanatory gap moves from why neural computation feels like anything to why the geometry of inference does, a strategy shared with Russellian neutral monism [169,170], panpsychism [171,172], and the identity claim of integrated information theory [173, 174], to which the framework adds vocabulary, not an explanatory mechanism.

The author's position in the panpsychism taxonomy, stated in the interpretive stance of Section 1, is panprotopsychist in Chalmers' sense [172,175]: agents at level zero carry frame structure and take part in variational dynamics, and no phenomenal property is ascribed to them. Strawson's argument against brute emergence [170] presses on constitution from protophenomenal structure as well, and the stance accepts the resulting combination problem as its central open question (status C).

The one conjecture offered is that conscious systems are meta-agents with active cross-level coupling that represent their constituents. It is a necessary condition that does not discriminate: meta-agent formation is triggered by coherence alone, so a highly ordered crystal qualifies, and the representation requirement is layered on the formation rule, not derived from it. The conjecture therefore does not compete with the integrated-information measure Φ [176,177] as a test, and the framework draws no boundary between conscious and non-conscious systems.

23 Gauge Invariance as Consensus and the Philosophy of Science

The thesis of this section, that the gauge structure observed in physics reflects human cognitive architecture, has status C and may be unfalsifiable: any observed physics can be reinterpreted as reflecting cognitive constraints, and any disappearance of gauge structure as a change in cognitive architecture. The formalism is consistent with the thesis; it does not establish it.

23.1 Model Alignment, Covariance, and the Selection Criterion

The model-alignment term $\mathcal{F}_{\text{model}} = \sum_{ij} \int_{\mathcal{C}} \gamma_{ij}(x) D_{\text{KL}}(s_i(x) \| \tilde{\Omega}_{ij}[s_j](x)) dx$ drives agents toward shared generative models, and consensus formation requires that scientific models be statable frame-independently, so that their content survives transport under $\tilde{\Omega}_{ij}$ (register entry P10); smallness of a pairwise divergence does not by itself establish a covariant transformation law. A century-old objection bounds what such a criterion can deliver. Kretschmann argued in 1917 that general covariance alone carries no empirical content, since any theory can be rewritten in generally covariant form by introducing auxiliary geometric objects [36], and the literature surveyed by Norton [37]

has sharpened the distinction between physically substantive and merely notational covariance, and between coordinate invariance and gauge invariance proper, the latter constraining the form of the action, for example by forbidding explicit gauge-boson mass terms and fixing the universality of gauge couplings.

Remark 1 (Kretschmann bound)**.** *Cognitive shareability, as an intersubjectivity constraint on the form of theoretical statements, at most motivates general covariance and does not deliver gauge invariance proper; it is a candidate constraint on the formal scaffolding within which theories are stated, not an explanation of why nature exhibits the particular gauge groups it does.*

The $\mathrm{GL}(K_q)$ invariance of the divergence is already present for a single agent, so it is a consistency requirement that consensus must respect, not a symmetry that consensus generates. No Standard-Model factor is derived: $\mathrm{GL}(K_q, \mathbb{C})$ contains $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, but every compact Lie group has a faithful finite-dimensional unitary representation and so embeds in some $\mathrm{U}(K_q)$, and the containment carries no selection content. The gauge group of physics is an empirical input, and whether free-energy minimization dynamically selects any subgroup is open. The historical progression from Galilean to Lorentz to general covariance to gauge invariance is consistent with the candidate cognitive-shareability constraint, which offers an interpretive reading of that history, not a derivation of physical law.

23.2 Altered States, Counterfactual Architectures, and Informational Compatibility

The cognitive reading re-describes altered states of consciousness as temporary shifts of one agent's frame relative to the consensus, so that the transport $\tilde{\Omega}_{\text{altered,normal}}$ lies far from the identity and the residual disagreement is large; this is a frame mismatch, not curvature, since $F = 0$ in Regime I [178–180]. The consistency is not confirmatory (status S). Counterfactually, a population whose frames shifted coherently would pull back different metrics and construct different physics, the predictive-processing reading of perception [149–151, 181] made geometric; this is a thought experiment, not a prediction, since no population with a substantially different cognitive architecture is accessible. Read this way, physics is a constitutive but revisable constraint on shareable description, in Friedman's relativized sense of the a priori [134]; what the Discussion disavows is the apodictic a priori, fixed and universal, not this constitutive role. The resulting objectivity is that of robust consensus constrained by sensory evidence, and whether this reading explains or merely relabels remains open, since a realist account on which nature is simply uniform accommodates the same data at least as well.

23.3 Scientific Knowledge as Collective Free-Energy Minimization

On this reading scientific success is collective free-energy minimization under several constraints: fitting the empirical term $-\mathbb{E}_q[\log p(o \mid x)]$, reaching intersubjective consensus through the alignment term, and keeping the covariance needed for shareability. This is a pragmatic reconception in the spirit of constructive empiricism [182] and of Laudan's problem-solving account of scientific rationality [183]. It applies reflexively: the acceptance of this manuscript would itself be consensus-optimal free-energy minimization within a research community, and we offer the framework as a stance in van Fraassen's sense [184].

Kuhnian revolutions [185] read as collective frame changes $\phi_{\text{old}} \to \phi_{\text{new}}$ with accompanying model shifts $s_{\text{old}} \to s_{\text{new}}$. A uniform change of frame is a symmetry of $\mathcal{F}$, so the incommensurability analog lives in the model shift: the global transport $\Omega_{\text{old,new}}$ stays invertible, compatible with

the local incommensurability of Hoyningen-Huene [186], and the formal content is communication breakdown, the collapse of inter-community attention when transported old-paradigm beliefs sit far from the new model's free-energy minima. Quinean underdetermination [187, 188] is engaged in its weaker form: models of equal empirical adequacy, $D_{\mathrm{KL}}(p_{\mathrm{emp}}\|p_{s_1}) = D_{\mathrm{KL}}(p_{\mathrm{emp}}\|p_{s_2})$, are distinguished by further free-energy criteria (alignment, simplicity, unification), which break ties pragmatically. Whether such criteria are evidential [189] or pragmatic [182] is left open, and the framework is compatible with scientific pluralism [190].

The framework implies epistemic limits of a structural kind. The base $\mathcal{C}$ is a presupposed smooth index manifold (Definition 1) carrying no physical metric; every metric feature that represents perceived geometry is agent-induced through the pullback, so the formalism contains no agent-independent geometry to be known, and what is achievable is intersubjective agreement within an aligned community (Section 20.3). The presupposed continuum is a departure from Wheeler, who answers regress with a loop, not a floor [1]. Wigner's unreasonable effectiveness [191] is read as effective physics being the physics that achieves stable consensus, and Nagel's view from nowhere [192] is the unattainable limit. What it would take to convert this structure into physics is the subject of Section 24.

24 Critical Open Problems and Future Directions

Several problems must be resolved before the framework can claim to explain physical reality, not merely supply structural analogies. Taken constructively, they form an ordered program: five mathematical problems, C1 through C5, and one derivation target, C6, each stated with what it would deliver and what it depends on, followed by open items that do not gate the program.

C1 (estimating loop holonomy). At a single base point the independent links on the interaction graph of Section 4.6.5 are the general case, and the frame-coboundary links of Regime I are the graph-flat special case. A normalized population record whose law separates two link configurations that differ only in the conjugacy class of a loop holonomy exists [30, Ch. 10] (the operational loop-holonomy theorem), so detectability is settled at the level of the model. The mathematical problem is to construct a consistent estimator of loop-holonomy conjugacy classes from finitely many normalized population records under a stated noise model, with a finite-sample error bound. Its empirical application to a named target system would further require a held-out statistic and a tolerance fixed before the data, together with a control arm in which the links are frozen at a flat or random value. Under Regime I every loop holonomy is trivial, so no such test has content there. A base-connection version, with holonomy along base loops, is separate and deferred [31].

C2 (consensus geometry). Section 20.1 shows that the presence-weighted average of covariant Fisher pullbacks is gauge invariant without orbit averaging. What remains open is quantitative: a bound on the residual disagreement $G_i - G_j$ among similar agents in terms of their transported divergences at a consensus stationary point, which would give the intersubjective objectivity of Section 20.3 quantitative content. A candidate mechanism is the slow-channel alignment $\gamma_{ij} D_{\mathrm{KL}}(s_i\|\tilde{\Omega}_{ij} s_j)$, with each agent's prior inheriting alignment through the lagged predictive prior of Section 10. A regulated orbit average is needed only for the frame-twist block (Proposition 11), and only if the signature constructions of C3 and C4 are kept.

C3 (the four-generator construction beyond the commuting case). Section 19 exhibits an all-real frame-twist form of signature $(-,+,+,+)$ from one skew and three commuting

diagonal generators. The signature is fixed by that choice (register entry P4), so its availability is immediate from the indefinite trace form on $\mathfrak{gl}(K_q, \mathbb{R})$, and the construction is coordinate-flat; in Regime I the frame-twist block is also pure gauge (Proposition 11), so the construction is a property of a declared reference section. This item asks for a curved four-dimensional candidate beyond the abelian-exact regime, with quantitative control of the non-abelian corrections, built on a gauge-invariant form. The natural candidate, the relational trace form on agent pairs built from $\Omega_{ij}^{-1}\partial_\mu\Omega_{ij}$ (Section 19), needs base derivatives of frames and is deferred [31]. The Lorentzian signature problem remains the central physical obstacle: Fisher-Rao geometry is Riemannian, and no tensor in this paper is shown to be both Lorentzian and the carrier of the clock and the sectors.

C4 (a free-energy selection mechanism for the 1+3 split). Exhibit a mechanism by which free-energy dynamics compare frame configurations and select one compact and three non-compact commuting frame directions, the sharpened open problem of Section 19. This item depends on C3 for the configuration space over which the selection would operate.

C5 (the density-operator level-0 fiber). Wheeler's first question, how come the quantum, is outside the scope of this paper, and this item names the first object a quantum extension would need. It would replace the real Gaussian belief fiber by a non-commutative statistical object, such as density matrices with a quantum relative entropy, for which the $\mathrm{GL}(K_q, \mathbb{C})$ enlargement of Section 19 supplies mathematical tools but not a quantum theory. It would further require inter-agent unitary transport with attention coupling under the subgroup $\mathrm{U}(K_q) \subset \mathrm{GL}(K_q, \mathbb{C})$ that preserves the Hermitian structure, with tensor-product bundles carrying entangled beliefs, a demonstration that the higher levels reduce to the classical Gaussian dynamics used here, and a choice of monotone (Petz-family) metric on the density-operator manifold [193]. The metric and transport tools exist in the quantum-information literature, and a free-energy principle for generic quantum systems has been proposed [46]; the open part is the multi-agent coupling.

C6 (a dimensionless-ratio derivation). Derive at least one dimensionless ratio of physical constants from information geometry alone, for example the fine-structure constant at a stated momentum-transfer scale (it runs from about 1/137 in the Thomson limit to about 1/128 at the Z mass). Such a test is symmetric only under advance specification: the statistical manifolds, the comparison connection, and the Čencov scale fixings, each unique only up to a positive scalar so that predictions must be scale-free ratios, must be fixed before the measured value is consulted, together with a completion criterion that certifies the construction independently of whether the derivation succeeds. No such criterion is available, so C6 is a derivation target, not a decidable problem, and failure under an incomplete or post-hoc construction is non-diagnostic. It depends on C2, because the hierarchy across which dimensionful quantities would be fixed presupposes a consensus geometry.

Three clusters of open items sit outside the ordered program. First, physical content. Matter fields and fundamental interactions are absent, and the construction makes no quantitative prediction for particle masses or coupling strengths; whether they could enter as excitations or topological features of the induced geometry is open, and the gravitational reading remains the postulate P8 (Section 21.1). Two natural extensions are recorded without development. One is a cross-level field strength built from the variation of frames across the level index ℓ as well as across base position; because coarse-graining steps are not invertible and compose as a one-parameter semigroup indexed by ℓ, such an object would pair that semigroup with the structure group G, and no

construction of it is attempted. The other is a pair of cross-bundle morphisms $\Phi : \mathcal{E}_{\text{model}} \to \mathcal{E}_{\text{state}}$ and $\tilde{\Phi} : \mathcal{E}_{\text{state}} \to \mathcal{E}_{\text{model}}$, read as the instantiation of a model in a state and as memory consolidation, which the companion theory defines at one base point [30, Ch. 2] but which do not enter the free-energy functional (49).

Second, the scale question. No result on how the hierarchy behaves with growing agent count or depth is available; the exact coarse-graining statements and the nested bound (Table 2) hold at fixed finite size. Its precise statements are the companion theory's open problems on a persistent multiscale hierarchy [30, Ch. 12] and on a family in which attention, learning, or holonomy selects the components of a collective [30, Ch. 13]. The one decidable test of the participatory reading of the formalism, the companion theory's pairing-selection conjecture [30, Ch. 14], is stated in Section 2.1. A finite-coupling transition remains conjectural and requires the prospective study of Section 13, including its frozen-attention-prior arm.

Third, open mathematical items pointed to from the body. On the noncompact $\mathrm{GL}^+(K_q)$ the frame mean of Eq. (123) needs a declared convention. Under any left-invariant Riemannian metric the Riemannian center of mass exists and is unique below Afsari's radius but is only left-equivariant [95], while the exponential barycenter of the canonical Cartan-Schouten connection is left-, right-, and inversion-equivariant and exists uniquely for sufficiently concentrated frames [97]; the additive form $\exp(\sum_i w_i \phi_i)$ is the first fixed-point iterate of the latter from the identity. What is open is the choice of convention, which Proposition 5 (Section 9.1) identifies as a gauge choice. The kinetic-metric postulate of Section 8.7 needs its own second-order equation of motion before any frequency test can be posed, since the dynamics used in this paper are first order, with the Fisher metric as friction tensor. A separate timescale for each level of the hierarchy (Section 12) is not constructed.

25 Conclusion

This paper has given a classical, multi-agent, information-geometric model of parts of Wheeler's participatory program, and Table 1 records the status of each part. For registered change as a clock (W1), each agent's clock is the Fisher length of its belief history, with an exact dissipation identity along an autonomous single-rate natural-gradient flow, and a binary registration channel gives Wheeler's bit its meaning as at most one bit of mutual information per yes-no registration (Section 17, Proposition 6). For observer-dependent geometry (W2), metrics on a base that carries no physical metric are pulled back from belief sections; with the covariant derivative the pullback is gauge invariant, so its observer-dependence is dependence on belief (Sections 18 and 20.1). For one world out of many observer-participants (W3), intersubjective agreement is a stationary point of an explicit consensus functional, a formal counterpart of Wheeler's adoption of Føllesdal's dictum, with the residual transported divergence as the frame-independent disagreement (Section 20). For participation across scales (W4), exact coarse-graining builds parents from their members, and a parent's belief returns to its members as a one-episode-lagged predictive prior; the hierarchy closes at an energy-level apex that is, in any generative reading, a declared root, the formal residue of Wheeler's refusal of a tower of turtles (Sections 10 and 12). These are constructions and derivations under stated assumptions; their identification with Wheeler's notions has status S.

What the model does not deliver (W5) is as definite. It contains no discrete irreversible registration with complementarity, so it does not model Wheeler's elementary quantum phenomenon; its functional, belief family, and divergence are fixed in advance, so it does not model law without law; it presupposes a smooth index manifold and an ordering parameter, so it keeps a continuum; its feedback relaxes to stationarity unless observations drive it, so it is not a self-excited circuit; and

how come the quantum is outside its scope, with the density-operator fiber of C5 as the first object an extension would need. The signature constructions exhibit $(-,+,+,+)$ forms for postulated generators without selecting them, and no construction supplies physical units.

The companion theory [30] proves the single-point mathematics in greater generality and declines, without refuting, each physical identification postulated in Part II. Items C1 through C5 of Section 24 are mathematical problems and C6 is a derivation target. Solving them would decide questions about the formalism, not the participatory reading, whose one stated decidable test is the pairing-selection conjecture of the companion theory [30, Ch. 14].

26 Methods

26.1 Prospective Multi-Agent Validation Procedure

No multi-agent simulation result is reported or claimed in this manuscript. The prospective procedure for testing the multi-scale claims, with its three arms (among them the frozen-attention-prior control), predeclared formation event, persistence window, primary statistic, seed count, and pass rules, is stated in Section 13; it has not been executed.

26.2 Computational Details

No numerical result of this manuscript depends on simulation. The analytic gradient formulas of Appendix A were checked against central finite differences in double precision, with the tolerances stated there. Several appendices (Appendices A, D, F, G, and I) and the binary-registration example of Section 17.2 report small deterministic numerical witnesses. Each is a direct double-precision evaluation of the stated closed forms or iterations on the stated instance, and none is a simulation of the multi-agent dynamics.

26.3 Large Language Model Assistance

Claude Sonnet 4.5 was used as a programming and typesetting assistant. It generated and refactored Python implementations under author direction, drafted and edited prose, and assisted with bibliographic lookup. Claude Opus 5.5 assisted with the revision of this version of the manuscript under author direction, including numerical checks of stated identities. Neither was used to generate experimental results or accepted as an authority for mathematical correctness. Mathematical derivations and citations remain the author's responsibility.

27 Statements and Declarations

Competing Interests

The author declares no competing interests, either financial or non-financial, related to the work submitted for publication.

Author Contributions

R.C.D. conceived the theoretical framework, performed the mathematical derivations, designed and wrote the multi-agent reference implementation, and wrote the manuscript. Large language model assistance was limited to programming support, fact checking, numerical checking, and drafting and editing prose under author direction, as detailed in the Methods section.

Data Availability

This manuscript reports no empirical dataset and no simulation output; the validation procedure of Section 26.1 has not been executed, so no simulation data exist for it.

Code Availability

A reference implementation of the multi-agent dynamics is publicly available at `https://github.com/cdenn016/MAgent_Model`; no argument in this manuscript depends on it.

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A Explicit Gradient Expressions for $\mathrm{GL}(K_q)$ Gaussian Agents

This appendix collects the gradients and natural gradients of the consensus free-energy functional $\mathcal{F}$ of Section 5 for Gaussian beliefs with $\mathrm{GL}^+(K_q)$ transport, the connected component containing the transports $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ of Eq. (14). The compact restriction $\mathrm{SO}(K_q)$ is the special case in which the frame coefficients ϕ_i are skew-symmetric, so that each factor $\exp(\phi_i)$ and hence every transport Ω_{ij} is a rotation. The Gaussian-sector operations are covariant under the congruence action of $\mathrm{GL}(K_q)$ on the covariance fiber; the frame sector is discussed with the update equations.

Setup and Notation

Agent i maintains primitive belief $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and primitive generative model $s_i = \mathcal{N}(\mu_{s,i}, \Sigma_{s,i})$, with derived prior $p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i})$ on the state fiber and derived hyper-prior $r_i = \mathcal{N}(\mu_{r,i}, \Sigma_{r,i})$ on the model fiber, supplied by the enveloping meta-agent through the lagged predictive prior of Eq. (131). The frame of agent i is the group element $U_i = \exp(\phi_i)$. The state-fiber transport $\Omega_{ij} = U_i U_j^{-1} \in G$ acts via the representation $\rho_{\text{state}} : G \to \mathrm{GL}(K_q)$, and the model-fiber transport $\tilde{\Omega}_{ij}$ acts via ρ_{model}. This appendix computes the gradients on the state-fiber sector (q_i, p_i, U_i); the model-fiber sector (s_i, r_i) has parallel structure under the substitution $(\beta, \Omega, q, p, \tau) \to (\gamma, \tilde{\Omega}, s, r, \tau_\gamma)$.

The attention temperatures are τ on the belief fiber and τ_γ on the model fiber, as in Eq. (45); with the parameterization $\tau = \kappa\sqrt{K_q}$ and $\tau_\gamma = \kappa\sqrt{K_m}$, the dimensionless $\kappa > 0$ is a declared constant, not a variational parameter of $\mathcal{F}$. The computation is made at the base point c_* with every agent present, so the receiver-presence weight of each row equals one. Condition numbers are written $\mathrm{cond}(\cdot)$ to avoid a collision with κ. The pair energies and the attention rows are

$$E_{ij} := D_{\mathrm{KL}}(q_i \| \Omega_{ij}[q_j]), \qquad \beta_{ij} = \frac{\pi_{ij} \exp\left(-E_{ij}/\tau\right)}{\sum_m \pi_{im} \exp\left(-E_{im}/\tau\right)}, \tag{155}$$

so that $\beta_{ij} = \beta_{ij}(\{q\}, \{U\})$ depends on the beliefs and frames of the receiver i and of every source m in its row. The scalar α_i is the per-agent prior precision of Eq. (54), with optimal value $\alpha_i^* =$

$c_0/(b_0 + D_{\mathrm{KL}}(q_i\|p_i))$ from Eq. (58); the constant-precision limit $\alpha_i = 1$ recovers the unit-weight self term of Eq. (49). On the model fiber a fixed hyper-prior weight λ_h plays the analogous role. A diagonal extension would replace the scalar α_i coordinate-wise by Eq. (62); it is not developed here.

The Canonical Envelope Gradient

The dynamics of the main text descend the reduced functional $\mathcal{F}_{\mathrm{red}}$, in which every attention row has been replaced by its Gibbs value $-\tau \log Z_i$ (Section 6.2). At the Gibbs row the derivative of the row block with respect to its weights vanishes, so the gradient of the row value is the gradient taken at frozen weights; this is the attention-reduced potential proposition of [30, Ch. 6], whose potential $-\log Z_i$ differs from the row value here by the factor τ. The reduced slow descent is then the natural-gradient flow of the reduced functional by the envelope identity of [30, Ch. 8]. The adaptive-precision block has the same cancellation (see the discussion following Eq. (64)), so α_i enters with its optimal value α_i^*. A state variable x_i of agent i appears in the row of agent i and, as the sender, in the row of every agent k that attends to i. For $x_i \in \{\mu_i, \Sigma_i, U_i\}$ the complete gradient is therefore

$$\nabla_{x_i}\mathcal{F}_{\mathrm{red}} = \alpha_i^*\nabla_{x_i}D_{\mathrm{KL}}(q_i\|p_i) + \sum_j \beta_{ij}\nabla_{x_i}E_{ij} + \sum_k \beta_{ki}\nabla_{x_i}E_{ki} + \nabla_{x_i}\mathcal{F}_{\mathrm{obs},i}, \tag{156}$$

with every weight at its Gibbs value, $\mathcal{F}_{\mathrm{obs},i}$ the observation term of agent i, and the model channel contributing terms of the same form, with $(\gamma_{ij}, \tilde{\Omega}_{ij}, s_i, r_i, \lambda_h)$ in place of $(\beta_{ij}, \Omega_{ij}, q_i, p_i, \alpha_i^*)$, to the gradients with respect to the model variables and, when the model-fiber transport is generated by the same frames, with respect to U_i. The observation and self terms carry no frame dependence.

Belief mean and covariance. Write $A_{ij} := \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ for the transported covariance of the source. Differentiating the Gaussian relative entropy of Appendix C.2, the self and receiver pieces are

$$\begin{aligned} \frac{\partial D_{\mathrm{KL}}(q_i\|p_i)}{\partial \mu_i} &= \Sigma_{p,i}^{-1}(\mu_i - \mu_{p,i}), & \frac{\partial D_{\mathrm{KL}}(q_i\|p_i)}{\partial \Sigma_i} &= \frac{1}{2}\left(\Sigma_{p,i}^{-1} - \Sigma_i^{-1}\right), \\ \frac{\partial E_{ij}}{\partial \mu_i} &= A_{ij}^{-1}(\mu_i - \Omega_{ij}\mu_j), & \frac{\partial E_{ij}}{\partial \Sigma_i} &= \frac{1}{2}\left(A_{ij}^{-1} - \Sigma_i^{-1}\right). \end{aligned} \tag{157}$$

Differentiating $E_{ki} = D_{\mathrm{KL}}(q_k\|\Omega_{ki}[q_i])$ with respect to the sender's mean and covariance, using the precision transport of Eq. (89), gives

$$\frac{\partial E_{ki}}{\partial \mu_i} = \Lambda_{q_i}\Omega_{ki}^{-1}\left(\Omega_{ki}\mu_i - \mu_k\right), \tag{158}$$

$$\frac{\partial E_{ki}}{\partial \Sigma_i} = \frac{1}{2}\Omega_{ki}^\top\left[\tilde{\Lambda}_{q_i}^{(k)} - \tilde{\Lambda}_{q_i}^{(k)}\left(\Sigma_k + d_{ki}d_{ki}^\top\right)\tilde{\Lambda}_{q_i}^{(k)}\right]\Omega_{ki}, \tag{159}$$

where $\Lambda_{q_i} = \Sigma_i^{-1}$, $\tilde{\Lambda}_{q_i}^{(k)} := (\Omega_{ki}\Sigma_i\Omega_{ki}^\top)^{-1} = \Omega_{ki}^{-\top}\Lambda_{q_i}\Omega_{ki}^{-1}$ is agent i's precision transported into agent k's frame, and $d_{ki} := \Omega_{ki}\mu_i - \mu_k$ is the mean residual in agent k's frame. The rank-one piece $-\tilde{\Lambda}_{q_i}^{(k)}d_{ki}d_{ki}^\top\tilde{\Lambda}_{q_i}^{(k)}$ vanishes at consensus and contributes a stiffness aligned with d_{ki} away from it. The inverse-transpose form of Eq. (89) is what makes these expressions valid for general $\Omega_{ki} \in \mathrm{GL}^+(K_q)$, where Ω_{ki}^{-1} does not reduce to $\Omega_{ki}^\top$. Substituting into Eq. (156) gives the complete mean and

covariance gradients,

$$\begin{aligned}\nabla_{\mu_i}\mathcal{F}_{\mathrm{red}} &= \alpha_i^*\Sigma_{p,i}^{-1}(\mu_i-\mu_{p,i}) + \sum_j \beta_{ij}A_{ij}^{-1}(\mu_i-\Omega_{ij}\mu_j)\\ &\quad + \sum_k \beta_{ki}\Lambda_{q_i}\Omega_{ki}^{-1}\left(\Omega_{ki}\mu_i-\mu_k\right) + \nabla_{\mu_i}\mathcal{F}_{\mathrm{obs},i},\end{aligned} \tag{160}$$

$$\begin{aligned}\nabla_{\Sigma_i}\mathcal{F}_{\mathrm{red}} &= \frac{\alpha_i^*}{2}\left(\Sigma_{p,i}^{-1}-\Sigma_i^{-1}\right) + \frac{1}{2}\sum_j \beta_{ij}\left(A_{ij}^{-1}-\Sigma_i^{-1}\right)\\ &\quad + \frac{1}{2}\sum_k \beta_{ki}\Omega_{ki}^{\top}\left[\tilde{\Lambda}_{q_i}^{(k)} - \tilde{\Lambda}_{q_i}^{(k)}\left(\Sigma_k + d_{ki}d_{ki}^{\top}\right)\tilde{\Lambda}_{q_i}^{(k)}\right]\Omega_{ki} + \nabla_{\Sigma_i}\mathcal{F}_{\mathrm{obs},i},\end{aligned} \tag{161}$$

plus the model-channel terms. Eq. (160) is Eq. (100) with the sender index relabeled $k \to j$, and Eq. (100) gives the observation terms for a Gaussian identity readout.

Frame sector: the transported relative entropy. The frames enter only through the links. For Gaussians write $A := \Omega_{ij}\Sigma_j\Omega_{ij}^{\top}$, $r := \mu_i - \Omega_{ij}\mu_j$, and $\Lambda := A^{-1}$. The Gaussian relative entropy has three Ω_{ij}-dependent pieces ($\log|A|$, $\mathrm{tr}(\Lambda\Sigma_i)$, $r^{\top}\Lambda r$), and the group is $\mathrm{GL}^+(K_q)$, so the directional derivative lives in the full Lie algebra $\mathfrak{gl}(K_q)$ with no skew-symmetry constraint. Along $X \in \mathfrak{gl}(K_q)$ acting by $\Omega_{ij} \to e^{tX}\Omega_{ij}$, with $\partial_X A = XA + AX^{\top}$ and $\partial_X r = -X\Omega_{ij}\mu_j$, the derivative is

$$\partial_X E_{ij} = \mathrm{tr}\left[X^{\top}\Big(I - \Lambda\Sigma_i - \Lambda r(\Omega_{ij}\mu_j)^{\top} - \Lambda r r^{\top}\Big)\right], \tag{162}$$

where the leading identity is the log det contribution $\partial_X \log|A| = 2\,\mathrm{tr}(X)$ carried at the half weight of the Gaussian relative entropy, and the quadratic term uses $\Lambda A = I$ to reduce $\Lambda r r^{\top}\Lambda A$ to $\Lambda r r^{\top}$. Writing $N_{ij} := I - \Lambda\Sigma_i - \Lambda r(\Omega_{ij}\mu_j)^{\top} - \Lambda r r^{\top}$ for the bracket of Eq. (162), the unconstrained matrix derivative is

$$\frac{\partial E_{ij}}{\partial \Omega_{ij}} = N_{ij}\Omega_{ij}^{-\top} = \Omega_{ij}^{-\top} - \Lambda\Sigma_i\Omega_{ij}^{-\top} - \Lambda(\mu_i - \Omega_{ij}\mu_j)\mu_j^{\top} - \Lambda r r^{\top}\Lambda\Omega_{ij}\Sigma_j, \tag{163}$$

where the mean term carries a negative sign because r depends on Ω_{ij} through $-\Omega_{ij}\mu_j$, and the simplifications $(\Omega_{ij}\mu_j)^{\top}\Omega_{ij}^{-\top} = \mu_j^{\top}$ and $\Omega_{ij}^{-\top} = \Lambda\Omega_{ij}\Sigma_j$ reduce the remaining factors. At consensus, $q_i = \Omega_{ij}[q_j]$, the bracket N_{ij} vanishes. The compact subgroup $\mathrm{SO}(K_q)$ is the special case $X \in \mathfrak{so}(K_q) \subset \mathfrak{gl}(K_q)$ ($X^{\top} = -X$), in which the log det piece vanishes because $\mathrm{tr}(X) = 0$ and the gradient is obtained by projecting onto $\mathfrak{so}(K_q)$ through antisymmetrization.

Frame sector: gradient in the group variable. Since $\Omega_{ij} = U_i U_j^{-1}$, a variation of U_i moves the receiver link by $\delta\Omega_{ij} = (\delta U_i)U_j^{-1}$ and the sender link by $\delta\Omega_{ki} = -\Omega_{ki}(\delta U_i)U_i^{-1}$. Combining with Eq. (163) and the identities $\Omega_{ij}^{-\top}U_j^{-\top} = U_i^{-\top}$ and $\Omega_{ki}^{-\top}U_i^{-\top} = U_k^{-\top}$ gives

$$\nabla_{U_i}E_{ij} = N_{ij}U_i^{-\top}, \qquad \nabla_{U_i}E_{ki} = -\Omega_{ki}^{\top}N_{ki}U_k^{-\top}, \tag{164}$$

with N_{ki} the bracket of Eq. (162) for the pair (k,i). The complete frame gradient is therefore

$$\nabla_{U_i}\mathcal{F}_{\mathrm{red}} = \sum_j \beta_{ij}N_{ij}U_i^{-\top} - \sum_k \beta_{ki}\Omega_{ki}^{\top}N_{ki}U_k^{-\top}, \tag{165}$$

plus the model-channel terms of the same form when the model-fiber transport $\tilde{\Omega}_{ij}$ is generated by the same frames through ρ_{model} (tied frames); with independent model-fiber frames the model-channel gradient is taken with respect to those frames instead. Every term vanishes at consensus. In the frame of the receiving agent of each row, the body covector $U_i^\top \nabla_{U_i} \mathcal{F}_{\text{red}}$ reads $\sum_j \beta_{ij} U_i^\top N_{ij} U_i^{-\top} - \sum_k \beta_{ki} U_k^\top N_{ki} U_k^{-\top}$, so each pair contributes its bracket conjugated into the frame of the agent whose row contains it.

Frame sector: chart gradients. In the chart $U_i = \exp(\phi_i)$ with generators G_a, the chain rule gives $\partial \mathcal{F}_{\text{red}} / \partial \phi_i^a = \operatorname{tr}[(D \exp_{\phi_i}[G_a])^\top \nabla_{U_i} \mathcal{F}_{\text{red}}]$. For the receiver link the exact Fréchet derivative of the matrix exponential is

$$\frac{\partial \Omega_{ij}}{\partial \phi_i^a} = \left(\int_0^1 e^{(1-s)\phi_i} G_a e^{s\phi_i} \mathrm{d}s \right) e^{-\phi_j}, \tag{166}$$

and for the sender link, where ϕ_i enters through the right factor $\exp(-\phi_i)$,

$$\frac{\partial \Omega_{ki}}{\partial \phi_i^a} = -\exp(\phi_k) J_a, \qquad J_a := \int_0^1 e^{-(1-s)\phi_i} G_a e^{-s\phi_i} \mathrm{d}s, \tag{167}$$

so that

$$\frac{\partial E_{ij}}{\partial \phi_i^a} = \operatorname{tr}\left[\left(\frac{\partial \Omega_{ij}}{\partial \phi_i^a} \right)^\top N_{ij} \Omega_{ij}^{-\top} \right], \qquad \frac{\partial E_{ki}}{\partial \phi_i^a} = -\operatorname{tr}\left[J_a^\top (e^{\phi_k})^\top N_{ki} \Omega_{ki}^{-\top} \right]. \tag{168}$$

When G_a commutes with ϕ_i the integrals collapse, $\partial \Omega_{ij} / \partial \phi_i^a = G_a \Omega_{ij}$ and $\partial \Omega_{ki} / \partial \phi_i^a = -\Omega_{ki} G_a$, and the sender derivative becomes

$$\frac{\partial E_{ki}}{\partial \phi_i^a} = -\operatorname{tr}\left[G_a^\top \left(\Omega_{ki}^\top N_{ki} \Omega_{ki}^{-\top} \right) \right] \qquad \text{(abelian approximation)}, \tag{169}$$

while the receiver derivative becomes the projection of Eq. (162) along $X = G_a$. Used as an approximation for noncommuting frames, the abelian form is local and does not rely on compactness of G (the group $\mathrm{GL}^+(K_q)$ is noncompact); its relative error in $\partial \Omega / \partial \phi$ is about $0.37 \|\phi_i\|_F$ for small frames, reaching 1 percent at $\|\phi_i\|_F \approx 0.03$, 10 percent at $\|\phi_i\|_F \approx 0.3$, and about 40 percent at $\|\phi_i\|_F \approx 1.4$ (median over generic frames in $\mathfrak{gl}(3)$). The group-variable form Eq. (165) requires no Fréchet integral and is exact at every frame.

Natural Gradients and Frame Covariance

The natural gradient raises the index of the gradient with the Fisher metric of Appendix C.3: $\tilde{\nabla}_{\mu_i} \mathcal{F} = \Sigma_i \nabla_{\mu_i} \mathcal{F}$ on the mean and $\tilde{\nabla}_{\Sigma_i} \mathcal{F} = 2\Sigma_i (\nabla_{\Sigma_i} \mathcal{F}) \Sigma_i$ on $\mathcal{S}_{++}^{K_q}$.

Frame covariance. Under the congruence action of $g \in \mathrm{GL}(K_q)$, $\Sigma_i \mapsto g \Sigma_i g^\top$, which preserves the cone $\mathcal{S}_{++}^{K_q}$ for any invertible g, the Euclidean gradient $\nabla_{\Sigma_i} \mathcal{F}$ is a covector on the symmetric matrices and transforms by the inverse-transpose congruence, whereas the natural gradient transforms like Σ_i itself:

$$\Sigma_i \mapsto g \Sigma_i g^\top, \tag{170}$$

$$\nabla_{\Sigma_i} \mathcal{F} \mapsto g^{-\top} (\nabla_{\Sigma_i} \mathcal{F}) g^{-1}, \tag{171}$$

$$\tilde{\nabla}_{\Sigma_i} \mathcal{F} \mapsto g (\tilde{\nabla}_{\Sigma_i} \mathcal{F}) g^\top. \tag{172}$$

The two laws coincide if and only if g is orthogonal, $g^{-\top} = g$, that is $g \in \mathrm{O}(K_q)$, so for general g only the natural gradient is covariant on $\mathcal{S}_{++}^{K_q}$; the analogous statement for a constant-metric

Gaussian mean flow, that the Euclidean metric is equivariant only under orthogonal reframings, is proved in [38, Ch. G8]. Riemannian metrics on $\mathcal{S}_{++}^{K_q}$ invariant under the full congruence action form the two-parameter family $a\,\mathrm{tr}(\Sigma^{-1}U\Sigma^{-1}V)+b\,\mathrm{tr}(\Sigma^{-1}U)\,\mathrm{tr}(\Sigma^{-1}V)$ with $a>0$ and $a+K_qb>0$ [94]. The Gaussian Fisher metric on the covariance block is the member $a=\frac{1}{2}$, $b=0$ [38, Ch. G5], and $2\Sigma(\cdot)\Sigma$ is its gradient structure; the member $a=1$, $b=0$ would give $\Sigma(\cdot)\Sigma$. On the unit-determinant directions $\mathrm{tr}(\Sigma^{-1}U)=0$, which form the $\mathfrak{sl}(K_q)$ sector after whitening, the second term vanishes and all members coincide up to scale. The gradient structure $2\Sigma(\cdot)\Sigma$ is therefore one invariant choice among many, singled out by the Fisher metric.

Exponential-Map Update on the SPD Manifold

After computing $\tilde{\nabla}_{\Sigma_i}\mathcal{F}$, the covariance is updated through a retraction, a smooth map from the tangent bundle to the manifold whose differential at the zero vector is the identity [194], so that Σ_i remains positive definite:

$$\Sigma_i^{\mathrm{new}}=\mathcal{R}_{\Sigma_i}(-\eta_\Sigma\tilde{\nabla}_{\Sigma_i}\mathcal{F}). \tag{173}$$

The retraction used is the affine-invariant exponential map,

$$\mathcal{R}_\Sigma(\Delta)=\Sigma^{1/2}\exp\left(\Sigma^{-1/2}\Delta\Sigma^{-1/2}\right)\Sigma^{1/2}, \tag{174}$$

where $\exp(\cdot)$ is the matrix exponential. It preserves positive-definiteness, since the exponential of a symmetric matrix is SPD and congruence preserves the cone, and it is equivariant under congruence, $\mathcal{R}_{A\Sigma A^\top}(A\Delta A^\top)=A\mathcal{R}_\Sigma(\Delta)A^\top$ for any invertible A. With the eigendecomposition $\Sigma=VDV^\top$,

$$\mathcal{R}_\Sigma(\Delta)=VD^{1/2}\exp\left(D^{-1/2}V^\top\Delta VD^{-1/2}\right)D^{1/2}V^\top. \tag{175}$$

This is the Riemannian exponential map of the affine-invariant metric, so the update follows the geodesic from Σ in direction Δ for unit parameter.

The Entropy-Suppressed Surrogate

Replacing each row block by $S_k:=\sum_m\beta_{km}E_{km}$, with the Gibbs weights attached but the attention entropy removed, defines a different scalar S_{surr}. Applying the covariance identity of Section 5.6 to every row in which x_i appears gives

$$\nabla_{x_i}S_{\mathrm{surr}}=\nabla_{x_i}\mathcal{F}_{\mathrm{red}}-\frac{1}{\tau}\sum_k\mathrm{Cov}_{\beta_k}\left(E_{k\cdot},\nabla_{x_i}E_{k\cdot}\right), \tag{176}$$

where the sum runs over the receiver row $k=i$ and every sender row containing i. In a sender row only E_{ki} depends on x_i, and the covariance reduces to $\beta_{ki}(E_{ki}-\langle E\rangle_{\beta_k})\nabla_{x_i}E_{ki}$. In the receiver row the surrogate contribution is $\sum_j\beta_{ij}[1-(E_{ij}-\langle E\rangle_{\beta_i})/\tau]\nabla_{x_i}E_{ij}$. It reduces to the frozen-weight gradient whenever $\mathrm{Cov}_{\beta_i}(E,\nabla E)=0$, for example when all E_{ij} are equal; it tends to the π-weighted mean gradient as $\tau\to\infty$; and it tends to ∇E_{ij^*} as $\tau\to0$ when the minimizer j^* is unique, since for an energy gap $\Delta>0$ the response term is bounded by a multiple of $(\Delta/\tau)e^{-\Delta/\tau}$. The response is largest near ties, where energy differences are of order τ. The split of the same quantity into a direct piece weighted by $(1-E_{ij}/\tau)$ and a reweighting piece $\tau^{-1}\langle E\rangle_{\beta_i}\sum_m\beta_{im}\nabla E_{im}$ is algebraically equal but not centered: a common shift of all energies moves weight between the two pieces, each grows like E/τ as $\tau\to0$ while their sum stays bounded, and evaluating them separately cancels catastrophically at small τ. The centered form above avoids this. On the model fiber the same

statements hold with $(\gamma, \tilde{\Omega}, s, \tau_\gamma)$. For the precision weight, differentiating $\alpha_i^*(\mu_i, \Sigma_i) D_{\mathrm{KL}}(q_i \| p_i)$ without the regularizer of Eq. (54) gives the prefactor $(\alpha_i^*)^2 b_0 / c_0$ in place of α_i^* (Eq. (64)).

Discarding the sender rows defines a further object, the receiver-only field $V_i^{\mathrm{rec}}(x) := \partial_{x_i} S_i(x)$, obtained by dropping $\sum_{k \neq i} \partial_{x_i} S_k$. It is not the gradient of S_{surr} and is generically nonintegrable; the receiver and sender terms together form ∇S_{surr}, whose Hessian is symmetric. Neither the surrogate nor the receiver-only field is the update of the main text, which descends $\mathcal{F}_{\mathrm{red}}$ with the gradients of Eqs. (156)–(165).

Complete Update Equations

The update uses a Fisher-Rao natural-gradient step on the Gaussian sector and a left-trivialized group retraction on the frame sector (cf. Eqs. (82)–(85)). Every derivative below is the complete receiver-plus-sender envelope gradient of $\mathcal{F}_{\mathrm{red}}$:

$$\mu_i^{t+1} = \mu_i^t - \eta_\mu \Sigma_i^t \nabla_{\mu_i} \mathcal{F}_{\mathrm{red}}, \tag{177}$$

$$\Sigma_i^{t+1} = \left(\Sigma_i^t\right)^{1/2} \exp\left[\left(\Sigma_i^t\right)^{-1/2} \left(-\eta_\Sigma \tilde{\nabla}_{\Sigma_i} \mathcal{F}_{\mathrm{red}}\right) \left(\Sigma_i^t\right)^{-1/2}\right] \left(\Sigma_i^t\right)^{1/2}, \tag{178}$$

$$\xi_i^t = \left(U_i^t\right)^{-1} \nabla_{U_i} \mathcal{F}_{\mathrm{red}}, \tag{179}$$

$$U_i^{t+1} = U_i^t \exp\left(-\eta_\phi \xi_i^t\right). \tag{180}$$

Here ξ_i is the left-trivialized frame velocity of Eq. (83): the chart natural-gradient vector $\mathcal{G}_{\mathrm{pull}}^{-1} \nabla_{\phi_i} \mathcal{F}$ pushes forward to $\nabla_{U_i} \mathcal{F}$, so its left-trivialized velocity is exactly $U_i^{-1} \nabla_{U_i} \mathcal{F}$ when the generators span $\mathfrak{gl}(K_q)$, and its tangent projection on a subalgebra (Section 7), with $\nabla_{U_i} \mathcal{F}_{\mathrm{red}}$ given by Eq. (165). Evaluated in this form the velocity costs one $K_q \times K_q$ solve per agent; forming and inverting the $K_q^2 \times K_q^2$ pullback metric explicitly would cost $O(K_q^6)$ per agent and square its condition number. The alternative left-invariant Frobenius preconditioner replaces ξ_i by the body covector $U_i^\top \nabla_{U_i} \mathcal{F}_{\mathrm{red}}$. The body-covector update is equivariant under per-agent left translations with the beliefs co-transformed, and under right translations only by orthogonal g; the ambient-pullback update above is equivariant only under orthogonal translations. Neither descends to the quotient by the global right redundancy described below. A right-trivialized update would descend to that quotient at the cost of left equivariance, and no Riemannian metric on $\mathrm{GL}^+(K_q)$ is invariant under both translations [82]. The exponential map in the last line keeps $U_i \in G$ exactly. The learning rates follow the separation declared in Section 7, with $\eta_\mu \equiv \eta_q$ the belief-fiber rate written for the separately updated mean block.

Computational Complexity

Per gradient step with N agents and dimension K_q, forming the links costs $O(N K_q^3)$ for the frames and their inverses plus $O(N^2 K_q^3)$ for the pairwise products; the transported relative entropies require factorizations of the $O(N^2)$ transported covariances, $O(N^2 K_q^3)$ in total; and the natural-gradient steps and exponential-map updates, with the frame velocity evaluated as $U_i^{-1} \nabla_{U_i} \mathcal{F}_{\mathrm{red}}$, cost $O(K_q^3)$ per agent, $O(N K_q^3)$ in total. The overall complexity is $O(N^2 K_q^3)$ per step, dominated by the pairwise coupling.

Numerical Stability and Verification

Positive-definiteness in finite precision. In exact arithmetic the exponential-map update maps into $\mathcal{S}_{++}^{K_q}$ for any symmetric Δ, so $\Sigma_i \succ 0$ is preserved by construction. In single precision, with

unit roundoff $u = 2^{-24} \approx 5.96 \times 10^{-8}$, a symmetric matrix whose condition number exceeds about $u^{-1} \approx 1.68 \times 10^7$ can already be indefinite as stored, before any eigendecomposition is computed; an eigenvalue floor on the stored matrices, or storage of a factor, is what restores definiteness, and a higher-precision eigendecomposition does not. The same bound applies to the transported covariances $A_{ij} = \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ as well as to Σ_i, and the bound

$$\operatorname{cond}(A_{ij}) \le \operatorname{cond}(\Omega_{ij})^2 \operatorname{cond}(\Sigma_j) \tag{181}$$

holds, where $\operatorname{cond}(\Omega_{ij})$ grows exponentially with the frame norms, so a bound on the frame norms is needed alongside the eigenvalue floor. Cholesky factorization of an $n \times n$ matrix is backward stable independently of conditioning: the computed factor satisfies $R^\top R = \Sigma + \delta\Sigma$ with $|\delta\Sigma| \le \gamma_{n+1}|R^\top||R|$ componentwise, $\gamma_{n+1} = (n+1)u/(1-(n+1)u)$ [195, Thm. 10.3]. What depends on conditioning is completion and forward error. The factorization runs to completion if $\lambda_{\min}(H) > n\gamma_{n+1}/(1-\gamma_{n+1})$, where $H = D^{-1}\Sigma D^{-1}$ with $D = \operatorname{diag}(\Sigma)^{1/2}$ is the unit-diagonal scaling of Σ [195, Thm. 10.7]; since $\lambda_{\max}(H) \ge 1$, this holds in particular when $\operatorname{cond}_2(H)$ is below about $1/(n(n+1)u)$, which for $n = K_q = 10$ in single precision is about 1.5×10^5.

Right-translation redundancy. Under the global right translation $U_i \mapsto U_i g$ by a single $g \in G$ shared by all agents (Section 4.6.2), every link $\Omega_{ij} = U_i U_j^{-1} \mapsto U_i g g^{-1} U_j^{-1} = \Omega_{ij}$ is unchanged. The free energy and the belief updates, which depend on the frames only through the links, are therefore unchanged. The frame update is not: since $\nabla_{U_i g}\mathcal{F} = (\nabla_{U_i}\mathcal{F})g^{-\top}$, the velocity becomes $g^{-1}\xi_i g^{-\top}$, and $U_i g \exp(-\eta_\phi g^{-1}\xi_i g^{-\top})$ agrees with $U_i \exp(-\eta_\phi \xi_i) g$ for generic ξ_i only when $g^{-\top} = g$, that is for orthogonal g; the body-covector update behaves in the same way. Initial frames related by a nonorthogonal right translation, which describe the same links, therefore evolve to different links. In the chart, $U_i g$ corresponds to $\log(e^{\phi_i} e^{\zeta}) = \phi_i + \psi(\operatorname{ad}_{\phi_i})(\zeta) + O(\|\zeta\|^2)$ with $\zeta = \log g$ and $\psi(z) = z/(1-e^{-z}) = 1 + z/2 + z^2/12 + \cdots$. The additive shift $\phi_i \mapsto \phi_i + \zeta$ is therefore correct only to first order in both ϕ_i and ζ, or exactly in the abelian sector, and the group-level statement $U_i \mapsto U_i g$ is the one to use for the nonabelian redundancy.

Gradient verification. Every formula of this appendix was checked against central finite differences with step 10^{-6} in double precision, for $K_q \in \{2, 3\}$, three agents, and generic $\mathrm{GL}^+(K_q)$ frames with $\|\phi_i\|_F \approx 1.5$. The receiver, sender, and self gradients of the mean and covariance, the matrix derivative Eq. (163), the Fréchet integrals Eqs. (166) and (167), the exact chart and group-variable frame gradients, the complete envelope gradients, and the surrogate gradients with both precision conventions agree with the finite differences to relative error at most 10^{-7}; the directional derivative Eq. (162) agrees to 2.3×10^{-7}. The abelian approximation carries the error quantified above, not machine-precision error.

B Intuitive Examples and Framework Extensions

The derivations of Section B.1.1 are exact consequences of the Gaussian consensus functional under stated limits (status D); their reading as statements about rocks and observers has status S (Table 2). None of the quantities is identified with relativistic proper time, inertial mass, or the quantum-mechanical measurement formalism.

B.1 Macroscopic Objects as Meta-Agents: The Rock Example

To illustrate how the framework describes familiar objects, consider a rock. In the formalism a rock is not a primitive entity but a meta-agent with hierarchical structure.

A typical rock comprises approximately 10^{23} atomic constituents. Each atom is read as a level-0 agent maintaining beliefs about its local environment, primarily the positions and momenta of

neighboring atoms. Atoms in a solid are strongly coupled to their neighbors through electromagnetic interactions, which the framework represents by large attention weights $\beta_{\text{atom,neighbor}} \approx 1$. This strong coupling drives belief alignment: atoms in a crystalline lattice maintain highly coherent beliefs about their relative spatial configuration.

When these atomic agents achieve sufficient coherence (small divergences between transported beliefs, in the sense of the threshold detector of Section 9.3, which is a proxy for the licensing criterion of Eq. (125)), they form a level-1 meta-agent, the rock itself. This meta-agent carries a collective belief q_{rock} representing the consensus configuration of its constituents, a collective prior p_{rock} encoding the stable lattice structure of the material, and a collective frame U_{rock}, which by Proposition 5 is a declared convention, not an observable. The rock admits an induced metric G_{rock} by pullback of its collective belief section (Section 18.1); the commitment to universal agency of Section 1 does not ascribe phenomenal properties to this geometry.

The rock's Fisher information is large, and the derivation below shows that the pull of a low-precision neighbor on its mean is suppressed by the rock's own small covariance. Its relaxation rate toward its own prior is η_q, independent of its precision, because the Fisher metric and the stiffness of the prior term scale together; the insensitivity is a property of the coupling, not slow dynamics. In the reading of Section 8, large precision is large stiffness. Near its stationary state the rock's Fisher clock ϑ_{rock} (Section 17.1) advances little, because its belief changes little.

When a human observes the rock, the coupling $\beta_{\text{human,rock}}$ becomes nonzero. The coupling is not direct but is mediated by photons, electromagnetic quanta that couple strongly to both rock atoms (scattering) and human photoreceptors (absorption). The photons carry information about the rock's configuration to the observer, and the human agent's beliefs q_{human} update toward the transported rock beliefs $\Omega_{\text{human,rock}}[q_{\text{rock}}]$ through free-energy minimization. Within the framework, "seeing the rock" is read as agent-agent information coupling mediated by intermediate channels.

Observation is bidirectional: the rock's state also updates in response to the photons that scattered from it. The derivation below shows that its belief mean barely moves, while coupling to any agent lowers its stationary precision by a factor that does not tend to one in the high-precision limit.

B.1.1 Derivations: insensitivity to low-precision neighbors, cross-observer consensus, and back-action

The statements of the previous paragraphs follow from the natural-gradient flow on $\mathcal{F}$ together with the Gaussian relative-entropy gradients. Three derivations make this explicit.

Insensitivity to low-precision neighbors. Consider the single-agent free energy

$$\mathcal{F}_i = D_{\text{KL}}(q_i \| p_i) + \beta_{ij} D_{\text{KL}}(q_i \| \Omega_{ij} q_j), \tag{182}$$

with Gaussian $q_i = \mathcal{N}(\mu_i, \Sigma_i)$, $p_i = \mathcal{N}(\bar{\mu}_i, \bar{\Sigma}_i)$, and transported neighbor $\Omega_{ij} q_j = \mathcal{N}(\tilde{\mu}_j, \tilde{\Sigma}_{q_j})$. Differentiating the Gaussian relative entropy of Appendix C.2 with respect to μ_i gives

$$\nabla_{\mu_i} \mathcal{F}_i = \bar{\Lambda}_{p_i}(\mu_i - \bar{\mu}_i) + \beta_{ij} \tilde{\Lambda}_{q_j}(\mu_i - \tilde{\mu}_j), \tag{183}$$

where $\bar{\Lambda}_{p_i} = \bar{\Sigma}_i^{-1}$ and $\tilde{\Lambda}_{q_j} = \tilde{\Sigma}_{q_j}^{-1}$. The natural-gradient flow with Fisher preconditioner $g_{\mu\mu} = \Sigma_i^{-1}$ (Appendix C.3) gives the mean update

$$\dot{\mu}_i = -\eta_q \Sigma_i \nabla_{\mu_i} \mathcal{F}_i = -\eta_q \Sigma_i \left[\bar{\Lambda}_{p_i}(\mu_i - \bar{\mu}_i) + \beta_{ij} \tilde{\Lambda}_{q_j}(\mu_i - \tilde{\mu}_j) \right]. \tag{184}$$

For a rock with $\Sigma_i = \bar{\Sigma}_i = \delta I$, where $\delta \ll 1$ is the dimensionless precision scale of Section 8, to which no physical units are assigned ($\bar{\Lambda}_{p_i} = \delta^{-1} I$), and a low-precision observer agent j with $\tilde{\Lambda}_{q_j} = O(1)$, substitution yields

$$\dot{\mu}_i = -\eta_q(\mu_i - \bar{\mu}_i) - \eta_q \delta \beta_{ij} \tilde{\Lambda}_{q_j}(\mu_i - \tilde{\mu}_j) = -\eta_q(\mu_i - \bar{\mu}_i) + O(\delta). \tag{185}$$

The prior-anchor term is $O(1)$ and survives; the coupling to the low-precision observer is $O(\delta)$ and vanishes as $\delta \to 0$. The rock's mean is pinned to its own sharp prior at leading order, while its relaxation rate toward that prior remains η_q: the Fisher preconditioner Σ_i in Eq. (184) converts the large precision of the rock into a small response to coupling terms whose own precision is of order one [83].

Cross-observer consensus via photon-mediated coupling. Consider three kinds of agent: the rock R with precision $\Lambda_R = \delta^{-1}$, a photon channel ph, and human observers h_k. Under the scattering event the channel couples to R with weight $\beta_{\mathrm{ph},R}\tilde{\Lambda}_{q_R} \gg \bar{\Lambda}_{\mathrm{ph}}$; setting $\dot{\mu}_{\mathrm{ph}} = 0$ in Eq. (183) gives

$$\mu_{\mathrm{ph}} = \big(\bar{\Lambda}_{\mathrm{ph}} + \beta_{\mathrm{ph},R}\tilde{\Lambda}_{q_R}\big)^{-1}\big(\bar{\Lambda}_{\mathrm{ph}}\bar{\mu}_{\mathrm{ph}} + \beta_{\mathrm{ph},R}\tilde{\Lambda}_{q_R}\Omega_{\mathrm{ph},R}\mu_R\big) \to \Omega_{\mathrm{ph},R}\mu_R, \tag{186}$$

the limit holding when $\beta_{\mathrm{ph},R}\tilde{\Lambda}_{q_R} \gg \bar{\Lambda}_{\mathrm{ph}}$. Each human observer h_k subsequently equilibrates against the channel in the absorption-dominated regime $\beta_{h_k,\mathrm{ph}}\tilde{\Lambda}_{q_{\mathrm{ph}}} \gg \bar{\Lambda}_{h_k}$, giving $\mu_{h_k} \to \Omega_{h_k,\mathrm{ph}}\mu_{\mathrm{ph}}$. Composing through the channel,

$$\mu_{h_k} \to \Omega_{h_k,\mathrm{ph}}\Omega_{\mathrm{ph},R}\mu_R = \Omega_{h_k R}\mu_R, \tag{187}$$

where the last equality is the transport composition rule $\Omega_{ik} = \Omega_{ij}\Omega_{jk}$, which follows from $\Omega_{ij} = e^{\phi_i}e^{-\phi_j}$ by cancellation of the middle factor $e^{-\phi_{\mathrm{ph}}}e^{\phi_{\mathrm{ph}}} = I$ (cf. Eq. (14)). This cancellation is exact for frame-generated links, consistent with the cocycle identity of Lemma 1; loop holonomy enters only for the independent links on the interaction graph of Section 4.6.5 and does not affect this equation. Two human observers h_1, h_2 therefore both equilibrate to the same transported rock state, each expressed in its own frame. This is the framework analog of cross-observer agreement on the rock's appearance. Classically the mechanism is precision-weighted opinion pooling [129]; its closest relative in quantum foundations is quantum Darwinism, in which objectivity arises from redundant records of a system carried by its environment, such as scattered photons [45].

Back-action in the rock limit. The rock's update under coupling to a low-precision human contributes through the coupling term only:

$$\dot{\mu}_R^{(\mathrm{back})} = -\eta_q \Sigma_R \beta_{Rh}\tilde{\Lambda}_{q_h}(\mu_R - \Omega_{Rh}\mu_h) = O(\delta), \tag{188}$$

since $\Sigma_R = \delta I$ and the remaining factor is $O(1)$. For the covariance sector the natural gradient is $\tilde{\nabla}_{\Sigma_R}\mathcal{F} = 2\Sigma_R(\nabla_{\Sigma_R}\mathcal{F})\Sigma_R$ (Appendix C.3). The coupling contribution

$$\nabla_{\Sigma_R}\mathcal{F}_{\mathrm{coupling}} = \tfrac{1}{2}\beta_{Rh}\big(\tilde{\Lambda}_{q_h} - \Lambda_R\big) \tag{189}$$

produces

$$\begin{aligned}\dot{\Sigma}_R^{(\mathrm{back})} &= -2\eta_q\Sigma_R \cdot \tfrac{1}{2}\beta_{Rh}\big(\tilde{\Lambda}_{q_h} - \Lambda_R\big)\cdot\Sigma_R \\ &= -\eta_q\beta_{Rh}\delta^2\big(\tilde{\Lambda}_{q_h} - \delta^{-1}I\big) = \eta_q\beta_{Rh}\delta\big(I - \delta\tilde{\Lambda}_{q_h}\big).\end{aligned} \tag{190}$$

The two back-actions have different sizes in the rock's own geometry. The mean back-action is $O(\delta)$ in absolute terms and $O(\delta^{1/2})$ in the rock's Fisher metric, since a mean displacement $\Delta\mu$ has Fisher

length $|\Delta\mu|/\delta^{1/2}$. The covariance back-action is $O(\delta)$ in absolute terms but of order one relative to Σ_R: $\dot{\Sigma}_R^{(\mathrm{back})}\Sigma_R^{-1} \to \eta_q\beta_{Rh}I$ as $\delta \to 0$. At stationarity of $\mathcal{F}_R = D_{\mathrm{KL}}(q_R\|p_R) + \beta_{Rh}D_{\mathrm{KL}}(q_R\|\Omega_{Rh}q_h)$, the covariance condition $\frac{1}{2}(\bar{\Lambda}_{p_R} - \Lambda_R) + \frac{1}{2}\beta_{Rh}(\tilde{\Lambda}_{q_h} - \Lambda_R) = 0$ gives

$$\Lambda_R^* = \frac{\bar{\Lambda}_{p_R} + \beta_{Rh}\tilde{\Lambda}_{q_h}}{1 + \beta_{Rh}}, \tag{191}$$

so with $\bar{\Lambda}_{p_R} = \delta^{-1}I$ the stationary precision is $\delta^{-1}(I + \delta\beta_{Rh}\tilde{\Lambda}_{q_h})/(1 + \beta_{Rh})$, reduced relative to the prior precision by a factor that tends to $1/(1+\beta_{Rh})$ as $\delta \to 0$. The Fisher-Rao distance between the uncoupled and the coupled rock covariance tends to $\sqrt{K_q/2}\log(1 + \beta_{Rh})$, which does not vanish in the high-precision limit. The cause is that each relative-entropy term contains $-H[q_R]$, so coupling to any agent reweights the rock's entropy. Precision dominance therefore suppresses belief-mean back-action but not precision back-action.

Asymptotic reading. The derivations above are stated in the $\delta \to 0$ rock limit. At finite δ the coupling pull on μ_R and the mean back-action are nonzero at $O(\delta)$, while the precision back-action remains of order one at every δ. The claim is that this scaling sets the regime in which insensitivity to low-precision neighbors, consensus, and negligible mean back-action appear, not that these quantities vanish at finite δ.

Within this interpretive extension the example supports three readings: observation is ordinary agent-agent coupling, not a special process; the observer-observed distinction is conventional, not ontological, since both are agents coupled through information exchange; and macroscopic objects are meta-agents formed from microscopic constituents through consensus. The derivation adds one framework-internal result, that precision dominance protects a high-precision agent's belief mean but not its precision.

C Mathematical Details

C.1 Variational Free Energy Preliminaries

Given a generative model $p(o, x)$ over observations o and latent states x, exact inference by Bayes' rule $p(x \mid o) = p(o, x)/p(o)$ is intractable because the evidence $p(o) = \int p(o, x)\mathrm{d}x$ requires integrating over all latent states. For any approximate posterior $q(x)$ the variational free energy is an upper bound on the surprisal,

$$\begin{aligned} F[q] &= \mathbb{E}_{q(x)}[\log q(x) - \log p(o, x)] \\ &= D_{\mathrm{KL}}(q(x)\|p(x \mid o)) - \log p(o) \geq -\log p(o), \end{aligned} \tag{192}$$

the inequality following from non-negativity of the relative entropy; minimizing F over q simultaneously tightens the evidence lower bound $\log p(o) \geq -F[q]$ and drives $D_{\mathrm{KL}}(q\|p(\cdot \mid o)) \to 0$. The measure-theoretic form of this identity, with its integrability conditions, is the classical finite evidence identity of [30, Ch. 5]. The free energy decomposes into a complexity term and an accuracy term,

$$F[q] = D_{\mathrm{KL}}(q(x)\|p(x)) - \mathbb{E}_{q(x)}[\log p(o \mid x)], \tag{193}$$

the first penalizing departure from the prior and the second rewarding fit to observations [17, 19].

C.2 Gaussian KL Divergence

For multivariate Gaussian distributions $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$ in $\mathbb{R}^{K_q}$, the Kullback-Leibler divergence is

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left[\log\frac{|\Sigma_p|}{|\Sigma_q|} + \mathrm{tr}(\Sigma_p^{-1}\Sigma_q) + (\mu_p - \mu_q)^\top \Sigma_p^{-1}(\mu_p - \mu_q) - K_q\right], \tag{194}$$

the Gaussian reverse relative entropy of [38, Ch. G4]. This formula is used throughout for the self and alignment terms of the consensus functional.

C.3 Fisher Information Matrix for Gaussians

The Fisher information matrix for Gaussian distributions parametrized by $\theta = (\mu, \Sigma)$ has the block structure

$$F = \begin{pmatrix} \Sigma^{-1} & 0 \\ 0 & \frac{1}{2}(\Sigma^{-1} \otimes \Sigma^{-1}) \end{pmatrix}, \tag{195}$$

and every affine map $(\mu, \Sigma) \mapsto (Q\mu + v, Q\Sigma Q^\top)$ with $Q \in \mathrm{GL}(K_q)$ is an isometry of it [38, Ch. G5, Fisher metric in the moment chart]. The covariance block is to be read as the Fisher quadratic form restricted to symmetric perturbations, $g(d\Sigma, d\Sigma) = \frac{1}{2}\,\mathrm{tr}(\Sigma^{-1}d\Sigma\Sigma^{-1}d\Sigma)$. As an operator on symmetric matrices, $X \mapsto \frac{1}{2}\Sigma^{-1}X\Sigma^{-1}$ has inverse $X \mapsto 2\Sigma X\Sigma$, so the natural gradient is

$$\tilde{\nabla}_\mu \mathcal{F} = \Sigma(\nabla_\mu \mathcal{F}), \tag{196}$$

$$\tilde{\nabla}_\Sigma \mathcal{F} = 2\Sigma(\nabla_\Sigma \mathcal{F})\Sigma, \tag{197}$$

where $\nabla_\mu \mathcal{F}$ and $\nabla_\Sigma \mathcal{F}$ are the standard Euclidean gradients, with $\nabla_\Sigma \mathcal{F}$ the symmetric matrix gradient. For the non-redundant $\mathrm{vech}(\Sigma)$ parameterization the matrix representation of this block is $\frac{1}{2}\mathcal{D}^\top(\Sigma^{-1} \otimes \Sigma^{-1})\mathcal{D}$ with $\mathcal{D}$ the duplication matrix, and its matrix inverse is not simply $2(\Sigma \otimes \Sigma)$; the inverse as an operator on symmetric matrices, and hence the displayed natural gradient, is unchanged.

C.4 SO(3) Generators and Transport Operators

For the compact group $G = \mathrm{SO}(3)$ acting through its irreducible representation of spin j, the belief dimension is $K_q = 2j + 1$, and the generators $\{G_x, G_y, G_z\}$ are real skew-symmetric $K_q \times K_q$ matrices satisfying

$$[G_a, G_b] = \epsilon_{abc} G_c. \tag{198}$$

Transport operators are computed by the matrix exponential,

$$\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j), \tag{199}$$

where $\phi_i = \phi_i^x G_x + \phi_i^y G_y + \phi_i^z G_z \in \mathfrak{so}(3)$ in the spin-j representation.

The product $\exp(\phi_i)\exp(-\phi_j)$ may be replaced by the first-order Baker-Campbell-Hausdorff truncation $\exp(\phi_i - \phi_j)$, with error controlled by the leading commutator, $\Omega_{ij} = \exp\big(\phi_i - \phi_j - \frac{1}{2}[\phi_i, \phi_j] + \mathcal{O}(\|\phi\|^3)\big)$. The truncated transport is not of the form $U_i U_j^{-1}$: it breaks the cocycle identity, so its holonomy around a triangle of agents is of the order of $\|[\phi_i, \phi_j]\|$, and it is not invariant under the right translation $U_i \mapsto U_i g$. The vanishing field strength of Regime I (Lemma 1) holds for the exact two-exponential product only. The truncation is acceptable only when the frames are small enough that this holonomy is negligible for the purpose at hand; otherwise the exact product is used.

C.5 Numerical Stability and Cholesky Parametrization

Covariance matrices Σ_i must remain positive definite throughout optimization. Two parameterizations of the covariance sector achieve this, and they differ in their dynamics. The first is the Cholesky parametrization

$$\Sigma_i = L_i L_i^\top, \tag{200}$$

where L_i is lower triangular with positive diagonal entries and gradients are taken with respect to L_i. Any lower-triangular L_i gives a positive-semidefinite Σ_i, but Σ_i is positive definite only when every diagonal entry of L_i is nonzero, and a Euclidean gradient step on L_i can drive a diagonal entry through zero. Strict positive-definiteness is therefore enforced by writing the diagonal through a diffeomorphism of the real line onto the positive half-line, such as softplus or the exponential, so that the stored degrees of freedom are unconstrained while the resulting diagonal entries cannot cross zero in exact arithmetic. The absolute value is not admissible: it vanishes at zero, is not differentiable there, and is two-to-one, so the map from stored parameters to $\mathcal{S}_{++}^{K_q}$ would not be a diffeomorphism. The second parameterization is the Fisher-Rao natural-gradient step with the affine-invariant exponential retraction of Eq. (82) and Appendix C.3.

With a diffeomorphic positive map on the diagonal, the map from the stored parameters to $\mathcal{S}_{++}^{K_q}$ is a diffeomorphism, so the chain-rule gradient with respect to L_i vanishes exactly when $\partial\mathcal{F}/\partial\Sigma_i = 0$ and the two schemes share stationary points. They do not share trajectories, since the Euclidean step on L_i induces a different metric on the covariance sector, and they do not share discrete stability. For the model free energy $\frac{1}{2}[\mathrm{tr}(M\Sigma) - 2\log\det\Sigma]$ with stationary point $\Sigma^* = 2M^{-1}$, the whitened linearization of the exponential-map update is multiplication by $1-2\eta_\Sigma$, so the update is linearly stable for $\eta_\Sigma < 1$ independently of conditioning; a Euclidean step on Σ is stable only for $\eta_\Sigma < 2\lambda_{\min}(\Sigma^*)^2$, and a Euclidean step on L, with Σ^* normalized to $\lambda_{\max}(\Sigma^*) = 1$, only for a step of order $\lambda_{\min}(\Sigma^*)$. Positive-definiteness, not step-size stability, is what the Cholesky parametrization guarantees. Under the Cholesky-Euclidean scheme the covariance relaxation rate also depends on conditioning, so a fixed ratio of learning rates does not fix the ratio of physical relaxation times. Trajectory-level natural-gradient statements therefore apply to the retraction path only; results obtained under the Cholesky-Euclidean scheme are stationarity-equivalent but follow different descent paths.

D Covariance Dynamics and Equilibrium Analysis

Scope of this appendix. The analysis below is a local, fixed-attention, receiver-only surrogate. We treat the attention weights β_{ij} as exogenous constants when differentiating with respect to Σ_i, vary only the receiver covariance, and never differentiate the sender covariances Σ_j or the weights β_{ij} themselves. The full reduced functional, obtained by envelope-eliminating β_{ij} via its entropy-regularized softmax optimum and including the complete sender-row contributions, is not derived here; no explicit first-derivative $\partial\beta_{ij}$ response survives in that canonical functional. The fixed point characterized below is therefore a partial Euler-Lagrange stationarity condition, not a stationary point of the fully reduced free energy.

The covariance fixed point is also a property of the consensus potential of Section 5.2, which weights the belief entropy of agent i by $1 + B_i$ with $B_i := \sum_j \beta_{ij}$. Minimizing $\mathbb{E}_q[U] - (1+B_i)H(q)$ over laws q gives $q \propto e^{-U/(1+B_i)}$, so each belief is a Gibbs state at temperature $1+B_i = 2$ in units in which Bayesian updating has temperature one. In a normalized Gaussian joint the entropy enters once, and the mean-field covariance is the inverse diagonal precision block, fixed in one step and

independent of the neighbors' covariances [38, Ch. G4, block optimum]. The factor $\frac{1}{2}$ in Eq. (206) and the homogeneous limit $\Sigma_\infty = \Sigma_0$ below both come from this doubled entropy weight.

D.1 Covariance Gradient of the Per-Agent Block

Here we derive the gradient of agent i's block $\mathcal{F}_i$ of the consensus functional with respect to its covariance Σ_i, a standard computation for Gaussian families [83, 196].

The block is

$$\mathcal{F}_i = D_{\mathrm{KL}}(q_i \| p_i) + \sum_{j \neq i} \beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)], \tag{201}$$

where $q_i = \mathcal{N}(\mu_i, \Sigma_i)$, $p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i})$, and $\sum_j \beta_{ij} = 1$ by construction.

D.1.1 Gaussian relative entropy and its derivative

Differentiating the Gaussian relative entropy of Appendix C.2, with first argument $\mathcal{N}(\mu_1, \Sigma_1)$ and second argument $\mathcal{N}(\mu_2, \Sigma_2)$, with respect to Σ_1 at fixed μ_1, μ_2, Σ_2 gives

$$\frac{\partial D_{\mathrm{KL}}}{\partial \Sigma_1} = \frac{1}{2}\left[-\Sigma_1^{-1} + \Sigma_2^{-1}\right]. \tag{202}$$

Applying this to each term in $\mathcal{F}_i$:

$$\frac{\partial}{\partial \Sigma_i} D_{\mathrm{KL}}(q_i \| p_i) = \frac{1}{2}\left[-\Sigma_i^{-1} + \Sigma_{p,i}^{-1}\right], \tag{203}$$

$$\frac{\partial}{\partial \Sigma_i} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) = \frac{1}{2}\left[-\Sigma_i^{-1} + (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1}\right]. \tag{204}$$

The observation term, for the Gaussian likelihood $-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)] = \frac{1}{2}(o_i - \mu_i)^\top \Lambda_{o_i}(o_i - \mu_i) + \frac{1}{2}\mathrm{tr}(\Lambda_{o_i}\Sigma_i) + \text{const}$, contributes a constant precision correction $\partial[-\mathbb{E}_{q_i} \log p(o_i \mid k_i)]/\partial \Sigma_i = \frac{1}{2}\Lambda_{o_i}$, independent of Σ_i. Including it,

$$\boxed{\frac{\partial \mathcal{F}_i}{\partial \Sigma_i} = \frac{1}{2}\left[-2\Sigma_i^{-1} + \Sigma_{p,i}^{-1} + \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} + \Lambda_{o_i}\right].} \tag{205}$$

Because $\sum_j \beta_{ij} = 1$, the prefactor $(1 + \sum_j \beta_{ij})$ in front of Σ_i^{-1} reduces to the constant 2, as displayed in the boxed gradient.

D.2 Fixed-Point Equation and Symmetric Solution

At equilibrium, we set $\partial \mathcal{F}_i / \partial \Sigma_i = 0$, giving

$$\Sigma_i^{-1} = \frac{1}{2}\left[\Sigma_{p,i}^{-1} + \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} + \Lambda_{o_i}\right]. \tag{206}$$

The sensory precision Λ_{o_i} enters with the same prefactor as the other precision contributions; setting $\Lambda_{o_i} = 0$ gives the observation-free case used in the homogeneous limit below. Equation (206) couples all agents: each agent's precision is half the sum of its prior and sensory precision and the β-weighted transported neighbor precisions.

Precision coordinates. Write $\Lambda_i := \Sigma_i^{-1}$ for the belief precision, $A_i := \Sigma_{p,i}^{-1} + \Lambda_{o_i} \succ 0$ for the prior-and-sensory anchor, and define the linear map

$$\mathcal{L}(\Lambda)_i := \sum_j \beta_{ij} \Omega_{ij}^{-\top} \Lambda_j \Omega_{ij}^{-1} \tag{207}$$

on the space $\mathrm{Sym}(K)^N$ of N-tuples of symmetric matrices. Since $(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} = \Omega_{ij}^{-\top}\Lambda_j\Omega_{ij}^{-1}$, Eq. (206) is linear in precision coordinates,

$$2\Lambda = A + \mathcal{L}(\Lambda), \tag{208}$$

and the covariance fixed-point map

$$T(\Sigma)_i = 2\big(\Sigma_{p,i}^{-1} + \sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} + \Lambda_{o_i}\big)^{-1} \tag{209}$$

is, in precision coordinates, the affine map $\Lambda \mapsto \frac{1}{2}(A+\mathcal{L}(\Lambda))$. Each term of $\mathcal{L}$ is a congruence with a nonnegative weight, so $\mathcal{L}$ carries the product cone of positive semidefinite tuples into itself. Under a change of frames, $\Lambda_i \mapsto g_i^{-\top}\Lambda_i g_i^{-1}$ and $\Omega_{ij} \mapsto g_i\Omega_{ij}g_j^{-1}$, the map $\mathcal{L}$ transforms by similarity, so its spectral radius $\rho(\mathcal{L})$ is frame independent.

Proposition 12 (Frozen-attention covariance fixed point)**.** *Fix $\beta_{ij} \geq 0$ and $\Omega_{ij} \in \mathrm{GL}(K)$, and assume $\Sigma_{p,i} \succ 0$ and $\Lambda_{o_i} \succeq 0$.*

(i) Equation (206) has a solution with every $\Sigma_i \succ 0$ if and only if $\rho(\mathcal{L}) < 2$. The solution is then unique, $\Lambda^* = \frac{1}{2}\sum_{k\geq 0}(\mathcal{L}/2)^k A$, and the iteration of T converges to it geometrically with asymptotic ratio $\rho(\mathcal{L})/2$.

(ii) If $\rho(\mathcal{L}) < 2$, the simultaneous receiver-only natural-gradient flow of all agents, $d\Sigma_i/dt = -2\eta_\Sigma\Sigma_i(\partial\mathcal{F}_i/\partial\Sigma_i)\Sigma_i$ with $\eta_\Sigma > 0$, remains in the SPD cone and converges to Λ^* from every SPD initial condition, at the exponential rate $2\eta_\Sigma(1-\rho(\mathcal{L})/2)$.

(iii) If $\rho(\mathcal{L}) \geq 2$, no SPD fixed point exists, and along the same flow the precision grows without bound, so that some covariance direction collapses.

(iv) $\rho(\mathcal{L}) \leq \max_i \lambda_{\max}(C_i)$ with $C_i := \sum_j \beta_{ij}(\Omega_{ij}\Omega_{ij}^\top)^{-1} = \mathcal{L}(I)_i$. If every row sums to one and a change of frames makes every link orthogonal, in particular if they are graph-flat, $\Omega_{ij} = U_iU_j^{-1}$, then $\rho(\mathcal{L}) = 1$.

Proof. With the boxed gradient, $\partial\mathcal{F}_i/\partial\Sigma_i = \frac{1}{2}(A_i + \mathcal{L}(\Lambda)_i - 2\Lambda_i)$, and $d\Lambda_i/dt = -\Lambda_i(d\Sigma_i/dt)\Lambda_i$, so the flow of (ii) is the linear system

$$\frac{d\Lambda}{dt} = \eta_\Sigma\big(A + \mathcal{L}(\Lambda) - 2\Lambda\big). \tag{210}$$

For (i), if $\rho(\mathcal{L}) < 2$ the Neumann series converges, $2I - \mathcal{L}$ is invertible, and $\Lambda^* \succeq A/2 \succ 0$ because every term of the series is positive semidefinite; the iteration error is $(\mathcal{L}/2)^n(\Lambda_0 - \Lambda^*)$. Conversely, let $\Lambda \succ 0$ solve Eq. (208), and choose $\epsilon > 0$ with $A_i \succeq \epsilon\Lambda_i$ for every i. Then $\mathcal{L}(\Lambda) = 2\Lambda - A \preceq (2-\epsilon)\Lambda$. In the order-unit norm $\|X\|_\Lambda := \min\{s \geq 0 : -s\Lambda \preceq X \preceq s\Lambda\}$, which is a norm because Λ is interior to the cone, cone preservation gives $\|\mathcal{L}(X)\|_\Lambda \leq (2-\epsilon)\|X\|_\Lambda$, hence $\rho(\mathcal{L}) \leq 2 - \epsilon < 2$. The same argument with $\Lambda = I$ and $\mathcal{L}(I) \preceq \max_i \lambda_{\max}(C_i)I$ proves the bound in (iv). For (ii), variation of constants gives

$$\Lambda(t) = e^{-2\eta_\Sigma t}e^{\eta_\Sigma t\mathcal{L}}\Lambda(0) + \eta_\Sigma\int_0^t e^{-2\eta_\Sigma s}e^{\eta_\Sigma s\mathcal{L}}(A)\mathrm{d}s.$$

The exponential $e^{s\mathcal{L}} = \sum_k (s\mathcal{L})^k/k!$ preserves the cone and satisfies $e^{s\mathcal{L}}A \succeq A$, so $\Lambda(t) \succeq (1 - e^{-2\eta_\Sigma t})A/2$ plus a positive semidefinite term, and $\Lambda(t)$ stays SPD. The deviation is $\Lambda(t) - \Lambda^* = e^{-\eta_\Sigma(2I-\mathcal{L})t}(\Lambda(0) - \Lambda^*)$. Every eigenvalue of $2I - \mathcal{L}$ has real part at least $2 - \rho(\mathcal{L})$, with equality at $2 - \rho(\mathcal{L})$ itself, because the spectral radius of a cone-preserving map is an eigenvalue [197]. For (iii), the same theorem applied to the adjoint $\mathcal{L}^*(W)_j = \sum_i \beta_{ij}\Omega_{ij}^{-1}W_i\Omega_{ij}^{-\top}$, which preserves the positive semidefinite cone (self-dual under $\langle X, Y\rangle = \sum_i \mathrm{tr}(X_iY_i)$), supplies $W \succeq 0$, $W \neq 0$, with $\mathcal{L}^*W = \rho(\mathcal{L})W$. Pairing a fixed point with W gives $(2 - \rho(\mathcal{L}))\langle W, \Lambda\rangle = \langle W, A\rangle > 0$, which is impossible for $\rho(\mathcal{L}) \geq 2$ and $\Lambda \succ 0$. Along Eq. (210), $d\langle W, \Lambda\rangle/dt = \eta_\Sigma(\langle W, A\rangle + (\rho(\mathcal{L}) - 2)\langle W, \Lambda\rangle) \geq \eta_\Sigma\langle W, A\rangle > 0$, since $\Lambda(t)$ stays positive semidefinite, so $\langle W, \Lambda(t)\rangle$ grows at least linearly in t. For the second part of (iv), if frames g_i make every link orthogonal, then in those frames $\mathcal{L}(I)_i = \sum_j \beta_{ij}I = I$, so I is an interior eigenvector with eigenvalue one and the bound gives $\rho(\mathcal{L}) = 1$; similarity carries this back to the original frames. For graph-flat links the frames $g_i = U_i^{-1}$ make every link the identity. □

Frame dependence and holonomy. A Thompson-metric contraction argument for the map T [198,199] yields the same fixed point under the hypothesis $\lambda_{\max}(C_i) < 2$ for every i. By part (iv) that hypothesis is sufficient, but it is not frame independent, because C_i picks up the factor $g_j^{-1}g_j^{-\top}$ under a change of frames. For graph-flat $\mathrm{GL}(K)$ links it can fail by a large margin while $\rho(\mathcal{L}) = 1$: for three agents with frames $U = \mathrm{diag}(10, 1)$, I, and $\left(\begin{smallmatrix} 1 & 3 \\ 0 & 1/2 \end{smallmatrix}\right)$, one finds $\max_i \lambda_{\max}(C_i) \approx 610$, while the fixed point exists and is globally stable. The value of $\rho(\mathcal{L})$ departs from one only through loop holonomy whose holonomy group is not conjugate into $\mathrm{O}(K)$, and such holonomy can push it past two. For $K = 1$ and two agents with $\beta_{12} = \beta_{21} = 1$ and independent links $w = \Omega_{12}$, $v = \Omega_{21}$, the map is $\mathcal{L} = \left(\begin{smallmatrix} 0 & w^{-2} \\ v^{-2} & 0 \end{smallmatrix}\right)$ with $\rho(\mathcal{L}) = 1/h$, where $h = wv$ is the backtrack holonomy; no SPD fixed point exists for $h \leq 1/2$, and the iterated covariances decay to zero. With orientation reversal $\Omega_{ji} = \Omega_{ij}^{-1}$ imposed, as for the independent links of Regime II, a two-cycle carries no holonomy, but the obstruction reappears on a triangle: with rows $\beta_{ij} = 1/2$ and holonomy h around the loop, the characteristic polynomial of $\mathcal{L}$ is $\lambda^3 - \frac{3}{4}\lambda - (h^2 + h^{-2})/8$, and $\rho(\mathcal{L}) \geq 2$ exactly when $h^2 + h^{-2} \geq 52$.

D.2.1 Homogeneous limit

In the homogeneous limit we assume that (i) all agents are identical, so $\Sigma_i = \Sigma_\infty$ for all i; (ii) every link preserves the shared prior covariance, $\Omega_{ij}\Sigma_0\Omega_{ij}^\top = \Sigma_0$ (for instance $\Omega_{ij} = I$); (iii) $\Sigma_{p,i} = \Sigma_0$; and (iv) $\Lambda_{o_i} = 0$, the observation-free case. Under (i)-(iv), (206) becomes

$$\Sigma_\infty^{-1} = \frac{1}{2}\left[\Sigma_0^{-1} + \Sigma_\infty^{-1}\right] \quad \Longrightarrow \quad \Sigma_\infty = \Sigma_0, \tag{211}$$

and the solution is unique by Proposition 12, since links preserving Σ_0 are orthogonal in the common frame $\Sigma_0^{-1/2}$. With a common sensory precision $\Lambda_{o_i} = \Lambda_o > 0$ and links that also preserve $\Sigma_0^{-1} + \Lambda_o$, the same computation gives $\Sigma_\infty = (\Sigma_0^{-1} + \Lambda_o)^{-1}$.

Hence, in a symmetric observation-free population, the equilibrium covariance reproduces the shared prior. This reflects the doubled entropy weight of the consensus potential: a normalized Gaussian joint with prior precision Σ_0^{-1} and edge precision Σ_0^{-1} gives the mean-field covariance $\Sigma_0/2$, which is precision addition [38, Ch. G4, block optimum].

D.2.2 Alignment-dominated regime

Although $\sum_j \beta_{ij} = 1$, one might expect the effective strength of alignment to be controlled by the attention temperature τ in

$$\beta_{ij} = \frac{\exp\left[-\frac{1}{\tau} D_{\mathrm{KL}}\big(q_i \| \Omega_{ij} q_j\big)\right]}{\sum_k \exp\left[-\frac{1}{\tau} D_{\mathrm{KL}}\big(q_i \| \Omega_{ik} q_k\big)\right]}. \tag{212}$$

As $\tau \to 0$, β_{ij} becomes sharply peaked on $j^\star = \arg\min_j D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$. Since $\sum_j \beta_{ij} = 1$ for every $\tau > 0$, sharpening β by itself does not amplify the coefficient on the alignment block: in the limit, $\sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \to (\Omega_{ij^\star} \Sigma_{j^\star} \Omega_{ij^\star}^\top)^{-1}$, which is bounded. To make the alignment-dominated regime precise we place an explicit coupling strength $a > 0$ on the alignment block,

$$\mathcal{F}_i^{(a)} = D_{\mathrm{KL}}(q_i \| p_i) + a \sum_{j \neq i} \beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)], \tag{213}$$

of which the derivation above is the case $a = 1$. Repeating the differentiation term by term, the stationarity condition becomes

$$(1 + a)\Sigma_i^{-1} = \Sigma_{p,i}^{-1} + a \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} + \Lambda_{o_i}, \tag{214}$$

which reduces to (206) at $a = 1$. In precision coordinates it reads $(1 + a)\Lambda = A + a\mathcal{L}(\Lambda)$, and the argument of Proposition 12(i), applied to $a\mathcal{L}/(1 + a)$, shows that an SPD solution exists if and only if $\rho(\mathcal{L}) < 1 + 1/a$. The high-coupling limit $a \to \infty$ therefore exists only if $\rho(\mathcal{L}) \leq 1$. Dividing by a gives formally

$$\boxed{\Sigma_i^{-1} \longrightarrow \sum_j \beta_{ij} (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} = \big\langle (\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1} \big\rangle_\beta \qquad \text{(alignment-dominated)},} \tag{215}$$

where $\langle \cdot \rangle_\beta$ denotes a β-weighted average over neighbors. This is the eigenvalue equation $\Lambda = \mathcal{L}(\Lambda)$. If $\rho(\mathcal{L}) < 1$, the finite-coupling solution is $\Lambda_a = (I + a(I - \mathcal{L}))^{-1} A$, which tends to zero like $1/a$, so the covariances diverge. The boxed relation has a nonzero SPD limit only when $\rho(\mathcal{L}) = 1$ with an SPD Perron eigenvector. Graph-flat links are the basic case. In the frames $g_i = U_i^{-1}$ the map acts as $\tilde{\Lambda}_i \mapsto \sum_j \beta_{ij} \tilde{\Lambda}_j$ with $\tilde{\Lambda}_i := U_i^\top \Lambda_i U_i$. For an irreducible row-stochastic β with stationary distribution π, contracting the finite-coupling equation with π gives $\sum_i \pi_i \tilde{\Lambda}_i = \sum_i \pi_i \tilde{A}_i$ for every a, with $\tilde{A}_i := U_i^\top A_i U_i$, while the components off the Perron direction decay like $1/a$. The limit is therefore the common precision $\tilde{\Lambda}_\infty = \sum_k \pi_k \tilde{A}_k$, and the limit covariances $\Sigma_i = U_i \tilde{\Lambda}_\infty^{-1} U_i^\top$ satisfy $\Sigma_i = \Omega_{ij} \Sigma_j \Omega_{ij}^\top$ on every edge.

Per-edge covariance alignment is thus exact in the flat-link, high-coupling limit, and small τ alone does not produce it. At finite coupling, or with holonomy, the equilibrium satisfies only the β-averaged relation (206), and with several heterogeneous neighbors the per-edge mismatch is generically nonzero, since Σ_i cannot equal every transported neighbor covariance at once.

D.2.3 Stability

Proposition 12(ii) gives global stability of the simultaneous receiver-only natural-gradient flow whenever $\rho(\mathcal{L}) < 2$. The own-block Hessian is also exact,

$$\frac{\partial^2 \mathcal{F}_i}{\partial \Sigma_i \partial \Sigma_i}[X] = \Sigma_i^{-1} X \Sigma_i^{-1}, \tag{216}$$

since each KL term contributes $\frac{1}{2} \Sigma_i^{-1} X \Sigma_i^{-1}$ from its log-determinant, the self and alignment weights sum to $2 \times \frac{1}{2} = 1$, and the trace and observation terms are linear in Σ_i. It is positive definite for every

$\Sigma_i \succ 0$, so each agent's problem with its neighbors frozen is strictly convex in Σ_i. Neither statement covers the complete canonical objective, whose Hessian carries sender cross-blocks and the reduced-Hessian softmax covariance correction; stability of that coupled equilibrium, and existence of an equilibrium of the live-softmax dynamics, are not established here. A Euclidean flow $d\Sigma_i/dt = -\eta_\Sigma \partial \mathcal{F}_i/\partial \Sigma_i$ has the same fixed point, but its linearization depends on the conditioning of the fixed point, and we have no general stability theorem for it.

Mean update where covariances align. The natural-gradient mean update $\Sigma_i \nabla_{\mu_i} \mathcal{F}_i$ carries the per-edge alignment term $\beta_{ij}\Sigma_i(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}(\mu_i - \Omega_{ij}\mu_j)$. Where per-edge alignment $\Sigma_i = \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ holds, as in the flat-link, high-coupling limit above, the precision factors cancel and

$$\beta_{ij}\Sigma_i(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}(\mu_i - \Omega_{ij}\mu_j) \longrightarrow \beta_{ij}(\mu_i - \Omega_{ij}\mu_j), \tag{217}$$

so the step is an attention-weighted sum of transported residuals with no covariance factor. This is again a property of the consensus potential: in the exact receiver update of a normalized comparison-record model the corresponding step keeps a Kalman-type gain set by the receiver's posterior covariance [30, Ch. 6, exact receiver update].

E Relating the Quadratic Forms to Transported Relative Entropies

We now show that the pairwise quadratic expectations can be written as transported relative entropies. The direction of the divergence follows from building the coupling precision from the transported sender covariance, as shown below; it is not supplied by the representation theorem of Appendix F.

E.1 Exact expansion for Gaussian beliefs

For independent Gaussians $q_i = \mathcal{N}(\mu_{q,i}, \Sigma_{q,i})$ and $q_j = \mathcal{N}(\mu_{q,j}, \Sigma_{q,j})$ and any symmetric matrix M, the expectation of a quadratic form is

$$\mathbb{E}_{q_i q_j}[\delta^\top M \delta] = \mathrm{tr}\big(M\mathrm{Cov}(\delta)\big) + \bar{\delta}^\top M \bar{\delta}, \tag{218}$$

where $\delta = k_i - \Omega_{ij}k_j$, $\bar{\delta} = \mathbb{E}[\delta] = \mu_{q,i} - \Omega_{ij}\mu_{q,j}$, and

$$\mathrm{Cov}(\delta) = \Sigma_{q,i} + \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top. \tag{219}$$

Applying (218) with $M = W_{ij}^{(q)}$:

$$\begin{aligned}\mathbb{E}_{q_i q_j}[(k_i - \Omega_{ij}k_j)^\top W_{ij}^{(q)}(k_i - \Omega_{ij}k_j)] = \mathrm{tr}\big[W_{ij}^{(q)}(\Sigma_{q,i} + \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)\big] \\ + (\mu_{q,i} - \Omega_{ij}\mu_{q,j})^\top W_{ij}^{(q)}(\mu_{q,i} - \Omega_{ij}\mu_{q,j}).\end{aligned} \tag{220}$$

An identical expansion holds for the model channel with $W_{ij}^{(s)}$ and $\tilde{\Omega}_{ij}$. For a fixed coupling $W_{ij} \succ 0$ the mean-dependent part is exactly a relative entropy at common covariance: $\frac{1}{2}\mathbb{E}[\delta^\top W_{ij}\delta] = D_{\mathrm{KL}}\big(\mathcal{N}(\mu_{q,i}, W_{ij}^{-1}) \| \mathcal{N}(\Omega_{ij}\mu_{q,j}, W_{ij}^{-1})\big) + \frac{1}{2}\mathrm{tr}\big(W_{ij}(\Sigma_{q,i} + \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^\top)\big)$ [38, Ch. G5, edge relative entropy at fixed covariance].

To obtain the transported relative entropy of the beliefs themselves we choose state-dependent couplings,

$$W_{ij}^{(q)} := c_{ij}^{(q)}(\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^{\top})^{-1}, \qquad W_{ij}^{(s)} := c_{ij}^{(s)}(\tilde{\Omega}_{ij}\Sigma_{s,j}\tilde{\Omega}_{ij}^{\top})^{-1}, \tag{221}$$

where $c_{ij}^{(q)}, c_{ij}^{(s)} > 0$ are dimensionless coupling strengths. Because $W_{ij}^{(q)}$ is built from the current variational covariance $\Sigma_{q,j}$ (and $W_{ij}^{(s)}$ from $\Sigma_{s,j}$), the identification below is a value-level matching at the current variational state, not a property of a fixed generative model, consistent with the status of the construction in Section 5.2. No dynamics in this manuscript are obtained by differentiating the quadratic form with $W_{ij}^{(q)}$ held fixed; all gradient flows differentiate the KL coupling directly.

Inserting the Gaussian KL closed form of Appendix C.2 for $q_i = \mathcal{N}(\mu_{q,i}, \Sigma_{q,i})$ and $\Omega_{ij}q_j = \mathcal{N}(\Omega_{ij}\mu_{q,j}, \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^{\top})$ into (220) gives the exact identity

$$\begin{aligned}\frac{1}{4}\mathbb{E}_{q_iq_j}[(k_i - \Omega_{ij}k_j)^{\top}W_{ij}^{(q)}(k_i - \Omega_{ij}k_j)] = \frac{c_{ij}^{(q)}}{2}D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j) + \frac{c_{ij}^{(q)}K_q}{2} \\ + \frac{c_{ij}^{(q)}}{4}\log\det\big((\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^{\top})^{-1}\Sigma_{q,i}\big).\end{aligned} \tag{222}$$

The identification with the transported relative entropy is thus exact up to the constant and the single log-determinant term. The direction comes from the transported sender covariance in $W_{ij}^{(q)}$: with $W = c\Sigma_{q,i}^{-1}$ the same expansion yields $D_{\mathrm{KL}}(\Omega_{ij}q_j\|q_i)$. The log-determinant term vanishes when $\det\Sigma_{q,i} = \det(\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^{\top})$. Expanded in the covariance mismatch $\Delta := \Sigma_{q,i} - \Omega_{ij}\Sigma_{q,j}\Omega_{ij}^{\top}$, its leading contribution is $\frac{c_{ij}^{(q)}}{4}\mathrm{tr}\big((\Omega_{ij}\Sigma_{q,j}\Omega_{ij}^{\top})^{-1}\Delta\big)$, so the correction is $O(\|\Delta\|)$ in general and $O(\|\Delta\|^2)$ only along determinant-preserving deviations. With several heterogeneous neighbors the per-edge mismatch Δ_{ij} is generically nonzero at the stationary point of Eq. (206) (Appendix D), and the first-order correction persists there. The model channel satisfies the same identity with $(m, s, \tilde{\Omega}_{ij}, W_{ij}^{(s)}, c_{ij}^{(s)}, K_m)$ in place of $(k, q, \Omega_{ij}, W_{ij}^{(q)}, c_{ij}^{(q)}, K_q)$.

E.2 Defining raw alignment weights

To connect with the main text we define the raw alignment weights

$$\tilde{\beta}_{ij} := \frac{c_{ij}^{(q)}}{2}, \qquad \tilde{\gamma}_{ij} := \frac{c_{ij}^{(s)}}{2}, \tag{223}$$

the bare belief- and model-alignment strengths between agents i and j. In the limit $\tilde{\beta}_{ij} \to \infty$ with fixed $\tilde{\gamma}_{ij}$, agents' beliefs are forced to agree after transport while their models may still differ; $\tilde{\gamma}_{ij} = 0$ decouples model alignment entirely. The normalized weights β_{ij}, γ_{ij} of the main text, which sum to one over each row, are not identified with the raw couplings. They arise by composition: at a uniform coupling $c_{ij}^{(q)} \equiv c^{(q)}$, the identity (222) supplies the per-edge energies $E_{ij} = D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ up to the common prefactor, and minimizing the entropy-regularized row objective of Section 5.2 (Eqs. (42) and (50)) over the row simplex returns the softmax weights. The quadratic form reproduces the canonical functional only through this composition, and only up to the log-determinant term of Eq. (222).

F Conditional Representation Theorem for the Transported Relative Entropy

This appendix characterizes the belief coupling $D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$, with the receiver law in the first slot (the variational or I-projection direction), within a class of divergences and for one problem: agent i's best response with its neighbors' beliefs and its attention weights held fixed. The companion theory declares this transported divergence and does not derive it [30, Chs. 6 and 12], and nothing below fixes the divergence of the joint scalar functional; the scope paragraph at the end of this appendix shows why the closure criterion cannot. Within the class of local f-divergences that enter the free energy linearly in β_{ij}, the relative entropy is the only divergence whose best response has the geometric-mean Boltzmann form for all priors and neighbor beliefs; richness of the attainable likelihood ratios, which the converse needs, holds automatically in the Gaussian fiber (Lemma 4). We assume matched fibers, a common dimension K_q and representation ρ; the argument uses only the density of the transported law $\Omega_{ij}q_j$ and is unchanged when an intertwiner produces it.

The best-response problem

With the neighbor beliefs q_j and the weights $\beta_{ij} \geq 0$ held fixed, agent i solves

$$F_i[\beta_i] = \min_{q_i} \Big\{ D_{\mathrm{KL}}(q_i\|p_i) + \sum_{j\neq i} \beta_{ij}\mathcal{D}(q_i, \Omega_{ij}q_j) \Big\}, \qquad \mathcal{D}(q,r) = \int r(c) f\left(\frac{q(c)}{r(c)}\right) dc, \tag{224}$$

with the second-argument density as the integration measure. The weights are in turn chosen by minimizing $\sum_j \beta_{ij}C_{ij} + \tau \sum_j \beta_{ij}\log\beta_{ij}$ over the row simplex, where $C_{ij} := \partial F_i/\partial\beta_{ij}$ is the marginal cost of attending to agent j.

Lemma 3 (KL-sum identity)**.** *Let p and $r_1, \ldots, r_n$ be probability densities with overlapping supports, let $\beta_j \geq 0$ with $B := \sum_j \beta_j$, and define*

$$g := Z_g^{-1} p^{1/(1+B)} \prod_j r_j^{\beta_j/(1+B)}, \qquad Z_g := \int p^{1/(1+B)} \prod_j r_j^{\beta_j/(1+B)} dc. \tag{225}$$

Then for every density q with $D_{\mathrm{KL}}(q\|p)$ and every $D_{\mathrm{KL}}(q\|r_j)$ finite,

$$D_{\mathrm{KL}}(q\|p) + \sum_j \beta_j D_{\mathrm{KL}}(q\|r_j) = (1+B)D_{\mathrm{KL}}(q\|g) - (1+B)\log Z_g. \tag{226}$$

Proof. The left-hand side is $\int q[(1+B)\log q - \log p - \sum_j \beta_j \log r_j]dc = (1+B)\int q[\log q - \log g - \log Z_g]dc$, which is the right-hand side. □

With $r_j = \Omega_{ij}q_j$ and $\mathcal{D} = D_{\mathrm{KL}}$, the identity solves the best-response problem (224) at once: the unique minimizer is $q_i^* = g$, the geometric-mean form (227) below, and the minimum value is $F_i = -(1+B)\log Z_g \geq 0$, nonnegative because $Z_g \leq 1$ by Hölder's inequality (the exponents sum to one). The exponents $1/(1+B)$ are the reciprocal of the entropy weight discussed in Appendix D: the best response is a Gibbs state at temperature $1+B$ relative to Bayesian updating. In a normalized joint the analogous coordinate update is a product of the prior and pair factors with unit exponents, and its stationarity includes each agent's appearance in its neighbors' factors [30, Ch. 5, local Gibbs bound and exact block E-coordinate].

Conditional representation theorem

Theorem 2 (Conditional representation for the frozen-neighbor best response)**.** *Let* $\mathcal{D}(q,r) = \int r(c) f(q(c)/r(c)) dc$ *with* f real-analytic on $(0,\infty)$*, convex,* $f(1) = 0$*, and, without loss of generality*[1]*,* $f'(1) = 0$*. Fix agent* i*'s neighbor beliefs* $\{q_j\}$ *and weights* $\beta_{ij} \geq 0$ *with* $B := \sum_j \beta_{ij} > 0$*, and let* $\mathcal{D}$ *enter the best-response problem* (224) *linearly. Then the stationary best response has the geometric-mean Boltzmann form*

$$q_i^*(c) \propto p_i(c)^{1/(1+B)} \prod_j \left[\Omega_{ij}(c) q_j(c)\right]^{\beta_{ij}/(1+B)} \tag{227}$$

for all choices of prior p_i *and neighbor beliefs* $\{q_j\}$ *if, and only if,* $\mathcal{D}$ *is the relative entropy (up to the affine equivalence above):*

$$\mathcal{D}(q,r) = D_{\mathrm{KL}}(q\|r).$$

For $B = 1$*, as when* β_i *is a probability vector, the exponents are* $1/2$*.*

Corollary 1 (Closure forces the KL generator among nondegenerate divergences)**.** *Within the same class, with real-analytic generator and nondegeneracy* $f''(1) > 0$*, exponential-family closure of the best response for all Gaussian priors and neighbor beliefs already forces* $f'(t) = \gamma \log t$ *with* $\gamma > 0$*, that is,* $\mathcal{D} = \gamma D_{\mathrm{KL}}$*, where the positive multiple* γ *is absorbable into the coupling weights* β_{ij}*; the geometric-mean exponents of Eq.* (227) *normalize* $\gamma = 1$*. The nondegeneracy hypothesis cannot be dropped:* $f \equiv 0$ *is real-analytic and convex, satisfies the normalization, and is closed for every Gaussian prior and neighbor set, since stationarity then reduces to* $q_i^* = p_i$*; it yields* $\gamma = 0$ *and the identically zero divergence. Because* $f'(t) = \gamma \log t$ *gives* $f''(1) = \gamma$*, the hypothesis* $f''(1) > 0$ *is exactly the statement* $\gamma > 0$ *and excludes that degenerate solution alone.*

Proof sketch. Stationarity gives $\log(q_i/p_i) + 1 + \sum_j \beta_{ij} f'(r_{ij}) = \lambda$. If q_i^* is required to be log-quadratic for all Gaussian $(p_i, \{q_j\})$, then $h(x) := f'(e^x)$ must map every attained quadratic in c to a quadratic in c; restricting a nondegenerate attained quadratic to lines and using the $t \mapsto -t$ symmetry of the restriction forces h to be affine on the range of the quadratic, and real-analyticity extends h affinely to all of $\mathbb{R}$, so $f'(t) = \alpha + \gamma \log t$. The normalization $f'(1) = 0$ kills α, convexity gives $\gamma \geq 0$, and nondegeneracy gives $\gamma > 0$. Finally γD_{KL} with weights β_{ij} is identical to D_{KL} with weights $\gamma\beta_{ij}$. □

Proof of Theorem 2. Step 1. Stationarity for a general f-divergence. With ratio $r(c) = q_i(c)/\left[\Omega_{ij}(c) q_j(c)\right]$, the first-slot functional derivative is $\delta\mathcal{D}/\delta q_i(c) = f'(r(c))$, so the stationarity condition $\delta F_i/\delta q_i(c) = \lambda$, with Lagrange multiplier λ for normalization, is

$$\log \frac{q_i(c)}{p_i(c)} + 1 + \sum_j \beta_{ij} f'\left(\frac{q_i(c)}{\Omega_{ij}(c) q_j(c)}\right) = \lambda. \tag{228}$$

Step 2. Sufficiency. For $\mathcal{D} = D_{\mathrm{KL}}$ the generator is $f(t) = t \log t - t + 1$ with $f'(t) = \log t$, and Lemma 3 gives the minimizer (227) directly.

Step 3. Necessity. Assume the best response has the form (227) for all priors p_i and neighbor beliefs $\{q_j\}$, with general B. Taking logarithms, $(1+B)\log q_i^* = \log p_i + \sum_j \beta_{ij} \log(\Omega_{ij} q_j) + C$, and splitting $(1+B)\log q_i^* = \log q_i^* + \sum_j \beta_{ij} \log q_i^*$ gives

$$\log \frac{q_i^*}{p_i} + \sum_j \beta_{ij} \log \frac{q_i^*}{\Omega_{ij} q_j} = C', \tag{229}$$

[1] Any f-divergence is invariant under $f(t) \mapsto f(t) + c(t-1)$ when q and r are normalized, so we may always choose the representative with $f'(1) = 0$.

with $C' := -C$. It remains to pass from Eqs. (228) and (229) to the per-term identity

$$f'\left(\frac{q_i}{\Omega_{ij}q_j}\right) = \log\frac{q_i}{\Omega_{ij}q_j} + k, \tag{230}$$

with a single constant k, and from there to $f'(t) = \log t + k$ on all of $(0,\infty)$. The Lagrange multiplier and the normalizer vary from configuration to configuration, and the geometric-mean form constrains the minimizer, not the neighbor beliefs. Both points are handled by fixing one configuration and sweeping the base point c within it, where the multiplier and the normalizer are constants.

Lemma 4 (Per-term separation and richness)**.** *Let f be real-analytic on $(0,\infty)$ with $f'(1) = 0$, and suppose the stationary solution of* (228) *has the geometric-mean form* (227) *for all priors and neighbor beliefs. Then $f'(t) = \log t$ as a functional identity on $(0,\infty)$.*

Proof. Subtracting the rearranged target identity (229) from the stationarity condition (228) gives, for each fixed configuration $(p_i, \{q_j\}, \beta_i)$,

$$\sum_j \beta_{ij}\left[f'\big(r_{ij}(c)\big) - \log r_{ij}(c)\right] = \text{const}, \tag{231}$$

with $r_{ij}(c) = q_i^*(c)/\big(\Omega_{ij}(c)q_j(c)\big)$ and a constant that depends on the configuration but not on c. To separate the terms, take a single-neighbor configuration: one neighbor j with coupling $\beta_{ij} = B > 0$ and all other couplings zero. Then (231) reads $f'(r(c)) - \log r(c) = k$ directly, with one constant k for the whole configuration; the Lagrange multiplier and the normalizer Z_i are single constants within this fixed configuration and are absorbed into k. Under the geometric-mean form the single-neighbor ratio is $r(c) = Z_i^{-1}\big(p_i(c)/(\Omega_{ij}(c)q_j(c))\big)^{1/(1+B)}$, and choosing p_i and q_j so that the pointwise ratio $p_i(c)/(\Omega_{ij}(c)q_j(c))$ varies with c (two non-identical Gaussians suffice) makes $r(c)$ sweep an open subinterval of $(0,\infty)$. The function $g(t) := f'(t) - \log t$ is real-analytic on $(0,\infty)$ and constant on that open subinterval, hence constant on all of $(0,\infty)$ by analytic continuation; this is the only point at which real-analyticity is used, and without it the identification holds on the attainable subinterval only. Thus $f'(t) = \log t + k$ on $(0,\infty)$, which is the per-term identity (230) as a functional identity, and $f'(1) = 0$ forces $k = 0$. □

Integrating $f'(t) = \log t$ gives $f(t) = t\log t - t + C_0$, and $f(1) = 0$ gives $C_0 = 1$. Therefore $f(t) = t\log t - t + 1$, the generator of $D_{\mathrm{KL}}(q\|r)$. □

Scope: the joint scalar functional

Theorem 2 concerns the best response alone. In the canonical scalar functional of Section 5.2 each belief q_i also sits in the second slot of every row $\beta_{ki}\mathcal{D}(q_k, \Omega_{ki}q_i)$ in which agent i is a sender. The second-slot functional derivative of $\int rf(q/r)dc$ with respect to r is $f(t) - tf'(t)$ at $t = q/r$, which for the relative entropy is $1 - t$. Free-form exponential-family closure of the joint stationarity condition would require both slot derivatives to be affine in $\log t$ on an open interval, $f'(t) = \alpha + a\log t$ and $f(t) - tf'(t) = \alpha' + c'\log t$. Differentiating, $f''(t) = a/t$ and $-tf''(t) = c'/t$, so $-a = c'/t$ on that interval, which forces $a = c' = 0$; then $f'' \equiv 0$, and the normalization with analyticity gives $f \equiv 0$. No nonzero f-divergence therefore gives free-form exponential-family closure once an agent is also a sender. A two-agent witness makes this concrete. In one dimension with $\Omega_{12} = \Omega_{21} = 1$, $\beta_{12} = \beta_{21} = 1$, $p_1 = \mathcal{N}(0,1)$ and $p_2 = \mathcal{N}(3,1/4)$ (second argument the variance), the geometric-pool laws $q_1 = \mathcal{N}(2,1/2)$ and $q_2 = \mathcal{N}(8/3,1/3)$ are mutual best responses, yet the gradient of the joint

scalar at that point has norm 2.59, and the joint Gaussian minimizer is $q_1 \approx \mathcal{N}(1.88, 0.83)$, $q_2 \approx \mathcal{N}(2.53, 0.60)$. In a Gaussian implementation the family imposes closure, so the closure criterion cannot select the divergence of the joint functional; the geometric-mean form (227) describes the frozen-neighbor best response only.

Dual cost identification

By the envelope theorem applied to the β_{ij} optimization, the marginal attention cost is

$$C_{ij} \equiv \frac{\partial F_i}{\partial \beta_{ij}}\bigg|_{q_i^*} = \mathcal{D}(q_i^*, \Omega_{ij} q_j), \tag{232}$$

which for the relative entropy identifies the cost of attention with the divergence between the agent's best response and the neighbor's transported law. The identity holds for every divergence that enters linearly in β_{ij}, so it is a consistency property and does not select the divergence.

Relation to probabilistic opinion pooling

The best response (227) carries exponents $\beta_{ij}/(1+B)$ together with the prior exponent $1/(1+B)$, which sum to one over the prior-and-neighbor set, so it is the normalized weighted geometric (logarithmic) opinion pool, whose Gaussian case gives the precision-weighted aggregate $\Sigma_*^{-1} = \sum_k \lambda_k \Sigma_k^{-1}$. Dietrich and List [200] characterize this geometric pool axiomatically as externally Bayesian and unanimity/zero-preserving; Bordley [201] and Genest and Zidek [202] supply the multiplicative, log-linear precursor, which coincides with the normalized geometric pool in the continuous Gaussian fiber used here. Theorem 2 re-derives this operator, for the frozen-neighbor best response, by an f-divergence route within its hypothesis class. The meta-agent barycenter of Section 9.1 is the Gaussian M-projection of the linear pool that Dietrich and List characterize from eventwise independence. The axiomatic properties belong to the pool, not to its projection, and a form that drops the between-mean dispersion approximates the projection only in the high-coherence regime.

Interpretation and summary

Gauge covariance fixes what is compared, the law q_i and the transported law $\Omega_{ij} q_j$ in one frame, while the theorem above constrains how it is compared, and only for the best response. On the information-geometric side, the relative entropy is the Bregman divergence of the negative-entropy potential; among f-divergences on normalized distributions it is the only Bregman divergence [203], and it is the canonical divergence of the dually flat e/m structure, whence the Pythagorean relation [136]. The Fisher metric, by contrast, is the local Hessian of every smooth f-divergence, up to the factor $f''(1)$.

In Theorem 2 the competing f-divergences (the reverse relative entropy, χ^2, squared Hellinger, and the Amari α-divergences) are excluded already on the attainable subinterval of ratios, without invoking analyticity; a merely smooth generator agreeing with the relative entropy on that interval is observationally equivalent within the model. For Gaussian beliefs, exponential-family closure together with $f''(1) > 0$ already forces the KL generator up to a positive multiple absorbable into the coupling weights (Corollary 1), so the geometric-mean target serves to normalize that multiple. By the $\mathrm{GL}(K_q)$ invariance theorem (Theorem 1), the divergence is invariant under the full general linear group $\mathrm{GL}(K_q, \mathbb{R})$, which contains $\mathrm{SO}(K_q)$ and, after the realification $\mathrm{GL}(N, \mathbb{C}) \hookrightarrow \mathrm{GL}(2N, \mathbb{R})$ with $K_q = 2N$, the images of $\mathrm{U}(N)$ and $\mathrm{SU}(N)$.

G Augmented Hierarchical Model, the Rigid-Link Predictive Density, and Mean-Field Fixed Points

The main text reads the cross-scale shadow $p_i^{(\ell)} = \Omega_{i,I}[q_I^{(\ell+1)}]$ of Section 10 as the one-episode-lagged predictive prior of an episode-indexed joint. When a parent forms at episode n, the prior of its children at episode $n+1$ is the filtered parent law pushed through a declared transition, and for a transported-copy transition this is the shadow formula exactly [30, Ch. 12]. This appendix records what a same-time hierarchical joint does and does not supply. We define an augmented hierarchical generative model with Gaussian link kernels on the tree of agent latents, prove that it is a proper joint, state its mean-field fixed-point equations, and compute the variational predictive density whose rigid-link limit reproduces the shadow formula. That density is not a message of the joint: evaluated against the full parent posterior it reuses each constituent's own evidence, and the functional built from it is not an evidence bound (Remark 2). The general statements behind this are the typing prohibition of the companion theory, under which no generative kernel reads a recognition law, and its result that an energy appended to a joint without the corresponding normalizer is not an evidence bound [30, Ch. 4]. The companion theory's downward kernels and conditional cross-scale divergence give the evidence-consistent form of top-down influence [30, Ch. 12].

G.1 The Augmented Joint

Let the hierarchy be a finite tree of depth $\ell_{\max}$ and finite breadth, with I_ℓ the set of agents at level ℓ and $\pi(i)$ the parent of agent i. Each agent $i \in I_\ell$ carries a latent variable $k_i^{(\ell)} \in \mathbb{R}^{K_q}$. The construction is made at a single base point, so it is a finite-dimensional probability law on the product space $\mathbb{K} := \prod_{\ell, i \in I_\ell} \mathbb{R}^{K_q}$.

Definition 7 (Augmented hierarchical joint)**.** *For each parent-child edge $(i, \pi(i))$ with $i \in I_\ell$, define the Gaussian link kernel*

$$\mathcal{K}_{i,\pi(i)}^{(\ell)}\big(k_i^{(\ell)} \mid k_{\pi(i)}^{(\ell+1)}\big) = (2\pi\sigma^2)^{-K_q/2} \exp\left[-\frac{1}{2\sigma^2}\big\|k_i^{(\ell)} - \Omega_{i,\pi(i)} k_{\pi(i)}^{(\ell+1)}\big\|^2\right], \tag{233}$$

where $\Omega_{i,\pi(i)} = \exp(\phi_i^{(\ell)}) \exp(-\phi_{\pi(i)}^{(\ell+1)}) \in G$ is the inter-level transport of Section 4.7 and $\sigma^2 > 0$ is a link-stiffness parameter. The kernel is a normalized transition density, a propagator from parent to child, and not a Boltzmann weight. The top-level prior $p_{\mathrm{top}}(k^{(\ell_{\max})})$ is a declared root, fixed as a boundary condition (Section 12). The augmented joint over all agent latents is

$$p_\sigma\big(\{k_i^{(\ell)}\}, o\big) = p(o \mid \{k_i^{(0)}\}) \prod_{\ell=0}^{\ell_{\max}-1} \prod_{i \in I_\ell} \mathcal{K}_{i,\pi(i)}^{(\ell)}\big(k_i^{(\ell)} \mid k_{\pi(i)}^{(\ell+1)}\big) p_{\mathrm{top}}\big(k^{(\ell_{\max})}\big). \tag{234}$$

Lemma 5 (Well-definedness)**.** *Assume the observation likelihood $p(o \mid \cdot)$ is bounded above by a finite constant (unity in the discrete-observation case, the normalization constant in the Gaussian case). Then for every finite $\sigma^2 > 0$ and every finite hierarchical tree, the augmented joint* (234) *is integrable over $\mathbb{K}$ for every fixed o, with normalization equal to the evidence $p(o)$; the prior portion, obtained by omitting the likelihood factor, is a proper probability density on $\mathbb{K}$.*

Proof. Each kernel $\mathcal{K}_{i,\pi(i)}^{(\ell)}$ is a Gaussian density on $k_i^{(\ell)}$ with mean $\Omega_{i,\pi(i)} k_{\pi(i)}^{(\ell+1)}$ and covariance $\sigma^2 I$, and it integrates to unity for every value of $k_{\pi(i)}^{(\ell+1)}$. The factor graph of (234) is the hierarchical tree itself. For tree-structured products of proper conditional densities, the integral over the leaves factorizes through the tree [204, §3.2] and leaves the top-level prior p_{top}, which is normalized, so

the prior portion is a proper probability density on $\mathbb{K}$. The likelihood enters as a multiplicative factor at the leaves and is bounded above by hypothesis, so it does not affect integrability, and integrating it against the prior portion yields $p(o)$. □

G.2 Mean-Field Fixed-Point Equations

Mean-field variational inference on the augmented joint factorizes the variational posterior across agent latents.

Lemma 6 (Mean-field fixed-point equations)**.** *The variational free energy*

$$\mathcal{F}_\sigma[q] = \sum_{\ell,i} \mathbb{E}_{q_i^{(\ell)}}\big[\log q_i^{(\ell)}\big] - \sum_{\ell<\ell_{\max},i} \mathbb{E}_{q_i^{(\ell)} q_{\pi(i)}^{(\ell+1)}}\big[\log \mathcal{K}_{i,\pi(i)}^{(\ell)}\big] - \mathbb{E}_{q^{(\ell_{\max})}}\big[\log p_{\mathrm{top}}\big] - \mathbb{E}_{q^{(0)}}\big[\log p(o \mid k^{(0)})\big] \tag{235}$$

is stationary under the mean-field factorization $q(\{k_i^{(\ell)}\}) = \prod_{\ell,i} q_i^{(\ell)}(k_i^{(\ell)})$ when each factor satisfies

$$\begin{aligned}\log q_i^{(\ell)}(k_i) = \mathit{const} + \mathbb{E}_{q_{\pi(i)}^{(\ell+1)}}\big[\log \mathcal{K}_{i,\pi(i)}^{(\ell)}(k_i \mid k_{\pi(i)})\big] + \sum_{j:\pi(j)=i} \mathbb{E}_{q_j^{(\ell-1)}}\big[\log \mathcal{K}_{j,i}^{(\ell-1)}(k_j \mid k_i)\big] \\ + \delta_{\ell,0}\mathbb{E}_{q_{\setminus i}^{(0)}}\big[\log p(o \mid k^{(0)})\big] + \delta_{\ell,\ell_{\max}} \log p_{\mathrm{top}}(k_i),\end{aligned} \tag{236}$$

where $q_{\setminus i}^{(0)}$ is the product of the other leaf factors. The first expectation is the top-down message from the parent of agent i (absent at $\ell = \ell_{\max}$), the sum collects the bottom-up messages from its children (absent at $\ell = 0$), and the data likelihood enters only at the leaves.

Proof. Equation (236) is the standard coordinate-ascent mean-field update [205, §5.3], [87, §10.1]. It follows by varying (235) with respect to $q_i^{(\ell)}$ at fixed values of the other factors, imposing $\int q_i^{(\ell)} = 1$ by a Lagrange multiplier, and collecting the terms that depend on $k_i^{(\ell)}$. □

G.3 The Rigid-Link Variational Predictive Density

Lemma 7 (Rigid-link variational predictive density)**.** *Let the variational posterior of the parent be Gaussian, $q_{\pi(i)}^{(\ell+1)} = \mathcal{N}(\mu_{\pi(i)}^{(\ell+1)}, \Sigma_{\pi(i)}^{(\ell+1)})$. The variational predictive density of agent i, the marginal of the link kernel against the parent's variational posterior, is*

$$\hat{p}_i^{(\ell),\sigma}(k_i) := \int \mathcal{K}_{i,\pi(i)}^{(\ell)}(k_i \mid k_{\pi(i)}) q_{\pi(i)}^{(\ell+1)}(k_{\pi(i)}) dk_{\pi(i)} = \mathcal{N}\big(k_i; \Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}, \Omega_{i,\pi(i)}\Sigma_{\pi(i)}^{(\ell+1)}\Omega_{i,\pi(i)}^\top + \sigma^2 I\big), \tag{237}$$

and as $\sigma^2 \to 0$ it converges to the pushforward $(\Omega_{i,\pi(i)})_* q_{\pi(i)}^{(\ell+1)}$, which is the shadow formula of Eq. (131)*. The lemma computes a density. That density is neither the mean-field factor of Eq.* (236) *nor a message of the joint* (234) *(Remark 2).*

Proof. Conditional on $k_{\pi(i)}$, the kernel (233) is Gaussian with mean $\Omega_{i,\pi(i)}k_{\pi(i)}$, linear in the parent, and covariance $\sigma^2 I$. The linear-Gaussian marginalization rule [87, §2.3.3] gives a Gaussian marginal with mean $\Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}$ and covariance $\Omega_{i,\pi(i)}\Sigma_{\pi(i)}^{(\ell+1)}\Omega_{i,\pi(i)}^\top + \sigma^2 I$. As $\sigma^2 \to 0$ the kernel converges weakly to $\delta(k_i - \Omega_{i,\pi(i)}k_{\pi(i)})$, and the marginal converges to

$$\int \delta\big(k_i - \Omega_{i,\pi(i)}k_{\pi(i)}\big) q_{\pi(i)}^{(\ell+1)}(k_{\pi(i)}) dk_{\pi(i)} = \big(\Omega_{i,\pi(i)}\big)_* q_{\pi(i)}^{(\ell+1)}. \tag{238}$$

□

Remark 2 (The predictive density is not a message of the joint)**.** *Four facts fix the status of Lemma 7. First, the strict mean-field factor is a different density. Substituting the kernel into the top-down term of Eq.* (236) *gives*

$$\mathbb{E}_{q_{\pi(i)}^{(\ell+1)}}\big[\log \mathcal{K}_{i,\pi(i)}^{(\ell)}\big] = const - \frac{1}{2\sigma^2}\Big[\big\|k_i - \Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}\big\|^2 + \operatorname{tr}\big(\Omega_{i,\pi(i)}\Sigma_{\pi(i)}^{(\ell+1)}\Omega_{i,\pi(i)}^{\top}\big)\Big],$$

so in isolation from children and data the mean-field factor is $\mathcal{N}(\Omega_{i,\pi(i)}\mu_{\pi(i)}^{(\ell+1)}, \sigma^2 I)$*: the parent covariance enters only the constant, and in the rigid limit the factor collapses to a point mass. Second, the predictive density* (237) *equals the prior marginal of* k_i *under* p_σ *when* $q_{\pi(i)}^{(\ell+1)}$ *is the prior marginal of the parent. A parent posterior, by contrast, has absorbed the evidence of every leaf below the parent, including the leaves below agent* i*.*

Third, evaluated against the full parent posterior, the shadow reuses each constituent's own evidence; the evidence-consistent downward message uses the parent posterior with that constituent's contribution removed. On a tree, belief propagation is exact [87, §8.4]: the exact posterior of k_i *is proportional to the bottom-up evidence of agent* i *times the cavity predictive density* $\int \mathcal{K}_{i,\pi(i)}^{(\ell)}(k_i \mid k_{\pi(i)}) q_{\pi(i)}^{\setminus i}(k_{\pi(i)}) dk_{\pi(i)}$*, where* $q_{\pi(i)}^{\setminus i}$ *is the parent posterior computed without the message from agent* i*. Multiplying the full-posterior predictive density by agent* i*'s likelihood counts that likelihood twice. In a Gaussian instance with one parent, two children,* $K_q = 2$*,* $\mathrm{GL}^+(2)$ *transports, and Gaussian observations under each child, the cavity construction reproduces the exact child posterior to* 10^{-16} *at* $\sigma^2 = 0.3$*, while the full-posterior shadow scheme errs in the child mean by* 0.13*; at* $\sigma^2 = 10^{-8}$ *its precision exceeds the exact child precision by exactly the precision of the child's own observation.*

Fourth, the functional built from the predictive density is not an evidence bound. Replacing the mean-field cross term $\mathbb{E}_{q_i q_{\pi(i)}}[-\log \mathcal{K}_{i,\pi(i)}]$ *by the predictive cross-entropy* $\mathbb{E}_{q_i}[-\log \hat{p}_i]$ *can only lower the objective, since* $-\log$ *is convex and Jensen's inequality gives* $\mathbb{E}_{q_i}[-\log \int \mathcal{K}_{i,\pi(i)} q_{\pi(i)}] \le \mathbb{E}_{q_i q_{\pi(i)}}[-\log \mathcal{K}_{i,\pi(i)}]$*. The resulting same-time shadow functional* $D_{\mathrm{KL}}(q_{\mathrm{top}} \| p_{\mathrm{top}}) + \sum_i [D_{\mathrm{KL}}(q_i \| \hat{p}_i) - \mathbb{E}_{q_i} \log p(o_i \mid k_i)]$ *can therefore fall below the surprisal* $-\log p_\sigma(o)$*, which no variational free energy of* p_σ *can do. In a scalar instance with* $p_{\mathrm{top}} = \mathcal{N}(0,1)$*,* $\sigma^2 = 0.05$*, and observation variance* 0.2*, its minimum over Gaussian factors lies* 0.14 *nats below the surprisal for the concordant data* $o = (1,1)$ *and* 12.98 *nats below for* $o = (2,-2)$*. In the rigid limit the child-parent term becomes* $D_{\mathrm{KL}}(q_i \| (\Omega_{i,\pi(i)})_* q_{\pi(i)})$ *with both factors varying. Its mixed third variation* (244) *is nonzero, so Theorem 3 applies to it exactly as to the peer terms, and the same-time shadow edge has no fixed state-level evidence-bound representation.*

G.4 Evidence-Consistent Readings of Top-Down Influence

Lemmas 5–7 do not make the same-time shadow a derived prior. Four readings of top-down influence are consistent with one normalized joint. The first is the conditional coupling $\mathbb{E}_{q_{\pi(i)}} D_{\mathrm{KL}}(q_i \| \mathcal{K}_{i,\pi(i)}(\cdot \mid k_{\pi(i)}))$, the mean-field cross term of Eq. (235); it is the conditional cross-scale divergence of the companion theory's downward-coupling proposition [30, Ch. 12], and its rigid limit diverges as $1/\sigma^2$ unless the factors degenerate. The second is the cavity predictive density of Remark 2, the exact downward message on a tree. The third is the random-effects, or empirical-Bayes, reading: the parent is a point hyperparameter $\theta = (m, S)$, the constituents are conditionally independent draws $k_i \sim \mathcal{N}(\Omega_{i,I} m, \Omega_{i,I} S \Omega_{i,I}^{\top})$, and the shadow is then an honest prior. Fitting θ is type-II maximum likelihood, the barycenter of Eq. (120) with its dispersion term is the exact update of θ at fixed constituent factors [103], and the parent carries no belief of its own; Σ_I is then a population covariance and not a posterior uncertainty, so this is a different model from Definition 7. The fourth is the one-episode-lagged reading of the main text, stated at the opening of this appendix,

with the transported-copy transition $k_i^{(n+1)} = \Omega_{i,I} k_I^{(n)}$; Eq. (131) is its formula with the variational parent standing in for the filtered law. The first three readings are same-time; the fourth carries the participatory reading of Section 10.

In hierarchical generative models of cortical inference [19,110] the level-ℓ prior is the conditional $p(x^{(\ell)} \mid x^{(\ell+1)})$ of the generative model; the link kernel (233) plays that role here, with a gauge-equivariant mean and an isotropic variance. It is also a special case of the top-down Gaussian conditional of hierarchical variational autoencoders [206], whose conditional variances are learned per dimension. Their bound evaluates this conditional at parent samples, which is the conditional coupling above and not the predictive density of Lemma 7, and the rigid limit $\sigma^2 \to 0$ is singular for that variational problem.

The shadow formula of the main text is the rigid-link limit $\sigma^2 = 0$. The finite-σ^2 density (237) inflates the transported parent covariance by $\sigma^2 I$ and can regularize an ill-conditioned $\Omega_{i,I} \Sigma_I \Omega_{i,I}^\top$. It defines a one-parameter family of softer shadows for future study; no simulation at small positive σ^2 is reported.

The lemmas concern the cross-scale terms alone. Reinstating the within-scale alignment terms gives a fixed-point system in which each level-ℓ posterior balances the transported parental pull against the within-scale alignment energy. Because the softmax weights β_{ij} are row-stochastic, a "small β_{ij}" is not a small parameter. The dimensionless coupling strengths are the raw alignment weights $\tilde{\beta}_{ij}, \tilde{\gamma}_{ij}$ of Section E.2, and the within-scale terms vanish from the functional in the limit $\tilde{\beta}_{ij}, \tilde{\gamma}_{ij} \to 0$. Perturbative control of the within-scale corrections is open (Section G.5).

G.5 Status of the Mean-Field Fixed-Point Question

At zero within-scale coupling, mean field on the augmented joint is exact and uniquely solvable (Section G.5.2). The shadow dynamics of Section 10 are not this mean field (Remark 2) and inherit none of these guarantees. The within-scale question has three separate answers: a general obstruction for a fixed joint on the original state variables, a self-consistent auxiliary model that becomes fixed only after an equilibrium is selected, and an exact Gibbs lift whose random variable is an entire belief configuration (Theorem 6, Appendix I).

G.5.1 Fixed Representation on the State Variables: General Obstruction

Let (X_i, ν_i) be the state space of agent i, and let T_{ij} denote the fixed mass-preserving positive linear transport induced by $(\Omega_{ij})_*$. A family $\mathcal{Q}_i$ of strictly positive densities is called mixture-open here when every zero-mass tangent h_i under consideration with $h_i/q_i \in L^\infty(q_i)$ generates $q_i(t) = q_i + t h_i \in \mathcal{Q}_i$ for all sufficiently small $|t|$. All differentiated integrals below are assumed finite.

Theorem 3 (State-level mean-field ELBO obstruction)**.** *Let the factors $q_i \in \mathcal{Q}_i$ vary independently, let at least one nonself coefficient $\beta_{ij} > 0$, and consider*

$$\mathcal{C}_{\mathrm{cons}}(q) = \mathcal{F}_{\mathrm{rest}}(q) + \sum_{a \neq b} \beta_{ab} D_{\mathrm{KL}}(q_a \| \mathsf{T}_{ab} q_b), \tag{239}$$

where $D^3_{q_i q_j q_j} \mathcal{F}_{\mathrm{rest}} = 0$ for $i \neq j$. Assume that for one active edge (i,j) there is an admissible tangent h_j such that

$$g_{ij} := \left(\frac{\mathsf{T}_{ij} h_j}{\mathsf{T}_{ij} q_j} \right)^2 \tag{240}$$

is bounded and nonconstant q_i-almost everywhere. Then there is no density p_θ on $\prod_i \mathsf{X}_i$, independent of q, and no constant C_0 such that

$$\mathcal{C}_{\mathrm{cons}}(q) = D_{\mathrm{KL}}\left(\bigotimes_i q_i \middle\| p_\theta\right) + C_0, \tag{241}$$

throughout the open product family.

Proof. Write $Q_q = \bigotimes_i q_i$. The standard fixed-model mean-field decomposition [205] gives

$$D_{\mathrm{KL}}(Q_q \| p_\theta) = \sum_i \int_{\mathsf{X}_i} q_i \log q_i d\nu_i - \int_{\prod_i \mathsf{X}_i} \left(\prod_i q_i\right) \log p_\theta d\nu_{1:N}. \tag{242}$$

The entropy is separable, while the fixed-model energy is affine in each factor. Therefore

$$D^3_{q_i q_j q_j} D_{\mathrm{KL}}(Q_q \| p_\theta)[a_i, h_j, h_j] = 0, \qquad i \neq j. \tag{243}$$

For the active consensus edge, take $q_i(s) = q_i + s a_i$ and $q_j(t) = q_j + t h_j$. Direct differentiation gives

$$\frac{\partial^3}{\partial s \partial t^2} D_{\mathrm{KL}}(q_i(s) \| \mathsf{T}_{ij} q_j(t))\Big|_{s=t=0} = \int_{\mathsf{X}_i} a_i g_{ij} d\nu_i. \tag{244}$$

Choose $a_i = q_i(g_{ij} - \mathbb{E}_{q_i}[g_{ij}])$, which has zero mass and satisfies $a_i/q_i \in L^\infty(q_i)$ because g_{ij} is bounded; it is therefore admissible for a sufficiently small mixture path. Equation (244) becomes $\mathrm{Var}_{q_i}(g_{ij}) > 0$, contradicting Eq. (243). The terms collected in $\mathcal{F}_{\mathrm{rest}}$ cannot cancel this mixed variation by hypothesis. Every other ordered-edge term also has zero q_i, q_j, q_j variation: an unrelated edge omits one varied factor, while the reverse edge is affine in q_j outside its entropy and that entropy's second variation is independent of q_i. □

Remark 3 (Scope of the obstruction)**.** *The theorem applies directly to the entropy-retaining functional by holding its independently varied attention rows fixed. The self terms, likelihood terms with fixed observation models, and categorical attention entropies lie in $\mathcal{F}_{\mathrm{rest}}$. The same argument applies to the child-parent edge of a same-time shadow (Remark 2). The result concerns an ordinary mean-field factorization on the original agent-state variables. It does not exclude frozen source templates, restricted quadratic families, compatible auxiliary variables, a model selected after a fixed point, or a probability law over belief configurations. If the attention optimum $\beta^*(q)$ has already been substituted, its response derivatives define a different reduced functional and require a separate representation test.*

G.5.2 Exact Cross-Scale Gaussian Exception

At zero raw within-scale coupling ($\tilde{\beta}_{ij} = \tilde{\gamma}_{ij} = 0$, so the alignment sectors are absent), the augmented joint of Definition 7 is a closed variational object. Assume the top-level prior p_{top} is a nondegenerate Gaussian with full-rank precision Λ_{top}, and take the leaf observation term absent or Gaussian; this is needed both for the negative log joint to be a quadratic form in the stacked latents and for the positive-definiteness below. The precision matrix of that quadratic form is block-sparse along the hierarchical tree, with nonzero blocks only on the diagonal and on parent-child edges. The diagonal block of agent i is $\sigma^{-2}\big(I + \sum_{j:\pi(j)=i} \Omega_{j,i}^\top \Omega_{j,i}\big)$, where the identity contribution is absent at the top level (the prior precision Λ_{top} takes its place) and the leaf blocks additionally carry any Gaussian likelihood curvature; the off-diagonal block on each parent-child edge is $-\sigma^{-2}\Omega_{i,\pi(i)}$

together with its transpose. The link terms alone are only positive semidefinite, with kernel the consistent-propagation subspace on which $k_i = \Omega_{i,\pi(i)} k_{\pi(i)}$ along every edge. The full-rank prior precision at the top removes this kernel, so for finite $\sigma^2 > 0$ the stacked precision is symmetric positive definite, the joint is a proper Gaussian, and the coordinate-ascent mean-field updates of Lemma 6 are Gauss-Seidel sweeps on a symmetric positive-definite system, which converge to the unique global optimum. In this regime the evidence-bound interpretation is exact and the fixed point exists and is unique. For a single parent and its constituents, an anchored Gaussian star, the companion supplement gives the exact fixed point and the contraction rate of this coordinate-ascent schedule, with Gaussian observations of the constituents included [38, Ch. G10].

G.5.3 Self-Consistent Equilibrium Representation

Theorem 4 (Existence of a self-consistent source model)**.** *Let the latent alphabet X and the agent set be finite. Let $p_i(k) > 0$, let $\pi_{ij} > 0$ with $\sum_j \pi_{ij} = 1$, and let $a_i > 0$, $\lambda > 0$, and $\tau > 0$. Assume every stochastic transport has a uniform floor: there is $\epsilon > 0$ such that $(\mathsf{T}_{ij} q)(k) \geq \epsilon$ for every probability vector q, state k, and active edge. Define*

$$E_{ij}(q) := D_{\mathrm{KL}}(q_i \| \mathsf{T}_{ij} q_j), \qquad \beta_{ij}(q) := \frac{\pi_{ij} \exp[-E_{ij}(q)/\tau]}{\sum_m \pi_{im} \exp[-E_{im}(q)/\tau]}, \tag{245}$$

and $\bar{a}_i := a_i + \lambda$. The geometric-pooling response map

$$B_i(q)(k) := \frac{p_i(k)^{a_i/\bar{a}_i} \prod_j [\mathsf{T}_{ij} q_j(k)]^{\lambda \beta_{ij}(q)/\bar{a}_i}}{\sum_{u \in \mathsf{X}} p_i(u)^{a_i/\bar{a}_i} \prod_j [\mathsf{T}_{ij} q_j(u)]^{\lambda \beta_{ij}(q)/\bar{a}_i}} \tag{246}$$

has at least one receiver-response equilibrium $q^\star = B(q^\star)$. Write $\beta^\star := \beta(q^\star)$.

At any such fixed point, define normalized row distributions

$$P_i^\star(k,j) := \pi_{ij} [\mathsf{T}_{ij} q_j^\star](k), \qquad Q_i^\star(k,j) := \beta_{ij}^\star q_i^\star(k). \tag{247}$$

They satisfy the exact identity

$$D_{\mathrm{KL}}(Q_i^\star \| P_i^\star) = \sum_j \beta_{ij}^\star D_{\mathrm{KL}}(q_i^\star \| \mathsf{T}_{ij} q_j^\star) + D_{\mathrm{KL}}(\beta_i^\star \| \pi_i). \tag{248}$$

The identity is algebraic: it holds with $(q^\star, \beta^\star)$ replaced by any (q, β) in which each q_i and each row β_i is a probability vector, and in particular at the equilibrium.

Proof. The uniform floor makes every energy in Eq. (245) continuous on the closed product simplex. Hence $\beta(q)$ and $B(q)$ are continuous, and B maps that compact convex simplex into itself. Brouwer's theorem gives a receiver-response equilibrium. For fixed source templates and fixed β_i, the Euler equation for

$$a_i D_{\mathrm{KL}}(r \| p_i) + \lambda \sum_j \beta_{ij} D_{\mathrm{KL}}(r \| \mathsf{T}_{ij} q_j), \tag{249}$$

under $\sum_k r(k) = 1$ gives $r = B_i(q)$, so the fixed point is self-consistent with the frozen-row optimum. Expanding $\log[Q_i/P_i] = \log(\beta_{ij}/\pi_{ij}) + \log[q_i(k)/\mathsf{T}_{ij} q_j(k)]$ and summing against Q_i over (k,j), using $\sum_k q_i(k) = \sum_j \beta_{ij} = 1$, proves Eq. (248) for every such (q, β). □

Remark 4 (What the equilibrium construction proves)**.** *The distributions $P_i^\star$ are fixed and normalized only after $q^\star$ has been selected. Their product is a proper auxiliary model over independent*

row variables, but it duplicates the transported source templates. It is not automatically a joint over shared latent variables $(k_1, \dots, k_N)$. Such a joint additionally requires compatible conditionals, an acyclic factorization, or an undirected normalized potential. Equation (248) *matches the ordinary row KL at unit attention temperature; arbitrary τ gives the tempered loss-based update of Eq.* (245)*, a generalized-Bayesian construction and not one unscaled KL [207]. Because the response holds incoming source roles fixed, its equilibrium is not thereby a stationary point of the global smoothing scalar. The theorem proves existence, not uniqueness or convergence of a particular update schedule.*

G.5.4 Scalar Objectives and Update Fields

The within-scale construction requires a distinction between three objects. First, the canonical entropy-retaining row objective is $\mathcal{F}_{i,\mathrm{full}} = \sum_j \beta_{ij} E_{ij} + \tau \sum_j \beta_{ij} \log(\beta_{ij}/\pi_{ij})$. At its simplex optimum it is exactly $-\tau \log Z_i$, and the envelope theorem gives $\mathrm{d}\mathcal{F}_{i,\mathrm{red}} = \sum_j \beta^*_{ij} \mathrm{d}E_{ij}$. Summing over rows therefore gives one scalar potential with a symmetric Hessian. Second, the entropy-suppressed quantity $S = \sum_{ij} \beta^*_{ij}(x) E_{ij}(x)$ is a different scalar. Its response terms change its gradient but do not change the symmetry of its Hessian. Third, the deliberately asymmetric field

$$V_i^{\mathrm{rec}}(x) := \partial_{x_i} \left(\sum_j \beta^*_{ij}(x) E_{ij}(x) \right), \qquad \text{with all } \partial_{x_i} \left(\sum_m \beta^*_{\ell m} E_{\ell m} \right),\ \ell \neq i,\ \text{discarded}, \tag{250}$$

is not the gradient of either scalar and is generically nonintegrable because cross-row mixed derivatives need not match. This nonpotential field is an optional update approximation, not a property of the softmax, the noncompact group, or differentiation of a scalar. The consensus functional remains a composite scalar potential and not a fixed state-level generative-model evidence bound; that distinction does not require a nonsymmetric Hessian.

Existence for the full Gaussian system is more delicate than the compact finite-state result of Theorem 4. A restricted stationary point exists under a symmetry hypothesis: the mean-only consensus state $\mu_i \equiv 0$ is stationary for every prior strength $\alpha > 0$, temperature $\tau > 0$, and finite N when the prior means vanish and observation forcing is absent, because every pairwise alignment mean-gradient $\Lambda_{t,ij}(\mu_i - \Omega_{ij}\mu_j)$ and the prior pull vanish there. This does not prove a fixed point for the covariance and frame coordinates. For live state-dependent softmax weights on $\mathrm{GL}^+(K_q)$, local stability, global uniqueness, and global convergence remain open, and noncompact frame directions prevent a direct compactness argument unless one imposes a gauge slice, a bounded invariant container, or a coercive frame prior. A Banach guarantee holds in a restricted setting. Freeze covariances and attention weights, restrict transports to the orthogonal group, and take isotropic couplings $\Lambda_{t,ij} = \lambda_{ij} I$ with an arbitrary symmetric positive-definite Λ_p. The mean sub-map $\mu_i \leftarrow (\alpha\Lambda_p + b_i I)^{-1}(\alpha\Lambda_p \mu_{p,i} + \sum_j \beta_{ij}\lambda_{ij}\Omega_{ij}\mu_j)$, with $b_i = \sum_j \beta_{ij}\lambda_{ij}$, then has row norm $\sum_j \|(\alpha\Lambda_p + b_i I)^{-1}\beta_{ij}\lambda_{ij}\Omega_{ij}\|_2 = b_i/[\alpha\lambda_{\min}(\Lambda_p) + b_i] < 1$, so it is a contraction in the norm $\max_i \|\mu_i\|_2$. The isotropy hypothesis cannot be dropped. For anisotropic $\Lambda_{t,ij}$ that do not commute with their weighted sum, the operator $(\alpha\Lambda_p + \sum_j \beta_{ij}\Lambda_{t,ij})^{-1}\beta_{ij}\Lambda_{t,ij}$ can have norm above $\beta_{ij}\lambda_{\max}(\Lambda_{t,ij})/\lambda_{\max}(\sum_j \beta_{ij}\Lambda_{t,ij})$. In an instance with $N = 3$, $K_q = 2$, $\alpha = 0.05$, $\Lambda_p = I$, anisotropic $\Lambda_{t,ij}$, and rotations Ω_{ij}, the Jacobian of the map has spectral radius 1.173, while the bound $\max_i \lambda_{\max}(M_i)/[\alpha + \lambda_{\max}(M_i)]$ with $M_i = \sum_j \beta_{ij}\Lambda_{t,ij}$ equals 0.9993; iterating the map amplifies the error by a factor of about 10^3 in 60 steps, so it is not a contraction in any norm. Relaxing the restrictions removes the contraction proof, but it does not remove the canonical scalar potential. Strong monotonicity can instead provide a sufficient uniqueness criterion for the receiver-only

field (250) on a bounded invariant set; that statement concerns the optional asymmetric approximation and not descent on the canonical scalar.

For the prospective protocol of Section 13, the consequence is that the frozen-model-channel arm ($\gamma_{ij} = 0$) and the arm with declared nonzero slow coupling both retain the fast-channel belief alignment, so neither reduces to the exactly solvable cross-scale mean field. The open items are a perturbative bound on how the within-scale corrections move the Gaussian cross-scale fixed point, compatibility of cyclic row models, and uniqueness and convergence for the fully coupled, state-dependent $\mathrm{GL}^+(K_q)$ system.

H Beyond Mean Field: Correlated Posteriors, the Nested Bound, and Fluctuation Corrections

Mean field enters the present construction in four places. The factorization $Q = \prod_i q_i$ suppresses state-posterior correlations. Deterministic minimization of one population functional selects a single configuration and suppresses configuration fluctuations. The exchangeable large-N description replaces a finite labeled population by an empirical law. The operational coarse-graining construction projects an exact pushforward back into a selected multi-agent functional class. These are four different approximations, and improving one does not remove the other three.

The configuration-space Gibbs lift of Theorem 6 places a probability law on whole belief configurations and introduces the distinct configuration temperature T_{cfg}, which is a bookkeeping scale of that ensemble and not a physical temperature. The Laplace term in Eq. (277) also retains a local Gaussian fluctuation determinant when internal cluster variables are eliminated. Those results do not yet define a correlated posterior over the latent states of different agents, a generating functional for connected configuration correlations, or a self-consistent two-point theory. This appendix states the nested bound that organizes these objects, the correlated state families that preserve evidence semantics, and an outlook on fluctuation corrections, without reclassifying the peer-KL functional as an ordinary state-level evidence bound.

H.1 A Normalized Nested Variational Model

Let X denote a slow generative configuration of prior and transition parameters, supports, graph or source priors, precision fields, and frame or edge variables. It does not contain the variational posterior densities q_i or s_i. Let $Y = (Y_1, \dots, Y_N)$ denote the state-level and model-level latent variables, and let o denote the observations. The state model is a fixed normalized conditional law

$$p_\theta(dY, o \mid X) := p_\theta(dY \mid X) p_\theta(o \mid Y, X), \tag{251}$$

whose factors do not depend on the variational posterior used to approximate them. For a possibly correlated conditional posterior $Q_X(dY)$, define the state variational free energy in nats by

$$\mathcal{F}_{\mathrm{state}}[Q_X; X, o] := D_{\mathrm{KL}}\big(Q_X \| p_\theta(\cdot \mid X)\big) - \mathbb{E}_{Q_X} \log p_\theta(o \mid Y, X). \tag{252}$$

It obeys

$$\mathcal{F}_{\mathrm{state}}[Q_X; X, o] = -\log p_\theta(o \mid X) + D_{\mathrm{KL}}\big(Q_X \| p_\theta(\cdot \mid o, X)\big), \tag{253}$$

so correlated inference changes the approximation family without changing the evidence semantics.

Theorem 5 (Nested state and configuration VFE)**.** *Let $P_0(dX)$ be a proper configuration prior, let Eq. (251) be normalized for P_0-almost every X, and define $p_\theta(o) := \int p_\theta(o \mid X) P_0(dX) \in (0, \infty)$.*

Let $R \ll P_0$ be a configuration posterior and let $X \mapsto Q_X$ be a measurable probability kernel with $Q_X \ll p_\theta(\cdot \mid X)$ for R-almost every X. Assume the terms below are finite, or interpret the identity in the extended-real sense without an indeterminate $\infty - \infty$. Then

$$\begin{aligned}\mathcal{J}[R, Q] &:= T_{\mathrm{cfg}} D_{\mathrm{KL}}(R \| P_0) + T_{\mathrm{cfg}} \int \mathcal{F}_{\mathrm{state}}[Q_X; X, o] R(dX) \\ &= -T_{\mathrm{cfg}} \log p_\theta(o) + T_{\mathrm{cfg}} D_{\mathrm{KL}}\big(R(dX) Q_X(dY) \| P_\theta(dX, dY \mid o)\big). \end{aligned} \tag{254}$$

Consequently, $-\mathcal{J}/T_{\mathrm{cfg}}$ is an evidence lower bound for one normalized hierarchical model, with equality if and only if the joint variational law equals the exact posterior almost everywhere.

Proof. Expand the joint KL divergence by the chain rule, use $P_\theta(dX, dY, o) = P_0(dX) p_\theta(dY, o \mid X)$, and collect the conditional terms into Eq. (252). □

The quantity $\mathcal{J}/T_{\mathrm{cfg}}$ is the two-level case, with latent (X, Y), of the tower variational free energy of the companion theory [30, Ch. 12]. It is stated here because the configuration temperature and the vacuum prior below are specific to this paper. If P_0 is itself defined by a vacuum functional,

$$P_0(dX) := Z_0^{-1} \exp[-\mathcal{F}_{\mathrm{vac}}(X)/T_{\mathrm{cfg}}] \nu_0(dX), \tag{255}$$

then the first term of Eq. (254) is, up to the constant $T_{\mathrm{cfg}} \log Z_0$, the sum of $\mathbb{E}_R \mathcal{F}_{\mathrm{vac}}$ and $T_{\mathrm{cfg}} D_{\mathrm{KL}}(R \| \nu_0)$. This nests state-level evidence inside the configuration ensemble. The vacuum functional may contain only observation-free structural energies in the declared variables X; it cannot contain live q_i, s_i, or their peer-KL terms without promoting separate measure-valued fields to generative variables and specifying their relation to Q_X. The held-out likelihood is excluded from $\mathcal{F}_{\mathrm{vac}}$ and enters once through Eq. (252). Environmental variables may be represented as ordinary latent nodes or observed boundary conditions; the construction does not treat a variational belief of another agent as a fixed generative factor.

Both terms of $\mathcal{J}$ carry T_{cfg}, so at the optimal state posterior the optimal configuration posterior is $R^*(dX) \propto \exp[-\mathcal{F}_{\mathrm{vac}}(X)/T_{\mathrm{cfg}}] p_\theta(o \mid X) \nu_0(dX)$. As $T_{\mathrm{cfg}} \to 0$ it concentrates on the minimizers of $\mathcal{F}_{\mathrm{vac}}$ independently of o: for $\mathcal{F}_{\mathrm{vac}}(X) = X^2/2$ and $p_\theta(o \mid X) = \mathcal{N}(o; X, 1)$ one finds $\mathbb{E}_{R^*}[X] = oT_{\mathrm{cfg}}/(1 + T_{\mathrm{cfg}})$. With this scaling the data enter the configuration law only at order T_{cfg}, the same order as the fluctuation corrections below, so the limit $T_{\mathrm{cfg}} \to 0$ does not recover data-driven minimization of the multi-agent functional.

H.2 Correlated State Posteriors and Region Free Energies

The first extension beyond mean field preserves the normalized model above while enlarging Q_X. Suppose the conditional state prior has a gauge-covariant pairwise factorization on a graph $\mathcal{G} = (V, E)$,

$$p_\theta(Y \mid X) := Z_X^{-1} \prod_{i \in V} \psi_i(Y_i; X) \prod_{(i,j) \in E} \psi_{ij}(Y_i, Y_j; X), \tag{256}$$

where each edge factor is a fixed model potential and not a function of live variational densities. Introduce node beliefs b_i and edge beliefs b_{ij} satisfying normalization and local marginal consistency. Their Bethe entropy is

$$H_{\mathrm{B}}[b] := \sum_{(i,j) \in E} H[b_{ij}] - \sum_{i \in V} (d_i - 1) H[b_i]. \tag{257}$$

The corresponding region free energy is the expectation of the fixed node and edge energies $-\log \psi_i$ and $-\log \psi_{ij}$ minus H_{B}. On a tree the local marginal polytope equals the marginal polytope [205,

Prop. 4.1], and minimizing this functional over it returns the exact marginals and $-\log Z_X$. On a loopy graph it is the Bethe approximation. Larger regions give the cluster-variation method [208] and its region-graph generalizations, a family of more expressive approximations whose accuracy depends on the choice of regions and is not guaranteed to improve [209, Sec. VIII.C]. Generic loopy pseudomarginals need not be the marginals of any global Q_X, and their region objective is not a generic upper or lower evidence bound. Bound-preserving constructions require additional structure, such as a globally realizable variational family or an appropriate tree-reweighted relaxation. Ordinary mean field is recovered by the restriction $b_{ij} = b_i b_j$ on every edge when those beliefs arise from a global product law. It is therefore a lower-complexity variational subfamily of the correlated theory, not the definition of the theory.

Gauge covariance acts on the region beliefs as a joint pushforward. If $Y_i \mapsto G_i Y_i$ and $Y_j \mapsto G_j Y_j$, then b_{ij} is pushed forward by $G_i \times G_j$, while the edge potential is transformed contragrediently so that its scalar energy is unchanged. Entropies must be defined relative to a covariantly transformed or quasi-invariant reference measure, including its Jacobian; a finite invariant reference is not generally available for noncompact $\mathrm{GL}^+(K_q)$. A Regime I link $\Omega_{ij} = U_i U_j^{-1}$ has trivial loop holonomy and is removed by a passive frame re-expression. A correlated theory with nontrivial holonomy must instead retain the independent links on the interaction graph of Regime II (Section 4.6.5), with orientation reversal imposed; integrating only vertex frames cannot generate nontrivial holonomy.

For the Gaussian sector, a computationally direct correlated family is

$$Q_X(Y) = \mathcal{N}(\bar{Y}, C), \qquad C^{-1} = \Lambda_Q, \tag{258}$$

with a sparse block precision Λ_Q supported on the state interaction graph. The off-diagonal blocks C_{ij} carry inter-agent correlations, and if $Y_i = (k_i, m_i)$ contains fast state and slow model variables, the within-node block $C_i^{(km)}$ carries cross-channel correlations that the separate q_i and s_i factors discard. Under a local linear frame change, $C_{ij} \mapsto \rho_i(G_i) C_{ij} \rho_j(G_j)^\top$, where ρ_i is block diagonal if the state and model channels use distinct representations, so the covariance family is closed under the group action. The block-diagonal restriction $C_{ij} = 0$ for $i \neq j$ is the Gaussian mean-field limit. At a Gaussian target of precision Λ, its exact cost at the optimum is the determinant gap $\frac{1}{2}[\sum_b \log\det \Lambda_{bb} - \log\det \Lambda] \geq 0$, which vanishes exactly when all cross-block precisions vanish [38, Ch. G4]. Sparse block-Gaussian inference is therefore the smallest extension that exposes a measurable connected two-point state function while retaining tractable linear algebra.

The live alignment term $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ does not become an exact instance of Eq. (256) merely because edge beliefs have been introduced: its target still moves with another variational factor, which is the obstruction proved by Theorem 3. It has two coherent placements. It may remain a configuration-level consensus potential, or it may be replaced by a fixed state-transition or pair potential from which the correlated posterior is inferred. Using both for the same interaction would count that edge twice.

Remark 5 (No-double-counting conditions)**.** *An observation likelihood enters the state free energy once and is not also placed in the vacuum action; a physical interaction is represented either by a fixed edge factor of the state model or by a consensus potential on configurations, not both; attention rows are either integrated as random simplex variables or eliminated at their envelope optimum β^*, and the row entropy is not added again after elimination; and marginal Gaussian entropies and the correlation correction are components of one state-posterior entropy, while attention-row entropy and configuration meta-entropy belong to separate sample spaces. These are instances of two general statements of the companion theory: an energy appended to a joint without its normalizer is not an evidence bound [30, Ch. 4], and the complexity of a correlated recognition law splits exactly into its total correlation and its marginal divergences [30, Ch. 5].*

H.3 Outlook: Fluctuation Corrections and the Effective Action

After the state family has been specified, define a state-reduced configuration action

$$\mathcal{A}_o[X] := \mathcal{F}_{\mathrm{vac}}[X] + T_{\mathrm{cfg}} \inf_{Q_X \in \mathcal{Q}_{\mathrm{corr}}} \mathcal{F}_{\mathrm{state}}[Q_X; X, o]. \tag{259}$$

With the unrestricted posterior family the second term is $-T_{\mathrm{cfg}} \log p_\theta(o \mid X)$; with a normalized family such as a sparse block-Gaussian Q_X it is a variational upper approximation to that conditional surprisal, while a Bethe or Kikuchi pseudomarginal optimization does not inherit this evidence-bound statement. The live multi-agent functional cannot serve as $\mathcal{F}_{\mathrm{vac}}$ in Theorem 5, because it depends on the variational q_i, s_i and contains the observation term. The peer-KL sector is a gauge-covariant consensus energy and, when its belief-configuration partition function is finite, an exact generalized-Bayesian energy on that separate sample space; inside $\mathcal{A}_o$ it is only a candidate truncation pending an explicit summary map from Q_X to belief fields and quantitative energy and force residuals.

Let $\mathfrak{X}_{\mathrm{phys}}$ be the configuration space after quotienting only the transformations declared to be redundancies; active frame changes remain coordinates of X. Around a background $\bar{X}$, write $X = \mathrm{Exp}_{\bar{X}} \xi$ in normal coordinates on the quotient and define

$$Z_{\bar{X}}[J] := \int_{\mathfrak{X}_{\mathrm{phys}}} \exp\left[-\frac{\mathcal{A}_o[\mathrm{Exp}_{\bar{X}} \xi] - J_A \xi^A}{T_{\mathrm{cfg}}}\right] d\nu_{\mathrm{phys}}(\xi), \qquad W_{\bar{X}}[J] := T_{\mathrm{cfg}} \log Z_{\bar{X}}[J]. \tag{260}$$

The mean displacement is $\varphi^A = \delta W / \delta J_A$, and the Legendre transform $\Gamma_{\bar{X}}[\varphi] := J_A \varphi^A - W_{\bar{X}}[J]$ is the one-particle-irreducible (1PI) effective action in the chosen background chart. With this sign convention its value at the source-free mean is $-T_{\mathrm{cfg}} \log Z_{\bar{X}}[0]$, the configuration free energy, and its Hessian gives the connected configuration covariance $G^{AB} = T_{\mathrm{cfg}}[(\Gamma^{(2)})^{-1}]^{AB}$ on the physical modes, a covariance across whole configurations distinct from the state covariance C_{ij} of Eq. (258). The 1PI action is not the coarse energy $\mathcal{F}^{\mathrm{exact}}_{\ell+1}$ of Appendix I, which is a conditional partition function of eliminated variables (a Wilsonian effective energy in the block-variable sense, without rescaling), and it is background- and base-point-dependent until compatible constructions on overlapping quotient charts are supplied. Write $d\nu_{\mathrm{phys}} = \rho_{\mathrm{phys}} \, \mathrm{vol}_{g_{\mathrm{phys}}}$ and assume that the ambient metric and reference measure are invariant under the quotiented redundancy. When $\bar{X}$ is a nondegenerate saddle on the quotient and the covariant Hessian H_{phys} of $\mathcal{A}_o$ on the physical tangent modes is positive, the Laplace expansion gives

$$\Gamma[\bar{X}] = \mathcal{A}_o[\bar{X}] + \frac{T_{\mathrm{cfg}}}{2} \operatorname{Tr} \log\left(\frac{g^{-1}_{\mathrm{phys}} H_{\mathrm{phys}}}{2\pi T_{\mathrm{cfg}}}\right) + \Gamma_{\mathrm{measure}}[\bar{X}] + O(T_{\mathrm{cfg}}^2), \qquad \Gamma_{\mathrm{measure}} = -T_{\mathrm{cfg}} \log \rho_{\mathrm{phys}}(\bar{X}). \tag{261}$$

The factor $2\pi T_{\mathrm{cfg}}$ is part of the result; replacing it by a fixed scale leaves an $O(T_{\mathrm{cfg}} \log T_{\mathrm{cfg}})$ remainder. For a free action of the redundancy group by isometries of the ambient metric g, with ν the Riemannian volume of g, that volume factors locally into the quotient volume and the orbit volume, so $\rho_{\mathrm{phys}} = (\det \mathcal{G}_{ab})^{1/2}$ with $\mathcal{G}_{ab} = g(V_a, V_b)$ the Gram matrix of the redundancy generators, and $\Gamma_{\mathrm{measure}} = -\frac{T_{\mathrm{cfg}}}{2} \log \det \mathcal{G}$. The Frobenius frame metric of Eq. (84) is not invariant under the right redundancy $U_i \mapsto U_i g$, so for $\mathrm{GL}^+(K_q)$ frames g_{phys} is defined only after a right-invariant metric is adopted. An equivalent redundant-field calculation instead imposes a gauge condition $\Phi_{\mathrm{gf}} = 0$ and includes one Faddeev–Popov factor [210]: in coordinates orthonormal for g at $\bar{X}$, the one-loop and measure terms together equal $\frac{T_{\mathrm{cfg}}}{2} \log \det(H_{\mathrm{slice}}/2\pi T_{\mathrm{cfg}}) - T_{\mathrm{cfg}} \log |\det M_{\mathrm{FP}}|$, where H_{slice} is the Hessian on the slice and $M_{\mathrm{FP}} = \partial \Phi_{\mathrm{gf}} / \partial \theta$, because $\det M^2_{\mathrm{FP}} / \det H_{\mathrm{slice}} = \det \mathcal{G} / \det H_{\mathrm{phys}}$. The

direct-quotient and gauge-fixed prescriptions are alternatives and must not be multiplied together. A background-field construction is needed if the redundant-field expansion is to retain covariance under background gauge transformations [211–213]. Raw Haar integration over noncompact $\mathrm{GL}^+(K_q)$ is not admissible: redundant noncompact directions require a quotient or gauge fixing, and noncompact active directions require a proper confining reference measure, so that Eq. (260) exists. Equation (261) extends the cluster-specific log determinant in Eq. (277) to the full physical configuration Hessian; it is a local asymptotic expansion, not a nonperturbative definition when the Hessian has additional soft or unstable modes.

The 1PI action closes equations for the mean configuration but treats G as the inverse Hessian at the selected truncation. A two-particle-irreducible (2PI) extension would add a symmetric bilocal source K_{AB} and perform a double Legendre transform to $\Gamma_{2\mathrm{PI}}[\varphi, G]$, whose stationarity condition $\delta\Gamma_{2\mathrm{PI}}/\delta G = 0$ gives a self-consistent configuration covariance [214]. The gauge-consistent 2PI construction is a proposed completion, not a result derived from the five-term functional; it would matter when soft collective modes make the one-loop determinant large or when correlation feedback moves the saddle or changes its stability.

As Appendix I records, the exact pushforward generally generates every operator allowed by the retained symmetries, here including higher-order region interactions, nonlocal kernels, covariance couplings, curvature terms, and measure contributions [112]; closure in the original five-term family is therefore a truncation hypothesis, to be tested by the calibrated energy residual $\delta_{I,0}$ and the force residual $\delta_{I,1}$ of Eq. (271), not assumed from gauge covariance. A complete calculation would compare at least three nested truncations: the present mean-field saddle, a correlated state family with one-loop configuration corrections, and a 2PI or region-enlarged theory in which the two-point functions are varied self-consistently; the change in held-out evidence, connected correlations, spectra, and closure residuals decides whether the extra operators earn their place. This ordering defines a research program, not a controlled approximation ladder. Mean field is formally recovered by factorizing the state posterior. Exchangeable empirical-measure closure and projection of the coarse-grained image back to the five-term family are formal limits or truncations until propagation-of-chaos and quantitative closure-residual hypotheses are established.

I Exact Coarse-Graining for Meta-Agent Formation

The free-energy improvement criterion (125), the M-projection barycenter (120), and the frame Karcher mean (123) fit into an exact coarse-graining construction whose structure this appendix records. In the vocabulary of statistical physics the construction is a Kadanoff block-variable map [27] applied to a posited configuration ensemble and followed by a projection onto an ansatz; Kadanoff's variational renormalization group [28, 29] is the closest precedent for that combination. No rescaling is defined, no coupling space is identified across levels, and no flow or fixed point is constructed. On a finite network none can be: with blocking ratio at least $b > 1$ the tower has at most $\log_b N$ levels, and for finite agent spaces the interaction spaces before and after a reducing step have different dimensions and admit no comparison identification [30, Ch. 11]. The companion theory accordingly constructs no renormalization semigroup with rescaling, fixed points, or beta functions at one base point [30, Ch. 1], and treats that apparatus, for infinite networks, in its base-transport supplement [31, Chs. B5 and B6]. The one coupling that the construction does renormalize is the parent's entropy weight, which integrating out the internal modes shifts at one loop in proportion to cluster size (Eq. (278)). Throughout, ℓ is the level index of the hierarchy, and the coarse-graining map $\mathcal{R}_\ell$ carries the state at level ℓ to the state at level $\ell + 1$.

Three claims are kept separate. The pushforward step is exact: any measurable coarse-graining

map carries the level-ℓ Gibbs measure to a level-$(\ell+1)$ measure with the same partition function and the same retained observables, and in the companion theory's conditional-partition form it needs no coarea density and no separately chosen parent reference [30, Ch. 11]. Closure, meaning that the coarse energy again lies in the multi-agent functional class, is proved here only for the mean sector at fixed covariances, frames, and couplings (Lemma 8); elsewhere it is an ansatz whose error is measured by the residuals (271), and exact closure generically generates hyperedges [30, Ch. 11]. The threshold detector is a candidate filter, and the conditional theorem that links a strengthened detector to a positive retention gain rests on a hypothesis with no identified source (Remark 8). Standard results are cited in place: Laplace's method for the saddle-point step [215], the Schur complement [216], Fenichel's theorem for the slow manifold [90], and the Riemannian center of mass [93, 95]. The status of each claim is recorded in Section 11 and in Table 2.

Status of the Gibbs ensemble. The pushforward construction is stated for a configuration-space Gibbs measure $d\mathbb{P}_\ell = Z_\ell^{-1} \exp[-\mathcal{F}_\ell/T_{\mathrm{cfg}}] d\nu_\ell$. This is a hypothetical thermal ensemble. The framework's deterministic natural-gradient dynamics descend $\mathcal{F}_\ell$ along a single trajectory and do not sample it, and we do not posit a Langevin or maximum-entropy mechanism that would make this Gibbs law the distribution of agent states. The partition-function and observable-preservation results below are exact statements about the posited ensemble when it is normalizable, and contact with the deterministic dynamics is made only in the zero-configuration-temperature limit $T_{\mathrm{cfg}} \to 0$. The symbol T_{cfg} is distinct from the attention temperature τ.

Theorem 6 (Belief-configuration Gibbs lift)**.** *Let $(\mathfrak{X}, \mathcal{A}, \nu_0)$ be a measurable configuration space with proper reference probability measure ν_0, let $\mathcal{F} : \mathfrak{X} \to (-\infty, +\infty]$ be measurable, and let $T_{\mathrm{cfg}} > 0$. Define*

$$Z_{\mathcal{F}} := \int_{\mathfrak{X}} \exp[-\mathcal{F}(X)/T_{\mathrm{cfg}}] d\nu_0(X). \tag{262}$$

A probability measure $\mathbb{P}_{\mathcal{F}} \ll \nu_0$ with density

$$\frac{d\mathbb{P}_{\mathcal{F}}}{d\nu_0}(X) = Z_{\mathcal{F}}^{-1} \exp[-\mathcal{F}(X)/T_{\mathrm{cfg}}] \tag{263}$$

exists if and only if $0 < Z_{\mathcal{F}} < \infty$. For every probability measure $R \ll \nu_0$ for which the terms are finite,

$$\mathbb{E}_R[\mathcal{F}] + T_{\mathrm{cfg}} D_{\mathrm{KL}}(R\|\nu_0) = T_{\mathrm{cfg}} D_{\mathrm{KL}}(R\|\mathbb{P}_{\mathcal{F}}) - T_{\mathrm{cfg}} \log Z_{\mathcal{F}}. \tag{264}$$

Consequently,

$$-T_{\mathrm{cfg}} \log Z_{\mathcal{F}} = \inf_{R \ll \nu_0} \left\{ \mathbb{E}_R[\mathcal{F}] + T_{\mathrm{cfg}} D_{\mathrm{KL}}(R\|\nu_0) \right\}, \tag{265}$$

with unique minimizer $R = \mathbb{P}_{\mathcal{F}}$. If $\mathcal{F}$ is bounded below ν_0-almost surely and finite on a set of positive ν_0-measure, then $0 < Z_{\mathcal{F}} < \infty$.

Proof. The condition $0 < Z_{\mathcal{F}} < \infty$ is necessary and sufficient to normalize Eq. (263). When it holds,

$$\log \frac{dR}{d\mathbb{P}_{\mathcal{F}}} = \log \frac{dR}{d\nu_0} + \frac{\mathcal{F}}{T_{\mathrm{cfg}}} + \log Z_{\mathcal{F}}. \tag{266}$$

Integration under R proves Eq. (264). Nonnegativity of relative entropy gives the infimum and its unique minimizer. If $\mathcal{F} \geq m$ almost surely, then $Z_{\mathcal{F}} \leq \exp(-m/T_{\mathrm{cfg}}) < \infty$; finiteness of $\mathcal{F}$ on a positive-measure set gives $Z_{\mathcal{F}} > 0$. □

Remark 6 (Probabilistic level and gauge volume)**.** *Theorem 6 proves the existence of a generative model over belief configurations X, not over only the original state latents k_i. The exact density relation is $\mathcal{F}(X)/T_{\mathrm{cfg}} = -\log[d\mathbb{P}_{\mathcal{F}}/d\nu_0](X) - \log Z_{\mathcal{F}}$, so $\mathcal{F}$ is the configuration energy, equivalently the temperature-scaled negative log Radon-Nikodym density up to normalization. Equation* (264) *is the resulting exact meta-level variational free energy; it does not convert the pointwise population functional into the state-level ELBO excluded by Theorem 3. A hierarchical model may treat the beliefs q_i themselves as random latent measures, draw $X \sim \mathbb{P}_{\mathcal{F}}$, then draw $k_i \sim q_i$ and observations from a fixed likelihood. Its ELBO necessarily contains the configuration entropy $T_{\mathrm{cfg}} D_{\mathrm{KL}}(R\|\nu_0)$ in addition to $\mathbb{E}_R[\mathcal{F}]$, so the original functional is the prior energy or MAP objective inside the enlarged model. This is also the loss-based generalized-Bayesian reading of the Gibbs law [207]. The companion theory lists five conditions that a normalized prior on law points must meet [30, Ch. 3]: a σ-finite reference, a finite normalizer, the transformation law under the active gauge action, record kernels that read law points, and compatibility of the law-point posterior with the sample-level recognition marginals. For finite-dimensional Gaussian presentations the lift supplies the first two, the next paragraph addresses the third, and the last two remain open.*

A proper reference probability makes the existence condition transparent. Raw invariant volume on a noncompact gauge orbit does not. If $\mathcal{F}$ is gauge-invariant and ν_0 disintegrates into a quotient measure times Haar measure on $G = \mathrm{GL}^+(K_q)$, the Haar factor has infinite volume. A proper construction must therefore quotient or gauge-fix with the appropriate Jacobian, impose a compact container, add a coercive symmetry-breaking prior, or choose a proper non-Haar reference measure. No unregulated $\mathrm{GL}^+(K_q)$ Haar ensemble is claimed below.

Exact pushforward. Let $X_\ell = \{q_i, s_i, U_i, \chi_i\}_{i\in\mathcal{I}_\ell}$ denote the state at level ℓ, consisting of beliefs, model sections, frames, and presence masks, and write the Gibbs measure as $d\mathbb{P}_\ell = Z_\ell^{-1}\exp[-\mathcal{F}_\ell/T_{\mathrm{cfg}}]d\nu_\ell$ with a proper reference probability ν_ℓ, or a normalized quotient or gauge-fixed reference, on the Gaussian parameters, masks, and frames. For each admissible cluster $I \subset \mathcal{I}_\ell$, a connected component of the pairwise overlap graph, the parent state $Y_I = (q_I, s_I, U_I, \chi_I)$ is defined by the M-projection barycenter (120), the Karcher frame (123), and the multi-child overlap mask $\chi_I(c) = \mathbb{1}\{\sum_{i\in I}\chi_i(c) \geq 2\}$; the cluster maps assemble into the coarse-graining map $\mathcal{R}_\ell$. The pushed reference $\nu^c_{\ell+1} := (\mathcal{R}_\ell)_*\nu_\ell$ is again a probability measure. The pushforward $\widetilde{\rho}_{\ell+1} = (\mathcal{R}_\ell)_*\big(e^{-\mathcal{F}_\ell/T_{\mathrm{cfg}}}d\nu_\ell\big)$ of the unnormalized Gibbs measure is absolutely continuous with respect to it, with density given by the conditional partition function

$$\exp[-\mathcal{F}^c_{\ell+1}(Y)/T_{\mathrm{cfg}}] = \frac{d\widetilde{\rho}_{\ell+1}}{d\nu^c_{\ell+1}}(Y) = \mathbb{E}_{\nu_\ell}\big[e^{-\mathcal{F}_\ell/T_{\mathrm{cfg}}}\big|\mathcal{R}_\ell = Y\big]. \tag{267}$$

This is the companion theory's conditional-partition formula for the deterministic kernel of $\mathcal{R}_\ell$ [30, Ch. 11]. It preserves the partition function, $Z_{\ell+1} = Z_\ell$, and every observable that depends only on retained variables, $\mathbb{E}_{\ell+1}[A(Y)] = \mathbb{E}_\ell[A(\mathcal{R}_\ell(X))]$, and it composes along any chain of levels. If a different proper parent reference $\nu_{\ell+1}$ with $\nu^c_{\ell+1} \ll \nu_{\ell+1}$ is declared, the coarse energy acquires a reference-change term, $\mathcal{F}_{\ell+1} = \mathcal{F}^c_{\ell+1} - T_{\mathrm{cfg}}\log(d\nu^c_{\ell+1}/d\nu_{\ell+1})$, and we write $\mathcal{F}^{\mathrm{exact}}_{\ell+1}$ for the coarse energy against whichever parent reference is declared. When ν_ℓ and $\nu_{\ell+1}$ have smooth strictly positive densities relative to Riemannian volume on each fixed-dimensional positive-weight stratum, and $\mathcal{R}_\ell$ is almost everywhere a C^1 submersion on the mean, covariance, and frame sector, the coarea formula writes the same density as a fiber integral,

$$\exp[-\mathcal{F}_{\ell+1}(Y)/T_{\mathrm{cfg}}] = \frac{d\widetilde{\rho}_{\ell+1}}{d\nu_{\ell+1}}(Y) = \int_{\mathcal{R}_\ell^{-1}(Y)} e^{-\mathcal{F}_\ell/T_{\mathrm{cfg}}}d\mathcal{H}_Y, \tag{268}$$

where $d\mathcal{H}_Y$ denotes the conditional fiber measure, including the normal-Jacobian factor and the ratio of the source and target reference densities. The fiber form is used below only for the Laplace step. In the zero-configuration-temperature limit the coarse energy reduces to the fiber profile

$$\mathcal{F}_{\mathrm{eff}}(Y) := \inf_{X:\mathcal{R}_\ell(X)=Y} \mathcal{F}_\ell(X), \tag{269}$$

up to terms of order $T_{\mathrm{cfg}} \log T_{\mathrm{cfg}}$, the leading one of which is displayed in Eq. (277). None of these identities claims that $\mathcal{F}_{\ell+1}^{\mathrm{exact}}$ has multi-agent form.

Gauge actions at one base point. Every statement of this appendix is pointwise in the base point. A finite grid of base points enters only through the presence fraction of the detector, and no base transport is used. At a fixed base point three operations must be kept apart [30, Ch. 2]. A passive re-expression replaces (U_i, q_i) by $(g_i U_i, (g_i)_\# q_i)$ for an arbitrary tuple (g_i); it sends $\Omega_{ij} \mapsto g_i \Omega_{ij} g_j^{-1}$ and $\Omega_{ij} q_j \mapsto (g_i)_\#(\Omega_{ij} q_j)$, so every transported divergence is unchanged by Theorem 1, and in Regime I the frames can be removed from the functional altogether. A change of reference, $U_i \mapsto U_i g$ for one g shared by all agents with the beliefs held fixed, leaves every $\Omega_{ij} = U_i U_j^{-1}$ unchanged. The active gauge group at one base point is a single copy of G, not the product of per-agent copies [30, Ch. 2]. The equivariance statement proved below concerns the diagonal action of one $g \in G$, $U_i \mapsto g U_i$ and $q_i \mapsto g_\# q_i$ for every agent, which is the passive re-expression with all $g_i = g$. It transports the parent frame as $U_I \mapsto g U_I$ and acts on transports and beliefs as

$$\Omega_{ij} \mapsto g \Omega_{ij} g^{-1}, \qquad q_j \mapsto g_\# q_j, \qquad \Omega_{ij} q_j \mapsto g_\#(\Omega_{ij} q_j), \tag{270}$$

so KL-based observables transform by common pushforward and are invariant. Under this action the M-projection barycenter and the Karcher frame commute with the pushforward, $\mathcal{R}_I(g \cdot X) = g \cdot \mathcal{R}_I(X)$ (Theorem 9), and $\mathcal{F}_{\ell+1}^{\mathrm{exact}}(g \cdot Y) = \mathcal{F}_{\ell+1}^{\mathrm{exact}}(Y)$ when the reference measure ν_ℓ is gauge-invariant. For compact $G = \mathrm{SO}(K_q)$ the normalized bi-invariant Haar measure provides such a reference. For the noncompact $\mathrm{GL}^+(K_q)$, Haar measure is still bi-invariant, since the group is unimodular, but it has infinite total mass, and the Lebesgue factors of the reference measure on the Gaussian parameters are only quasi-invariant under the action, with per-agent Jacobian density $|\det g|^{K_q+2}$ ($|\det g|$ from the mean, $\mu \mapsto g\mu$, and $|\det g|^{K_q+1}$ from the covariance congruence $\Sigma \mapsto g \Sigma g^\top$), so a gauge slice or Radon-Nikodym correction is required.

Closure ansatz on the multi-agent class. The pushforward construction defines $\mathcal{F}_{\ell+1}^{\mathrm{exact}}$ exactly. Whether $\mathcal{F}_{\ell+1}^{\mathrm{exact}}$ retains the multi-agent functional form of $\mathcal{F}_\ell$ is a separate claim. Only the normalized Gibbs probability $Z^{-1} e^{-\mathcal{F}/T_{\mathrm{cfg}}} d\nu$ is invariant under $\mathcal{F} \mapsto \mathcal{F} + c$: the unnormalized density is multiplied by $e^{-c/T_{\mathrm{cfg}}}$ and its partition function changes by the same factor. We therefore separate quotient closure, partition-function calibration, uniform energy error, and force error. On a parent neighborhood $\mathcal{N}_I$, let Z_I^{exact} and Z_I^{agent} be the corresponding local partition functions and define

$$\begin{aligned}
\delta_{I,0}^{\mathrm{quot}} &= \inf_{c \in \mathbb{R}} \left\| \mathcal{F}_{\ell+1}^{\mathrm{exact}} - \mathcal{F}_{\ell+1}^{\mathrm{agent}} - c \right\|_{L^\infty(\mathcal{N}_I)}, \\
c_I^Z &= T_{\mathrm{cfg}} \log \frac{Z_I^{\mathrm{agent}}}{Z_I^{\mathrm{exact}}}, \qquad \widehat{\mathcal{F}}_{\ell+1}^{\mathrm{agent}} = \mathcal{F}_{\ell+1}^{\mathrm{agent}} + c_I^Z, \\
r_I &= \mathcal{F}_{\ell+1}^{\mathrm{exact}} - \widehat{\mathcal{F}}_{\ell+1}^{\mathrm{agent}}, \qquad \delta_{I,0} = \|r_I\|_{L^\infty(\mathcal{N}_I)}, \\
r_I^{(0)} &= \mathcal{F}_{\mathrm{eff}} - \widehat{\mathcal{F}}_{\ell+1}^{\mathrm{agent}}, \qquad \delta_{I,1} = \|dr_I^{(0)}\|_{L^\infty(\mathcal{N}_I; G_{\mathrm{slow}}^{-1})}.
\end{aligned} \tag{271}$$

The choice c_I^Z makes the calibrated agent partition function equal to the exact one. Writing $D = \mathcal{F}_{\ell+1}^{\mathrm{exact}} - \mathcal{F}_{\ell+1}^{\mathrm{agent}}$, one has $c_I^Z = T_{\mathrm{cfg}} \log \mathbb{E}_{\mathcal{N}_I}^{\mathrm{exact}}[e^{D/T_{\mathrm{cfg}}}] \in [\inf D, \sup D]$ under the normalized local exact Gibbs law, while $\delta_{I,0}^{\mathrm{quot}} = \frac{1}{2} \operatorname{osc}_{\mathcal{N}_I} D$; hence the two energy residuals satisfy $\delta_{I,0}^{\mathrm{quot}} \le \delta_{I,0} \le 2\delta_{I,0}^{\mathrm{quot}}$. The force residual $\delta_{I,1}$ is measured against the zero-temperature profile (269), because the deterministic flow descends that profile; the finite-temperature energy $\mathcal{F}_{\ell+1}^{\mathrm{exact}}$ differs from it by the fluctuation determinant of Eq. (277), whose gradient the deterministic flow does not feel (see the discussion after Eq. (282)). The calibrated residual $\delta_{I,0}$ is used for the static Gibbs statements. For reporting across levels, choose a fixed energy scale $E_{\mathrm{ref}} > 0$, by default $E_{\mathrm{ref}} = T_{\mathrm{cfg}}$, and use the shift-invariant dimensionless diagnostic

$$\varepsilon_I = \frac{\delta_{I,0}}{E_{\mathrm{ref}} + \operatorname{osc}_{\mathcal{N}_I}(\mathcal{F}_{\ell+1}^{\mathrm{exact}})}, \qquad \operatorname{osc}_{\mathcal{N}_I}(\mathcal{F}) = \sup_{\mathcal{N}_I} \mathcal{F} - \inf_{\mathcal{N}_I} \mathcal{F}. \tag{272}$$

The dimensionful residual $\delta_{I,0}$ controls calibrated scalar-energy estimates, while $\delta_{I,1}$ controls perturbations of the gradient flow; ε_I is a reporting diagnostic and is not added to an energy or a vector field. Conditions (i) Gaussian children, (ii) coherent cluster, $V_I = \sum_i w_i^I D_{\mathrm{KL}}(\widetilde{q}_i^I \| q_I) \ll 1$, (iii) positive constrained weighted gap $\lambda_{I,w} > 0$ defined below, (iv) edge-marginal compatibility of cluster weights, and (v) compact gauge group with the constituent frames in a geodesic ball of radius below the convexity radius r_{cx} of the coarse-graining-map paragraph, or explicit gauge fixing or regulation for noncompact G, support the schematic C^0 quotient estimate of Proposition 13. They do not control $\delta_{I,1}$: a small force residual is an independent local C^1 closure hypothesis, because a small uniform energy error alone gives no derivative bound. The scalar approximations below are therefore C^0 closure statements, and the reduced-flow estimate (282) is conditional on separately establishing or measuring a small $\delta_{I,1}$. Deriving such a force-level estimate from explicit derivative-regularity assumptions remains open.

Working in the native mean and covariance coordinates of the Gaussian family around q_I, write the transported child belief as $\widetilde{q}_i^I = q_I + \xi_i$ with the barycentric constraint $\sum_i w_i^I \xi_i = 0$, which is exact in the mean sector and holds to leading order in ξ in the covariance sector. When the surrounding functional carries fixed-parameter couplings β_{ij}, the intra-cluster alignment energy expands as

$$\mathcal{F}_I^{\mathrm{int}} = \tfrac{1}{2} \xi^\top H_I^\perp(q_I) \xi + O(\|\xi\|^3), \qquad H_I^\perp(q_I) \approx L_I \otimes F(q_I), \tag{273}$$

where L_I is the weighted graph Laplacian of the symmetrized internal couplings $(\beta_{ij} + \beta_{ji})/2$ and $F(q_I)$ is the Fisher information at the parent in these coordinates; in the mean sector $F(q_I) = \Sigma_I^{-1}$. When instead β_{ij} has been envelope-eliminated as the softmax-optimized attention $\beta_{ij}^* = \pi_{ij} \exp(-E_{ij}/\tau) / \sum_k \pi_{ik} \exp(-E_{ik}/\tau)$, the reduced free energy $F_{\mathrm{red},i} = -\tau \log \sum_j \pi_{ij} \exp(-E_{ij}/\tau)$ has gradient $\nabla F_{\mathrm{red},i} = \sum_j \beta_{ij}^* \nabla E_{ij}$ and full Hessian

$$\nabla^2 F_{\mathrm{red},i} = \sum_j \beta_{ij}^* \nabla^2 E_{ij} - \tau^{-1} \mathrm{Cov}_{\beta^*}\big(\nabla E_{ij}, \nabla E_{ij}\big), \tag{274}$$

where the covariance is taken under the optimized attention; this is τ times the curvature of the companion theory's attention-reduced potential [30, Ch. 6]. The first term is the β^*-weighted Hessian of the edge energies; the second is a Fisher-type correction that lowers curvature along directions in which the per-edge gradients ∇E_{ij} disagree across $j \in I$. The block decomposition $H_I^\perp \approx L_I \otimes F$ is therefore a coherent-cluster leading-order approximation, valid when $\tau^{-1} \|\mathrm{Cov}_{\beta^*}(\nabla E_{ij})\| \ll \|\sum_j \beta_{ij}^* \nabla^2 E_{ij}\|$, that is, when the within-cluster gradient dispersion is small relative to the temperature-scaled curvature. The covariance correction vanishes identically when all ∇E_{ij} for $j \in I$ are equal, the regime targeted by condition (ii). The stiffness of the internal

modes on the constraint subspace is

$$\lambda_{I,w} = \inf_{\xi\neq 0 \text{ with } \sum_i w_i\xi_i=0} \frac{\xi^\top L_I \xi}{\xi^\top \xi}, \qquad m_I = \lambda_{I,w}\lambda_{\min}\big(F(q_I)\big), \tag{275}$$

so that m_I bounds from below the fiber Hessian $L_I \otimes F(q_I)$ on the constraint subspace in these Euclidean fiber coordinates. It is an inverse susceptibility of the internal modes: it bounds the inverse fiber Hessian, $\|H^{-1}\| \le m_I^{-1}$, and controls the Laplace remainder. The constrained gap is comparable to the algebraic connectivity $\lambda_2(L_I)$,

$$\frac{\lambda_2(L_I)}{|I|\cdot\|w\|^2} \le \lambda_{I,w} \le \lambda_2(L_I), \tag{276}$$

applied componentwise in the fiber index. For the lower bound, write $\xi = a\mathbf{1}/\sqrt{n}+\eta$ with $\eta \perp \mathbf{1}$ and $n = |I|$; the constraint and $\mathbf{1}^\top w = 1$ give $a = -\sqrt{n}(w-\mathbf{1}/n)^\top\eta$, so $\xi^\top\xi \le (1+n\|w-\mathbf{1}/n\|^2)|\eta|^2 = n\|w\|^2|\eta|^2$, while $\xi^\top L_I\xi = \eta^\top L_I\eta \ge \lambda_2|\eta|^2$. The upper bound is the Courant-Fischer principle, since the constraint cuts out a subspace of codimension one. Equality on the right holds for uniform weights and also for some nonuniform ones, and concentrated weights lower the constrained gap by at most a factor $|I|$. The internal relaxation rate of the dynamics is a different quantity. Under the natural-gradient flow with mobility η and child metric blocks $g_iF(q_I)$, the internal modes relax at the nonzero generalized eigenvalues of $L_Iv = \lambda W_g v$ with $W_g = \mathrm{diag}(g_i)$, which do not depend on the barycentric weights, and the slowest internal rate is $t_{\mathrm{int}}^{-1} = \eta\lambda_2(L_I; W_g)$ (for equal Fisher blocks, $\eta\lambda_2(L_I)$). The two quantities coincide when $w_i^I \propto g_i$ and the fiber is measured in the g-weighted norm. Internal disagreement modes are stiff when $m_I > 0$, and the parent variable is then the surviving collective coordinate.

Integrating out the internal modes by Laplace's method [215] yields

$$\mathcal{F}_{\ell+1}^{\mathrm{exact}}(Y_I) = \mathcal{F}_{\mathrm{eff}}(Y_I) + \tfrac{T_{\mathrm{cfg}}}{2}\log\det{}'H_I^\perp(Y_I) + \mathrm{const} + \mathcal{E}_{\mathrm{anh}}, \tag{277}$$

where $\det'$ is the product of the nonzero eigenvalues of $H_I^\perp$ on the full internal space, whose kernel is the common translation $\mathbf{1}\otimes\mathbb{R}^d$, with d the dimension of one child's parameter block; with the normal-Jacobian factor of $d\mathcal{H}_Y$ this pseudo-determinant is independent of the barycentric weights. The anharmonic remainder $\mathcal{E}_{\mathrm{anh}} = O(T_{\mathrm{cfg}}^2\|H^{-1}\|^3\|T_3\|^2 + T_{\mathrm{cfg}}^2\|H^{-1}\|^2\|T_4\|)$ is controlled by the cubic and quartic derivative tensors T_3, T_4 of $\mathcal{F}_\ell$ at the saddle (the cubic enters squared by parity, the quartic linearly). The one-loop term renormalizes a coupling. For the internal mean deviations at fixed couplings, $H_I^\perp \approx L_I \otimes \Sigma_I^{-1}$, whose nonzero eigenvalues are the products of the nonzero Laplacian eigenvalues $\lambda_2,\dots,\lambda_{|I|}$ with the eigenvalues of Σ_I^{-1}, so $\log\det'(L_I\otimes\Sigma_I^{-1}) = K_q\log\det' L_I - (|I|-1)\log\det\Sigma_I$ and

$$\frac{T_{\mathrm{cfg}}}{2}\log\det{}'H_I^\perp = \mathrm{const} - \frac{(|I|-1)T_{\mathrm{cfg}}}{2}\log\det\Sigma_I \tag{278}$$

as a function of the retained parent covariance. Integrating out the internal modes therefore shifts the coefficient of $\log\det\Sigma_I$, the entropy-like term of the parent's Gaussian relative entropies, by an amount proportional to $|I|-1$. This induced renormalization is absent at $T_{\mathrm{cfg}} = 0$, and at finite configuration temperature it contradicts any assumption that the coefficients of the functional are the same at every level; such coefficients can be held fixed across levels only by assumption.

Coarse inter-cluster couplings are pooled at the level of raw conductances, because β_{ij} is a row-normalized softmax distribution and summing it across cluster pairs would destroy that normalization. Define the unnormalized edge conductance $\kappa_{ij} := \pi_{ij}\exp(-E_{ij}/\tau)$, so that $\beta_{ij} = \kappa_{ij}/\sum_k\kappa_{ik}$

and the envelope-eliminated free energy is $F_{\mathrm{red},i} = -\tau \log \sum_k \kappa_{ik}$. Coarse-graining with cluster weights $w_i^I \geq 0$ subject to $\sum_{i\in I} w_i^I = 1$ gives

$$\kappa_{IJ}^R = \sum_{i\in I} w_i^I \sum_{j\in J} \kappa_{ij}, \qquad \beta_{IJ}^R = \frac{\kappa_{IJ}^R}{\sum_L \kappa_{IL}^R}, \tag{279}$$

where the sum over L includes $L = I$: the pooled intra-cluster conductance κ_{II}^R acts as an effective parent self-loop, which has no transported-divergence energy of its own in the five-term family. This pooling is the exact block row of the companion theory's single-edge label model, with receiver weights w_i^I and unit evidence factors; in its per-receiver model the exact block row is instead the selection-weighted average of normalized rows, and in general the object pushed from level to level is the joint edge-event law, under which events with both endpoints in I become meta-self events [30, Ch. 11]. The self-loop is required for the identical-children identity: when κ_{ij} does not depend on $i \in I$, the parent reduced free energy $F_{\mathrm{red},I} = -\tau \log \sum_L \kappa_{IL}^R$ equals the common child value $F_{\mathrm{red},i} = -\tau \log \sum_k \kappa_{ik}$ only when the term $L = I$ is kept. An unweighted row sum would instead introduce a spurious $-\tau \log |I|$ entropy-of-mixture offset, a convex weighting on the column index would shrink each partner conductance by a factor of order $1/|J|$ and bias the parent attention β_{IJ}^R toward small partner clusters, and summing β_{ij} directly would yield $\sum_J \beta_{IJ}^R = |I|$ in place of 1. The first-order energy residual in the linearized expansion of κ_{IJ}^R around the parent saddle vanishes on the row side when the cluster weights satisfy the edge-marginal compatibility $w_i^I = r_i / \sum_{i'} r_{i'}$, with row edge marginals $r_i = \sum_{j\in J} \kappa_{ij}$; on the column side the summed rule leaves a residual controlled by the mismatch between the partner's barycentric weights w_j^J and its conductance-weighted column profile $\sum_{i\in I} w_i^I \kappa_{ij}$, which contributes to the C^0 quotient closure residual, as does the partner-to-partner mismatch of row weights quantified in the edge-marginal paragraph of Section I.1. Controlling their derivative contribution to $\delta_{I,1}$ would require a separate uniform derivative estimate that is not supplied here.

Adiabatic elimination. The slow-manifold reduction uses the hypotheses that the companion theory states for its rate-separated flow [30, Ch. 8]: (a) $\mathcal{F}_\ell \in C^2$, and in a chart (Y, ξ) the metric is block diagonal, $G = G_{\xi\xi} \oplus G_{\mathrm{slow}}$, with symmetric positive-definite C^1 blocks and G_{slow} independent of ξ; (b) on a compact parent region the branch $\xi^*(Y)$ of conditional minimizers satisfies the uniform fast-curvature bound $\partial_{\xi\xi}^2 \mathcal{F}_\ell \succeq c_1 G_{\xi\xi}$ with $c_1 > 0$; and (c) the reduced flow stays in the interior of that region and the initial internal state lies in the basin of ξ^* for the frozen internal flow. Let $\epsilon_{\mathrm{ts}} = t_{\mathrm{int}}/t_{\mathrm{slow}}$ with $t_{\mathrm{int}}^{-1} = \eta\lambda_2(L_I; W_g)$. For a smooth one-parameter singularly perturbed family with $\epsilon_{\mathrm{ts}} \ll 1$, Fenichel's theorem [90] gives a locally invariant manifold $\mathcal{M}_{\epsilon_{\mathrm{ts}}}$ that is $O(\epsilon_{\mathrm{ts}})$-close to a compact normally hyperbolic piece of the critical manifold and can locally be written as a graph over Y. The companion theory records the application of these hypotheses to the collective free energy as open: no recognition family of the program is known to satisfy (b) on the slow states it visits, and (b) can fail after the attention rows are optimized even when every edge energy and every frozen-weight curvature is positive. In a general chart the metric has off-diagonal blocks, and the metric induced on the graph $\xi = \xi^*(Y)$ is

$$G_{\mathrm{slow}} = G_{YY} + G_{Y\xi} D\xi^* + (D\xi^*)^\top G_{\xi Y} + (D\xi^*)^\top G_{\xi\xi} D\xi^*. \tag{280}$$

The distinct quotient, or horizontal, metric obtained by minimizing the ambient quadratic form over an unconstrained internal velocity is

$$G_{\mathrm{quot}} = G_{YY} - G_{Y\xi} G_{\xi\xi}^{-1} G_{\xi Y}. \tag{281}$$

Completing the square gives $G_{\mathrm{slow}} - G_{\mathrm{quot}} = (D\xi^* - C^*)^\top G_{\xi\xi}(D\xi^* - C^*) \succeq 0$ with $C^* = -G_{\xi\xi}^{-1} G_{\xi Y}$, so the graph pullback equals the quotient metric exactly when $D\xi^* = C^*$; normal hyperbolicity alone does not impose that identity. This metric Schur complement is also distinct from the Hessian Schur complement A_{eff} of Lemma 8. In the barycentric chart with child metric weights g_i the metric is block diagonal, so that $G_{\mathrm{quot}} = G_{\mathrm{slow}}$ and the barycenter fibers coincide with the linearized Fenichel fibers, exactly when $w_i^I \propto g_i$. Because the internal energy is invariant under a common translation of the children, the fast flow conserves the g-weighted mean $\sum_i g_i x_i$ and relaxes a cluster to it. Under the edge-marginal weights of condition (iv), which are generically not proportional to g_i, the parent that $\mathcal{R}_\ell$ assigns to a cluster off the slow manifold therefore differs from the dynamically reached parent at first order in the internal displacement. Written with the calibrated agent ansatz, the parent flow has the form

$$\begin{aligned} \dot{Y}_I &= -\eta \mathrm{grad}_{G_{\mathrm{slow}}} \widehat{\mathcal{F}}^{\mathrm{agent}}_{\ell+1}(Y_I) + R_{\mathrm{ts}} + R_{\mathrm{cl}}, \\ \|R_{\mathrm{cl}}\|_{G_{\mathrm{slow}}} &\le \eta \delta_{I,1}, \qquad \|R_{\mathrm{ts}}\|_{G_{\mathrm{slow}}} = O(\epsilon_{\mathrm{ts}})\eta \|d\widehat{\mathcal{F}}^{\mathrm{agent}}_{\ell+1}\|_{G_{\mathrm{slow}}^{-1}}, \end{aligned} \tag{282}$$

with η the mobility of the natural-gradient flow. On the slow manifold the reduced vector field is $-\eta \mathrm{grad}_{G_{\mathrm{slow}}} \mathcal{F}_{\mathrm{eff}}$ by the envelope identity $d\mathcal{F}_{\mathrm{eff}} = \partial_Y \mathcal{F}_\ell(Y, \xi^*(Y))$, so $R_{\mathrm{cl}} = -\eta \mathrm{grad}_{G_{\mathrm{slow}}} r_I^{(0)}$ and its norm is at most $\eta\delta_{I,1}$. The sum $R_{\mathrm{ts}} + R_{\mathrm{cl}}$ is the semiconjugacy defect of the coarse map for the parent field $-\eta \mathrm{grad}_{G_{\mathrm{slow}}} \widehat{\mathcal{F}}^{\mathrm{agent}}_{\ell+1}$, the exact dynamical closure notion of the companion theory [30, Ch. 11]. A residual built from $\mathcal{F}^{\mathrm{exact}}_{\ell+1}$ in place of $\mathcal{F}_{\mathrm{eff}}$ would not bound R_{cl} at $T_{\mathrm{cfg}} > 0$. For $\mathcal{F}_\ell(Y, \xi) = bY^2/2 + a_0 e^{kY}\xi^2/(2\epsilon)$ one finds $\mathcal{F}^{\mathrm{exact}}_{\ell+1}(Y) = bY^2/2 + T_{\mathrm{cfg}} kY/2 + \mathrm{const}$; with the agent ansatz equal to this exact energy the finite-temperature force residual vanishes, while the deterministic flow relaxes ξ to zero and gives $\dot{Y} = -\eta b Y$, so the flow residual is $\eta T_{\mathrm{cfg}} k/2$ at every ϵ. Failure of (a)–(c) removes the Fenichel reduction and does not merely contribute an undefined additive error.

Retention. A Wilsonian relevance classification [111, 112] linearizes a flow at a fixed point $\theta^* = R(\theta^*)$ of a self-map of one coupling space. The present construction has no such self-map: $\mathcal{R}_\ell$ maps a state space with $|\mathcal{I}_\ell|$ agents to one with fewer, no comparison map between the parameter spaces of different levels is declared, a finite network admits only finitely many reducing steps, and for finite agent spaces no comparison identification of the interaction spaces exists across such a step [30, Ch. 11]. Relevance is therefore undefined here; an asymptotic classification needs infinite networks and is deferred to the companion supplement [31, Chs. B5 and B6]. The retention criterion used is the description-length gain [217, 218]

$$\Delta_I = \mathcal{L}_{\mathrm{micro}}(I) - \mathcal{L}_{\mathrm{parent}}(I), \qquad \mathcal{L}_{\mathrm{parent}}(I) = \widehat{\mathcal{F}}^{\mathrm{agent},*}_{\ell+1}(I) + C_{\mathrm{parent}}(I), \tag{283}$$

with complexity penalty $C(I)$ chosen as a Bayesian evidence prior $-T_{\mathrm{cfg}} \log P(I)$, a BIC penalty $\frac{T_{\mathrm{cfg}}}{2} k_I \log n_I$, or a variational complexity $D_{\mathrm{KL}}(Q_I \| P_I)$. The free-energy improvement criterion (125) of Section 9.2 is the special case $\Delta_I > 0$ at unspecified $C(I)$. Under an exact parent this criterion is degenerate: the fine variational free energy equals the coarse one plus the discarded conditional information, which is nonnegative, so a decrease under exact coarse-graining measures information discarded at unchanged evidence and neither model improvement nor evidence ascent [30, Ch. 9]. The criterion can discriminate between clusters only through the projected agent parent and the penalty $C(I)$, and no cost analyzed in the companion theory selects a partition [30, Ch. 9].

Detector as candidate-selection surrogate. Under conditions (i)–(v) and timescale separation, the exact coarse energy is expected to admit an agent-form approximation in the C^0 quotient

sense, with the contributions to $\delta_{I,0}^{\mathrm{quot}}$ listed in Proposition 13 and gauge-fixing corrections for non-compact groups. The threshold detector of Section 9.3 is a candidate-selection filter for these C^0 conditions of the closure ansatz, not a test of the exact pushforward. Energy retention further requires the calibrated residual $\delta_{I,0}$, checked after partition-function normalization, to be small, and force-level accuracy requires an independent bound on $\delta_{I,1}$, which neither the detector nor the residual proposition supplies.

I.1 Statements and Proofs

This subsection gives the statements and proofs behind the working forms of the preceding part of this appendix and behind Section 11, keeping the three claims of the opening (exact pushforward, closure, retention) separate. The construction is finite-dimensional, and no continuum limit is constructed. The statements are proved for a compact gauge group $G = \mathrm{SO}(K_q)$ with a bi-invariant Riemannian metric; the covariance theorem extends to any Lie group with a left-invariant Riemannian metric, and the noncompact $\mathrm{GL}^+(K_q)$ Gibbs ensemble requires a gauge slice or Radon-Nikodym correction, as stated in the gauge-action paragraph. The closure hypotheses used repeatedly below are collected once.

Assumption 1 (Closure regime)**.** *(A1)–(A5) are conditions (i)–(v) of the working-form part, with (A2) quantified as $V_I \le v_0 \ll 1$, the gap in (A3) positive uniformly on the parent neighborhood $\mathcal{N}_I$, (A4) the proportionality $w_i^I \propto r_i$, and the convexity radius r_{cx} of (A5) defined in the coarse-graining-map paragraph below.*

Two further hypotheses that a closure theorem for the full parent state might add are not used. Exact conditional Gaussianity on the complete fiber is either unsatisfiable for the map that retains Σ_I or satisfiable by any map in a suitably chosen chart (remark after Lemma 8), and flat transport across cluster pairs holds identically in Regime I (Eq. (293)).

State and reference measure. Fix a finite grid $\Lambda_\ell \subset \mathcal{C}$ of base points at level ℓ and an index set $\mathcal{I}_\ell$ of agents. The level-ℓ state is $X_\ell = \{x_i\}_{i\in\mathcal{I}_\ell}$ with $x_i = (q_i, s_i, U_i, \chi_i)$, where $q_i(c) = \mathcal{N}(\mu_i(c), \Sigma_i(c))$ is the belief, $s_i(c)$ a nondegenerate Gaussian model section on the model fiber, $U_i(c) \in G$ the frame, and $\chi_i(c) \in \{0,1\}$ the presence mask. Inter-agent transport is $\Omega_{ij} = U_i U_j^{-1}$ with $\Omega_{ij} q_j = \mathcal{N}(\Omega_{ij}\mu_j, \Omega_{ij}\Sigma_j\Omega_{ij}^\top)$, and $\widetilde{\Omega}_{ij}$ acts on the model fiber through its own representation of the same frames. The functional $\mathcal{F}_\ell : X_\ell \to \mathbb{R} \cup \{+\infty\}$ takes the multi-agent KL form of Section 4, and the Gibbs measure is $d\mathbb{P}_\ell = Z_\ell^{-1} \exp[-\mathcal{F}_\ell/T_{\mathrm{cfg}}] d\nu_\ell$, with ν_ℓ a proper probability reference or a normalized quotient or gauge-fixed measure (Remark 6).

Coarse-graining map. A candidate parent cluster $I \subset \mathcal{I}_\ell$ satisfies $|I| \ge 2$, connected overlap subgraph, nonempty multi-child support $\chi_I(c) = \mathbb{1}\{\sum_{i\in I}\chi_i(c) \ge 2\}$, constituent frames in an open geodesic ball of G of radius below r_{cx}, and finite belief and model dispersions. The selected partition is $\mathcal{P}_\ell = \{I_1, \dots, I_{N_{\ell+1}}\}$. For each I, the parent frame is the Riemannian center of mass

$$U_I(c) = \arg\min_{U\in G} \sum_{i\in I} w_i^I(c) d_G(U, U_i(c))^2, \tag{284}$$

with $G = \mathrm{SO}(K_q)$ and $d_G(U,V)^2 = \|\log(U^{-1}V)\|_F^2$ under the bi-invariant metric. It exists and is unique, as the global minimizer over G, whenever the weighted frames lie in an open geodesic ball of radius $\rho < r_{\mathrm{cx}} = \frac{1}{2}\min\{\mathrm{inj}\, G, \pi/\sqrt{\Delta}\}$, with Δ an upper bound on the sectional curvature [95, Thm. 2.1]. For $\mathrm{SO}(K_q)$ with this metric, $r_{\mathrm{cx}} = \pi/\sqrt{2}$ in d_G, a rotation angle of $\pi/2$; Kendall [96]

obtains the larger radius $\min\{\operatorname{inj} G, \pi/(2\sqrt{\Delta})\}$ only for the mean defined with the distance restricted to the ball, which is not the global minimizer used here. The weights and the detector are built from gauge-invariant divergences, so they cannot see how far apart the frames are, and the radius condition must be imposed separately. When it fails the frame mean can be set-valued, but the frame-free parent pair does not depend on the choice of U_I: with $\Omega_{Ii} = U_I U_i^{-1}$ one has $U_I^{-1}\mu_I = \sum_i w_i^I U_i^{-1}\mu_i =: m_I$ and $U_I^{-1}\Sigma_I U_I^{-\top} = \sum_i w_i^I [U_i^{-1}\Sigma_i U_i^{-\top} + (U_i^{-1}\mu_i - m_I)(U_i^{-1}\mu_i - m_I)^\top]$. Child-to-parent transport is $\Omega_{Ii}(c) = U_I(c)U_i(c)^{-1}$. The parent belief is the M-projection (moment-matching) barycenter of the transported children,

$$q_I(c) = \arg\min_{q \in \mathcal{G}_{K_q}} \sum_{i \in I} w_i^I(c) D_{\mathrm{KL}}\big(\Omega_{Ii} q_i \| q\big), \tag{285}$$

which by moment matching has $\mu_I = \sum_i w_i^I \widetilde{\mu}_i^I$ and

$$\Sigma_I = \sum_{i \in I} w_i^I \big[\widetilde{\Sigma}_i^I + (\widetilde{\mu}_i^I - \mu_I)(\widetilde{\mu}_i^I - \mu_I)^\top\big], \tag{286}$$

with the second term the between-child dispersion that must be retained in the exact Gaussian barycenter; dropping it is a high-coherence approximation, not an identity. The parent model section s_I is defined analogously. The coarse-graining map is $\mathcal{R}_\ell : X_\ell \mapsto \{Y_I = (q_I, s_I, U_I, \chi_I) : I \in \mathcal{P}_\ell\}$.

Exact pushforward. The working-form part of this appendix states the exact pushforward identities (267) and (268). Their structural content is recorded as two theorems, which are the deterministic-map cases of the companion theory's composition and functoriality statements [30, Chs. 9 and 11].

Theorem 7 (Partition function and observable preservation)**.** *Let $\mathcal{R}_\ell : X_\ell \to X_{\ell+1}$ be measurable, let $d\rho_\ell = e^{-\mathcal{F}_\ell/T_{\mathrm{cfg}}} d\nu_\ell$ denote the unnormalized Gibbs measure on X_ℓ with finite positive total mass $Z_\ell := \rho_\ell(X_\ell)$, and define the unnormalized pushforward $\rho_{\ell+1} := (\mathcal{R}_\ell)_* \rho_\ell$. Then $Z_{\ell+1} := \rho_{\ell+1}(X_{\ell+1}) = Z_\ell$, and the normalized pushforward $\mathbb{P}_{\ell+1} := (\mathcal{R}_\ell)_* \mathbb{P}_\ell$ of $\mathbb{P}_\ell = Z_\ell^{-1}\rho_\ell$ satisfies $\mathbb{E}_{\ell+1}[A(Y)] = \mathbb{E}_\ell[A(\mathcal{R}_\ell(X))]$ for every bounded measurable parent observable $A : X_{\ell+1} \to \mathbb{R}$.*

Proof. Pushforward preserves total mass: $\rho_{\ell+1}(X_{\ell+1}) = \rho_\ell(\mathcal{R}_\ell^{-1}(X_{\ell+1})) = \rho_\ell(X_\ell) = Z_\ell$, so $Z_{\ell+1} = Z_\ell$. The observable identity is the standard change of variables for the normalized pushforward, $\int A d\mathbb{P}_{\ell+1} = \int A \circ \mathcal{R}_\ell d\mathbb{P}_\ell$. □

Theorem 8 (Functoriality of the exact pushforward)**.** *For the exact pushforward, $(\mathcal{R}_{\ell+1})_*(\mathcal{R}_\ell)_* \mathbb{P}_\ell = (\mathcal{R}_{\ell+1} \circ \mathcal{R}_\ell)_* \mathbb{P}_\ell$. The maps $\mathcal{R}_\ell$ act between distinct state spaces, and the identity holds for any measurable maps between measure spaces; it expresses composability of the coarse-graining morphisms, not a semigroup acting on a coupling space Θ.*

Proof. For any measurable $A \subset X_{\ell+2}$,

$$\begin{aligned}
\big[(\mathcal{R}_{\ell+1})_*(\mathcal{R}_\ell)_* \mathbb{P}_\ell\big](A) &= (\mathcal{R}_\ell)_* \mathbb{P}_\ell\big(\mathcal{R}_{\ell+1}^{-1}(A)\big) \\
&= \mathbb{P}_\ell\big(\mathcal{R}_\ell^{-1}(\mathcal{R}_{\ell+1}^{-1}(A))\big) \\
&= \mathbb{P}_\ell\big((\mathcal{R}_{\ell+1} \circ \mathcal{R}_\ell)^{-1}(A)\big) \\
&= (\mathcal{R}_{\ell+1} \circ \mathcal{R}_\ell)_* \mathbb{P}_\ell(A).
\end{aligned}$$

□

On a chain of levels this composability imposes no condition [30, Ch. 9]. The statement with content is the semigroup property that two successive blockings equal one blocking with the merged partition, and it fails for $\mathcal{R}_\ell$ even for frozen partitions. The belief sector composes: given a common final frame, the two-stage M-projection equals the one-stage M-projection with product weights, because each stage matches the first two moments of a mixture and a mixture of mixtures is the mixture with product weights. The frame sector does not compose. The Karcher equation $\sum_i w_i \log(U^{-1}U_i) = 0$ is nonlinear in the data on a curved group, so the Karcher mean of the Karcher means of a partition is in general not the Karcher mean of the whole set; for six SO(3) frames within 0.5 rad of the identity, grouped into two triples, the two-stage and one-stage means differ by the geodesic distance 7.3×10^{-3}. For commuting data the Karcher equation reduces to averaging in the Lie algebra and does compose. Since the frames enter the belief transport, the full map inherits the failure.

A semigroup acting on a coupling space would require exact closure modulo constants ($\delta^{\mathrm{quot}}_{I,0} = 0$). It is available in one setting: on the Gaussian interaction family, aggregation of agents, which is conditioning on zero internal disagreement and not marginalization, maps couplings to couplings and satisfies the semigroup law under a declared rescaling, whereas marginalization in general leaves the family [38, Chs. G7 and G9]. If the partition $\mathcal{P}_\ell$ is selected adaptively by a threshold detector, the exact pushforward still exists whenever the selection rule is measurable, but the map is then state-dependent and not a homogeneous flow.

Gauge covariance. At one base point the equivariance of $\mathcal{R}_\ell$ concerns the diagonal action of one $g \in G$ described in the gauge-action paragraph. For compact G with a bi-invariant metric the change of reference $U_i \mapsto U_i g$ is also compatible with $\mathcal{R}_\ell$, sending $U_I \mapsto U_I g$ and leaving every transport unchanged.

Theorem 9 (Gauge covariance of $\mathcal{R}_\ell$ at one base point)**.** *Let G be a Lie group with a left-invariant Riemannian distance d_G, and assume that the constituent frames of each cluster lie in an open geodesic ball of radius below the convexity radius r_{cx} of that metric, that the weights satisfy $w_i^I(g \cdot X_\ell) = w_i^I(X_\ell)$ for the diagonal action of one $g \in G$ (as holds for weights built from transported divergences and support masks), and that the belief and model parent states are M-projection barycenters of the children transported to the parent frame. Then $\mathcal{R}_\ell(g \cdot X_\ell) = g \cdot \mathcal{R}_\ell(X_\ell)$.*

Proof. Left invariance gives $d_G(gU, gU_i) = d_G(U, U_i)$, so gU_I attains the minimum of the transformed Karcher objective. Left translation is an isometry, so it carries a ball of radius $\rho < r_{\mathrm{cx}}$ containing $\{U_i\}$ onto a ball of the same radius containing $\{gU_i\}$, and the uniqueness theorem [95, Thm. 2.1] makes gU_I the unique global minimizer that $\mathcal{R}_\ell$ returns. Child-to-parent transport transforms as $\Omega_{Ii} \mapsto g\Omega_{Ii}g^{-1}$, transported children transform by common pushforward, and the M-projection barycenter is equivariant under common invertible pushforward because $D_{\mathrm{KL}}(g_\# a \| g_\# b) = D_{\mathrm{KL}}(a\|b)$ and $g_\#$ preserves the Gaussian family. Uniqueness in the Gaussian family transfers $q_I \mapsto g_\# q_I$ and $s_I \mapsto g_\# s_I$. The support mask is unchanged. Hence $\mathcal{R}_\ell$ commutes with the diagonal action. □

Compactness enters only through the normalizability of the Gibbs reference measure, not through the equivariance of the map. Equivariance does not single out the M-projection barycenter and the Karcher frame: the precision-weighted product of experts and the barycenter in the opposite divergence direction are equally equivariant under common pushforward, so the theorem establishes these constructions as frame-consistent, not as uniquely correct. The companion theory takes kernel equivariance as a hypothesis for coarse channels [30, Ch. 9]; the theorem verifies that hypothesis for the present map. With independent links carrying loop holonomy, the parent law

must be fixed by the holonomy group, and for noncompact GL links this generically leaves no admissible Gaussian parent [30, Ch. 10].

Closure ansatz and projection. The exact $\mathcal{F}_{\ell+1}^{\mathrm{exact}}$ generally contains all interactions allowed by symmetry, including hyperedges among parents, non-Gaussian terms, long-range couplings, and memory; marginalizing an internal variable multiplies every incident factor and generically produces a factor on the union of the remaining scopes [30, Ch. 11]. Whether it lies in the multi-agent Gaussian KL functional class $\mathfrak{M} = \{\mathcal{F}_\theta : \theta \in \Theta\}$ is a separate claim. The projection back to $\mathfrak{M}$ is $\Pi_{\mathfrak{M}}(\mathcal{F}) = \arg\min_{\theta\in\Theta} \inf_{a\in\mathbb{R}} \|\mathcal{F} - \mathcal{F}_\theta - a\|_{\mathcal{B}}$, with quotient closure residual $\delta_{\ell+1,0}^{\mathrm{quot}}$ as in (271), and the projected coarse-graining map is $R_{\mathfrak{M}}(\theta_\ell) := \Pi_{\mathfrak{M}}(\mathcal{F}_{\ell+1}^{\mathrm{exact}})$. It maps the level-$\ell$ parameter space to the level-$(\ell+1)$ parameter space, a different space, so it is not a self-map. For a retained projection the companion theory gives an exact residual identity in place of a schematic bound [30, Ch. 11].

Exact closure in the mean sector.

Lemma 8 (Mean-sector Gaussian elimination)**.** *Consider the mean-only linear coarse-graining $Y = \mu_I = \sum_{i\in I} w_i^I \Omega_{Ii}\mu_i$ at fixed covariances, frames, and non-envelope couplings, with fiber coordinates $\xi \in \mathbb{R}^r$, $r = (|I|-1)K_q$, on the affine fiber, and Lebesgue reference in the native mean coordinates (the Gibbs density below is normalizable when the block matrix is positive definite). The mean-sector functional is then a quadratic form with constant blocks,*

$$\mathcal{F}_\ell(Y,\xi) = \frac{1}{2}\begin{bmatrix} Y \\ \xi \end{bmatrix}^\top \begin{bmatrix} A_{YY} & A_{Y\xi} \\ A_{\xi Y} & A_{\xi\xi} \end{bmatrix} \begin{bmatrix} Y \\ \xi \end{bmatrix} - \begin{bmatrix} h_Y \\ h_\xi \end{bmatrix}^\top \begin{bmatrix} Y \\ \xi \end{bmatrix} + c_0. \tag{287}$$

If the block matrix is positive definite, the exact pushforward of the mean-sector Gibbs density is Gaussian, and

$$\mathcal{F}_{\ell+1}^{\mathrm{exact}}(Y) = \tfrac{1}{2} Y^\top A_{\mathrm{eff}} Y - h_{\mathrm{eff}}^\top Y + c_{\mathrm{eff}}, \qquad A_{\mathrm{eff}} = A_{YY} - A_{Y\xi}A_{\xi\xi}^{-1}A_{\xi Y}, \tag{288}$$

with $h_{\mathrm{eff}} = h_Y - A_{Y\xi}A_{\xi\xi}^{-1}h_\xi$, $c_{\mathrm{eff}} = c_0 - \frac{1}{2}h_\xi^\top A_{\xi\xi}^{-1}h_\xi + \frac{T_{\mathrm{cfg}}}{2}\log\det A_{\xi\xi} - \frac{rT_{\mathrm{cfg}}}{2}\log(2\pi T_{\mathrm{cfg}})$, and saddle $\xi^*(Y) = A_{\xi\xi}^{-1}(h_\xi - A_{\xi Y}Y)$.

Proof. With fixed covariances and frames, each transported relative entropy between Gaussians is a quadratic function of the means plus a constant, and so are the self terms against fixed priors and the linear-Gaussian observation terms, which gives the displayed form; the map $(\mu_i) \mapsto \mu_I$ is linear and surjective, so its fibers are complete affine spaces and the normal-Jacobian factor is constant. The Gaussian integral

$$\int_{\mathbb{R}^r} \exp\left[-\frac{1}{T_{\mathrm{cfg}}}\left(\tfrac{1}{2}\xi^\top A_{\xi\xi}\xi + \xi^\top b\right)\right] d\xi = (2\pi T_{\mathrm{cfg}})^{r/2}(\det A_{\xi\xi})^{-1/2}\exp\left[\frac{b^\top A_{\xi\xi}^{-1}b}{2T_{\mathrm{cfg}}}\right], \tag{289}$$

with $b = A_{\xi Y}Y - h_\xi$, followed by $-T_{\mathrm{cfg}}\log$ and collection of terms in Y, gives Eq. (288) without a Laplace approximation. This is the normalized Schur marginal of the companion supplement [38, Ch. G7]. □

Remark 7 (Why closure is not claimed beyond the mean sector)**.** *A closure statement for the full parent state would need exact conditional Gaussianity on the complete fiber with a Y-dependent precision $A_{\xi\xi}(Y)$. For the map that retains the parent covariance that hypothesis cannot hold: fixing $\Sigma_I = \sum_i w_i^I \widetilde{\Sigma}_i^I + \sum_i w_i^I \xi_i\xi_i^\top$ with $\widetilde{\Sigma}_i^I \succ 0$ forces $\sum_i w_i^I \xi_i\xi_i^\top \prec \Sigma_I$ and bounds the child covariances,*

so every fiber is bounded. If instead an arbitrary fiber chart is allowed, the hypothesis is empty, since any smooth positive density on a fiber diffeomorphic to $\mathbb{R}^r$ is carried onto a standard Gaussian by the triangular change of variables built from its successive conditional distribution functions. A Y-dependent term $\frac{T_{\mathrm{cfg}}}{2}\log\det A_{\xi\xi}(Y)$ is moreover not quadratic (for example, $\partial_y^4\log(a_0+a_1y^2)\neq 0$), so the next-level fiber density is not Gaussian and the class would not be preserved under iteration. Only the constant-block case of the lemma iterates, and there each step is Gaussian decimation. In the operational state-dependent-precision regime the functional is only locally quadratic, Eq. (277) *applies, and $\mathcal{E}_{\mathrm{anh}}$ must be retained. Membership of A_{eff} in the multi-agent Gaussian KL class proper is not asserted, since whether it decomposes into transported-difference KL terms with nonnegative row-structured couplings remains open.*

Anharmonic remainder. Outside the mean sector, let $\xi^*(Y)$ be a nondegenerate internal saddle of the effective fiber action Φ_ℓ, including the coarea density, defined by $\nabla_\xi\Phi_\ell(Y,\xi^*(Y))=0$, and set $\zeta=\xi-\xi^*(Y)$. Under the standard uniform regularity and tail conditions, Laplace's method [215] gives the anharmonic remainder $\mathcal{E}_{\mathrm{anh}}$ of (277) with the schematic bound

$$|\mathcal{E}_{\mathrm{anh}}| = O\left(T_{\mathrm{cfg}}^2\|H^{-1}\|^3\|T_3\|^2 + T_{\mathrm{cfg}}^2\|H^{-1}\|^2\|T_4\|\right), \tag{290}$$

where T_3, T_4 are the cubic and quartic derivative tensors at the saddle. With $\|H^{-1}\|\le m_I^{-1}$ from (275), perturbative control requires $\sqrt{T_{\mathrm{cfg}}}\|T_3\|m_I^{-3/2}\ll 1$ and $T_{\mathrm{cfg}}\|T_4\|m_I^{-2}\ll 1$. This formalizes the high-coherence, spectrally gapped, weakly anharmonic regime in which the working-form approximation of the preceding part applies.

Parent-parent transport. Besides the pooled raw conductance κ_{IJ}^R of (279), closure also requires a single transport Ω_{IJ}^R for each cluster pair. Transport the microscopic edge into parent coordinates as $\Theta_{ij}^{IJ}=\Omega_{Ii}\Omega_{ij}\Omega_{jJ}$ and define the parent transport by the weighted Karcher mean

$$\Omega_{IJ}^R = \arg\min_{\Omega\in G}\sum_{i\in I}\sum_{j\in J} w_i^I\kappa_{ij}d_G(\Omega,\Theta_{ij}^{IJ})^2, \tag{291}$$

with weights matching the row-convex, column-summed rule (279), and the transport spread

$$\mathcal{H}_{IJ} = \sum_{i\in I}\sum_{j\in J} w_i^I\kappa_{ij}d_G(\Omega_{IJ}^R,\Theta_{ij}^{IJ})^2. \tag{292}$$

In Regime I, where every transport is built from the same frames, the spread vanishes identically:

$$\Theta_{ij}^{IJ} = U_IU_i^{-1}U_iU_j^{-1}U_jU_J^{-1} = U_IU_J^{-1} \quad \text{for all } i\in I,\ j\in J, \qquad \text{so} \qquad \mathcal{H}_{IJ}\equiv 0. \tag{293}$$

The state space X_ℓ of this appendix contains no independent links, so $\mathcal{H}_{IJ}$ is not a closure error here. It becomes one only with independent links on the interaction graph (Regime II), where it measures their failure to descend to a single parent-parent transport; the law-level treatment of that case is the companion theory's holonomy-conditioned parent [30, Ch. 10].

Edge-marginal compatibility. Let the row and column conductance marginals for a cluster pair (I,J) be $r_i^{I\to J}=\sum_{j\in J}\kappa_{ij}$ and $\bar r_j^{J\leftarrow I}=\sum_{i\in I}\kappa_{ij}$. The first-order energy residual identified in the working-form part vanishes when the cluster weights match the row marginals, $w_i^I = r_i^{I\to J}/\sum_{i'}r_{i'}^{I\to J}$. A single cluster typically interacts with several partners $J_1,\dots,J_m$, and one set of weights cannot satisfy every partner-specific marginal condition unless the marginals are colinear. The mismatch $\delta_I^{\mathrm{marg}}=\sum_J\rho_{IJ}\sum_{i\in I}|w_i^I - r_i^{I\to J}/\sum_{i'}r_{i'}^{I\to J}|$, with ρ_{IJ} normalized partner weights, contributes to the quotient C^0 closure residual.

Structural decomposition of the closure residual. The working-form quotient C^0 closure residual (271) admits a structural decomposition into the residuals identified above. The proposition below collects that zeroth-order decomposition; its multiplicative constants depend on the smoothness class, in the standard form of schematic asymptotic bounds [215], and a quantitatively tight version requires specifying the closure norm and the parent regularity class. It does not bound the derivative residual $\delta_{I,1}$.

Proposition 13 (Schematic closure-residual bound)**.** *Assume Assumption 1; small belief and model dispersions $V_I^{(q)} = \sum_i w_i^I D_{\mathrm{KL}}(\Omega_{Ii} q_i \| q_I)$ and $V_I^{(s)} = \sum_i w_i^I D_{\mathrm{KL}}(\widetilde{\Omega}_{Ii} s_i \| s_I)$; small inter-cluster transport spread $\mathcal{H}_{IJ}$; small edge-marginal mismatch δ_I^{marg}; uniformly positive stiffness m_I as in* (275)*; bounded anharmonic derivatives; small non-Gaussian closure error $\mathcal{E}_{\mathrm{nonG}}^{(I)}$, defined as the $L^\infty(\mathcal{N}_I)$ magnitude, modulo additive constants as in* (271)*, of the third- and higher-cumulant corrections to the transported child marginals $\Omega_{Ii} q_i$ (the failure of the Gaussian-children condition (i), distinct from $\mathcal{E}_{\mathrm{anh}}$, which records anharmonicity of $\mathcal{F}_\ell$ at the internal saddle); and slow parent modes separated from internal modes. Then the quotient closure residual obeys*

$$\delta_{\ell+1,0}^{\mathrm{quot}} \le C_1 \sum_I \big(V_I^{(q)} + V_I^{(s)}\big)^{3/2} + C_2 \sum_{I,J} \mathcal{H}_{IJ} + C_3 \sum_I \delta_I^{\mathrm{marg}} + C_4 \sum_I m_I^{-1} \|A_{Y\xi}^{(I)}\|^2 + C_5 \sum_I \mathcal{E}_{\mathrm{anh}}^{(I)} + C_6 \sum_I \mathcal{E}_{\mathrm{nonG}}^{(I)}, \quad (294)$$

with constants $C_1, \dots, C_6$ depending on the closure norm $\|\cdot\|_{\mathcal{B}}$ and the regularity of the parent ansatz. The C_2 term vanishes identically in Regime I by Eq. (293)*.*

This is stated as a schematic proposition because the constants C_k, the closure norm $\|\cdot\|_{\mathcal{B}}$, and the regularity class of the parent ansatz are not pinned down, and tightness is not established; its status is conjectural. The exponent $3/2$ of the dispersion term is the scaling suggested by the leading third-cumulant correction once the first two moments are matched; it is not derived here, and no explicit value of C_1 is claimed. The qualitative content is that small dispersion, compatible weights, large internal stiffness, weak anharmonicity, and approximate Gaussianity together are expected to imply a small quotient C^0 closure residual.

Proposition 14 (Two-loop anharmonic constants, one internal mode)**.** *Restrict the Laplace integral to one constrained internal mode, $r = 1$, a condition stronger than $K_q = 1$ that freezes all other internal belief, covariance, model, and frame directions. Suppose the fiber action is C^4 near a nondegenerate saddle with stiffness $m_I > 0$ and satisfies the standard tail conditions of Laplace's method. Then*

$$\limsup_{T_{\mathrm{cfg}} \to 0^+} \frac{|\mathcal{E}_{\mathrm{anh}}^{(I)}|}{T_{\mathrm{cfg}}^2} \le \frac{1}{8} \cdot \frac{\|T_4\|_{\mathrm{op}}}{m_I^2} + \frac{5}{24} \cdot \frac{\|T_3\|_{\mathrm{op}}^2}{m_I^3}. \quad (295)$$

Proof. For one mode, $-T_{\mathrm{cfg}} \log(Z/Z_0) = T_{\mathrm{cfg}}^2 \big(t_4/(8m_I^2) - 5t_3^2/(24 m_I^3)\big) + o(T_{\mathrm{cfg}}^2)$, where Z_0 is the Gaussian integral and t_3, t_4 are the third and fourth derivatives at the saddle [215, 219]; the coefficient $\frac{5}{24} = \frac{1}{12} + \frac{1}{8}$ collects the two two-loop contractions of the cubic vertex (the sunset and dumbbell diagrams), and $\frac{1}{8}$ is the single contraction of the quartic vertex. The triangle inequality gives the bound. □

The multiplier C_5 of Proposition 13 is then an absolute constant of order one. These are integrated free-energy coefficients of the Laplace expansion; the Edgeworth coefficients of the perturbed Gaussian density are different objects and do not enter the free energy. For $r > 1$, including

multi-mode clusters even when $K_q = 1$, the contractions $T_4 : (H^{-1} \otimes H^{-1})$ and $T_3 H^{-1} T_3$ carry dimension-dependent trace factors that the operator norms alone do not control, so the displayed values do not extend verbatim beyond $r = 1$.

Bounded exponential detector with a gap condition. The threshold detector of Section 9.3 is written here in bounded exponential form with the model channel. With $C_q(I) = \exp[-V_I^{(q)}/\tau_q]$, $C_s(I) = \exp[-V_I^{(s)}/\tau_s]$, and presence $P_I = |\Lambda_\ell|^{-1} \sum_{c \in \Lambda_\ell} \chi_I(c) \in [0,1]$, define $\Gamma_I = P_I C_q(I) C_s(I) \in [0,1]$. The strengthened detector selects I if $\Gamma_I > \Gamma_{\min}$, $|I| \geq N_{\min}$, and $m_I > m_{\min}$; the gap condition is what makes it test a closure condition and not only a similarity threshold, since the anharmonic constants of Proposition 14 scale as m_I^{-2} and m_I^{-3} and the remainder is uncontrolled as $m_I \to 0$. The detector of Section 9.3 tests only $\Gamma_I > \Gamma_{\min}$ and $|I| \geq N_{\min}$, without the gap condition. It is therefore a similarity filter, the clusters it admits carry no lower bound on m_I, and Theorem 10 below does not cover it.

Remark 8 (Status of the Lipschitz bound and of the saving term)**.** *The Lipschitz bound in the hypothesis of Theorem 10 is an assumption, not a result of this appendix. A first-order Taylor expansion of the parent free energy around the high-coherence configuration under conditions (i)–(v) motivates it, with local Lipschitz constants bounded by operator norms of the parent Fisher information [83], $L_q \leq \lambda_{\max}(F(q_I)) \cdot \max_i \|w_i^I\|$ and $L_s \leq \lambda_{\max}(F(s_I)) \cdot \max_i \|w_i^I\|$; these identifications are not derived here. The saving term $A_I > 0$, the assertion that coarse-graining lowers the optimized free energy at all, has no identified source. At Gibbs-optimal rows each row value is $-\tau \log Z_i$ with $Z_i = \sum_j \pi_{ij} e^{-E_{ij}/\tau}$, and deleting the coherent low-energy intra-cluster sources raises it: with the prior left unnormalized, removing positive terms lowers Z_i; with the prior normalized again over the surviving sources, Z_i falls exactly when the π-average of $e^{-E_{ij}/\tau}$ over the cluster exceeds its average over the other sources, which holds whenever the intra-cluster energies are the smaller ones. Any net gain must therefore come from the rows of agents outside the cluster, which couple to one parent in place of $|I|$ constituents, or from the parent's prior term, and neither is derived here. Under an exact parent, moreover, a gain in the variational free energy is discarded conditional information (paragraph on retention). The identification $A_I = T_{\text{cfg}} m_I$ in the leading-order Laplace approximation, which would tie the saving to the gap, is likewise not derived. The theorem is correct as a conditional statement, and its hypothesis $A_I > 0$ is open.*

Theorem 10 (Strengthened detector implies positive retention gain, conditionally)**.** *Suppose, for candidate cluster I, the calibrated agent parent satisfies the local Lipschitz bound $\widehat{\mathcal{F}}_{\ell+1}^{\text{agent},*}(I) \leq \mathcal{F}_{\text{micro}}^*(I) - A_I + L_q V_I^{(q)} + L_s V_I^{(s)} + \delta_{I,0}$, where $A_I > 0$ is the saving term, L_q, L_s are the local Lipschitz constants in belief and model dispersion, and $\delta_{I,0}$ is the single uniform calibrated energy error from* (271)*. Let $C_I = C_{\text{parent}}(I) - C_{\text{micro}}(I)$, with $\mathcal{L}_{\text{micro}}(I) = \mathcal{F}_{\text{micro}}^*(I) + C_{\text{micro}}(I)$. If*

$$\max(L_q \tau_q, L_s \tau_s) \log(1/\Gamma_{\min}) + \delta_{I,0} + C_I < A_I, \tag{296}$$

then $\Gamma_I > \Gamma_{\min}$ implies $\Delta_I = \mathcal{L}_{\text{micro}}(I) - \mathcal{L}_{\text{parent}}(I) > 0$.

Proof. The Lipschitz bound together with the definitions of Δ_I and $\mathcal{L}_{\text{parent}}$ in (283) gives $\Delta_I \geq A_I - L_q V_I^{(q)} - L_s V_I^{(s)} - \delta_{I,0} - C_I$. The detector condition $\Gamma_I > \Gamma_{\min}$, with $P_I \leq 1$, implies $V_I^{(q)}/\tau_q + V_I^{(s)}/\tau_s < \log(1/\Gamma_{\min})$, so $L_q V_I^{(q)} + L_s V_I^{(s)} \leq \max(L_q \tau_q, L_s \tau_s) \log(1/\Gamma_{\min})$. Substituting and applying (296) gives $\Delta_I > 0$. □

The gap condition $m_I > m_{\min}$ does not appear in the proof; it enters through the regime in which the Lipschitz bound and the calibrated residual $\delta_{I,0}$ are meaningful.

Slow manifold. The working-form reduction of the adiabatic-elimination paragraph is Fenichel's normal-hyperbolicity argument [90] under hypotheses (a)–(c), with $t_{\mathrm{int}}^{-1} = \eta\lambda_2(L_I; W_g)$ and $\epsilon_{\mathrm{ts}} = t_{\mathrm{int}}/t_{\mathrm{slow}} \ll 1$. Its metric is the graph pullback G_{slow} of (280), which exceeds the quotient metric in the Loewner order unless $D\xi^* = C^*$, and the closure perturbation of the reduced vector field is bounded by $\eta\delta_{I,1}$ in (282). Establishing a quantitative C^1 closure estimate, and applying the hypotheses to the collective free energy [30, Ch. 8], are open.

Finite-size scaling. A phase-transition or universality claim would require finite-size scaling tests, which in turn require a sequence of networks of growing size N; infinite networks belong to the companion supplement [31, Ch. B5]. Candidate observables include the mean cluster size $S_N = N^{-1}\sum_I |I|^2$, which is itself the susceptibility in the percolation convention (computed without the largest cluster above threshold), the correlation length on the overlap graph, the exponential coherence order parameter $M_N = N^{-2}\sum_{i,j} \exp[-D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)/\tau_q]$, and the largest meta-agent fraction $P_\infty = N^{-1}\max_I |I|$. On a graph without a dimension the scaling exponents in N must be declared as quantities in their own right, as in complete-graph scaling: a finite-size scaling ansatz takes the form $M_N(g) = N^{-a_M}\Phi_M((g - g_c)N^b)$ and $S_N(g) = N^{a_S}\Phi_S((g - g_c)N^b)$ for a control parameter g, with declared exponents a_M, a_S, and b; on a d-dimensional lattice with $N \sim L^d$ they would be $\hat{\beta}/(d\hat{\nu})$, $\hat{\gamma}/(d\hat{\nu})$, and $1/(d\hat{\nu})$ in terms of the standard exponents $\hat{\beta}, \hat{\gamma}, \hat{\nu}$. Evidence for universality requires curve collapse across N and robustness across microscopic details. This is a program for future work, not a delivered result, and the main text accordingly speaks of a formation event (Section 13), not of a phase transition in the statistical-mechanical sense.

Summary. The gauge-covariant map $\mathcal{R}_\ell$ carries the agent Gibbs measure to a parent measure with the same partition function and retained observables (Theorems 7, 8, and 9; status P), as deterministic cases of the companion theory's conditional-partition formula [30, Chs. 9 and 11]; the map is not a semigroup in the frame sector. Closure in the multi-agent class is proved only in the mean sector (Lemma 8; P). Beyond it the coarse energy is given by the Laplace formula (277), whose one-loop term shifts the parent's entropy weight (Eq. (278); D); the quotient C^0 residual bound of Proposition 13 is conjectural (C), its two-loop constants for one internal mode are proved (Proposition 14), and no C^1 force-residual estimate is proved. The strengthened detector implies a positive retention gain as a conditional statement (Theorem 10) whose saving hypothesis has no identified source, and the detector of Section 9.3, which lacks the gap condition, is not covered.

J Notation and Symbol Conventions

Several symbols are overloaded across the framework's distinct mathematical strata. Table 4 collects the conventions.

Table 4: Symbols used in this paper. Where a letter carries more than one meaning, the decoration or the section that uses it disambiguates.

Symbol	Meaning
$\mathcal{C}$, c, c_*	Base manifold, a presupposed smooth index manifold carrying no physical metric; a base point; the fixed base point of the single-point analysis.
G, $\mathfrak{g}$	Structure group ($\mathrm{GL}^+(K_q)$, or $\mathrm{SO}(K_q)$ where stated) and its Lie algebra.
U_i, ϕ_i; U_I, $\mathcal{U}_I$	Frame of agent i and its Lie-algebra coordinate; parent frame of cluster I and support region of the meta-agent (Eq. (124)).

Symbol	Meaning
Ω_{ij}, $\widetilde{\Omega}_{ij}$	Transport from agent j to agent i of Eq. (14), $U_i U_j^{-1}$ in Regime I and an independent link with $\Omega_{ji} = \Omega_{ij}^{-1}$ in Regime II; its action on the model fiber.
Ω_{Ii}, Ω_{IJ}^{R}	Child-to-parent transport $U_I U_i^{-1}$; parent-parent transport (Appendix I).
q_i, p_i, s_i, r_i	Belief on the state fiber, prior on the state fiber (in the hierarchy, the lagged predictive prior of Section 10), generative-model section on the model fiber, hyper-prior on the model fiber. The letters q, p, s, r refer to these fields throughout.
(q), (s)	Channel decorations, never level indices: belief channel and model channel, as in $V_I^{(q)}$, $V_I^{(s)}$, C_q, C_s, τ_q, τ_s, the coupling strengths $c_{ij}^{(q)}$, $c_{ij}^{(s)}$ and coupling precisions $W_{ij}^{(q)}$, $W_{ij}^{(s)}$ of the quadratic-form appendix (Section E.2), and the pullbacks $G_i^{(q)}$, $G_i^{(s)}$.
$\tilde{\beta}_{ij}$, $\tilde{\gamma}_{ij}$	Raw alignment weights $c_{ij}^{(q)}/2$, $c_{ij}^{(s)}/2$ (Section E.2), distinct from the row-normalized β_{ij}, γ_{ij}.
$G_i^{(q)}$; $\widetilde{G}_i^{(q)}$	Covariant Fisher block of the pullback; the pullback with horizontal block $\kappa\,\mathrm{tr}(A_\mu A_\nu)$ (Section 18).
κ	Unindexed, a declared dimensionless constant: the horizontal coupling of Section 18, or the temperature scale $\tau = \kappa\sqrt{K_q}$ of the gradient appendix; the section disambiguates.
L_0; $x^0, \ldots, x^3$; $\psi_0, \ldots, \psi_3$	Physical length scale of a line element; base coordinates and frame components in the signature examples (Section 19).
K_q, K_m	Dimensions of the state and model fibers.
$D_{\mathrm{KL}}(p\|q)$	Relative entropy, first argument the receiving or variational law.
β_{ij}, γ_{ij}	Attention weight on belief alignment and coupling on the model channel; both are row-normalized and not interchangeable.
π_{ij}; π	Attention prior; the bundle projection $\pi : \mathcal{N} \to \mathcal{C}$.
τ, τ_γ	Attention and entropy temperatures of the belief and model channels in Eq. (45); τ is never a clock.
τ_q, τ_s	Detector dispersion temperatures (Section 9.3).
T_{cfg}	Configuration temperature of the Gibbs ensemble over belief configurations (Appendix I).
ϑ_i	Information clock of agent i, the Fisher arc length of its belief history (Section 17).
t; t_q, t_s, t_ϕ	External flow parameter, and the coarse-grained shared update parameter of postulate P5; characteristic belief, model, and frame relaxation times (Section 7).
η_μ, η_Σ, η_ϕ, η	Mobilities of the natural-gradient flow; η for the parent flow of Appendix I.
α_i	Per-agent precision weight on $\alpha_i D_{\mathrm{KL}}(q_i\|p_i)$.
β_{W}	Wilson lattice coupling.
ℓ, $\ell_{\max}$	Index of the level in the coarse-graining hierarchy and top level; a superscript (ℓ) marks a level.
$\mathcal{R}_\ell$	Coarse-graining map from level ℓ to level $\ell + 1$.
$\mathcal{I}_\ell$; I, J	Agent index set at level ℓ; clusters of agents, with $\|I\|$ the cluster size.
w_i^I	Aggregation weight of constituent i in cluster I.
χ_i	Presence mask of agent i.
$\Gamma(I)$ or Γ_I, $\Gamma_{\min}$, $N_{\min}$	Consensus score of the detector, its threshold, and the minimum cluster size.
Γ, Γ_{2PI}	One-particle-irreducible effective action and two-particle-irreducible action of the configuration ensemble (Appendix H); distinct from the detector score.
$\mathcal{K}_{i,\pi(i)}^{(\ell)}$, σ^2, p_{top}	Gaussian link kernel of the augmented hierarchical joint, its link stiffness, and the declared top-level prior (Appendix G).
T_{ij}; $\mathcal{C}_{\mathrm{cons}}$	Fixed mass-preserving positive linear transport induced by $(\Omega_{ij})_*$; consensus functional of the state-level obstruction theorem (Appendix G).

Symbol	Meaning
Λ_i, A_i, a, $\mathcal{L}$	Belief precision Σ_i^{-1}, prior-and-sensory anchor, alignment coupling strength, and the linear precision-transport map on $\mathrm{Sym}(K)^N$ (Appendix D); the anchor A_i is distinct from the saving term A_I of Appendix I.
$\mathrm{cond}(\cdot)$; $\mathcal{D}$	Condition number, written so to avoid a collision with κ; duplication matrix (Appendix C).
ph; j	Photon channel of the rock example (Appendix B); spin label of an irreducible SO(3) representation (Appendix C), distinct from the agent index j.
κ_{ij}, κ^R_{IJ}	Raw edge conductances and pooled parent conductances (Appendix I); distinct from any unindexed κ.
m_I, $\lambda_{I,w}$	Stiffness of the internal modes of cluster I and the constrained weighted gap (Eq. (275)).
T_3, T_4	Cubic and quartic derivative tensors at an internal saddle; distinct from T_{cfg} and T_{ij}.
$M_{\mu\mu}$, $M_{\Sigma\Sigma}$, M_i; M_K	Stiffness (Hessian) blocks of Section 8; kinetic metric of the kinetic-metric postulate (Section 8.7).
ρ_{state}, ρ_{model}	Representations of G on the state and model fibers.
$B(X,Y)$	Killing form $2K_q\,\mathrm{tr}(XY) - 2\,\mathrm{tr}(X)\,\mathrm{tr}(Y)$ on $\mathfrak{gl}(K_q)$, indefinite and degenerate; the horizontal tensor of Part II uses the distinct trace form $\mathrm{tr}(XY)$, also indefinite on real $\mathfrak{gl}(K_q)$, and only $-\mathrm{tr}(XY)$ on a compact skew-symmetric subalgebra is positive. On $\mathfrak{so}(K_q)$, B denotes that algebra's own Killing form $(K_q-2)\,\mathrm{tr}(XY)$ (Section 8.7).

References

[1] John Archibald Wheeler. Information, physics, quantum: The search for links. In Wojciech H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 3–28. Addison-Wesley, Redwood City, CA, 1990.

[2] J.A. Wheeler. Law without law. In J.A. Wheeler and W.H. Zurek, editors, *Quantum Theory and Measurement*, pages 182–213. Princeton University Press, Princeton, 1983.

[3] W. K. Wootters. Statistical distance and Hilbert space. *Phys. Rev. D*, 23(2):357–362, 1981.

[4] Anton Zeilinger. A foundational principle for quantum mechanics. *Found. Phys.*, 29(4):631–643, 1999.

[5] Lucien Hardy. Quantum theory from five reasonable axioms, 2001.

[6] Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti. Informational derivation of quantum theory. *Phys. Rev. A*, 84(1):012311, 2011.

[7] Lluís Masanes and Markus P. Müller. A derivation of quantum theory from physical requirements. *New J. Phys.*, 13(6):063001, 2011.

[8] Philipp Andres Höhn. Toolbox for reconstructing quantum theory from rules on information acquisition. *Quantum*, 1:38, 2017.

[9] Markus P. Müller. Law without law: from observer states to physics via algorithmic information theory. *Quantum*, 4:301, 2020.

[10] Christopher A. Fuchs. On participatory realism. In Ian T. Durham and Dean Rickles, editors, *Information and Interaction: Eddington, Wheeler, and the Limits of Knowledge*, pages 113–134. Springer, 2017.

[11] Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8):1637–1678, 1996.

[12] Ariel Caticha. The information geometry of space and time. In *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, volume 803 of *AIP Conf. Proc.*, pages 355–365, 2005.

[13] Ariel Caticha. The information geometry of space-time. *Proceedings*, 33(1):15, 2019.

[14] A. Caticha. The entropic dynamics approach to quantum mechanics. *Entropy*, 21(10):943, 2019.

[15] Immanuel Kant. *Critique of Pure Reason*. Cambridge University Press, Cambridge, 1998. Originally published 1781/1787.

[16] H. von Helmholtz. *Handbuch der Physiologischen Optik*. Leopold Voss, Leipzig, 1867.

[17] K. Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11:127–138, 2010.

[18] K. Friston, T. FitzGerald, F. Rigoli, P. Schwartenbeck, and G. Pezzulo. Active inference: a process theory. *Neural Computation*, 29:1–49, 2017.

[19] Thomas Parr, Giovanni Pezzulo, and Karl J Friston. *Active inference: The free energy principle in mind, brain, and behavior*. MIT Press, 2022.

[20] Ted Jacobson. Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7):1260–1263, 1995.

[21] T. Padmanabhan. Thermodynamical aspects of gravity: new insights. *Reports on Progress in Physics*, 73:046901, 2010.

[22] R. Bousso. The holographic principle. *Reviews of Modern Physics*, 74:825–874, 2002.

[23] J. Maldacena. The large-N limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2:231–252, 1998.

[24] M. Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42:2323–2329, 2010.

[25] J. Maldacena and L. Susskind. Cool horizons for entangled black holes. *Fortschritte der Physik*, 61:781–811, 2013.

[26] B. Swingle. Entanglement renormalization and holography. *Physical Review D*, 86:065007, 2012.

[27] Leo P. Kadanoff. Scaling laws for Ising models near t_c. *Physics Physique Fizika*, 2(6):263–272, 1966.

[28] Leo P. Kadanoff. Variational principles and approximate renormalization group calculations. *Phys. Rev. Lett.*, 34(16):1005–1008, 1975.

[29] Leo P. Kadanoff, Anthony Houghton, and Mehmet C. Yalabik. Variational approximations for renormalization group transformations. *J. Stat. Phys.*, 14(2):171–203, 1976.

[30] Robert C. Dennis. Gauge-covariant variational free energy and exact coarse-graining. Manuscript in preparation, 2026.

[31] Robert C. Dennis. Gauge-covariant variational free energy, supplement b: Base transport, infinite networks, and the lattice cluster. Manuscript in preparation, 2026.

[32] Alfred North Whitehead. *Process and Reality: An Essay in Cosmology*. Free Press, New York, corrected, edited by david ray griffin and donald w. sherburne edition, 1978. Originally published 1929.

[33] James Ladyman, Don Ross, David Spurrett, and John Collier. *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press, Oxford, 2007.

[34] James Ladyman. What is structural realism? *Stud. Hist. Phil. Sci. A*, 29(3):409–424, 1998.

[35] M. Nakahara. *Geometry, Topology and Physics*. CRC Press, 2nd edition, 2003.

[36] Erich Kretschmann. Über den physikalischen sinn der relativitätspostulate, a. Einsteins neue und seine ursprüngliche relativitätstheorie. *Annalen der Physik*, 358(16):575–614, 1917.

[37] John D. Norton. General covariance and the foundations of general relativity: Eight decades of dispute. *Reports on Progress in Physics*, 56(7):791–858, 1993.

[38] Robert C. Dennis. Gauge-covariant variational free energy, supplement g: The multivariate-gaussian realization. Manuscript in preparation, 2026.

[39] Erik Verlinde. On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011(4):29, 2011.

[40] C. Cao, S.M. Carroll, and S. Michalakis. Space from Hilbert space: recovering geometry from bulk entanglement. *Physical Review D*, 95:024031, 2017.

[41] D. Deutsch and C. Marletto. Constructor theory of information. *Proceedings of the Royal Society A*, 471:20140540, 2015.

[42] Peter Salamon and R. Stephen Berry. Thermodynamic length and dissipated availability. *Phys. Rev. Lett.*, 51(13):1127–1130, 1983.

[43] Gavin E. Crooks. Measuring thermodynamic length. *Phys. Rev. Lett.*, 99(10):100602, 2007.

[44] Johanna Erdmenger, Kevin Grosvenor, and Ro Jefferson. Information geometry in quantum field theory: lessons from simple examples. *SciPost Phys.*, 8(5):073, 2020.

[45] Wojciech Hubert Zurek. Quantum Darwinism. *Nat. Phys.*, 5(3):181–188, 2009.

[46] Chris Fields, Karl Friston, James F. Glazebrook, and Michael Levin. A free energy principle for generic quantum systems. *Prog. Biophys. Mol. Biol.*, 173:36–59, 2022.

[47] Biswa Sengupta, Arturo Tozzi, Gerald K. Cooray, Pamela K. Douglas, and Karl J. Friston. Towards a neuronal gauge theory. *PLoS Biology*, 14(3):e1002400, 2016.

[48] Karl Friston. A free energy principle for a particular physics. *arXiv preprint arXiv:1906.10184*, 2019.

[49] Lancelot Da Costa, Karl Friston, Conor Heins, and Grigorios A. Pavliotis. Bayesian mechanics for stationary processes. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 477(2256):20210518, 2021.

[50] Maxwell J. D. Ramstead, Dalton A. R. Sakthivadivel, Conor Heins, Magnus Koudahl, Beren Millidge, Lancelot Da Costa, Brennan Klein, and Karl J. Friston. On bayesian mechanics: a physics of and by beliefs. *Interface Focus*, 13(3):20220029, 2023.

[51] Dalton A. R. Sakthivadivel. Towards a Geometry and Analysis for Bayesian Mechanics. *arXiv preprint*, 2022.

[52] Michael Kirchhoff, Thomas Parr, Ensor Palacios, Karl Friston, and Julian Kiverstein. The Markov blankets of life: autonomy, active inference and the free energy principle. *Journal of the Royal Society Interface*, 15(138):20170792, 2018.

[53] Jelle Bruineberg, Krzysztof Dolega, Joe Dewhurst, and Manuel Baltieri. The emperor's new markov blankets. *Behavioral and Brain Sciences*, 45:e183, 2022.

[54] Miguel Aguilera, Beren Millidge, Alexander Tschantz, and Christopher L. Buckley. How particular is the physics of the free energy principle? *Physics of Life Reviews*, 40:24–50, 2022.

[55] Martin Biehl, Felix A. Pollock, and Ryota Kanai. A technical critique of some parts of the free energy principle. *Entropy*, 23(3):293, 2021.

[56] Karl J. Friston and Christopher D. Frith. A Duet for one. *Consciousness and Cognition*, 36:390–405, 2015.

[57] Amit Singer and Hau-Tieng Wu. Vector diffusion maps and the connection Laplacian. *Commun. Pure Appl. Math.*, 65(8):1067–1144, 2012.

[58] Jakob Hansen and Robert Ghrist. Opinion dynamics on discourse sheaves. *SIAM J. Appl. Math.*, 81(5):2033–2060, 2021.

[59] Taco S. Cohen, Maurice Weiler, Berkay Kicanaoglu, and Max Welling. Gauge equivariant convolutional networks and the icosahedral CNN. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2019.

[60] Emily Adlam and Carlo Rovelli. Information is physical: cross-perspective links in relational quantum mechanics. *Philosophy of Physics*, 1(1):4, 2023. arXiv:2203.13342, 2022.

[61] Christopher A. Fuchs, N. David Mermin, and Rüdiger Schack. An introduction to QBism with an application to the locality of quantum mechanics. *American Journal of Physics*, 82(8):749–754, 2014.

[62] John B. DeBrota, Christopher A. Fuchs, and Rüdiger Schack. Respecting one's fellow: QBism's analysis of Wigner's friend. *Foundations of Physics*, 50(12):1859–1874, 2020.

[63] Nir Lahav and Zachariah A. Neemeh. A relativistic theory of consciousness. *Frontiers in Psychology*, 12:704270, 2022.

[64] Nir Lahav and Zachariah A. Neemeh. A relativistic theory of consciousness (shortened version). *arXiv preprint arXiv:2502.07247*, 2025.

[65] Nikolai Nikolaevich Čencov. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1982. Translated from Russian by Lev J. Leifman; originally published as *Statisticheskie reshayushchie pravila i optimalnye resheniya*, Moscow, Nauka, 1972.

[66] N. Ay, J. Jost, H. V. Lê, and L. Schwachhöfer. *Information Geometry*, volume 64 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, 2017.

[67] William Donnelly and Laurent Freidel. Local subsystems in gauge theory and gravity. *Journal of High Energy Physics*, 2016(9):102, 2016.

[68] Stephen D. Bartlett, Terry Rudolph, and Robert W. Spekkens. Reference frames, superselection rules, and quantum information. *Reviews of Modern Physics*, 79(2):555–609, 2007.

[69] Augustin Vanrietvelde, Philipp A. Höhn, Flaminia Giacomini, and Esteban Castro-Ruiz. A change of perspective: switching quantum reference frames via a perspective-neutral framework. *Quantum*, 4:225, 2020.

[70] Flaminia Giacomini, Esteban Castro-Ruiz, and Časlav Brukner. Quantum mechanics and the covariance of physical laws in quantum reference frames. *Nature Communications*, 10:494, 2019.

[71] Cédric Fournel, Jordan François, Serge Lazzarini, and Thierry Masson. Gauge invariant composite fields out of connections, with examples. *International Journal of Geometric Methods in Modern Physics*, 11:1450016, 2014.

[72] Tai Tsun Wu and Chen Ning Yang. Concept of nonintegrable phase factors and global formulation of gauge fields. *Phys. Rev. D*, 12(12):3845–3857, 1975.

[73] S. Kobayashi and K. Nomizu. *Foundations of Differential Geometry*, volume 1. Wiley, 1963.

[74] Theodore Frankel. *The Geometry of Physics: An Introduction*. Cambridge University Press, 3rd edition, 2011.

[75] Jordan François. Artificial versus substantial gauge symmetries: A criterion and an application to the electroweak model. *Philosophy of Science*, 86(3):472–496, 2019.

[76] C.N. Yang and R.L. Mills. Conservation of isotopic spin and isotopic gauge invariance. *Physical Review*, 96:191–201, 1954.

[77] Ryoyu Utiyama. Invariant theoretical interpretation of interaction. *Phys. Rev.*, 101(5):1597–1607, 1956.

[78] K.G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974.

[79] J. Kogut and L. Susskind. Hamiltonian formulation of Wilson's lattice gauge theories. *Physical Review D*, 11(2):395–408, 1975.

[80] M. Creutz. *Quarks, Gluons and Lattices*. Cambridge University Press, Cambridge, 1983.

[81] L. T. Skovgaard. A Riemannian geometry of the multivariate normal model. *Scandinavian Journal of Statistics*, 11(4):211–223, 1984.

[82] John Milnor. Curvatures of left invariant metrics on Lie groups. *Advances in Mathematics*, 21(3):293–329, 1976.

[83] S. Amari. *Information Geometry and Its Applications*. Springer, 2016.

[84] Karl J. Friston, Thomas Parr, Conor Heins, Axel Constant, Daniel Friedman, Takuya Isomura, Chris Fields, Tim Verbelen, Maxwell J. D. Ramstead, John Clippinger, and Christopher D. Frith. Federated inference and belief sharing. *Neuroscience & Biobehavioral Reviews*, 156:105500, 2024.

[85] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

[86] Robert C. Dennis. Gauge-covariant Gaussian attention through projected matrix links. Manuscript in preparation, 2026.

[87] C.M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[88] Paul Milgrom and Ilya Segal. Envelope theorems for arbitrary choice sets. *Econometrica*, 70(2):583–601, 2002.

[89] A. N. Tikhonov. Systems of differential equations containing small parameters in the derivatives. *Mat. Sb. (N.S.)*, 31(73)(3):575–586, 1952. In Russian. Zbl 0048.07101.

[90] N. Fenichel. Geometric singular perturbation theory for ordinary differential equations. *Journal of Differential Equations*, 31(1):53–98, 1979.

[91] M. Born and R. Oppenheimer. Zur Quantentheorie der Molekeln. *Annalen der Physik*, 389:457–484, 1927. English translation: On the quantum theory of molecules.

[92] Karsten Grove and Hermann Karcher. How to conjugate C^1-close group actions. *Math. Z.*, 132(1):11–20, 1973.

[93] Hermann Karcher. Riemannian center of mass and mollifier smoothing. *Communications on Pure and Applied Mathematics*, 30(5):509–541, 1977.

[94] Xavier Pennec. Statistical computing on manifolds: From Riemannian geometry to computational anatomy. *Emerging Trends in Visual Computing (Lecture Notes in Computer Science)*, 5416:347–386, 2009.

[95] Bijan Afsari. Riemannian L^p center of mass: existence, uniqueness, and convexity. *Proc. Amer. Math. Soc.*, 139(2):655–673, 2011.

[96] Wilfrid S. Kendall. Probability, convexity, and harmonic maps with small image I: uniqueness and fine existence. *Proc. London Math. Soc.*, s3-61(2):371–406, 1990.

[97] Xavier Pennec and Vincent Arsigny. Exponential barycenters of the canonical Cartan connection and invariant means on Lie groups. In Frank Nielsen and Rajendra Bhatia, editors, *Matrix Information Geometry*, pages 123–166. Springer, Berlin, Heidelberg, 2013.

[98] E. Andruchow, G. Larotonda, L. Recht, and A. Varela. The left invariant metric in the general linear group. *J. Geom. Phys.*, 86:241–257, 2014.

[99] Xavier Pennec, Pierre Fillard, and Nicholas Ayache. A Riemannian framework for tensor computing. *International Journal of Computer Vision*, 66(1):41–66, 2006.

[100] Peter Thestrup Waade, Christoffer Lundbak Olesen, Jonathan Ehrenreich Laursen, Samuel William Nehrer, Conor Heins, Karl Friston, and Christoph Mathys. As one and many: Relating individual and emergent group-level generative models in active inference. *Entropy*, 27(2):143, 2025.

[101] Conor Heins, Beren Millidge, Lancelot Da Costa, Richard P. Mann, Karl J. Friston, and Iain D. Couzin. Collective behavior from surprise minimization. *Proceedings of the National Academy of Sciences*, 121(17):e2320239121, 2024.

[102] Mahault Albarracin, Daphne Demekas, Maxwell J. D. Ramstead, and Conor Heins. Epistemic communities under active inference. *Entropy*, 24(4):476, 2022.

[103] Radford M. Neal and Geoffrey E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In Michael I. Jordan, editor, *Learning in Graphical Models*, volume 89 of *NATO ASI Series*, pages 355–368. Springer, Dordrecht, 1998.

[104] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, pages 368–377, 1999.

[105] William Bialek, Ilya Nemenman, and Naftali Tishby. Predictability, complexity, and learning. *Neural Computation*, 13(11):2409–2463, 2001.

[106] Gal Chechik, Amir Globerson, Naftali Tishby, and Yair Weiss. Information bottleneck for Gaussian variables. *Journal of Machine Learning Research*, 6:165–188, 2005.

[107] D. J. Strouse and David J. Schwab. The deterministic information bottleneck. *Neural Computation*, 29(6):1611–1630, 2017.

[108] Noam Slonim and Naftali Tishby. Agglomerative information bottleneck. In *Advances in Neural Information Processing Systems 12 (NIPS 1999)*, pages 617–623. MIT Press, 2000.

[109] B.C. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. Springer, 2nd edition, 2015.

[110] Karl Friston. Hierarchical models in the brain. *PLoS Computational Biology*, 4(11):e1000211, 2008.

[111] K.G. Wilson. Renormalization group and critical phenomena. I. Renormalization group and the Kadanoff scaling picture. *Physical Review B*, 4:3174–3183, 1971.

[112] K.G. Wilson and J. Kogut. The renormalization group and the ϵ expansion. *Physics Reports*, 12:75–199, 1974.

[113] J. Cardy. *Scaling and Renormalization in Statistical Physics*. Cambridge Lecture Notes in Physics. Cambridge University Press, Cambridge, 1996.

[114] Cédric Bény and Tobias J. Osborne. Information-geometric approach to the renormalization group. *Physical Review A*, 92(2):022330, 2015.

[115] Benjamin B. Machta, Ricky Chachra, Mark K. Transtrum, and James P. Sethna. Parameter space compression underlies emergent theories and predictive models. *Science*, 342(6158):604–607, 2013.

[116] David S. Berman, Marc S. Klinger, and Alexander G. Stapleton. Bayesian renormalization. *Machine Learning: Science and Technology*, 4(4):045011, 2023.

[117] Pankaj Mehta and David J. Schwab. An exact mapping between the variational renormalization group and deep learning. *arXiv preprint arXiv:1410.3831*, 2014.

[118] Florian Dörfler and Francesco Bullo. Kron reduction of graphs with applications to electrical networks. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 60(1):150–163, 2013.

[119] Andreas Loukas. Graph reduction with spectral and cut guarantees. *Journal of Machine Learning Research*, 20(116):1–42, 2019.

[120] Pablo Villegas, Tommaso Gili, Guido Caldarelli, and Andrea Gabrielli. Laplacian renormalization group for heterogeneous networks. *Nature Physics*, 19(3):445–450, 2023.

[121] Aernout C. D. van Enter, Roberto Fernández, and Alan D. Sokal. Regularity properties and pathologies of position-space renormalization-group transformations: Scope and limitations of Gibbsian theory. *J. Stat. Phys.*, 72(5–6):879–1167, 1993.

[122] Stanisław Łojasiewicz. Une propriété topologique des sous-ensembles analytiques réels. In B. Malgrange, editor, *Les Équations aux Dérivées Partielles (Paris, 1962)*, volume 117 of *Colloques Internationaux du CNRS*, pages 87–89. Éditions du Centre National de la Recherche Scientifique, Paris, 1963. Zbl 0234.57007.

[123] P.-A. Absil, R. Mahony, and B. Andrews. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization*, 16(2):531–547, 2005.

[124] Lancelot Da Costa, Thomas Parr, Biswa Sengupta, and Karl Friston. Neural dynamics under active inference: plausibility and efficiency of information processing. *Entropy*, 23(4):454, 2021.

[125] S. Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10:251–276, 1998.

[126] K. J. Friston, N. Trujillo-Barreto, and J. Daunizeau. DEM: A variational treatment of dynamic systems. *NeuroImage*, 41(3):849–885, 2008.

[127] K. Friston, L. Da Costa, N. Sajid, C. Heins, K. Ueltzhöffer, G. A. Pavliotis, and T. Parr. The free energy principle made simpler but not too simple. *Physics Reports*, 1024:1–29, 2023.

[128] Časlav Brukner. A no-go theorem for observer-independent facts. *Entropy*, 20(5):350, 2018.

[129] Morris H DeGroot. Reaching a consensus. *Journal of the American Statistical Association*, 69(345):118–121, 1974.

[130] Don N. Page and William K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27(12):2885–2892, 1983.

[131] Ernst Cassirer. *Substance and Function and Einstein's Theory of Relativity*. Open Court, Chicago, 1923. Translated by W. C. Swabey and M. C. Swabey; originally published as Substanzbegriff und Funktionsbegriff, 1910.

[132] J. Worrall. Structural realism: The best of both worlds? *Dialectica*, 43(1-2):99–124, 1989.

[133] M.J.D. Ramstead, M.D. Kirchhoff, and K.J. Friston. A tale of two densities: active inference is enactive inference. *Adaptive Behavior*, 28(4):225–239, 2020.

[134] Michael Friedman. *Dynamics of Reason*. CSLI Publications, Stanford, CA, 2001.

[135] Samuel L. Braunstein and Carlton M. Caves. Statistical distance and the geometry of quantum states. *Phys. Rev. Lett.*, 72(22):3439–3443, 1994.

[136] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, 2000.

[137] Gregory Bateson. *Steps to an Ecology of Mind*. Ballantine Books, New York, 1972.

[138] Ariel Caticha. Entropic dynamics, time and quantum theory. *J. Phys. A: Math. Theor.*, 44(22):225303, 2011.

[139] Alain Connes and Carlo Rovelli. Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories. *Classical and Quantum Gravity*, 11(12):2899–2917, 1994.

[140] S. Lloyd. Computational capacity of the universe. *Physical Review Letters*, 88:237901, 2002.

[141] C. Rovelli. *Quantum Gravity*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, Cambridge, 2004.

[142] Sosuke Ito. Stochastic thermodynamic interpretation of information geometry. *Phys. Rev. Lett.*, 121(3):030605, 2018.

[143] T.M. Cover and J.A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, 2nd edition, 2006.

[144] Miroslav Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2):298–305, 1973.

[145] S. W. Hawking, A. R. King, and P. J. McCarthy. A new topology for curved space-time which incorporates the causal, differential, and conformal structures. *J. Math. Phys.*, 17(2):174–181, 1976.

[146] David B. Malament. The class of continuous timelike curves determines the topology of spacetime. *J. Math. Phys.*, 18(7):1399–1404, 1977.

[147] Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Phys. Rev. Lett.*, 59(5):521–524, 1987.

[148] Hiroaki Matsueda. Emergent general relativity from Fisher information metric, 2013.

[149] Donald D. Hoffman. *The Case Against Reality: Why Evolution Hid the Truth from Our Eyes*. W. W. Norton & Company, New York, 2019.

[150] A. Clark. *Surfing Uncertainty: Prediction, Action, and the Embodied Mind*. Oxford University Press, Oxford, 2016.

[151] Anil Seth. *Being You: A New Science of Consciousness*. Faber & Faber, London, 2021.

[152] J. Greensite. Dynamical origin of the Lorentzian signature of spacetime. *Phys. Lett. B*, 300(1–2):34–37, 1993.

[153] Carlos Barceló, Stefano Liberati, and Matt Visser. Analogue gravity. *Living Rev. Relativ.*, 14(1):3, 2011.

[154] Elliott H. Lieb and Derek W. Robinson. The finite group velocity of quantum spin systems. *Commun. Math. Phys.*, 28(3):251–257, 1972.

[155] C. Marchioro, A. Pellegrinotti, M. Pulvirenti, and L. Triolo. Velocity of a perturbation in infinite lattice systems. *J. Stat. Phys.*, 19:499–510, 1978.

[156] Daniela Frauchiger and Renato Renner. Quantum theory cannot consistently describe the use of itself. *Nature Communications*, 9:3711, 2018.

[157] E. T. Jaynes. Information theory and statistical mechanics. *Phys. Rev.*, 106(4):620–630, 1957.

[158] S. Carlip. Challenges for emergent gravity. *Studies in History and Philosophy of Modern Physics*, 46:200–208, 2014.

[159] Robert J. Aumann. Agreeing to disagree. *Ann. Statist.*, 4(6):1236–1239, 1976.

[160] W. H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, 2003.

[161] A. Caticha. *Entropic Inference and the Foundations of Physics*. USP Press, São Paulo, Brazil, 2012. Monograph commissioned by the 11th Brazilian Meeting on Bayesian Statistics, EBEB-2012.

[162] D. T. Johnson and A. Caticha. Entropic dynamics and the quantum measurement problem. In *Bayesian Inference and Maximum Entropy Methods in Science and Engineering: 31st International Workshop*, volume 1443 of *AIP Conf. Proc.*, pages 104–111, 2012.

[163] Marcel Reginatto. Derivation of the equations of nonrelativistic quantum mechanics using the principle of minimum fisher information. *Physical Review A*, 58(3):1775–1778, 1998.

[164] B. Roy Frieden. *Physics from Fisher Information: A Unification*. Cambridge University Press, Cambridge, 1998.

[165] D. A. Lavis and R. F. Streater. Physics from Fisher information. *Stud. Hist. Phil. Mod. Phys.*, 33(2):327–343, 2002.

[166] Sydney Shoemaker. The inverted spectrum. *The Journal of Philosophy*, 79(7):357–381, 1982.

[167] D.J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2:200–219, 1995.

[168] David J. Chalmers. *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press, Oxford, 1996. May be relevant as foundational work, though earlier than 2013.

[169] Bertrand Russell. *The Analysis of Matter*. Kegan Paul, Trench, Trubner & Co., London, 1927. Reprinted 1954 by Dover Publications.

[170] Galen Strawson. Realistic monism: why physicalism entails panpsychism. In Anthony Freeman, editor, *Consciousness and its Place in Nature*, pages 3–31. Imprint Academic, Exeter, UK, 2006.

[171] Philip Goff. *Consciousness and Fundamental Reality*. Oxford University Press, Oxford, 2017.

[172] David J. Chalmers. Panpsychism and panprotopsychism. In Torin Alter and Yujin Nagasawa, editors, *Consciousness in the Physical World: Perspectives on Russellian Monism*, pages 246–276. Oxford University Press, Oxford, 2015. Originally circulated 2013; published 2015.

[173] G. Tononi. Consciousness as integrated information: a provisional manifesto. *Biological Bulletin*, 215:216–242, 2008.

[174] Giulio Tononi and Christof Koch. Consciousness: here, there and everywhere? *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1668):20140167, 2015.

[175] David J. Chalmers. The combination problem for panpsychism. In Godehard Brüntrup and Ludwig Jaskolla, editors, *Panpsychism: Contemporary Perspectives*, pages 179–214. Oxford University Press, Oxford, 2016.

[176] Giulio Tononi. An information integration theory of consciousness. *BMC Neuroscience*, 5(1):42, 2004.

[177] Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461, 2016.

[178] Robin L. Carhart-Harris, Robert Leech, Peter J. Hellyer, Murray Shanahan, Amanda Feilding, Enzo Tagliazucchi, Dante R. Chialvo, and David Nutt. The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs. *Frontiers in Human Neuroscience*, 8:20, 2014.

[179] Link R. Swanson. Unifying theories of psychedelic drug effects. *Frontiers in Pharmacology*, 9:172, 2018.

[180] R. L. Carhart-Harris and K. J. Friston. REBUS and the anarchic brain: toward a unified model of the brain action of psychedelics. *Pharmacological Reviews*, 71(3):316–344, 2019.

[181] Jakob Hohwy. *The predictive mind*. Oxford University Press, 2013.

[182] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, Oxford, 1980.

[183] Larry Laudan. *Progress and Its Problems: Towards a Theory of Scientific Growth*. University of California Press, Berkeley, CA, 1977. Alternative reference on pragmatic view of scientific progress.

[184] Bas C. van Fraassen. *The Empirical Stance*. Yale University Press, New Haven, 2002.

[185] Thomas S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, Chicago, 1962.

[186] Paul Hoyningen-Huene. *Reconstructing Scientific Revolutions: Thomas S. Kuhn's Philosophy of Science*. University of Chicago Press, Chicago, 1993. Originally published in German as *Die Wissenschaftsphilosophie Thomas S. Kuhns*, 1989.

[187] Willard Van Orman Quine. Two dogmas of empiricism. *The Philosophical Review*, 60(1):20–43, 1951.

[188] Willard Van Orman Quine. On empirically equivalent systems of the world. *Erkenntnis*, 9(3):313–328, 1975.

[189] Larry Laudan and Jarrett Leplin. Empirical equivalence and underdetermination. *The Journal of Philosophy*, 88(9):449–472, 1991.

[190] Helen E. Longino. *The Fate of Knowledge*. Princeton University Press, Princeton, NJ, 2002.

[191] E. P. Wigner. The unreasonable effectiveness of mathematics in the natural sciences. *Communications on Pure and Applied Mathematics*, 13(1):1–14, 1960.

[192] Thomas Nagel. *The View from Nowhere*. Oxford University Press, New York, 1986.

[193] Dénes Petz. Monotone metrics on matrix spaces. *Linear Algebra and its Applications*, 244:81–96, 1996.

[194] P.A. Absil, R. Mahony, and R. Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.

[195] N.J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, Philadelphia, 2nd edition, 2002.

[196] Kaare Brandt Petersen and Michael Syskind Pedersen. The matrix cookbook, 2012. Version 20121115, Technical University of Denmark.

[197] James S. Vandergraft. Spectral properties of matrices which have invariant cones. *SIAM J. Appl. Math.*, 16(6):1208–1222, 1968.

[198] Roger D. Nussbaum. Hilbert's projective metric and iterated nonlinear maps. *Memoirs of the American Mathematical Society*, 75(391), 1988.

[199] Jimmie D. Lawson and Yongdo Lim. The geometric mean, matrices, metrics, and more. *The American Mathematical Monthly*, 108(9):797–812, 2001.

[200] Franz Dietrich and Christian List. Probabilistic opinion pooling. In Alan Hájek and Christopher Hitchcock, editors, *The Oxford Handbook of Probability and Philosophy*. Oxford University Press, 2016.

[201] Robert F. Bordley. A multiplicative formula for aggregating probability assessments. *Management Science*, 28(10):1137–1148, 1982.

[202] C. Genest and J. V. Zidek. Combining probability distributions: A critique and an annotated bibliography. *Statistical Science*, 1(1):114–135, 1986.

[203] Shun-ichi Amari. α-divergence is unique, belonging to both f-divergence and Bregman divergence classes. *IEEE Trans. Inf. Theory*, 55(11):4925–4931, 2009.

[204] S.L. Lauritzen. *Graphical Models*. Oxford Statistical Science Series. Oxford University Press, 1996.

[205] M.J. Wainwright and M.I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008.

[206] C.K. Sønderby, T. Raiko, L. Maaløe, S.K. Sønderby, and O. Winther. Ladder variational autoencoders. In *Advances in Neural Information Processing Systems 29*, 2016.

[207] P. G. Bissiri, C. C. Holmes, and S. G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):1103–1130, 2016.

[208] Ryoichi Kikuchi. A theory of cooperative phenomena. *Phys. Rev.*, 81(6):988–1003, 1951.

[209] Jonathan S. Yedidia, William T. Freeman, and Yair Weiss. Constructing free-energy approximations and generalized belief propagation algorithms. *IEEE Transactions on Information Theory*, 51(7):2282–2312, 2005.

[210] L. D. Faddeev and V. N. Popov. Feynman diagrams for the Yang–Mills field. *Physics Letters B*, 25(1):29–30, 1967.

[211] Bryce S. DeWitt. Quantum theory of gravity. II. The manifestly covariant theory. *Phys. Rev.*, 162(5):1195–1239, 1967.

[212] L. F. Abbott. The background field method beyond one loop. *Nucl. Phys. B*, 185(1):189–203, 1981.

[213] Laurence F. Abbott. Introduction to the background field method. *Acta Physica Polonica B*, 13:33–50, 1982.

[214] John M. Cornwall, Roman Jackiw, and E. Tomboulis. Effective action for composite operators. *Physical Review D*, 10(8):2428–2445, 1974.

[215] R. Wong. *Asymptotic Approximations of Integrals*, volume 34 of *Classics in Applied Mathematics*. Society for Industrial and Applied Mathematics, 2001.

[216] R.A. Horn and C.R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.

[217] G. Schwarz. Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464, 1978.

[218] J. Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978.

[219] C. M. Bender and S. A. Orszag. *Advanced Mathematical Methods for Scientists and Engineers I: Asymptotic Methods and Perturbation Theory*. Springer, 1999.